

UÁKIRY REDD+ AUDD GROUPED PROJECT



Project title	Uákiry REDD+ AUDD Grouped Project
Project ID	4552
Crediting period	02-August-2021 to 01-August-2121
Project lifetime	02- August -2021 – 01- August -2061; 40-year lifetime for Project Activity Instances 02- August -2021 – 01- August -2121; 100-year total period for Grouped Project
(CCB) GHG accounting period	02- August -2021 – 01- August -2121; 100-year total period
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Version	Version 1.0
VCS Standard version	4.5
CCB Standards version	3.1
Project location	Brazil, Acre (AC) state, encompassing two municipalities: Sena Madureira and Rio Branco.
Project proponent(s)	BRCarbon Serviços Ambientais LTDA. CEO Mr. Marco Tuoto marco.tuoto@brcarbon.com.br Av. mtuoto@brcarbon.com.br Av. Cezira Giovanoni Moretti 655, sala 7, PwC AgTech Innovation, Reserva Jequitibá - Piracicaba, São Paulo +55 (19) 3424-3583
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1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or impact estimated by the end of project lifetime	Section reference
1) Digital inclusion: the project will install repeating towers to provide internet and an internet access point in schools. Thus, it is expected to reduce the community's vulnerability condition, favoring their access to information, educational processes, health and economics.	2.1.7
	2.1.8
	2.3.1
	2.3.8
	4.4.1
2) Technical assistance and rural extension: the project will provide to the beneficiary's instruction, knowledge, and inputs to make their areas more productive and sustainable, involving all the family members through courses, workshops, and lectures, it will work in different focus, as food security, income increasement, biodiversity, and improve on productivity.	2.1.18
	2.3.17
	2.3.18
	4.2.1
	4.4.1
3) Recognition and inclusion of women: the project will support the local women to create a group and strengthen their situation in the territory, aiming to improve their autonomy, production, rights, and other demands which can emerge.	2.1.18
	2.3.17
	2.3.18
	4.2.1
	4.4.1

1.2 Standardized Benefit Metrics

Category	Metric	Estimated by the end of project lifetime	Section reference
GHG emission reductions or carbon dioxide removals	Net estimated removals in the project area, measured against the without-project scenario	882,717.5 tCO ₂ e	2.1.13 3.2.4
Forest ¹ cover	For REDD ² projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario	85,739.82 hectares	2.1.1 2.1.13.6 2.1.16 3.1.4
Improved land management	Number of hectares of existing production forest land in which IFM ³ practices are expected to occur as a result of project activities, measured against the without-project scenario	Not applicable. This is a REDD+ project.	N/A
	Number of hectares of non-forest land in which improved land management practices are expected to occur as a result of project activities, measured against the without-project scenario	8,664.50 hectares	3.1.4.4
Training	Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities	200	2.1.18 2.3.17
	Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities	90	4.2.1 4.4.1

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (VCS Program Definitions)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (VCS Program Definitions)

³ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (VCS Program Definitions)

Category	Metric	Estimated by the end of project lifetime	Section reference
Employment	Total number of people expected to be employed in project activities ⁴ , expressed as number of full-time employees ⁵	20	2.1.18 2.3.18
	Number of women expected to be employed as a result of project activities, expressed as number of full-time employees	10	4.2.1 4.4.1
Livelihoods	Total number of people expected to have improved livelihoods ⁶ or income generated as a result of project activities	250	2.1.18 4.2.1
	Number of women expected to have improved livelihoods or income generated as a result of project activities	110	4.4.1
Health	Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	250	2.1.18 4.2.1
	Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	110	4.4.1
Education	Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	150	2.1.18 4.2.1
	Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	70	4.4.1
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water	250	2.1.18 4.2.1

⁴ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁵ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁶ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. The Sustainable Livelihood Approach to Poverty Reduction. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Estimated by the end of project lifetime	Section reference
	as a result of project activities, measured against the without-project scenario		4.4.1
	Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	110	
Well-being	Total number of community members whose well-being ⁷ is expected to improve as a result of project activities	250	2.1.18
	Number of women whose well-being is expected to improve as a result of project activities	110	4.2.1 4.4.1
Biodiversity conservation	Expected change in the number of hectares managed significantly better by the project for biodiversity conservation ⁸ , measured against the without-project scenario	Not applicable	
	Expected number of globally Critically Endangered or Endangered species ⁹ benefiting from reduced threats as a result of project activities ¹⁰ , measured against the without-project scenario	1	5.1.1

⁷ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

⁸ Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

⁹ Per IUCN's Red List of Threatened Species

¹⁰ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 PROJECT DETAILS

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

The **Uákiry REDD+ Grouped Project** (hereafter called GPD), aims to avoid unplanned deforestation in private properties at Acre State. Recent data available in PrevisIA platform¹¹, accomplished by the Instituto do Homem e Meio Ambiente da Amazônia (Imazon) reveals that state has the fourth highest index of area under risk of deforestation in 2023¹² in the Legal Amazon, with 1,269.34 km², behind only of the three states with the largest territory, Pará, Amazonas and Mato Grosso.

The area of deforestation involving three states in the north of Brazil has increased significantly in recent years, where before it was well preserved, the region known by the acronym AMACRO (where the states of Amazonas, Acre and Rondônia are located) has literally become no man's land. The lack of control caused cases of land grabbing and other crimes linked to the expansion of the agricultural frontier to explode. The region came to be known as the new Deforestation Frontier.

Acre's state covers approximately 164.000 km², which accounts for about 1,92% of the Brazilian territory, and about 4,26% of its northern region. Its territory borders the states of Amazonas and Rondônia, to the east. It also borders two countries: Peru, across the western portion, and Bolivia, to the southeast. At last, two years the deforestation rates were greater than 800 km² per year, according to PRODES data¹³.

Initially the first four project activity instance is composed by four properties with 85,739.82 ha of forest cover in project area, located in the Iaco River margins, one of Purus River affluents, in the Amazon basin. The occupation of the region mainly occurred between 1860 to 1915 at the first rubber cycle, through the expeditions guided by Manoel Urbano and João Gabriel de Melo, when people from northeast of Brazil migrated to there. The rubber crisis caused the loss of the centrality of cutting the syringe in the daily life of the placements. In the article "The contingent power of the Iaco River in the Federal Territory of Acre", Vital (2019) cites that in 1915 occurred a historical flood that deepen the rubber crisis in the region, with the abandon of rubbers tappers trails.

Through these projects, BrCarbon (hereafter known as BRC) and Yaco2 Gestão e Comércio de Carbono (Yaco2) seeks to promote integral forest conservation actions generating co-benefits not

¹¹ <https://previsia.org.br/>

¹² <https://g1.globo.com/ac/acre/noticia/2023/03/22/amazonia-legal-desmatamento-no-acre-mais-que-dobra-em-fevereiro-e-atinge-sete-quilometros-quadrados.ghtml>

¹³ http://terrabrasilis.dpi.inpe.br/app/dashboard/deforestation/biomes/legal_amazon/rates

only for the climate (reduction and removal of emissions), but also communities and local biodiversity. At the validation, the climate benefits are directed related with the emission reduction of 147,028 tCO₂e in average per year.

By joining BRC forest program, landowners can access funds from the carbon voluntary market to develop conservation and monitoring efforts in their forests. A partnership between BRC and landowners will result in the legal protection of forests, forest monitoring using satellite images, biodiversity survey (including fauna e flora species), and forest biomass inventory in a broad scale, using an upscaling approach based on field collection, drone-borne and satellite sensors such as LiDAR (light detection and ranging). In addition, PP will involve property surveillance, wildfire monitoring, fire prevention and fighting, threatened species protection and long-term social activities with local communities.

The project's name was inspired in the historic versions of the Acre's name, that arose even before became a state. The first version is that would be a derivation of Uwa'kürü, Uákiry, Aquiry, which means "river of alligators" in the native language of the Apurinã Indians, the original inhabitants of the region.

2.1.2 Audit History (VCS, 4.1)

The following table presents the audit chronogram for this project, from the Project Start Date to 1st Monitoring Period.

Table 1 – Audit chronogram of Uákiry REDD+ AUDD Grouped Project.

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	18-August-2023	VCS / CCB	EcoLance Private Limited	Project Start Date
1st Monitoring	02-August-2021 to 01- August-2023	VCS / CCB	EcoLance Private Limited	2+

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

Sectoral scope	14: Agriculture, forestry, and other land use
AFOLU project category ¹⁴	Reduced Emissions from Deforestation and Degradation (REDD)
Project activity type	Avoiding Unplanned Deforestation (AUD)

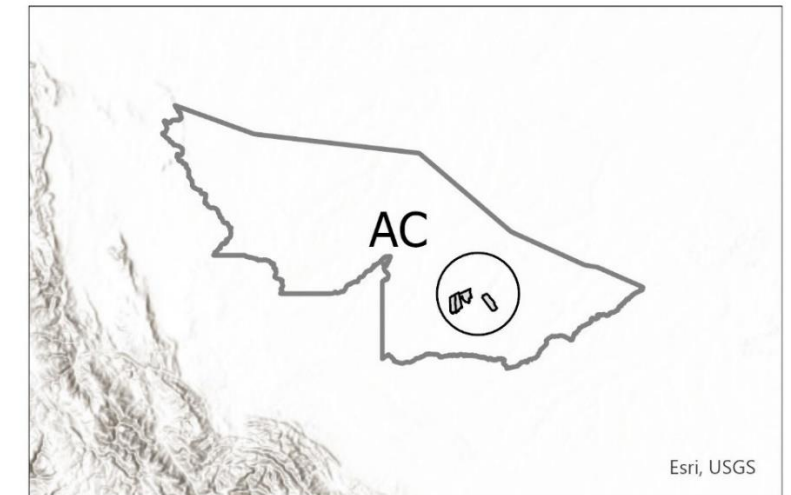
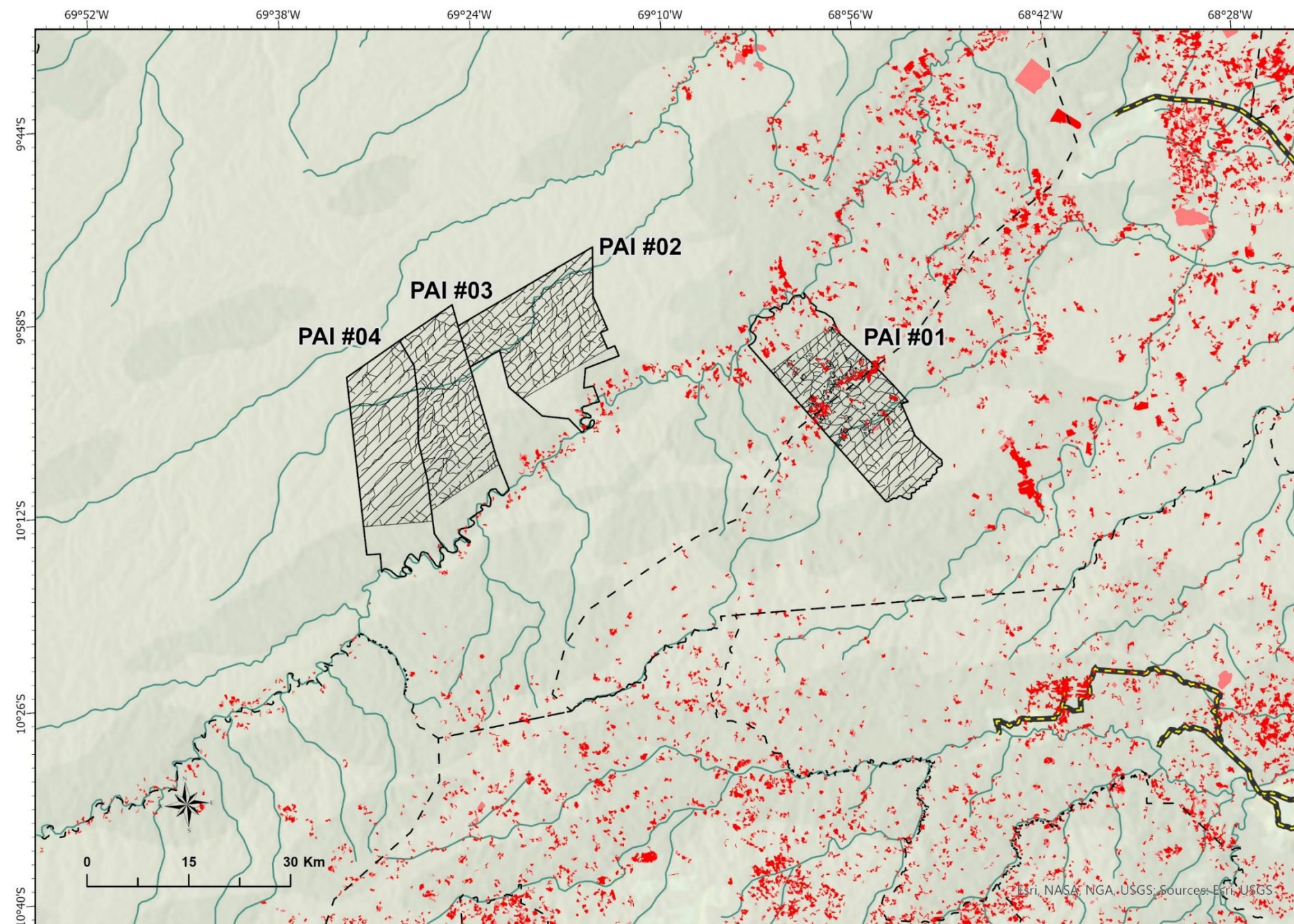
Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

According to VCS v4.5, REDD is an eligible AFOLU project category: “Eligible REDD activities are those that reduce net GHG emissions by reducing deforestation and/or degradation of forests. Deforestation is the direct, human-induced conversion of forest land to non-forest land. Degradation is the persistent reduction of canopy cover and/or carbon stocks in a forest due to human activities such as animal grazing, fuelwood extraction, timber removal or other such activities, but which does not result in the conversion of forest to non-forest land (which would be classified as deforestation), and qualifies as forests remaining as forests, such as set out under the IPCC 2003 Good Practice Guidance. The project area shall meet an internationally accepted definition of forest, such as those based on UNFCCC host-country thresholds or FAO definitions and shall qualify as forest for a minimum of 10 years before the project start date. The definition of forest may include mature forests, secondary forests, and degraded forests. Under the VCS Program, secondary forests are forests that have been cleared and have recovered naturally and that are at least 10 years old and meet the lower bound of the forest threshold parameters at the start of the project. Forested wetlands, such as floodplain forests, peatland forests and mangrove forests, are also eligible provided they meet the forest definition requirements mentioned above.”

The VCS v4.5 also states that: “Eligible REDD activities include Avoiding Unplanned Deforestation and/or Degradation (AUDD): This category includes activities that reduce net GHG emissions by stopping deforestation and/or degradation of degraded to mature forests that would have occurred in any forest configuration.”

This GPD meets all VCS v4.5 eligibility criteria, since the project area was defined in accordance with the forest benchmark map (Figure 1), considering the accumulated deforestation for 10-year period before project start (historical series from 2011 to 2021), in private areas in accordance with the law in force in Brazil. Section 3.4 provides additional information on the topic. Please refer to section 2.1.5.1 below for the eligibility criteria for the inclusion of new project activity instances.

¹⁴ See Appendix 1 of the VCS Standard



Deforestation (2011 - 2021)

- Highway
- Hydrography
- Degradation
- Deforestation
- Project Areas GPD #PAI
- Rural Properties
- Municipalities
- Acre State Boundaries (AC)

GEOGRAPHIC COORDINATE SYSTEM (GCS)
DATUM: SIRGAS 2000
(EPSG:4674)



Figure 1 – Accumulated deforestation from 2011 to 2021.

2.1.4 Transfer Project Eligibility (VCS, 3.23, Appendix 2)

Not applicable. This project is not a transfer project and/or CPAs seeking registration.

2.1.5 Project Design (VCS, 3.6)

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

2.1.5.1 Eligibility Criteria for Grouped Projects (VCS, 3.6; CCB, G1.14)

This project is designed as a Grouped Project, that allows the expansion of the project scope by inclusion of new project activity instances (PAIs) after its validation. The inclusion of new PAIs takes place at the time of each verification event. A set of eligibility criteria for the inclusion of new PAIs is described in the section below.

According to VCS v4.5, grouped projects shall meet the following criteria.

“3.6.8 The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other).”

In compliance with 3.6.8, it is incumbent upon the project proponent to amalgamate all instances of the project activity situated inside Project Zone under the same project proponent into a unified project.

“3.6.9 Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit (...).

Criteria 3.6.9 is not applicable. There is no capacity limit applicable to the project activity type (APD) considered in the grouped project, since the BRC team will be present in the Project Location and can add as many human resources as necessary to fill the area.

“3.6.10 Grouped projects shall specify one or more clearly defined geographic areas within which project activity instances may be developed. Such geographic areas shall be specified using geodetic polygons

as set out in Section 3.11 below. Geographic areas with no initial project activity instances shall not be included in the project unless it can be demonstrated that the same (or at least as conservative) baseline scenario and rationale for the demonstration of additionality is applicable to such an area as a geographic area that does include initial project activity instances.”

The GPD assumes only one wide geographic area known as the Acre state (164,173.429 km²). The project activity instances might encompass any private properties legally constituted, with forest cover on their Legal Reserve. The GPD can include any private properties with forest cover in Amazon biome or in Cerrado biome, considering all forest types as per Brazilian forest definition, including three categories of Cerrado (“Savana Florestada” and “Savana Arborizada”).

“3.6.11 Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances. The initial project activity instances are those that are included in the project description at validation and shall include all project activity instances currently implemented on the issue date of the project description. The initial project activity instances may also include any instances of the project activity that have been planned and developed to a sufficient level of detail to enable their assessment at validation.”

Regarding baseline and additionality, it is assumed that all private properties in this geographic area are subject to the same legal framework and similar deforestation agents and drivers as those identified for PAIs #1-4. All landowners within the Legal Amazon have the duty to protect and conserve their Legal Reserves (80% of property). In this context business-as-usual is livestock, sometimes followed by soybean, cotton or corn production. Specific assessment on baseline scenario and additionality will be done for each new activity instance included in the project scope after its validation.

“3.6.12 As with non-grouped projects, grouped projects may incorporate multiple project activities (see Section 3.6.1 – 3.6.3 for more information on multiple project activities). Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area.”

The only project activity considered in the project scope for which carbon credits generation is attributable and that is related to the methodology adopted in the project design in all project activity instances is avoiding unplanned deforestation.

To avoid unplanned deforestation BRC will sign long-term conservation agreements with landowners for new instances, like have signed long-term conservation agreements with landowners of actual instances. Complementary activities related to leakage mitigation, social and biodiversity related activities and monitoring activities are described in section 3, 4 and 5.

“3.6.13 The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project. Where a single baseline scenario cannot be determined for a project activity over the entirety of a geographic area, the geographic area shall be redefined or divided such that a single baseline scenario can be determined for the revised geographic area or areas.”

Only one wide geographic area is considered for this grouped project once the baseline scenario for avoided unplanned deforestation activity in Legal Amazon is the same, and all private properties are subjected to the same regulatory framework. Please refer to sections 2.1.16.

“3.6.14 The additionality of the initial project activity instances shall be demonstrated for each designated geographic area, in accordance with the methodology applied to the project. Where the additionality of the initial project activity instances within a particular geographic area cannot be demonstrated for the entirety of that geographic area, the geographic area shall be redefined or divided such that the additionality of the instances occurring in the revised geographic area or areas can be demonstrated.”

Only one geographic area is considered for this grouped project, as presented in section 2.1.16 – Project Zone Map and Project Location. Thus, the additionality approach for avoided unplanned deforestation activity is the same for all private properties encompassed by the grouped project once they are subjected to similar drivers and agents of illegal deforestation.

“3.6.15 Where factors relevant to the determination of the baseline scenario or demonstration of additionality require assessment across a given area, the area shall be, at a minimum, the grouped project geographic area. Examples of such factors include, inter alia, common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; determination of regional grid emission factors; and historical deforestation and degradation rates.”

The relevant factors for the baseline scenario determination and demonstration of additionality of a given project activity instance are the same for the entire geographic area. Any legally constituted private properties within Amazon biome, with more than 80% of forest cover (or 35% for Cerrado) is eligible for this grouped project.

“3.6.16: Grouped projects shall include one or more sets of eligibility criteria for the inclusion of new project activity instances. At least one set of eligibility criteria for the inclusion of new project activity instances shall be provided for each combination of project activity and geographic area specified in the project description. A set of eligibility criteria shall ensure that new project activity instances”:

Meet the applicability conditions set out in the methodology applied to the project: All PAIs under this Grouped Project must meet the applicability conditions set out in the VM0007 REDD+ Methodology Framework (REDD+MF), v1.7. Regarding the 35 initial project activity instances, the applicability conditions are met as demonstrated:

- Where the land use transition in the baseline scenario is forest land to non-forest land, meeting the definition of unplanned deforestation.
- Where the project involves activities aimed at avoiding UDef.
- Where agents of deforestation in the baseline scenario clear the land for tree harvesting, settlements, roads, settlements, unsanctioned expansion of roads and other infrastructure, agricultural crop production, ranching, or aquaculture.

Use the technologies or measures specified in the project description: sections 2.1.17 and 3.3 defines the technologies and measures available. The GHG emissions reductions will be caused by the signing of long-term forest conservation agreements with landowners in all PAIs. Complementary activities related to the reduction of unplanned deforestation, leakage mitigation, social and biodiversity related activities and monitoring activities are also described in the final Project Description and report.

Apply the technologies or measures in the same manner as specified in the project description: The current and future PAIs must apply the same technologies and measures specified in this document. Small adjustments area allowed to accommodate PAI specificities. Any adjustment will be reported, described, and must not overestimate the project climate benefits.

Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area: The grouped project assumes the Project Zone Map as a unique geographic area, taking into consideration that all properties within this biome are subject to the same legal framework. New PAIs must follow the same baseline approach.

Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area: Considering that the grouped project refers to avoidance unplanned deforestation and/or degradation (AUDD) in areas where the forest suppression is not permitted by law, all the subsequent project activity instances submitted under this GPD, must follow the same additionality approach for PAIs #1-35. It means that the plausible baseline scenarios, that will be prepared and provided by VERRA, will not differ from the three scenarios identified in the additionality analysis, as follow:

1. Forest cover maintenance in the legal reserve (80% of the property), i.e.: through conservation activities resulting from incentives other than the REDD project.

2. Legal deforestation of 20% of the property (deforestation with permits), where the land use conversion is allowed by law but is not legally authorized and documented. i.e.: pasture (cattle raising) and agriculture.
3. Illegal deforestation in the property (deforestation without permits), for pasture (cattle raising) and agriculture purposes or simply for real estate speculation.

In the case of the first thirty-five project instances, implemented only on the legal reserve of the properties, only scenario 1 is in accordance with current laws and regulations. Notwithstanding, in other cases, the entire property area can be accounted for the AUDD, including the area subject to legal suppression (20%), when the landowner has given up the legal right of vegetation suppression, in this situation scenario 1 and 2 will be in accordance with applicable laws and regulations.

2.1.6 Project Proponent (VCS, 3.7; CCB, G1.1)

This project has multiple proponents, as presented below.

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Organization name	Yaco2 Gestão e Comércio de Carbono
Contact person	Mr. Rogério Oliveira
Title	Technical Director
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Telephone	+55 (69) 9969 1707
Email	rogerio@yaco2.net

2.1.7 Other Entities Involved in the Project

Not applicable. There are no other entities involved in the project, besides the presented project owners and landowners. BRCarbon is the proponent of the project and is responsible for carrying out the project's activities. The project is being carried out on the private property of its partners. As such, they are the only two parties involved in the project, and there are no other entities directly involved in the project.

2.1.8 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

According to the VCS definitions, Uákiry REDD+ AUDD Grouped Project ownership is defined by item 6, section 3.7.1 of the VCS standard v.4.5, as follows:

“An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent.”

As shown before, the project ownership is held by the land ownership documents and by the agreements between BRC, YACO2 and landowners, as presented on section “2.1.7 – Project Proponent” and “2.1.8 – Other Entities Involved in the Project”.

2.1.9 Project Start Date (VCS, 3.8)

Table 2. Project Start Date

Project start date	02-August-2021
Justification	The first conservation agreement between BRCarbon and Landowner was celebrated on 02-August-2021, within the scope of the Avoided Planned Deforestation (APD) for project registered under Verra's ID number 2551. This date can be understood as a start date, as it involved the implementation of management activities across the property boundaries, such as protection plan actions such as command control actions and mapping of matrices of non-timber forest products were conducted in this context, this information can be accessed in ANNEX_START-DATE.

2.1.10 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Table 3. Benefits Assessment and Project Crediting Period

Crediting period	The GPD will generate GHG emission reductions eligible for issuance as VCUs for 100 years. 02 nd august 2021 and ends on 01 st August 2121. This long-term approach is considered due to the group project, that allows the inclusion of new PAIs in the project scope over the project lifetime. PAIs #1-4 will generate GHG emission reductions eligible for issuance as VCUs for 40 years (from 02 nd august 2021 until 01 st August 2061).
Start date of first or fixed crediting period	02-August-2021 to 01-August-2121
CCB benefits assessment period	This document is the project description to be validated. The next verification period expected to be between August 2nd, 2021, and August 2nd, 2023.

2.1.11 Differences in Assessment/Project Crediting Periods (CCB, G1.9)

The GPD start date is when the activities that led to the generation of GHG emission reductions of the first four (4) project activity instances (PAIs) took place. Thus, each new PAI will have its own start date.

The project crediting period is when GHG emission reductions by the project activity instances are eligible for issuance as VCUs and will typically match the first baseline period, provided by VERRA. Whenever necessary, the PP will renew the baseline, in accordance with VERRA, every 6 years to ensure that changes on the project's baseline scenario and/or regulatory surplus are taken into consideration throughout the project lifetime.

The 'project lifetime' is defined as the period when PP implements the project activities. The project lifetime starts when a given activity instance starts to generate climate, community or biodiversity impacts and will have a longevity of 40 years, as stated in the contract signed between the landowners and BRCARBON Serviços Ambientais LTDA.

2.1.12 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

The Uákiry REDD+ AUDD GPD is categorized as a Large Project considering the proposed group design and its larger objectives. Even if the GHG emission reductions or removals do not reach a Large Project amount (300,000+ tons of CO₂e per year) on this first approach, it is expected to achieve this GHG emission reductions or removals amount on first monitoring period, when new project instances should be added to project.

During the project, BRC will develop efforts to increase the protected forest area and increase project instances numbers.

- ☐ < 300,000 tCO₂e/year (project)
- ☒ ≥ 300,000 tCO₂e/year (large project)

Table 4. Estimated Reductions on first crediting period¹⁵.

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
02-August-2021 to 01-August-2121	882,171.5
Total estimated ERRs during the first or fixed crediting period	882,171.5
Total number of years	6
Average annual ERRs	147,028.58

2.1.13 Physical Parameters (CCB, G1.3)

The Uákiry REDD+ AUDD grouped project has four project activity instances located southeast of Acre State, more specifically between Sena Madureira and Rio Branco municipalities, and between the Purus and Baixo Acre regional division.

The table below lists the first four properties included in this grouped project and a centroid geographical coordinate by each project activity instance (Table 5). Coordinates are also submitted separately as a KML file (GIS database).

Table 5 – Project Instance centroid coordinates and municipality.

PAI #number	Property Name	Municipality / UF	Geodetic coordinates	
			X (Long)	Y (Lat)
PAI #01	Seringal Porongaba	Sena Madureira e Rio Branco/AC	-10.274	-68.801
PAI #02	Seringal Palmares	Sena Madureira/AC	-10.017	-69.259
PAI #03	Seringal Potiguar	Sena Madureira/AC	-10.109	-69.461
PAI #04	Seringal Katianã	Sena Madureira/AC	-10.136	-69.532

¹⁵ The estimated reduction and average are based on the baseline deforestation expected to occur in the reference region during the initial six-year baseline validity period

The project zone, which is defined by the boundaries of the state (Figure below), is in the expansion frontier of deforestation, between the new frontier and intermediate. The state is formed by two hydrographic basins, Juruá in the north and Acre/ Purus Valley in the south, which encompass the two existing mesoregion.

Next, to describe the physical parameters of the project, the location of the first instances of activities was considered, and the characterization of the region from the political limits of the municipalities in which they are located. The figure below illustrates the spatial boundaries of the project and the initial instances at the time of validation, that are in the Purus basin, on the banks of Iaco's river.

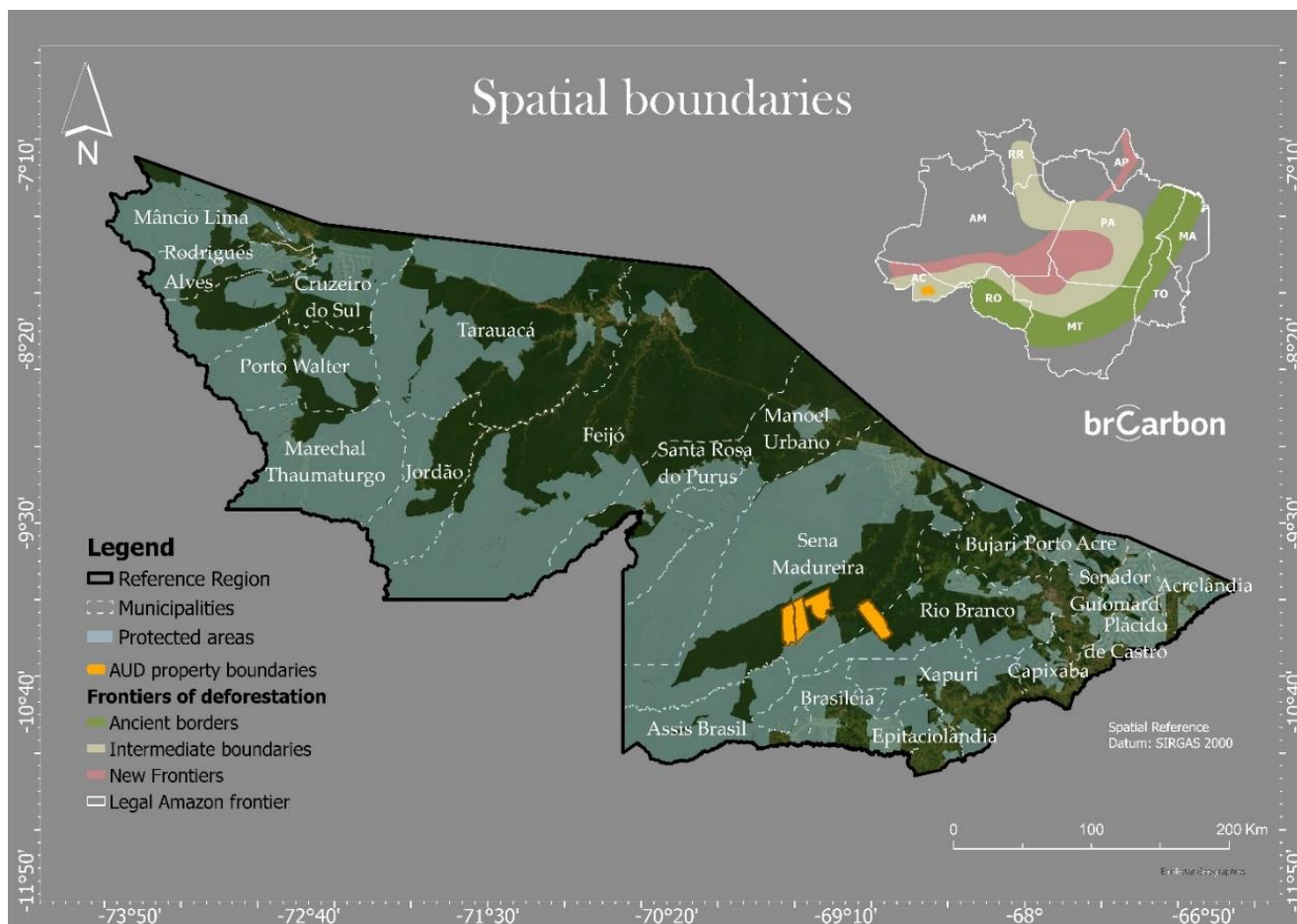


Figure 2 – Spatial Boundaries

2.1.13.1 Climate

Based on spatial data from global climate mapping at a spatial resolution of 0.5° , two climate types were identified in the project areas. In a more significant portion, the Am climate was identified and in a more punctual way the Af. The average precipitation in Rio Branco is 2022 mm per year while in Sena Madureira municipality it varies between 1500 to 1700 mm.

The humid tropical climate is a type of climate that corresponds to the "Am" category of the Köppen-Geiger climate classification. It has average monthly temperatures above 18 ° C in all months of the year and is considered a type of intermediate climate that is between the types Af (equatorial climate) and Aw (tropical savanna climate).

The tropical climate is usually found along the equator. Regions with this climate typically characterize tropical forests and are denoted by the Af group in the Köppen-Geiger climate classification. The equatorial climate is typically hot and humid. Places with this climate are typically located at a latitude of 10° of the Equator.

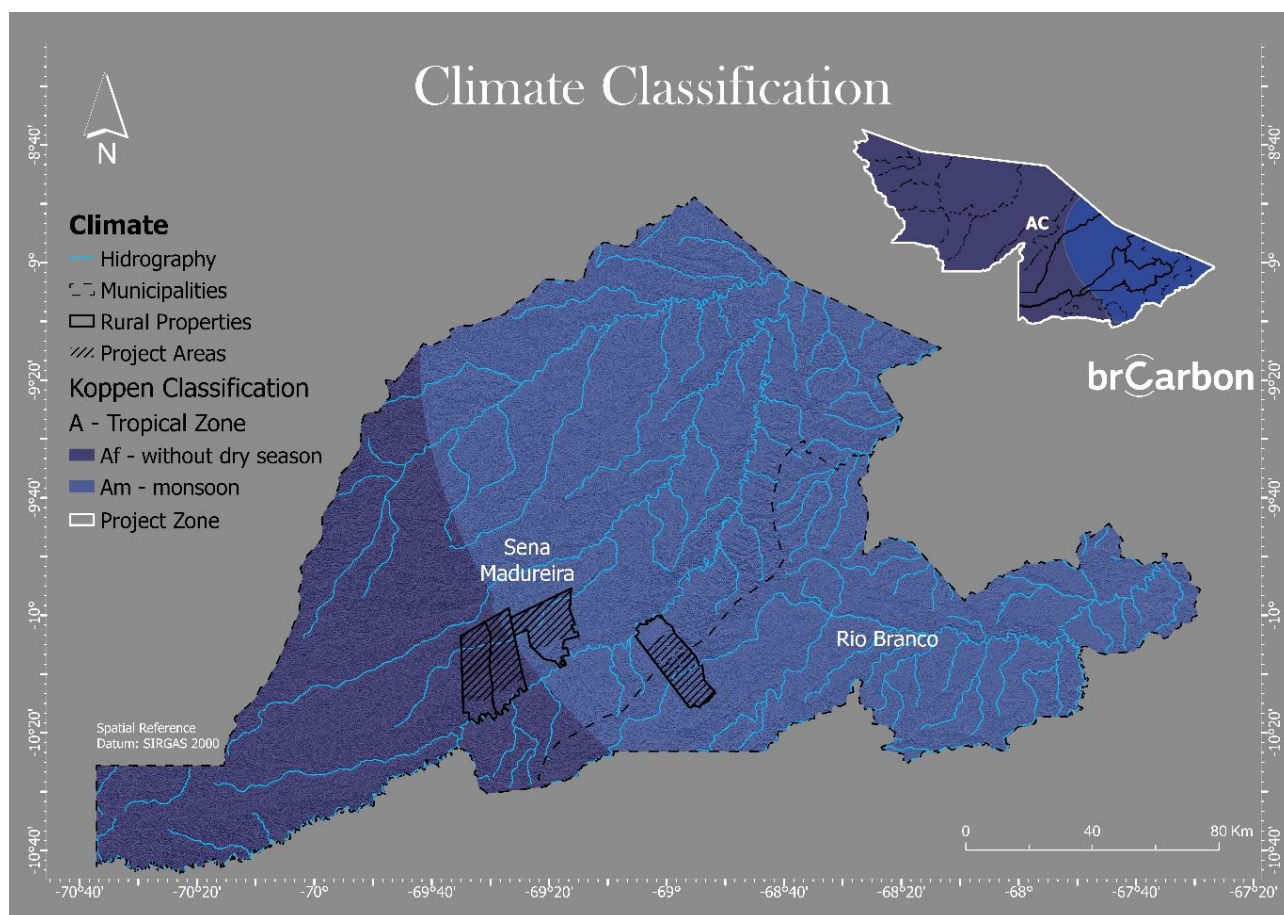


Figure 3 - Climate classification map.

2.1.13.2 Hidrology

Acre's hydrographic network plays an important role in transport, being divided between the Purus and Juruá basins. Rivers are an important means of transport for both people and goods. In isolated locations, the waterway transfer is the only way of access. The rivers with navigable sections of Acre are: Juruá, Tarauacá, Envira, Purus, Iaco and Acre, were identified.

The drainage network with the main bodies of water that make up the Project areas can be seen in the image below. The main basin found in the Project areas is the Purus basin. Acre state is subdivided by the Purus and Juruá basins.

The project areas are in the Purus basin, the main ones being the Purus, Iaco and Acre rivers. These rivers have an extreme flood and ebb regime, directly reflecting the regional rainfall regime. For example, the Iaco River in the city of Sena Madureira presents a cross-section in the rainy season equivalent to 50 times its value in the ebb, which occurs in the dry season. All these rivers are navigable, allowing the traffic of vessels at the time of the floods (December-April). The project areas are drained by three hydrographic networks: the Iaco River, Espalha River and the Macauã River, all contributors to the Iaco River, one of the main tributaries of the Purus River- drains along the right bank. Many small and intermittent igarapés located in the project area dry completely during the dry season and are extremely muddy at the beginning of the rainy season, indicating the low permeability of these soils and the high rate of erosion within the area of this basin.

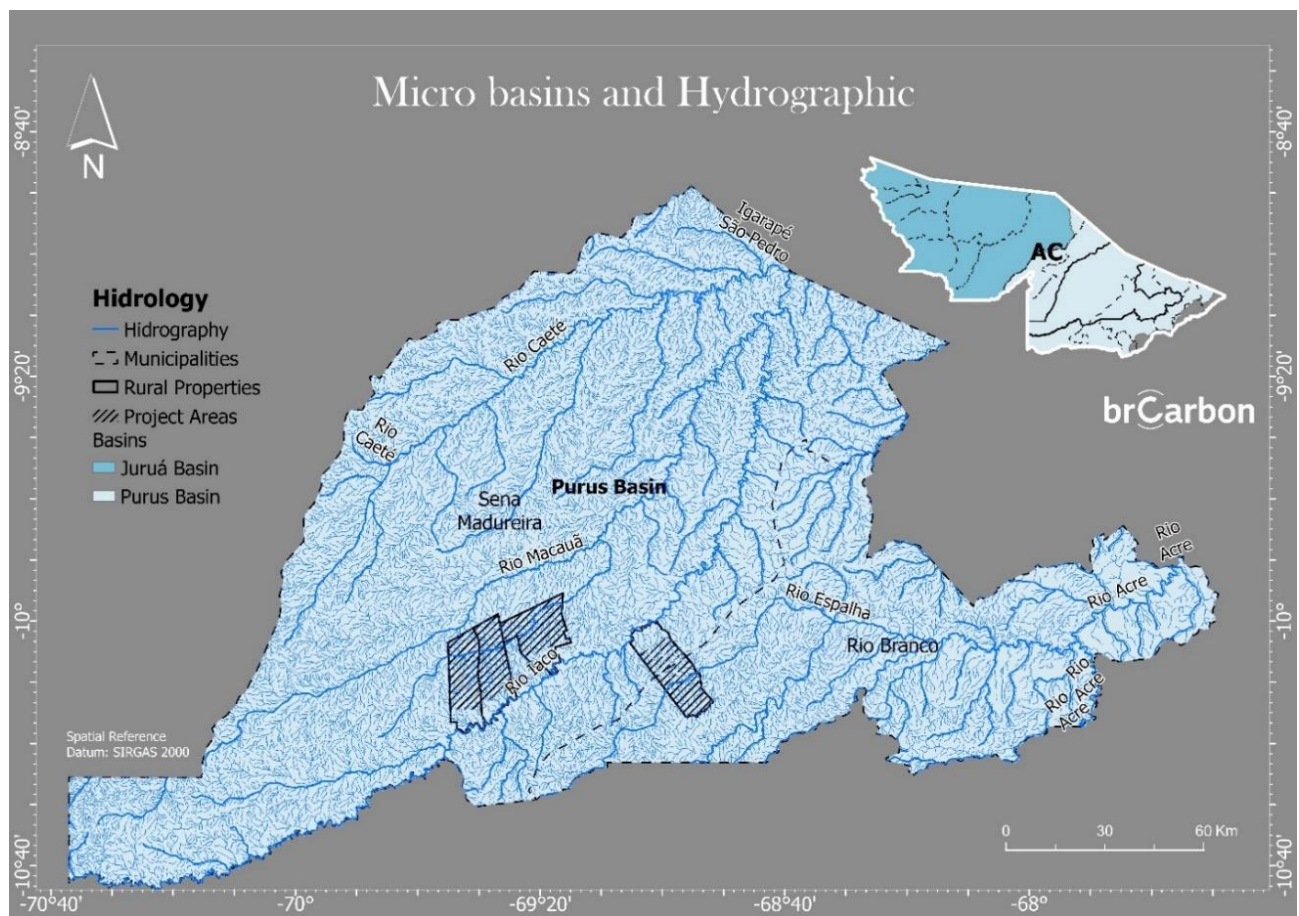


Figure 4 – Hydrography and basins at project region.

2.1.13.3 Soil

Three types of soils were identified in the project areas: Gleysols, Red Yellow Argisols and Luvisols, were identified (Figure 5). The soils were identified from thematic vectorial pedology base in the scale of 1:250,000 developed by IBGE (2018), generated from the refinement of the data of the RADAM Brazil Project (1979).

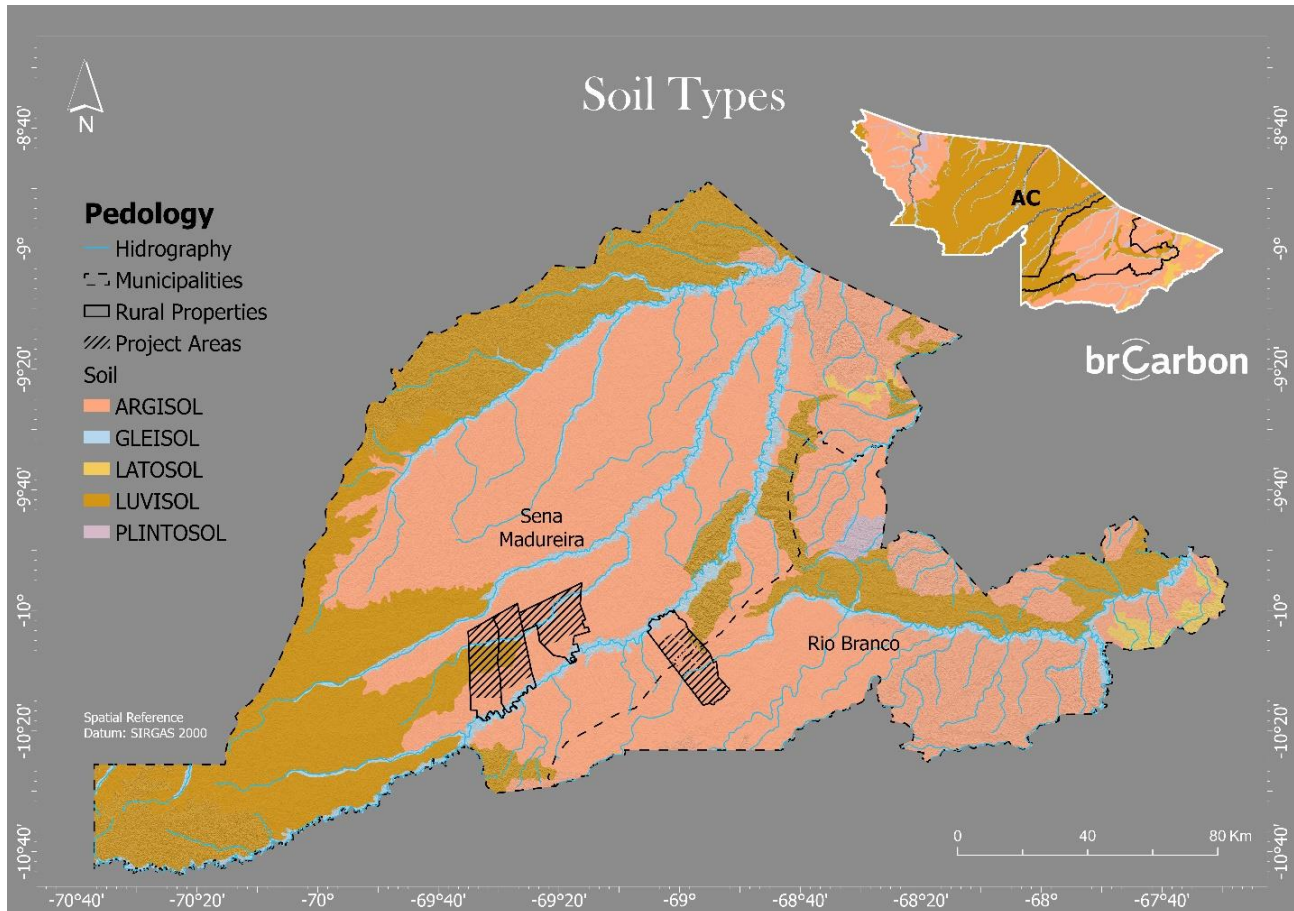


Figure 5 – Soil types at project activity instances

The Gleysols are found near the banks of rivers and subject to seasonal flooding of these areas following the regime of flood and drought of the Amazonian rivers, occurring in smaller numbers in the project areas.

The red- yellow Argisols occur in all instances of the project.it represents most of the soils found in the project areas, especially in the PAI#01and #02 area. Argisols is the most common soil order in Brazil and is widely used by agriculture, both for the practice of livestock, as well as agriculture due to its clay content that favors the retention of water and organic matter in the soil among other desirable characteristics for plant development. They are mineral soils, non-hydromorphic, with horizon A or E (horizon of loss of clay,

iron or organic matter, light color) followed by textural B horizon, with clear difference between horizons. Luvisols are found in the project areas, except at PAI#02.

They are little developed and have characteristics of the original material (rock) evidenced by the presence of primary minerals. They range from shallow to deep soils, typically low permeability.

The tree types of soils present in the Uákiry REDD+ GDP PAIs have in common an aspect related to the high concentration of clay in their composition and none of the classes identified has restrictions for pasture formation, are soils with agricultural aptitude and with support conditions in the project area for livestock activity. For example, in the vicinity of PAIs #1-4 areas, in the floodplains of the Iaco River, livestock farming is a common practice and occurs in the stains of the three different soils identified in the project area.

2.1.13.4 Topography

The state of Acre is in the extreme west of Brazil, in an area of transition between the Andes and the Amazonian lowlands. The southwestern Amazon is characterized by the presence of muddy and meandering rivers, with unstable courses, whose uneven topography reveals hills, rocks, waterfalls, and small mountains, due to the proximity to the Andes, as can be seen in the figure below.

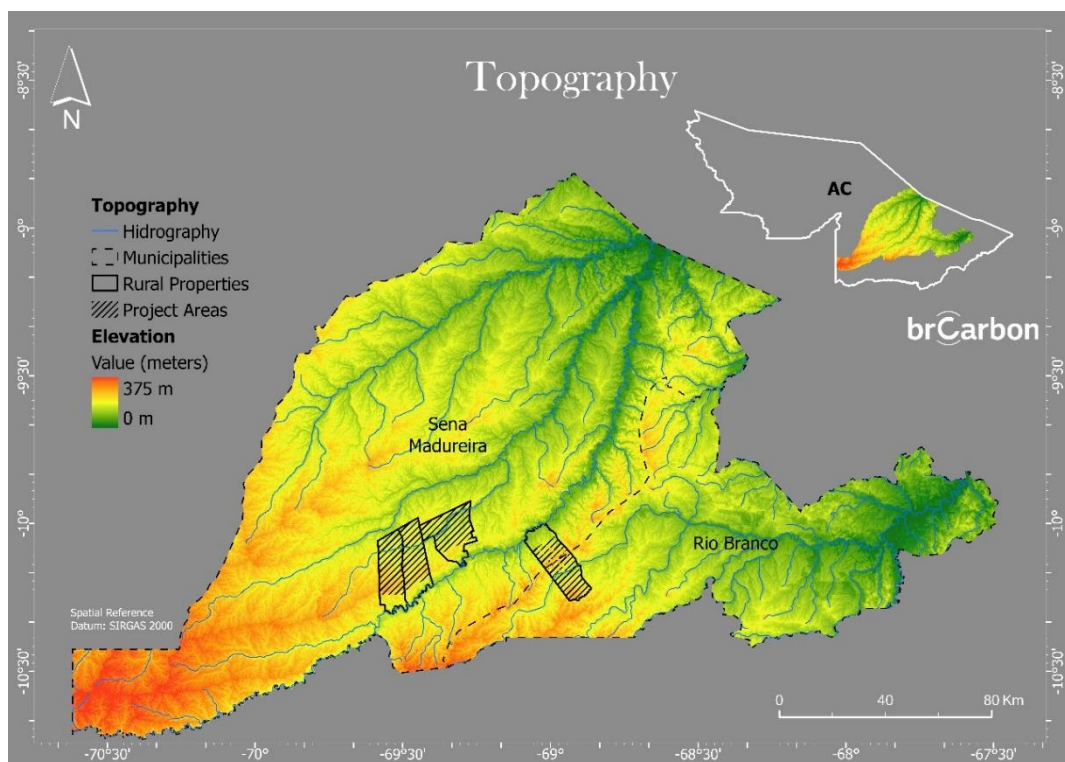


Figure 6 – Topography at project activity instances.

Topography data came from INPE's TOPODATA Program. The INPE data were generated in a resolution of 30m for the whole of Brazil from SRTM mission – Shuttle Radar Telematic Mapper data, with spatial resolution of 90m. Altitude data were used in relation to sea level, slope % and the land classification shape proposed by EMBRAPA for the entire national territory.

2.1.13.5 Slope

Slope data also came from INPE's TOPODATA Program. These data were generated from a 30m resolution for all of Brazil from data from the SRTM mission – Shuttle Radar Telematic Mapper. With a spatial resolution of 90m. This slope was reclassified as proposed by EMBRAPA for the entire national territory. Within the project areas it is possible to notice the greater volume of soft wavy to wavy frequency. Close to rivers there is a greater volume of flat to smooth wavy relief.

The relief on the first four PAIs varies from wavy to smooth wavy, with small patches of wavy relief according to EMBRAPA classification (1989), were identified that the average slope in project areas is 10,06%, ranging from 0.2% to 33.3% (figure below)

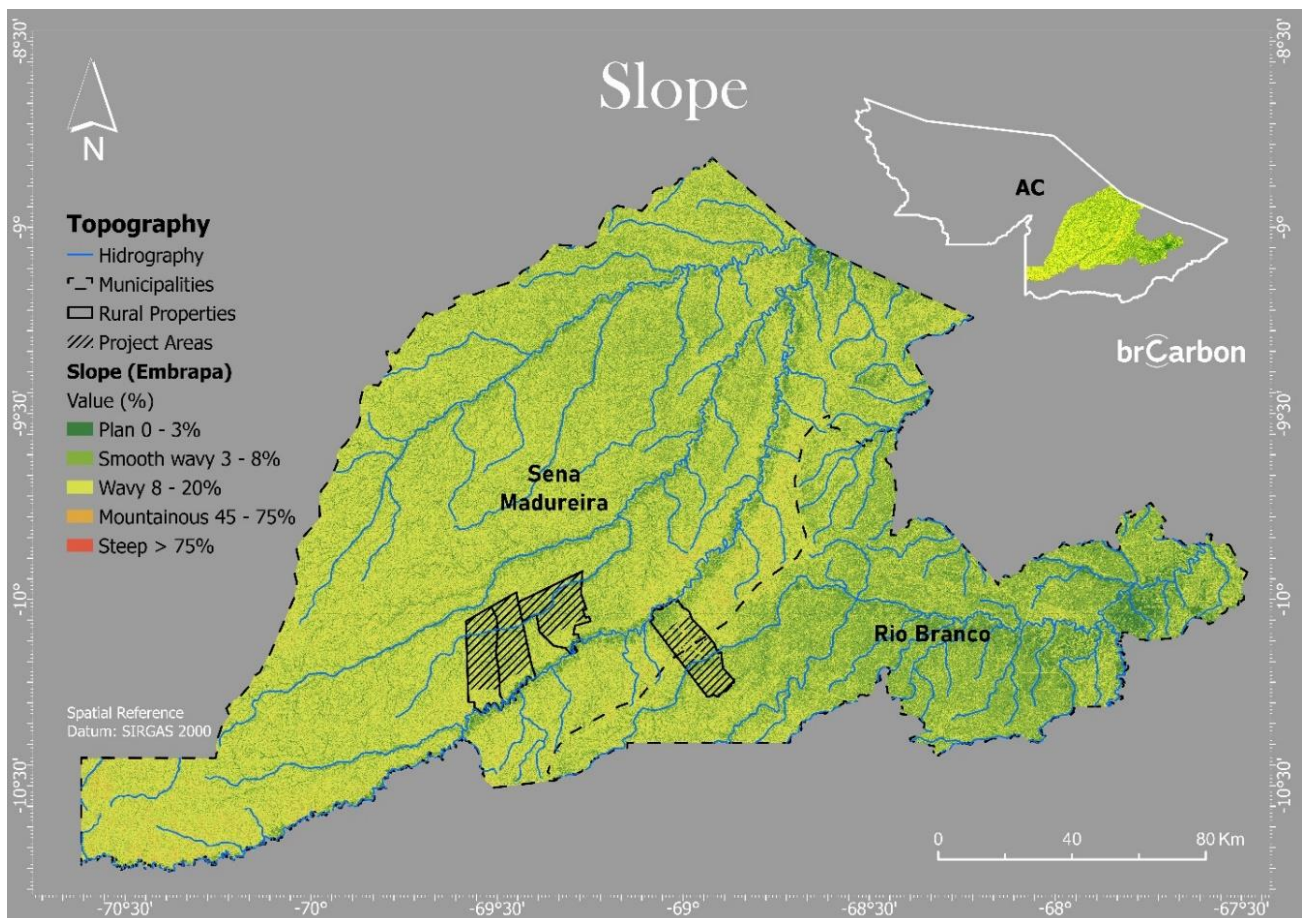


Figure 7 – Slope types at project activity instances.

2.1.13.6 Vegetation

The Uákiry REDD+ GPD PAIs are located in the Amazon biome, notable for its varied vegetation and edaphoclimatic diversity. It predominantly consists of open and dense Ombrophilous Forests, evergreen due to the stable high rainfall and solar radiation, common around the Equator.

Variations in Ombrophilous Forest types are directly linked to local edaphic and climatic conditions, influenced by weather, soil types, and proximity to large rivers. Along project areas, two forest types and three sub-formations were identified at Table 6, as follows: Dense Ombrophilous Forest: Lowland (Db) and Open Ombrophilous Forests: Alluvial (Aa) and Lowlands (Ab).

The Dense Ombrophilous Forests (Db), also known as tropical rain forests, has dense vegetation in all strata, and occurs in regions of the Amazon biomes and coastal zone of the Atlantic Forest where the biologically dry period is practically non-existent.

The Open Ombrophilous Forests are a variation of the Dense Ombrophilous Forest (Db), being a more open forest formation, where combinations of species are commonly observed in associations. It occurs in the transition regions between the Amazon biome and neighboring areas with more dry days than in the regions where dense anthropophilic forest occurs.

The Open Lowlands (Ab) does not vary topographically, always presenting repetitive environments on the alluvial terraces and margins of the watercourses. The Open Lowlands usually occurs in coastal plains in altitude ranging from 5 to 100m. According to Silveira (2001), among the eleven forest typologies identified in Acre, bamboo occurs in five. The open forest with bamboo dominating, the forest with bamboo plus forest with palm trees, the forest with bamboo plus dense forest and the bamboo forest in an alluvial area, represent 38% of the state's forest cover.

The vegetation of the project areas is composed of Open Alluvial, Open Lowland and Dense Ombrophilous, as presented at table Table 6.

Table 6 – Vegetation present in the first PAIs #1 – 4.

Forest Type	PAI #1 (ha)	PAI #2 (ha)	PAI #3 (ha)	PAI #4 (ha)	General (ha)
Open Alluvial (Aa)	-	2,442.58	699.77	758.07	3,900.42
Open Lowlands (Ab)	8,001.92	19,431.17	18,770.31	19,645.17	65,848.57
Dense Ombrophilous (Db)	14,471.10	-	-	-	14,459.54
Total	22,473.02	21,873.75	19,470.08	20,403.24	85,739.82

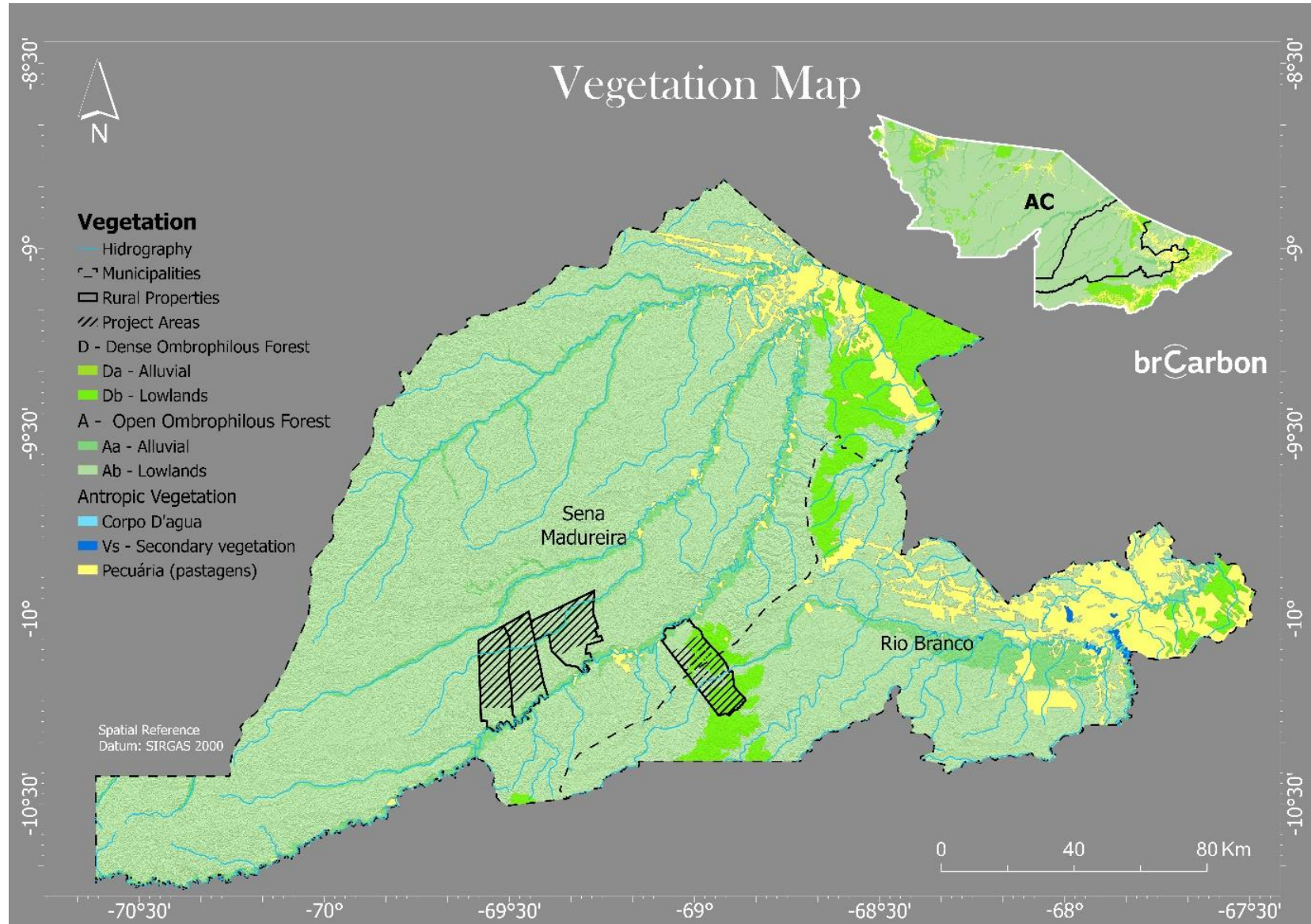


Figure 8 – Vegetation types at project activity instances.

2.1.14 Social Parameters (VCS, 3.18; CCB, G1.3)

The State of Acre is the fifth most populous in the northern region of Brazil, with 829,780 inhabitants distributed in 22 municipalities. The capital Rio Branco concentrates around 44% of the population of Acre and the other most populous cities are Cruzeiro do Sul, Sena Madureira, Tarauacá and Feijó with more than 30,000 inhabitants.

The Acre population is formed by descendants of different origins, such as indigenous, Europeans, Africans, and other ethnic groups present in Brazil. Indigenous peoples have a significant presence in the State, and several ethnic groups inhabit the region, including Ashaninka, Huni Kuin (Kaxinawá), Yawanawá, Shanenawa, Apurinã and others. In general terms, the diversity of Acre's population adds important aspects to the local social identity and reflects the historical process of occupation of this territory.

Originally, the region where is nowadays known as Acre state was occupied by various indigenous nations, among them were the Apurinãs, which, accord to various historians, gave the state's name. After the colonial period it became a part of the Bolivia and was occupied by foreign people. Later the discovery of the various rubber's utilities by the industry on late 19th century occurred a massive migration to the region by Brazilians seeking to explore all the commercial potential of the commodity, generating the first economic rubber cycle on Amazon. Which started a huge occupation on the region and turned the area in Brazilian part after the Treaty of Petrópolis on November 17th of 1903.

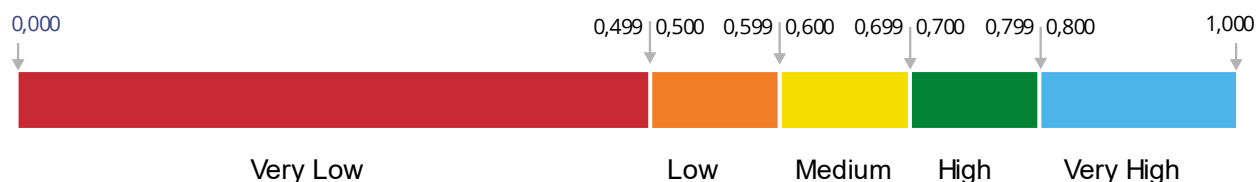
However, at the beginning of the 20th century, the crisis in rubber prices caused the debacle of Brazilian supremacy of this commodity. Part of the population which lived from this exploitation moved to the main urban centers and another part remained in the rubber farms, moving closer to the riverbanks, and surviving from the extraction of other products, the trade in animal skins, the small-scale agriculture, and livestock. During this long period of time, extractive exploitation contributed to the significant preservation of natural resources. On the other hand, the difficult access to a large part of this remaining population generated social inequalities that are reflected until today in terms of lack of social and productive infrastructure policies.

This history and its consequences to the huge social inequality generate a frame of the human development index as shown on Table 7 – it presents the reality of the state and of the two cities, Rio Branco, and Sena Madureira where the project is based, highlighting their dimensions of income, education, and longevity. According to these data, Sena Madureira has lower levels of IDHM when compared to the capital Rio Branco and the state of Acre.

Table 7 – IDHM of income, education and longevity of the project locations. Source: Adapted from Atlas Brasil, 2021¹⁶.

Locality	IDHM (2010)			
	IDHM	Income	Education	Longevity
Sena Madureira (AC)	0.603	0.593	0.456	0.81
Rio Branco (AC)	0.727	0.729	0.661	0.798
Acre State	0.663	0.671	0.559	0.777

Below there is an IDH scale reference:



In 2010, Acre State occupied the 21st position in the country's IDHM ranking, and in 2021 it moved to the 16th position, with the IDHM going from 0.663 (medium) to 0.710 (high). In addition, from 2020 the state occupies the 2nd position in the ranking of the northern region. This shows that the state has advanced in policies and actions related to income, education, and health, compared to other states in the region.

Acre also made progress in relation to the IDHM of women, which went from 0.669 (average) in 2010 to 0.739 (high) in 2021. However, there are still many challenges regarding the gender issue in the state, such as the actions of racial equality that still need to advance, since the state is mostly composed of brown and black people (83.09%).

¹⁶ <http://www.atlasbrasil.org.br/>

Table 8 – IDHM of women, men, black and white people of Acre. Source: Adapted from Atlas Brasil, 2021.

State	IDHM and ranking (2010)			
	Women (22°)	Men (15°)	Black (16°)	White (17°)
Acre	 0.669	 0.674	 0.651	 0.732
	HDI e ranking (2021)			
	Women (13°)	Men (22°)	Black (17°)	White (15°)
	 0.739	 0.650	 0.680	 0.759

Regarding the urban IDHM, in 2010, it was 0.710 (high), much higher than the rural IDHM, which was 0.506 (low), highlighting the need to develop actions related to education, health and income generation in rural areas. In reference to the age structure of the state, it is a younger population, with about 27% aged 0 to 14 years and 68% aged 15 to 64 years.

Another important index to consider is the Amazon Social Progress Index (IPS), hosted by IMAZON – Instituto do Homem e Meio Ambiente da Amazônia – which holistically and robustly measures the social and environmental performance of countries, states, and municipalities. It is relevant to highlight that this index is exclusively composed of social and environmental indicators, and the data is obtained from public repositories. It addresses the following dimensions: basic human needs, foundations for well-being and opportunities. In 2014, IPS was adapted for the Legal Amazon by Imazon, Social Progress Imperative (SPI) and Avina Foundation, with a particular method to the Amazonian reality. The table below presents the IPS values obtained in the years 2014, 2018 and 2021 for the project municipalities, state of Acre, Amazon region, and Brazil. It is observed that, except for Rio Branco's municipality, all the mentioned locations showed a reduction in the IPS in 2021 (the ranking showed is related to all municipalities of Brazilian amazon region).

Table 9 – Social Progress Index (IPS) of the project locations. Source: Adapted from IPS, 2021¹⁷

Locality	Social Progress Index Amazon (IPS)			
	Ranking 2021	2021	2018	2014
Sena Madureira	303°	55.22	55.64	52.09
Rio Branco	48°	61.16	59.83	59.15
Acre	-	53.55	54.18	54.09

¹⁷ <http://ipsamazonia.org.br/>

Locality	Social Progress Index Amazon (IPS)			
	Ranking 2021	2021	2018	2014
Amazon region	-	54.59	56.52	57.31
Brazil	-	63.29	67.18	67.73

Regarding education indicators, it is necessary to point out the age-grade adequacy of children and adolescents. In 2010 only 31.98% of young people aged between 18 to 20 graduated high school in the state of Acre, reducing even further to 25.11% in the case of the municipality of Sena Madureira. These numbers have improved since the 2010 census; however, they will possibly be impacted by the covid-19 pandemic, which directly affected the performance and school dropout of children and adolescents around the world, especially in developing countries like Brazil.

The covid-19 pandemic also directly impacted the state's monthly per capita income and income inequality, both of which had worse performances in 2021 when compared to 2020. According to Atlas Brasil, the monthly per capita income for 2021 was R\$ 471,54, with a reduction of 7.9%, while the Gini index increased from 0.515 to 0.539.

Equally important is the basic sanitation issues, once access to water supply services, sewage treatment and garbage collection are essential for minimum health conditions and hygiene. In Acre, around 55.2% of households have access to the general distribution network of clean water and 40.6% have sewage disposal in 2019. These data demonstrate that Acre's level is lower than other states in the North Region (58.8%) and the Brazilian average (85.5%) in 2019 (ZEE, 2021).

At last, is important to emphasize the Acre state have a fundamental importance, in the Amazonian context, can be summarized in its strategic geographic position, as an agricultural frontier, with edaphoclimatic conditions that present similarities to other regions of the Amazon. In Acre, as in a large portion of Amazon, mainly from the middle of 20th century, the occupation process accompanied the access roads, in the case, BR-364 and BR-317, where the implementation of large rural properties took place, which have pasture areas with extensive livestock exploitation (DO AMARAL, E. F.; BROWN, I. F.; ARAÚJO, E. A., 1999, as can be seen in the Figure 10:

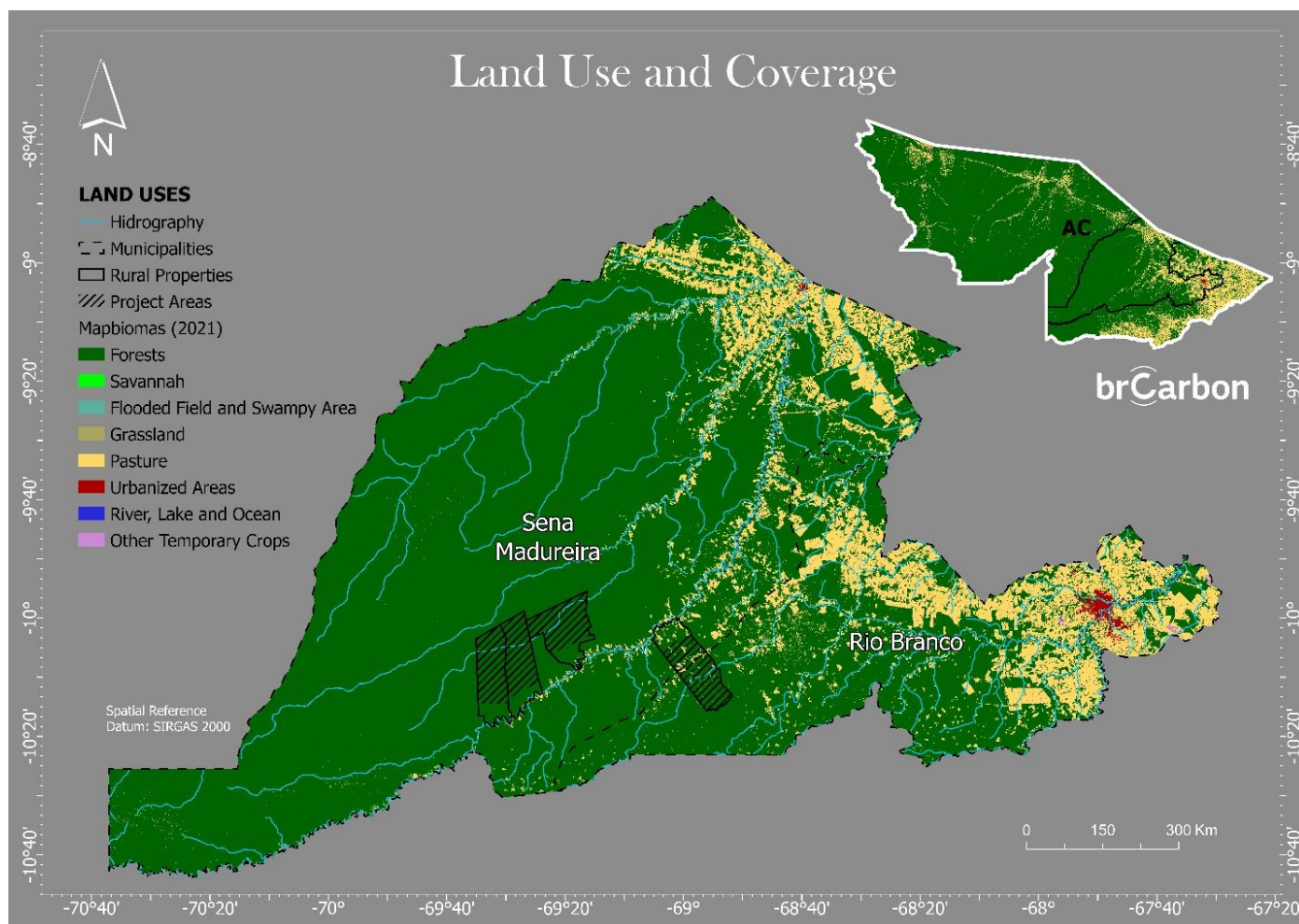


Figure 10 – Land Use at project region.

According to the ZEE summary report (2021), the main productive activities carried out in the State were: permanent banana crops and temporary cassava and corn crops. In monetary terms, animal production adds significant value to the economy of Acre, with emphasis on livestock, poultry, fish, and pig farming. Between 2008 and 2016, the cattle herd grew by an average of 2.67 million. The municipalities of Rio Branco and Sena Madureira concentrate most of this herd, reaching, respectively, 17% and 10% of beef production in the state. Regarding to extractivism, the main products are wood, rubber, Brazil nuts and açaí.

The most part of the presented data was obtained through the survey of secondary sources and is in line with the information obtained from the socioeconomic questionnaires applied to the families residing in the properties in the context of the APD Brazilian Amazon project. For example, the results of the questionnaires also found an age structure with a younger population, few people with complete primary and secondary education and other conditions that are related to the IDHM found. For a better understanding of the project's municipalities, a more detailed description is presented in Section 4 – Community.

2.1.15 Project Zone Map and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

The Uákiri REDD+ is a grouped project, defined as opened to additional instances which may be included after the initial validation, when meet pre-established eligibility criteria (please refer to section 2.1.5). The whole territory of Acre state was defined as the project zone, since this also covers potential areas where project activities may be implemented, including those related to the livelihood and development of communities.

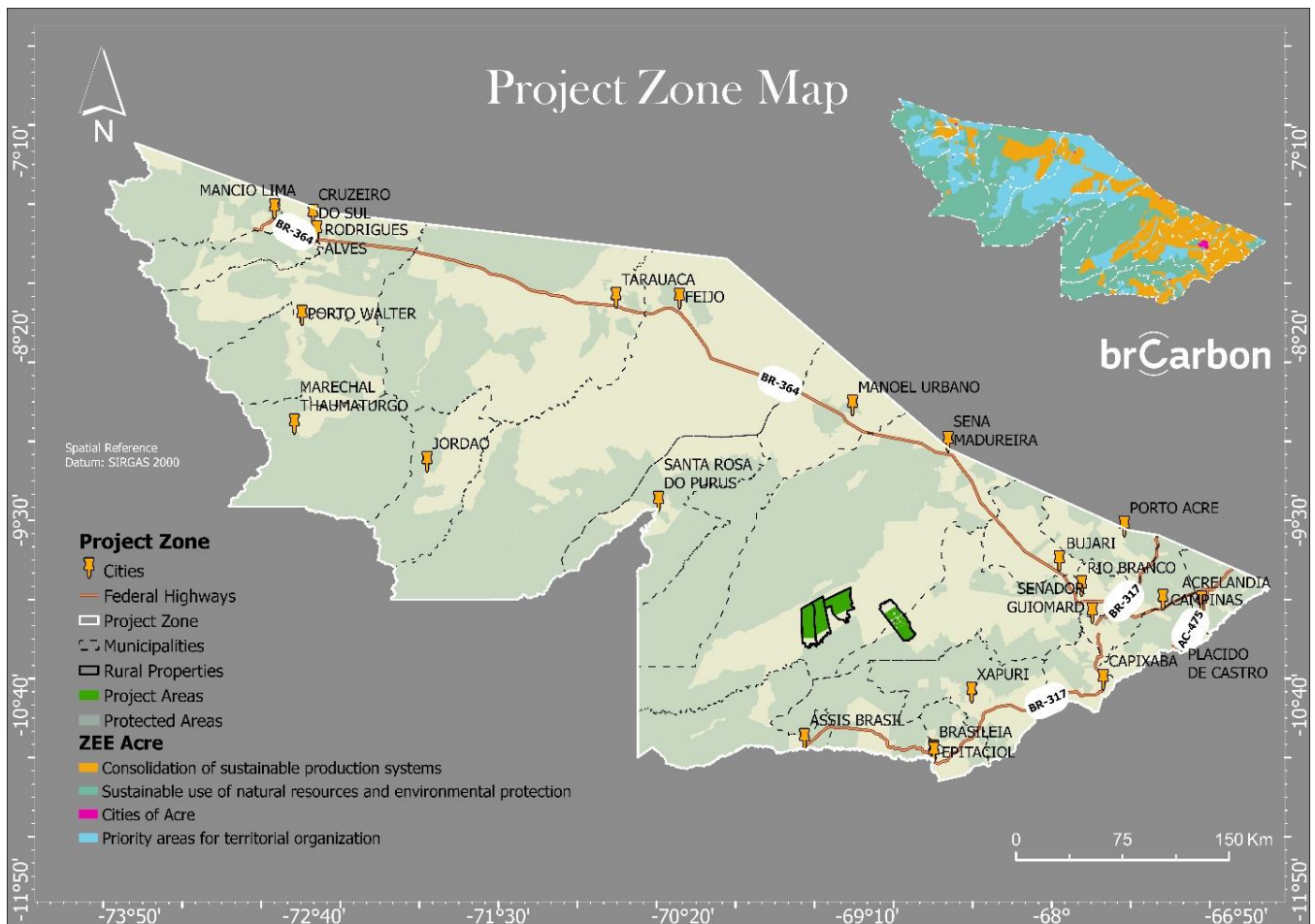


Figure 11 – Project Zone Map

For define the scalability limits of the project according to the territorial management map (ZEE, 2006), that the project areas and offsite climate impacts are predicted to scalable for zones of consolidation of sustainable productive systems and priority zones for territorial ordering. The project's zone of influence may also include the zone of sustainable use of natural resources and environmental protection conservation unities and urban zone, which encompasses the state's cities. The division of zones and the territorial management map of the Ecological Economic Zoning can be seen in the map on figure above.

Regarding the potential communities that may be included in project, and the selection to develop project activities, the BRC built a “Protocol of criteria for prioritizing communities and territories for

social activities” (APPENDIX_ELIGIBILITY-COMMUNITIES-PROTOCOL). The protocol defines five criteria:

- (1) Communities' relationship with the project area.
- (2) Geopolitical boundaries encompassing the project area.
- (3) Socioeconomic reality of communities.
- (4) Socio-environmental impact of productive activities.
- (5) Conflicts of interest.

Acre has the second largest rate of native forest coverage in the Legal Amazon, with 87% of native vegetation according to data from the III phase of the state ZEE (2020), about 50% of the territory is formed by protected natural areas. Being 05 Full Protection Conservation Units, 10 Sustainable Use Conservation Units such as RDS, FLONAS, RESEX, Area of relevant ecological interest, 32 Indigenous Lands, 129 Settlement Projects, 14 of which are agroextractivist state settlements.

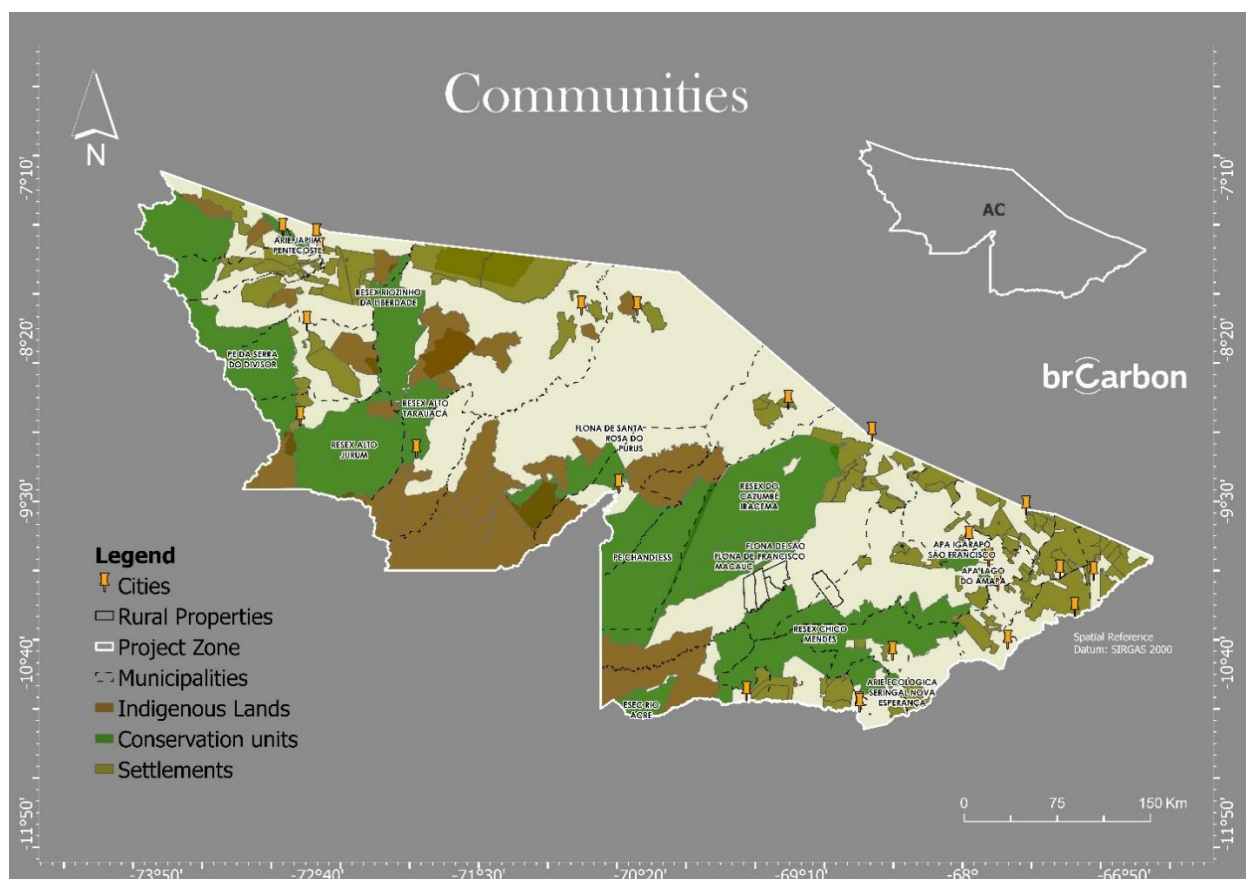


Figure 12 – Location of communities (conservation units, indigenous lands, and settlements) in Project Zone

Furthermore, the identified high conservation values identified in the project areas are related to species diversity, landscape-level ecosystem mosaics and Intact Forest Landscape (IFL) and ecosystems and habitats. First because PAI#02 to PAI#04 encompass the largest Important Bird

Area in Brazil, called Tabocais (BR059¹⁸), with importance to bird conservation. In addition, they are considered large intact forests, since together they have an area greater than 50,000 hectares. The HCVs related to community needs and cultural values will be discussed with the communities during the implementation of activities.

At the time of validation, brC identified communities living in the five initial instances (Seringal Porongaba, Palmares, Potiguar and Katianã) as shown on figure bellow.

PAI #1 is the area with the highest concentration of residents and where the access to road, makes Porongaba the only instance where the greatest part of the domestic groups lives in the center (forested areas) and not on the margins (riverside). In addition, it is located on the right bank of the Iaco River, where the neighboring Chico Mendes Extractive Reserve is located.

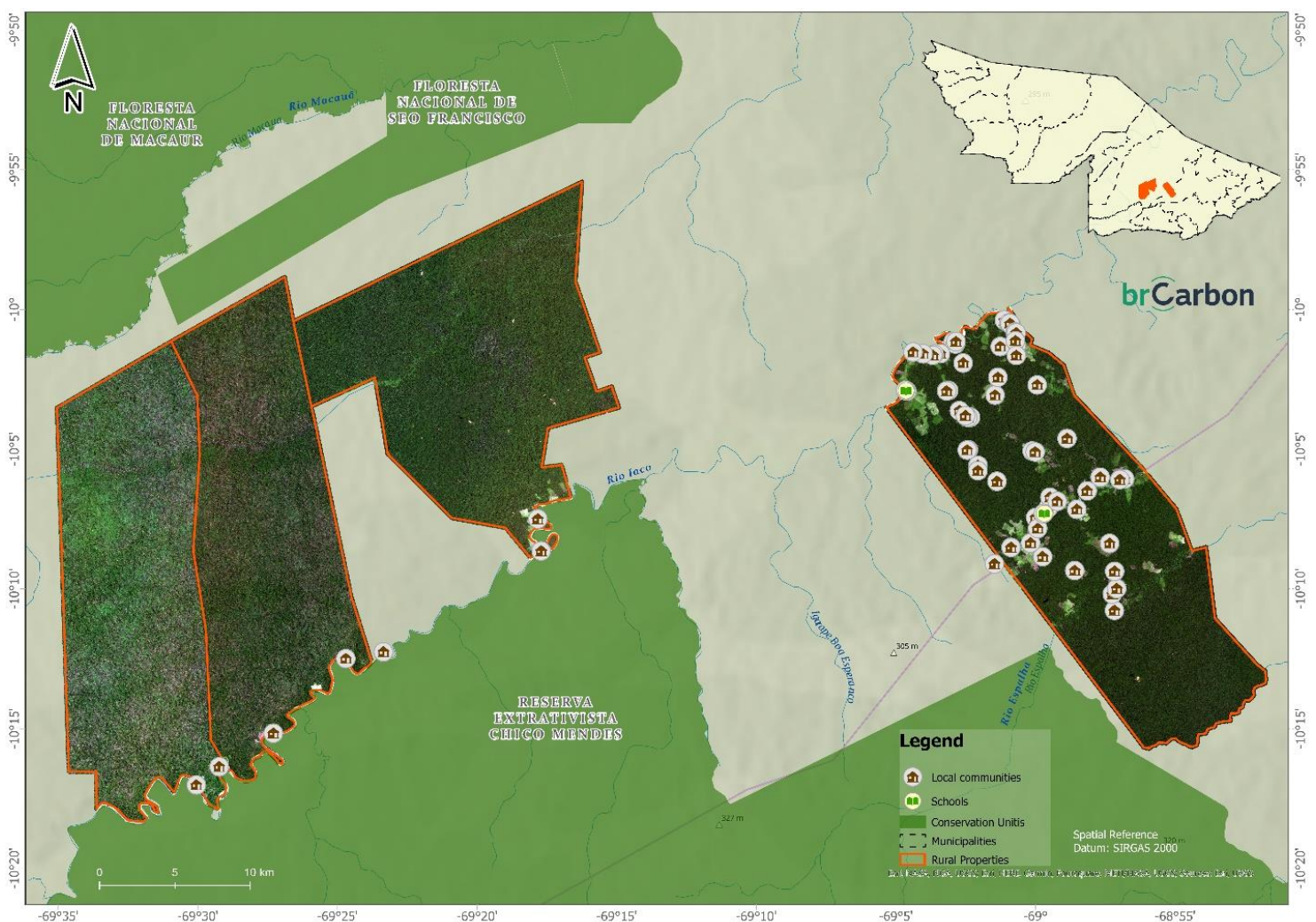


Figure 13 – Local communities at PAIs #1-4

¹⁸ <http://datazone.birdlife.org/userfiles/file/IBAs/AmCntryPDFs/Brazil.pdf>

2.1.16 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

BRC defines the project activities jointly with the communities, based on the results of the socio-economic diagnosis, public meetings with communities and interviews with other stakeholders. The activities described below represent a range of possible activities, and any changes that may be necessary will be presented and approved by their respective beneficiaries.

The activities' relevance to the project objectives depends on characteristics of each PAI. The outputs described represent verifiable targets for the first three years of the project, considering their current scope and may vary over time, with the inclusion of new PAIs.

The Outputs (short term) describe a period from 1 to 5 years. The Outcomes (medium term) describe the next 15 years of the project activities. The Impacts period (long term) describes the last 15 years until the end of the project instance lifetime (40 years). The Table 10 presents the activities description, Outputs, Outcomes, Impacts, and the relevance to project's objectives.

Table 10 – Theory of change

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
Installation of internet towers and solar energy equipment	05 Towers installed. 09 Solar panels deployed. Battery bank deployed. 70km of optic fiber installed	Internet services provided to communities at Porongaba, in special to children in scholar age	Digital inclusion and well-being	Very high. In many cases, existing communities within the project area lives in isolated conditions, without access to means of communication and travel days to the nearest city. This puts them in a vulnerable condition, negatively affecting their economy, health, education, and well-being.
Renovation and equipment of schools located in Seringal Porangaba.	02 Schools renovated and equipped with TV, Printer, and computer for better resources and distance learning	Improvement in basic education conditions and education quality.	Education and well-being	Very high. The percentage of illiteracy or functional illiteracy is high. Schools need repairs and better resources. Education is very important to change the condition of vulnerability they find themselves in.

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
Articulation with the Municipal / State / Federal Public Power and private initiative to support public policies and projects/program in the project area.	02 meetings per year with each public institution (health, education, social service secretaries) or private partner	Increase of public power presence in the territory and communities Improvement on public services in the territory	Education quality, health and well being	Very high. The articulation with the public power is fundamental to guarantee the implementation of education projects and programs in the schools.
Training and partnerships to strengthen social organization	02 Training sessions a year on social organization and associativism	Encouragement and support for creating and development of an association to organize work and represent the community.	Livelihood, well-being and trainings (ODS number 8)	High. Social organization is crucial for empowering communities and income generation, improvement their livelihoods and well-being.
Development of inclusive and low-carbon value chains	01 Mapping and selection participatory workshops for value chains with gender and youth issues, 01 Development of a participatory action plan, containing workshops, training, and exchanges Constant analyzes to the Monitoring and adaptative management of the action plan	Development of inclusive and low-carbon value chains, from production to commercialization Improvement of local governance through an inclusive, participatory, and transparent process Increase in family communities' income and improvement their livelihoods. Inclusion of young people and women in the value chains	Livelihood, trainings, and employment (ODS number 1, 5 and 8)	Very high. Inclusive and low-carbon value chains are essential for forest conservation and income generation in the community.
Carbon pedagogy lectures	03 Lectures per year about environment, forests, environmental services, biodiversity, environmental conservation, climate change,	Improving the quality of environmental education and including children and young people in debates on themes related to the project, generating	Education and well-being (ODS number 4)	High. Involving and engaging children and young people through educational processes is essential to ensure the effectiveness of the project, as it helps them to understand the project's themes and incorporate its importance for forest

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
	carbon cycle, environmental legislation, labor legislation, among others	appropriation, and belonging		conservation for the communities' livelihoods.
Socio-environmental monitor program	02 Public meetings per year to youth engagement for program participation 02 workshops per year to access knowledge, personal development, and collective skills to management and monitoring the actions	Structured and strengthened governance	Well-being, training, and livelihoods (ODS number 4)	High. This program is very important for the participatory development of the project, enabling the community itself to contribute to monitoring actions and activities throughout the project.
Technical Assistance and Rural Extension focused on low-carbon livestock and agroforestry systems	04 Workshops per year on good pasture management practices and implementation of agroforestry systems Constant individual technical assistance for each family of producers	Increase pasture productivity to reduce the necessity for deforestation of new areas. Improvement of production through the agroforestry systems	GHG Emission Reduction Livelihoods Education Training (ODS number 2, 4)	High. Most people in the community have livestock as their main productive activity and source of income. In the Amazon, it is a common practice to open new areas in order to increase the livestock production and income. In this way, increasing pasture productivity contributes to reducing deforestation pressure.
Recognition and inclusion of women in the productive and social activities	02 workshops per year to dialogue groups on topics related to female empowerment (women's rights, domestic violence, social division of labor, women's entrepreneurship)	Increased well-being and empowerment of women Increased autonomy	Education, well-being, and livelihood (ODS number 3, 4 and 5)	High. Many spaces in society are predominantly occupied by men. This becomes even more common in rural communities, depending on their culture. Encouraging women's self-organized spaces is essential to increasing women's participation in project activities, focused in their

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
				financial autonomy and empowerment.
Sustainable Forest Management	01 Training in forest management, reduced impact logging and work safety, focusing FSC certification, carried out. 01 development on sustainable forest management plans conform to the FSC standard 01 Safety equipment sets delivered	FSC certified forests	Improved land management and training	High. According to Walker et al. (2019), more than 20% of aboveground carbon stocks in the Amazon biome in Brazil were lost to forest degradation events between 2004 and 2016. According to data from INPE, the National Institute for Space Research in Brazil, 2,850 Km2 of forests were degraded in 2021 because of conventional forest exploitation.
Promotion of improvements in water quality and access to water	01 Water potability analysis 01 Water filter distribution 01 Chlorine distribution 01 Water pumps installed	Improvement in the water potability and access to water	Water and health	The water potability analysis performed initially indicated contamination by fecal coliforms in all tested samples. So, people have to bring water from the spring in cans or buckets to cook or wash their hands.
Hiring of community personnel	20 People employed	Improvement in household income	Employment and livelihood	High. The project will create job opportunities, training in different topics and generate complementary income in a place where options are limited and there are few prospects.
Biodiversity conservation	Continuous fauna monitoring	Promotion of species conservation through the maintenance of forest habitats and avoidance of hunt pressure.	Promotion of species conservation through the maintenance of forest habitats and avoidance of hunt pressure.	Very high. The project area is in HCV area, with the occurrence of rare, threatened, and endemic species.

Activity description	Expected climate, community, and/or biodiversity			Relevance to project's objectives
	Outputs (short term)	Outcomes (medium term)	Impacts (long term)	
Forest surveillance: deforestation, forest degradation and burn scars monitoring.	85,739.82 ha of forests annually monitored (PAI#1 to 2)	Maintenance of 85,739.82 ha of forest cover. Reduction of GHG emission due to deforestation.	GHG emission reductions and biodiversity conservation (ODS number 13 and 15)	Very high. The average GHG emission caused by land use change in Brazil is approximately 1 Gt CO₂e, representing half of the national emission profile. Deforestation in the Amazon biome in Brazil surpassed 11,000 km² in 2021 and shows an increasing trend.

All these activities from short to long term, as already mentioned, will be agreed with the population affected and BRC is aware that they can change during the project period for many reasons as cultural, political, social, or other reasons. Furthermore, as a grouped project, there might be different demands from different groups or communities. In this sense, the PP needs to be flexible to adjust previous accords or demands to build a robust relationship with the communitarians. Anyway, some activities will remain in a long term due to the needs to protect the forest and other project targets, among them can be listed the structuration of fire brigades and other activities related to environmental protection, which is the main target of BRC's work.

In addition, the activities proposed by this project aim to complement the activities that are already being developed by the APD Brazilian Amazon project in the territories. The two projects are developing activities with the same communities, some of them are focusing on infrastructure issues (solar panels, basic sanitation, equipment, and school renovation), and others are focusing on training, workshops, associations, and value chains, all of them turned to strategies to avoid unplanned deforestation, helping the success of the project's targets. Accordingly, the projects will complement each other so that it is possible to generate more benefits and social impacts, in an assertive and strategic way.

2.1.17 Sustainable Development Contributions (VCS, 3.17)

The project contributes to 11 (eleven) Sustainable Development Goals (SDG), considering 24 (twenty-four) specific objectives.

For each one of them, activities provided by the project were listed, as well as their respective indicators (Table 11). For the monitoring reports, impact and adherence analyses will be done for each of the activities and specific objectives proposed. In addition, each activity listed in the theory of change in the section above was related to the SDG.

Table 11 – Sustainable Development Goals and project activities

Sustainable Development Goals (SDG)	Specific objectives	Project activities
	1.1 - By 2030, end extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day	Increase in local income generation through incentives in low-carbon value chains
	1.4 - By 2030, ensure that all men and women, particularly the poor and vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technologies and financial services, including microfinance	Support for obtaining basic and essential services, such as electricity, water, and basic sanitation.
	2.3 - By 2030, double the agricultural productivity and income of small-scale food producers, particularly women, indigenous peoples, family farmers, pastoralists, and fishermen, including through secure and equal access to land, other productive resources and inputs, knowledge, financial services, markets and non-agricultural value-adding and employment opportunities	Training and supply of materials and inputs that contribute to increasing agricultural, NTFP and livestock productivity
	3.3 - By 2030, end epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases, and combat hepatitis, waterborne diseases, and other communicable diseases	Sensitization and awareness campaigns for the prevention of tropical and other diseases
	3.7 - By 2030, ensure universal access to sexual and reproductive health services, including family planning, information, and education, as well as the integration of reproductive health into national strategies and programs	Sensitization and awareness campaigns for the sexual and reproductive health
	3.8 - Achieve universal health coverage, including financial risk protection, access to quality essential health services and access to safe, effective, quality, and affordable essential medicines and vaccines for all	Articulation with the public sector for itinerant health actions and vaccination campaigns
	4.1 - By 2030, ensure that all girls and boys complete free, equitable, quality primary and secondary education that leads to relevant and effective learning outcomes	Schools' maintenance and supply of teaching materials and equipment
	4.4 - By 2030, substantially increase the number of young people and adults who have relevant skills, including technical and professional skills, for employment, decent work and entrepreneurship	Entrepreneurship and leadership development training for young people

Sustainable Development Goals (SDG)	Specific objectives	Project activities
	4.c - By 2030, substantially increase the pool of qualified teachers, including through international cooperation for teacher training, in developing countries, especially least developed countries and small island developing States	Training of schoolteachers on climate change and sustainable development
	5.1 - End all forms of discrimination against all women and girls everywhere	Creation of a women's group for dialogues on women's rights, domestic violence, social division of labor and women's empowerment
	5.5 - Ensure the full and effective participation of women and equal opportunities for leadership at all levels of decision-making in political, economic, and public life	Women's leadership and entrepreneurship workshops
	5.6 - Ensure universal access to sexual and reproductive health and reproductive rights, as agreed in accordance with the Program of Action of the International Conference on Population and Development and the Beijing Platform for Action and documents resulting from their review conferences	Sensitization and awareness campaigns for the sexual and reproductive health, focusing on women
	8.3 - Promote development-oriented policies that support productive activities, decent employment generation, entrepreneurship, creativity, and innovation, and encourage the formalization and growth of micro, small and medium-sized enterprises, including through access to financial services	Incentive to associativism through trainings and workshops on associativism, cooperativism and entrepreneurship, focusing on young people
	8.6 - By 2020, substantially reduce the proportion of young people without employment, education or training	
	13.2 - Integrate climate change measures into national policies, strategies and planning	Fostering good fire management and firefighting practices to communities to prevent the spread of wildfires in the project area
	13.3 - Improve education, raise awareness and human and institutional capacity on mitigation, adaptation, impact reduction and early warning of climate change	Carbon pedagogy lectures about climate change, carbon cycle and others
	15.1 - By 2020, ensure the conservation, recovery and sustainable use of terrestrial and inland freshwater ecosystems and their services, in	Avoided deforestation and reduced emissions

Sustainable Development Goals (SDG)	Specific objectives	Project activities
	particular forests, wetlands, mountains, and arid lands, in accordance with obligations arising from international agreements	
	15.2 - By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally	Encourage the sustainable extractivism of non-timber forest products, through training and supply of materials and equipment
	15.5 - Take urgent and significant measures to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species	Fauna (mammals, birds, reptiles, and amphibians) periodic monitoring for biodiversity conservation
	17.17 - Encourage and promote effective public, public-private and civil society partnerships, based on the experience of the resource mobilization strategies of these partnerships	Networking and partnerships with the state and municipal public sector, NGOs, and other private companies

2.1.18 Implementation Schedule (CCB, G1.9)

The Implementation Schedule is presented in the Table 12, that shows key dates and milestones for the four initial instances of the grouped project, as introductory meeting dates, start and end dates for each project activity, start and end dates for the GHG accounting period, actions taken by the owner to combat illegal deforestation, monitoring schedule, verification schedule, and past activities. It is worth noting that all project's activities are still underdeveloped, and the BRC staff is still planning the activities together with communities.

Table 12 – Project Implementation Schedule.

Date	Milestone(s) in the project's development and implementation
05 th to 18 th of April, 2021	First field visit in the Acre state Project Activity Instances (PAI#1-4) for socioeconomic diagnosis, to present the project concept to local communities and to make the first biodiversity survey
19 th July to 25 th July, 2021	Free prior and informed consent in the PAIs#1-4.
27 th June, 2021	Command and Control Action: Incident police report of land invaders in PAI#04
27 th July, 2021.	Meeting with Municipal Health Department and State Education department in Sena Madureira in Acre state
29 th July, 2021	Meeting with State Education Department in In Rio Branco, Acre state.
30 th July, 2021	Meeting with “Instituto de Mudanças Climáticas” (IMC) in Acre state.

Date	Milestone(s) in the project's development and implementation
02 nd August, 2021	Signature of the first conservation agreement between BRC and the landowner of the four project activity instances. The milestone that characterizes the project's start date.
04 th August, 2021	Water potability analysis
05 th August, 2021	Meeting with Municipal Education Department in Sena Madureira in Acre state.
30 th August to now	Digital inclusion: 8 towers (60m high), 1 tower (72m high), 1 tower (82m high) installed, 36 solar panels deployed, battery bank deployed, 70Km of optic fiber installed
8 th September, 2021	Incident police report of land invaders in PAI#01
09 th to 10 th October 2021	"Outubro Rosa": 6 health professionals from the municipal health department in Sena Madureira held two days of lectures and services focused on women's health in Seringal Porongaba.
05 th to 06 th October 2021	Sustainable Agriculture Management field day: 15 community members were taken to 02 field days in 02 different model farms where pasture management and sustainable production with a focus on soil organic carbon were discussed
09 th to 10 th October 2021	Photovoltaic systems installed at 02 schools in Porongaba.
11 th October 2021 to 12 th December 2021.	Survey, dimensioning and installation of photovoltaic and hydraulic systems for residents of Palmares (PAI#03), Potiguar (PAI#05) and Katiana (PAI#02) (on-going)
March 29 th , 2022	Beginning of chestnut tree mapping, with Avenzamaps app training.
09 th to 22 nd October 2021	Second biodiversity survey in PAIs #2 to 5 (dry season)
9 th May, 2023	Kick-off meeting between brCarbon and Mr. Eloy Moacir Corecetta Batista.
25 th May, 2023	Agreement signature between brCarbon and landowners.
30 th May, 2023	Submission of the draft VCS PD in the VERRA registry system.

2.1.19 Risks to the Project (CCB, G1.10)

In assessing the anticipated benefits for climate, community, and biodiversity over the project's lifespan, it is essential to acknowledge and confront both natural and human-induced risks.

This involves the identification of short and long-term risks, such as sustaining community involvement and adapting to climate variability.

The comprehensive assessment delineates measures for risk mitigation, ensuring the project's resilience. The risk table shown below (Table 13) presents a strategic approach to foresee,

address, and mitigate potential risks, thereby contributing to the project's overall success in realizing its intended benefits.

Table 13 – Risk table

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Mitigation or preventative measure(s) taken
Deforestation and degradation (Human-induced)	The communities around the project area can promote deforestation to expand pastures or new plantations or degradation to logging, or even just expand their areas. This is most often a reduced magnitude and low impact activities, however, over time, such activities can harm the expected benefits to the climate and biodiversity.	Constant presence on project areas; Remote and Field Monitoring; Placing of signs; Social activities. Workshops and lectures about the dangers and risks of the promotion of deforestation and degradation. Fostering and creation of other possibilities to livelihood
Political (Human Induced)	The complexity of any political frame makes it a risk to each activity, once it involves a lot of different interests, in this sense there can be organized groups (local and non-local stakeholders) working against the project activities, weakening the projects legitim in front of its public (specially the communitarians).	Constant presence in the area, throughout developing project activities to sustain and fortify social engagement, beyond keeping a communication channel open, and articulation with different kinds of stakeholders, mainly the representative institutions and the public power representants.
Disputes over land rights or access/use rights (Human Induced)	The low level of public governance all around the Amazonian region makes land rights a theme of dispute constantly, and the consequences of that can weaken not just the project, but the relations in the surrounding areas.	Constant monitoring of conflicts occurring in the area, implementation, and use of conflict resolution tools, including access, when necessary, to the public justice system.
Extreme droughts (Natural)	Climatic anomalies such as El-Nino could change the air currents over the Amazon biome, making the forest drier and more flammable, which could compromise the project's climate and biodiversity objectives. Under ongoing global warming, the frequency of severe droughts such as in 2005 and 2010 is projected to increase strongly, up to the point where these droughts may become the new climate normal in the second half of this century.	Promote sustainable project activities such as agroforestry systems which are productive systems resilient to climate changes.

2.1.20 Benefit Permanence (CCB, G1.11)

The project proponent (brCarbon) takes legal steps to ensure that the climate, community, and biodiversity benefits will be maintained beyond the project lifetime.

Each Project Activity Instance has a legal agreement (APPENDIX_LEGAL-AGREEMENTS) that was signed with the landowners to continue the management practices that will lead to the project area conservation for at least 40 years and guarantee the permanence of the carbon stocks in the post project scenario and, as well, the community and biodiversity benefits, as follow:

The GPD GHG accounting period is 100 years. Any PAI lifetime can be extended to 100 years.

2.1.21 Financial Sustainability (CCB, G1.12)

The Project Proponent operates with its own capital and will fund all activities planned under the GPD. The actual and projected revenues from GHG emissions reductions and the projected cashflow are considered as commercially sensitive information.

Detailed financial information about the project can be shared with relevant stakeholders upon signing a non-disclosure agreement.

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project (VCS, 3.13; CCB, G2.1)

The Acre state in Brazil is known for its rich and diverse ecosystem, characterized by vast expanses of Amazon rainforest, which hosts a variety of flora and fauna, including endemic species. Its rivers and wetlands play a crucial role in maintaining the ecological balance, supporting aquatic life, and serving as the lifeblood for many indigenous and local communities. Acre's ecosystem is a critical carbon sink, helping to mitigate climate change by absorbing significant amounts of CO₂ from the atmosphere. However, it faces threats from deforestation, illegal logging, and land-use changes, primarily driven by agricultural expansion.

Historically, the land use in Acre, was dominated by rubber extraction, which peaked at the turn of the 20th century but declined due to competition from Southeast Asian plantations. In recent decades, the economy and land use have shifted towards agriculture and cattle ranching, leading to significant deforestation and changes in the landscape. The state has also been at the forefront of environmental conservation efforts, pioneering initiatives for sustainable development and the establishment of extractive reserves to balance land use with forest preservation.

Information about the climate, hydrology, topography, relevant historic conditions, soils, vegetation, and ecosystems of the project area was described previously on section 0, 1.1.1 and 2.1.15.

According to Trase's publications¹⁹, Acre state is responsible for around 290 tons of beef trading (2020), which are mainly provenience from Project Instances region (Acre's south portion), as presented on the Figure 14.

¹⁹ https://trase.earth/explore/supply-chain/brazil/beef?chartType=map&year=2020&indicator=volume&dimension=region_production_1 and https://trase.earth/explore/supply-chain/brazil/soy?chartType=map&year=2020&indicator=commodity_deforestation_total_exposure&dimension=region_product_ion_1

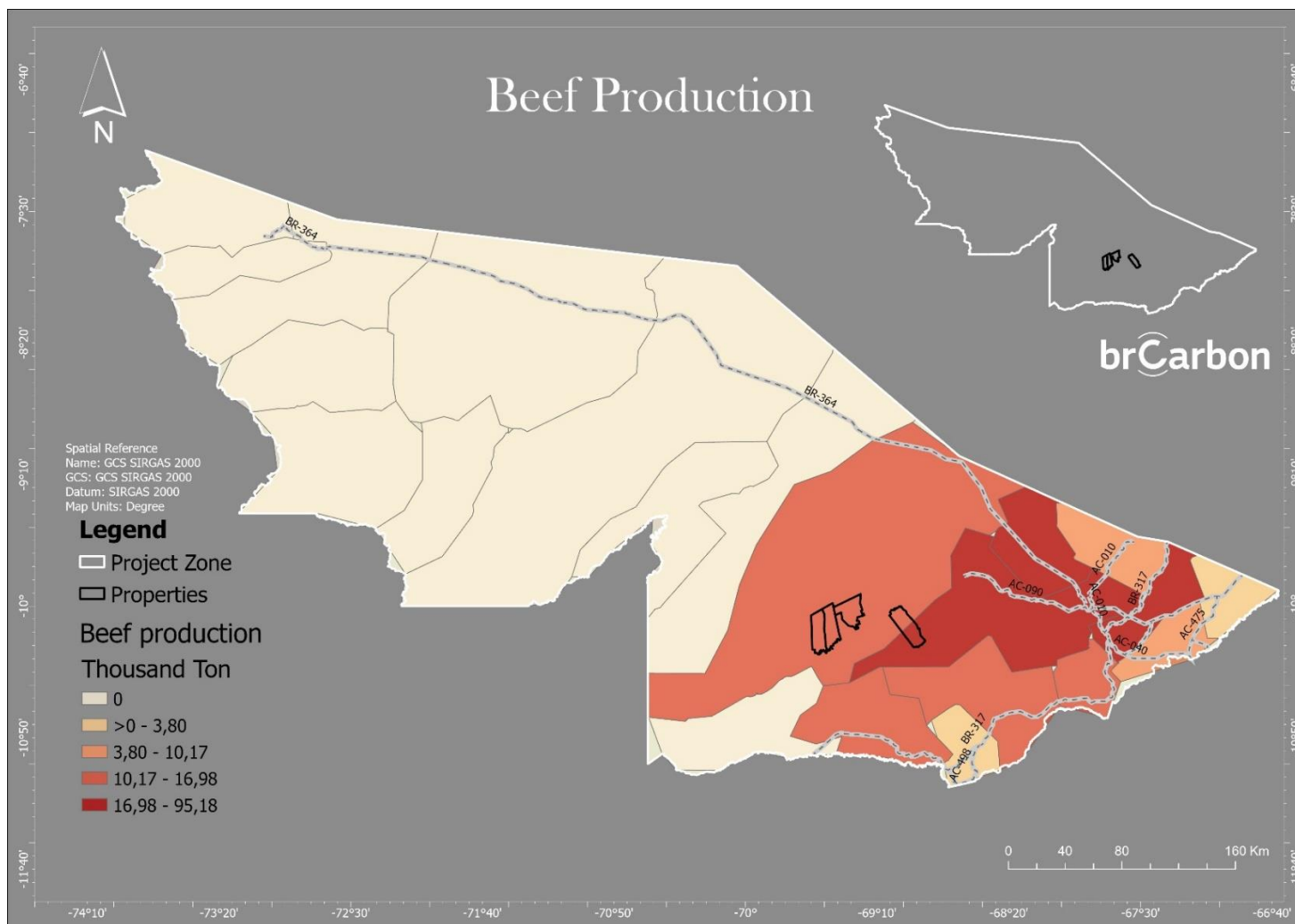


Figure 14 - Beef production in Acre's state.

2.2.2 Most-Likely Scenario Justification (CCB, G2.1)

The most likely scenario consists of the deforestation for cattle and beef production in the absence of the project.

As previously presented, in section 2.2.1, Acre state is responsible for beef production, answering for around 290 tons of beef (2020's data). Brazilian official deforestation data (PRODES/INPE) have already registered, from 2011 to 2021, more than 4,279 square kilometers of deforestation, as presented on Figure 15, on Acre's state.

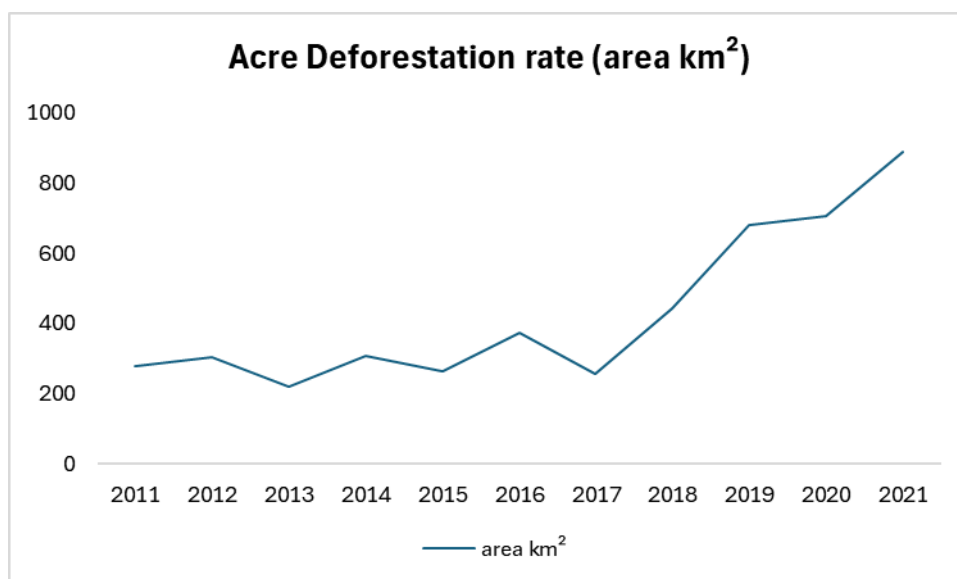


Figure 15 – Yearly deforestation on Acre (Terra brasiliis, 2024).

Considering what has been said in this section and in the previous section, it is understood that the most likely scenario, if the project were not proposed, would be deforestation with land use conversion to agricultural activities, mainly focused on beef cattle production.

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

The project proponent experience tells that public authorities play a very weak and/or restricted role in the state of Acre as a whole. There is a more expressive presence of public authorities in the state capital (Rio Branco, AC), but in places with difficult access, such as the region of the first instances, this presence is quite minimal.

The benefits that the project aims to achieve are already a reality in the region, which has benefited from the actions of the ID 2551 project that is already underway in the state. The aim is therefore to leverage and enhance the existing benefits, as well as bringing new benefits to the region, so that the projects produce complementary impacts in the same region.

Benefits for community, such as access to information and the internet, improved infrastructure, joint actions derived from partnerships between BRCarbon and the state and municipal governments - such as preventive health actions - already occur in the territory and can be boosted with other resources from project 4552.

2.2.4 Benefits to be used as Offsets (CCB, G2.2)

No distinct community and biodiversity benefits are intended to be used as an offset, beyond the VCUs generated by the current project.

2.3 Safeguards and Stakeholder Engagement

2.3.1 Stakeholder Identification (VCS, 3.18, 3.19; CCB G1.5)

The BRC implements a social engagement protocol which, in its first stage, involves stakeholder identification. This process is subdivided into four different phases. The first phase is the stakeholder identification, while subsequent phases comprise participatory workshops, monitoring and dissemination of results, as illustrated Figure 16. Detailed information on the social engagement protocol can be found in “APPENDIX_SOCIAL-ENGAGEMENT-PROTOCOL”.



Figure 16. Social Engagement Protocol

During the initial stakeholder identification phase, a detailed survey of the territory is carried out by researching articles, public data and drawing up maps. This process aims to recognize the territory and identify potential areas of interest, such as indigenous lands, conservation units, settlements and other communities close to the project. This information is fundamental for accurately identifying the stakeholders involved in the project.

Next, a list is drawn up to register the stakeholders, and letters, protocols and invitations are sent out to organize visits to introduce and recognize the stakeholders. Subsequently, specific key questions are developed for each social actor identified. An approach and relevance matrix are then created, considering data on the challenges of building partnerships, mapping local policies and understanding relationships with stakeholders. This process leads to the application of the prioritization protocol detailed in the (APPENDIX_TERRITORY-PRIORITIZATION-PROTOCOL).

2.3.2 Stakeholder Descriptions (VCS, 3.18, 3.19; CCB, G1.6, G1.13)

The table below presents the description of the stakeholders identified for the initial instances of the Uákiry REDD+ AUDD Grouped Project, describing, for each of them, the type of organization, sphere of activities, rights and interests and overall relevance to the project. The same approach will be applied for the future instances included in this grouped project, throughout its duration. Stakeholders are subdivided into communities, community groups and other stakeholders. In this case, the communities listed are those residing around the project area. According to the standard, community groups are community organizations (associations and cooperatives) that can represent communities within the project area or in the project zone.

The other stakeholders identified are the public sector, non-governmental organizations, and other companies, which may work in partnership on some project activities. In addition to these, as the project progresses and new instances are eligible, other new stakeholders can be identified, through workshops, public meetings, municipal liaisons, events, and other community projects.

Table 14. Stakeholder Description.

Stakeholders	Type of organization	Sphere	Rights, Interest and Overall Relevance to the Project
Communities			
Traditional communities of Porongaba	Residents of the placement	Local	Families residing in the rural properties, where project activities occur. This group has recognized tenure rights. Activities have already been carried out to understand the demands and interests of this group, which are already being considered in the project.
Traditional communities of Palmares			
Traditional communities of Potiguar			
Traditional communities of Katianã			
Community groups			
Agroextractivist Cooperative of Rural Producers of the Iaco River Valley - COOPERIACO	Social organization – Private association	Local	The cooperative is the main buyer of Brazil nuts produced in PAI#01. Being of relevant interest to understand and structure this value chain in the community.
Central Extractivist Commercialization Cooperative of the State of Acre - COOPERACRE	Social organization – Private association	State	Representative of the Central Cooperatives of the State of Acre. It is of relevant interest to the project, as they develop activities based on credibility, sustainability, and transparency in the commercialization of Amazonian products derived from the extractivism.

Stakeholders	Type of organization	Sphere	Rights, Interest and Overall Relevance to the Project
Agroextractivist Association Pôr-do-Sol do Riozinho do Rola	Social organization – Private association	Local	The association develops activities with latex extraction and sale of Brazil nuts. A group of PAI#01 residents make part the body of members and declare the institution as a partner.
Other stakeholders			
Recent Invaders, Land grabbers of Porongaba	Residents of the placement	Local	This group has no recognized tenure rights. Its main interest is occupying, deforesting, and selling land or starting cattle production.
Municipal Secretary of Health	Public sector	Municipal	Responsible for the Itinerant Health programs that bring medical, dental and vaccination services to the communities in the project areas. brC is building a public-private partnership to leverage this program and reach more families.
Secretary of State for Education, Culture and Sport	Public sector	State	It is currently responsible for managing the rural schools present in the area and in the project zone, making it necessary for joint articulations that have an impact on education. brC is building a public-private partnership to formalize jointly developed actions.
Municipal Secretary of Education	Public sector	Municipal	The management of schools in Porongaba will become the responsibility of the municipal secretary in the future, so an approximation will be necessary.
Municipal Secretary of State for the Environment and Indigenous Policies	Public sector	State	This secretariat works with extractivism, productive chains and environmental policies for the state. It is intended to initiate an articulation on possible actions in partnership, especially with the Institute of Climate Change and Regulation of Environmental Services of Acre – IMC.
Municipal Secretary of Environmental	Public sector	Municipal	It is intended to initiate an articulation on actions focused on the environmental education, environmental management and others.
Municipal Secretary of Social Assistance	Public sector	Municipal	This secretariat has a significant database about the socioeconomic reality of the municipality and communities encompassed by the project. These data contribute to the diagnosis and planning of subsequent actions.

Stakeholders	Type of organization	Sphere	Rights, Interest and Overall Relevance to the Project
Secretary of State for Production and Agribusiness of Acre – SEPA	Public sector	State	It is intended to initiate an articulation on possible actions in partnership regarding sustainable agriculture and livestock production, especially the Forest and Institute of Agricultural and Forestry Defense of Acre – IDAF.
Chico Mendes Institute for Biodiversity Conservation - ICMBio	Public sector	Federal	Managing body of federal conservation units, including RESEX Chico Mendes, which is within the project zone.
Brazilian Institute for the Environment and Renewable Natural Resources - IBAMA	Public sector	Federal	Body that oversees the use of natural resources and illegal activities such as deforestation. Considering that this is a Legal Reserve area, it is important to dialogue and articulate actions with IBAMA.
Acre Environmental Institute	Public sector	State	State autarchy which is responsible to implement the state environmental police.

2.3.3 Stakeholder Access to Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

All project documentation, including the description and monitoring reports, will be disseminated via the BRC and Verra websites, as well as on social media platforms and during public meetings held throughout the project's execution. The initial version containing the project description is already available on the BRC and Verra websites for public access.

The summary versions of the project and the monitoring reports containing the physical and social parameters, as well as other relevant aspects throughout the life cycle of the project are developed in the native language, specifically Brazilian Portuguese, and are delivered to representatives of the associations and institutions of interest to the project. In addition, BRC is committed to maintaining a continuous and open communication channel with all stakeholders. This commitment involves the use of various communication tools, such as campaigns, public meetings, protocols, cell phones, among others, to keep all relevant institutions, communities and community members informed and aware of the implementation of the project's activities and its monitoring, verification and reporting process.

2.3.4 Dissemination of Summary Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

The summarized documentation of the VCS and CCB Aggregated Project in Brazilian Portuguese, containing relevant information for criteria G1.1-9, will be distributed during public and institutional meetings, as well as at smaller-scale events, to the communities involved in the project and

stakeholders. Each project activity will be briefly described, highlighting the expected outcomes, consequences, and impacts. The project timeline, including the start date, duration, greenhouse gas accounting period, and community benefit evaluation period, will be outlined with explanations for any identified discrepancies.

2.3.5 Informational Meetings with Stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

The BRC implements a social engagement protocol divided into four different phases. In the second stage of this process, participatory workshops are conducted with local communities. During these sessions, questionnaires are administered to identify the needs and demands of the involved communities.

Following this protocol, the BRC organized five different informative meetings during the second phase of social engagement. These meetings took place between July 19th and 25th, 2021, in communities near the project areas Figure 18. The dissemination and communication of these meetings were done through local radios days prior, along with team mobilization to conduct door-to-door invitations to community residents Figure 17 and Figure 18.

The meetings are conducted in an accessible language, aiming to convey necessary and accurate information to the involved communities.



Figure 17. Invitation to information meeting.



Figure 18. Informational meeting.

2.3.6 Risks from the Project and No Net Harm (VCS, 3.18, 3.19)

BRC is committed to conducting the project in the best possible way. However, it acknowledges that there are risks involved, categorized into natural risks and risks induced by humans. Thorough assessment of the project's potential risks has been conducted, highlighting deforestation and degradation as consequences of human activity that can lead to significant impacts over time. To mitigate such actions, strategies such as remote area monitoring and constant team presence in the field help combat these risks, along with initiatives to raise awareness of the dangers and risks associated with deforestation.

Another identified risk is predatory hunting, a practice in the local culture. However, on a large scale, it can pose a threat to biodiversity and cause adverse climatic impacts on the project. It is crucial to raise awareness among communities and explore alternative exploitation methods that minimize significant impacts on biodiversity.

Land tenure issues and local political issues have also been identified as potential risks to the project. Such issues can trigger territorial disputes and compromise the integrity of the project and local communities. Emphasizes the need to guarantee land rights to ensure the success and long-term sustainability of the project.

These risks, classified as human-induced, have the potential to cause significant impacts to the project. Additionally, natural risks such as the extreme droughts that have occurred in recent years have been identified, which can result in significant losses of forests and biodiversity in the region.

2.3.7 Community Costs, Risks, and Benefits (CCB, G3.2)

The analysis of costs, risks, and benefits of any project involving the community requires dialogue between stakeholders and community members, making it not always easy to achieve these analyses at the project's outset. Measuring costs in the short, medium, or long term is simpler than assessing

benefits, such as improving healthcare or education quality. Risks, however, tend to become more apparent after the initial stages of the project implementation. Therefore, during the project's execution, it becomes possible to better understand the risks. It is through this process and community demands that the project can enhance its strategies to mitigate risks and unwanted impacts.

With the project's implementation, it is expected that there will be no risks to identified stakeholders, their resources, or property rights. On the contrary, the project is designed to promote net positive impacts for communities, in accordance with CCB requirements. However, risks were discussed with identified stakeholders during the initial consultation phase, due to doubts and concerns expressed by communities during interviews, of course, as already mentioned, to map the risks beforehand is not a easy task, in this sense, this mapping process is continuous during all the project lifetime.

To regard the initial and more clear risks, about resources and property rights and others, BRC addressed issues such as expropriation, involuntary relocation, logging, and threats to community livelihoods, all associated with the project's conservation objectives. BRC understands that both landowners and squatters have specific rights that must be respected and can be exercised harmoniously, as evidenced to communities during the initial stakeholder consultation phase as a measure to mitigate perceived risks. Additionally, BRC understands that community subsistence depends on clearing small areas for agriculture or pasture, and the project should not interfere with this local dynamic, also confirmed to communities as a mitigation measure.

Another specific mitigation measure ends in the second stage of the social engagement protocol, which concerns obtaining free, prior, and informed consent. At this stage, after a second clarification meeting and the development of participatory project design workshops, relevant communities must consent to the project's execution before project activities commence. Thus, through the stakeholder consultation process and BRC's social engagement protocols, risks perceived by communities are mitigated.

2.3.8 Information to Stakeholders on Validation and Verification Process (VCS, 3.18.6, 3.19; CCB, G3.3)

Communication regarding the project's verification and validation process encompasses various forms, including in-person and online meetings, social media interactions, WhatsApp messages, seminars, schematic diagrams, educational videos, and printed materials. The objectives are to present the project in a simple and didactic manner and to address any related queries. During meetings, Stakeholders are provided with advance information on the periodic verification procedures and, once the verification schedule has been established, communication with communities and stakeholders is maintained.

Before the validation and verification, the BRC team will visit key stakeholders in the project to brief them on the validation process and provide a summary of the project description.

2.3.9 Site Visit Information and Opportunities to Communicate with Auditor (VCS, 3.18.6; CCB, G3.3)

Before verification activities, communication with stakeholders, such as community leaders, is established through various channels, including phone calls, WhatsApp messages, email and, when necessary, through personal visits or radio broadcasts.

BRC has a team based in the Rio Branco and Sena Madureira region dedicated to providing logistical support and maintaining communication with local communities involved in the project. Therefore, BRC will be responsible for logistical organization and necessary travel to ensure the smooth progress of the audit, facilitating access to community representatives as required by validation/verification bodies.

2.3.10 Stakeholder Consultations (VCS, 3.18; CCB, G3.4)

Date of stakeholder consultation	21-07-2021
Stakeholder engagement process	The process of engaging all members of the Porongaba community began with sending invitations to participatory meetings in advance, followed by personal delivery of invitations. During the meetings, accessible language was used to ensure everyone understood the project's objectives. Additionally, photographic records were taken, and an attendance list was circulated during the meetings. These details can be found in (APPENDIX_ATTENDANCE-LIST-MEETING).
Consultation outcome	During the meeting, the potential risks, costs, and benefits of the project were discussed, along with a summary presented to all community members. It was emphasized that BRC fully acknowledges the communities' rights of ownership and that the project has no intention of relocating them. This specific concern was highlighted during the meeting and given due attention. Additionally, it was emphasized that the communities would incur no costs associated with the project and that areas used for cultivation would not be affected by the project.
Stakeholder input	During the meeting, the needs of the communities and their main concerns regarding the project were identified. All this information was gathered and considered to integrate the project's benefits according to budget constraints and activity prioritization. There were no changes to the project scope.

Date of stakeholder consultation	22-07-2021
Stakeholder engagement process	The process of engaging all members of the Katianã community began with sending invitations to participatory meetings in advance, followed by personal delivery of invitations. During the meetings, accessible language was used to ensure everyone understood the project's objectives. Additionally, photographic records were taken, and an attendance list was circulated during the meetings. These details can be found in (APPENDIX_ATTENDANCE-LIST-MEETING).
Consultation outcome	During the meeting, the potential risks, costs, and benefits of the project were discussed, along with a summary presented to all community members. It was emphasized that BRC fully acknowledges the communities' rights of ownership and that the project has no intention of relocating them. This specific concern was highlighted during the meeting and given due attention. Additionally, it was emphasized that the communities would incur no costs associated with the project and that areas used for cultivation would not be affected by the project.
Stakeholder input	During the meeting, the needs of the communities and their main concerns regarding the project were identified. All this information was gathered and taken into account to integrate the project's benefits according to budget constraints and activity prioritization. There were no changes to the project scope.

Date of stakeholder consultation	22-07-2021
Stakeholder engagement process	The process of engaging all members of the Palmares community began with sending invitations to participatory meetings in advance, followed by personal delivery of invitations. During the meetings, accessible language was used to ensure everyone understood the project's objectives. Additionally, photographic records were taken, and an attendance list was circulated during the meetings. These details can be found in (APPENDIX_ATTENDANCE-LIST-MEETING).
Consultation outcome	During the meeting, the potential risks, costs, and benefits of the project were discussed, along with a summary presented to all community members. It was emphasized that BRC fully acknowledges the communities' rights of ownership and that the project has no intention of relocating them. This specific concern was highlighted during the meeting and given due attention. Additionally, it was emphasized that the communities would incur no costs associated with the project and that areas used for cultivation would not be affected by the project.
Stakeholder input	During the meeting, the needs of the communities and their main concerns regarding the project were identified. All this information was gathered and taken into account to integrate the project's benefits according to budget constraints and activity prioritization. There were no changes to the project scope.

Date of stakeholder consultation	23-07-2021
Stakeholder engagement process	The process of engaging all members of the Potiguar community began with sending invitations to participatory meetings in advance, followed by personal delivery of invitations. During the meetings, accessible language was used to ensure everyone understood the project's objectives. Additionally, photographic records were taken, and an attendance list was circulated during the meetings. These details can be found in (APPENDIX_ATTENDANCE-LIST-MEETING).
Consultation outcome	During the meeting, the potential risks, costs, and benefits of the project were discussed, along with a summary presented to all community members. It was emphasized that BRC fully acknowledges the communities' rights of ownership and that the project has no intention of relocating them. This specific concern was highlighted during the meeting and given due attention. Additionally, it was emphasized that the communities would incur no costs associated with the project and that areas used for cultivation would not be affected by the project.
Stakeholder input	During the meeting, the needs of the communities and their main concerns regarding the project were identified. All this information was gathered and taken into account to integrate the project's benefits according to budget constraints and activity prioritization. There were no changes to the project scope.

Date of stakeholder consultation	27-07-2021
Stakeholder engagement process	An official letter presenting the project to the Honorable Coordinator of the Education Center in Sena Madureira/AC was drafted using appropriate language. This letter was sent via email, and subsequently, a meeting was scheduled. This letter can be reviewed in the (APPENDIX_ATTENDANCE-LIST-MEETING).
Consultation outcome	The meeting aimed to present the objectives and purpose of the project. During the gathering, there was a discussion on implementing communication channels at all levels of the executive government, as well as forming partnerships between institutions for project execution.
Stakeholder input	The considerations and contributions presented during the meeting were recorded and taken into account. All collected information will be integrated into the project according to the execution schedule and established priorities. This approach will ensure that the raised needs and concerns are adequately addressed and incorporated into the project development as planned.

Date of stakeholder consultation	27-07-2021
Stakeholder engagement process	An official letter presenting the project to the Honorable Municipal Secretary of Health in Sena Madureira/AC, Mrs. Nildete Lira do Nascimento, was drafted using appropriate language. This letter was sent via email, and subsequently, a meeting was scheduled. This letter can be reviewed in the (APPENDIX_ATTENDANCE-LIST-MEETING).
Consultation outcome	The meeting aimed to present the objectives and purpose of the project. During the gathering, there was a discussion on implementing communication channels at all levels of the executive government, as well as forming partnerships between institutions for project execution.
Stakeholder input	The considerations and contributions presented during the meeting were recorded and taken into account. All collected information will be integrated into the project according to the execution schedule and established priorities. This approach will ensure that the raised needs and concerns are adequately addressed and incorporated into the project development as planned.

Date of stakeholder consultation	29-07-2021
Stakeholder engagement process	An official letter presenting the project to the Honorable State Secretary of Education in Sena Madureira/AC, Mrs. Maria do Socorro Neri Medeiros de Souza, was drafted using appropriate language. This letter was sent via email, and subsequently, a meeting was scheduled. This letter can be reviewed in the (APPENDIX_ATTENDANCE-LIST-MEETING).
Consultation outcome	The meeting aimed to present the objectives and purpose of the project. During the gathering, there was a discussion on implementing communication channels at all levels of the executive government, as well as forming partnerships between institutions for project execution
Stakeholder input	The considerations and contributions presented during the meeting were recorded and taken into account. All collected information will be integrated into the project according to the execution schedule and established priorities. This approach will ensure that the raised needs and concerns are adequately addressed and incorporated into the project development as planned.

Date of stakeholder consultation	30-07-2021
Stakeholder engagement process	An official letter presenting the project to the Honorable President of IMC/AC, Mr. Érico Maurício Pires Barbosa, was drafted using appropriate language. This letter was sent via email, and subsequently, a meeting

	was scheduled. This letter can be reviewed in the (APPENDIX_LIST-OF-LETTERS).
Consultation outcome	The meeting aimed to present the objectives and purpose of the project. During the gathering, there was a discussion on implementing communication channels at all levels of the executive government, as well as forming partnerships between institutions for project execution.
Stakeholder input	The considerations and contributions presented during the meeting were recorded and taken into account. All collected information will be integrated into the project according to the execution schedule and established priorities. This approach will ensure that the raised needs and concerns are adequately addressed and incorporated into the project development as planned.

This project has not yet received public comments. The public consultation process will be opened following the submission of this project design document, which is expected by the end of March/ beginning of April.

2.3.11 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

Ongoing communication and consultation were established through a variety of channels, including face-to-face and digital meetings. Digital inclusion is part of the project development plan, and to ensure this, communication towers have been installed in community areas Figure 19. These towers provide frequent internet access for community members, enabling effective and regular contact with PP's team through various platforms such as WhatsApp, email, and virtual meetings. This approach ensures that communication is accessible and seamless, allowing all parties involved to stay connected and informed throughout the project lifecycle.

Additionally, annual public meetings are planned to communicate the project monitoring results and the implementation of activities. During these events, communities will have new opportunities to influence the project design and its performance during the monitoring period.



Figure 19. Communication tower.

2.3.12 Stakeholder Consultation Channels (CCB, G3.5)

The social engagement protocol developed by BRC ensured that all consultations and participatory processes were conducted directly with the communities and other stakeholders. The direct consultation and involvement of communities in participatory processes to date can be evidenced in this document through photographic records (Figs. 16, 17, 19).

2.3.13 Stakeholder Participation in Decision-Making and Implementation (VCS, 3.18, 3.19; CCB, G3.6)

During the verification phase, stakeholder participation in decision-making is ensured through the BRC Social Engagement Protocol (APPENDIX_SOCIAL-ENGAGEMENT-PROTOCOL). The protocol is divided into four distinct phases. In the second phase, a detailed diagnosis is conducted with stakeholders to identify and map their demands. During this process, participatory planning is developed to understand prioritized demands and enable their execution. The primary objective of this stage is to determine which actions can effectively benefit and support the territories and communities involved in the project.

Territorial analysis and participatory planning play a crucial role in decision-making regarding the necessary measures to reinforce and establish relationships with different groups in the territory. Among these groups, there are some specific ones, such as women and youth, who require special attention to effectively participate in decision-making. In this sense, the participatory planning intends to deepen its understanding of territorial reality to create strategies that ensure the deep participation of all community groups.

This approach aims not only to strengthen these groups individually but also to contribute to the overall empowerment of the community. By understanding the specific needs and challenges faced by women, youth, and others, it is possible to implement more effective strategies to promote inclusion, equality, and sustainable community development.

Based on these guidelines, meetings were held with communities and specific groups of young people and women, aiming to understand the specific demands of each group, as illustrated in Figure 20.



Figure 20. Information meetings and women's groups

2.3.14 Anti-Discrimination Assurance (VCS 3.19; CCB, G3.7)

BRC supports its anti-discrimination approach through internal policies and regulations, including the company's Bylaws (APPENDIX_BYLAWS), its Code of Ethics and Conduct (available in APPENDIX_ETHICS-AND-CONDUCT-CODE) and educational initiatives, such as activities and lectures on anti-discrimination for employees (see APPENDIX_HUMAN-RESOURCES-ACTIVITIES). In addition, the BRC Code of Ethics stipulates that all the company's employees, directors and partners comply with all relevant laws relating to the promotion of equality and the prevention of discrimination.

The owner partner must sign a Conservation Agreement (see APPENDIX_LEGAL-AGREEMENTS), which defines that both parties must adhere to all applicable laws in the implementation and management of the project. This agreement reinforces the joint commitment to avoid any form of segregation, including discrimination based on gender, race, religion, sexual orientation or other personal aspects.

Another strategy adopted to prevent such problems was the implementation of a direct channel between the communities and BRC's employees. This channel was established as a means of

facilitating communication to receive complaints and denunciations related to these challenges (see more on ALO-BRC_COMMUNICATION-CHANNEL).

2.3.15 Feedback and Grievance Redress Procedure (VCS, 3.18.4; CCB, G3.8)

Development process	For complaint resolution, BRC adopts a direct and continuous communication approach with stakeholders and communities, using phone calls and messages to ensure that the exchange of information, feedback, and resolution of complaints occurs smoothly and efficiently. BRC has developed the direct communication channel, (Alô brCarbon!), to handle complaints promptly. The flow of information received until resolution can be tracked in the system. (APPENDIX_FLOWCHART-COMPLAINTS).
Grievance redress procedure	Annually, the BRC team meets with each participating community of the project to gather feedback and suggestions on the activities conducted, as well as to collect requests for corrections or the introduction of new initiatives. During these meetings, objective questionnaires (x) are used to assess the effectiveness of the activities. The process of receiving, listening, responding, and resolving complaints is divided into three phases: Friendly negotiation: Complaints are carefully analyzed, and PP is commit to providing a response, either orally or in writing, within one week, using channels such as email, WhatsApp, or in-person meetings if necessary. Mediation by an impartial intermediary: If the BRC team cannot resolve a complaint in a reconciliatory manner, the dispute is referred to a third-party independent mediator specialized according to the specific nature of the conflict. In cases of land disputes, the responsible state or federal body is contacted. In conflicts between communities, mediation may be conducted by the BRC or by representatives of community associations, adapting to the particularities of the conflict. Arbitration: If mediation proves ineffective, the case is referred to the local jurisdiction of the project for judicial resolution. This conflict resolution can be seen in (APPENDIX_FEEDBACK-GRIEVANCES).

2.3.16 Accessibility of the Feedback and Grievance Redress Procedure (VCS, 3.19; CCB, G3.8)

When complaints and grievances are delivered to the PP, the strategy to turn it accessible and public to all stakeholders is multiple, but not ordered, depending upon the complexity on the issues presented. It can be a call to extraordinary meetings, where the BRC staff will try to make it clear and more understandable for everybody, or direct contact with the complainers, or through the strategy of publish periodically a newsletter about the whole project.

These strategies create conditions to through these strategic approaches allows new objectives to be defined together with the community, beyond to solve the problems pointed. Thus, these instruments will focus on turning the concerns expressed in the complaints into tangible opportunities, promoting a collaborative dialog to strengthen the partnership between the team and the community. The aim is not only to resolve the issues raised, but also to cultivate a deeper understanding of the

community's needs and aspirations, guiding the definition of goals and the implementation of sustainable solutions.

2.3.17 Worker Training (VCS, 3.19; CCB, G3.9)

The project adopts a comprehensive approach to identify and address the community's demands, starting with a social diagnosis that investigates local needs and dynamics. This diagnosis serves as the basis for providing training and capacity-building courses tailored to the specific demands identified. Moreover, it acts as a starting point for developing programs and initiatives aimed at improving the quality of life and promoting sustainable development within the community.

It is essential that all community members are informed about these procedures, adopting an inclusive approach in terms of gender, prioritizing the involvement of women and vulnerable or marginalized individuals. In addition to the training demanded by the communities, the project recognizes other needs such as field guides, cooks, boatmen, among others, and is willing to provide support to those who demonstrate interest and capability to meet the project's demands.

To ensure stability and continuity in the project's execution, periodic training programs will be implemented. These measures aim to ensure that all involved parties are fully informed and equipped to contribute to the community's and project's development. The project is committed not only to identifying the community's needs but also to empowering and strengthening it, fostering an environment of collaboration and sustainable growth that benefits all involved stakeholders.

2.3.18 Community Employment Opportunities (VCS, 3.19.13; CCB, G3.10)

Based on previously identified demands and the appreciation of qualified professional development, BRC commits not only to advertise available positions and their main requirements, but also to encourage active participation from the local community in the selection processes and job opportunities in the region.

In addition to advertising, BRC pledges to establish a list of qualified and available community members to be accessed and hired whenever there is a demand for project-related activities. This list will be a key tool to promote economic and social inclusion among community members.

Through this proactive approach, BRC not only provides job opportunities, but also invests in the sustainable growth of the region, strengthening community ties and contributing to the development of a more equitable and dynamic economy.

2.3.19 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

Risk analysis is fundamental to guaranteeing the safety of workers potentially affected by project activities, and this analysis was based on the activities carried out in the execution of the project.

This procedure is essential to guarantee the implementation of appropriate safety measures to protect workers during the execution of activities. Before each worker is hired, training, orientation and information talks will be held, highlighting the potential risks associated with the activity. In addition, they will be provided with the necessary safety equipment so that they can carry out the work with proper protection. Each job opportunity will undergo an individualized assessment, considering the specific characteristics of the job, which will guarantee a comprehensive assessment process adapted to the particularities of each role.

It is important to note that BRC provides all its employees with life insurance that covers the worker monthly, providing security not only for the workers, but also for their families. This insurance is an additional measure to ensure that workers are protected in the event of accidents or incidents during the performance of their duties.

By implementing these measures, BRC demonstrates its commitment to the safety and well-being of its employees, ensuring that they can carry out their activities with peace of mind and confidence, knowing that they are protected and supported by the company.

2.4 Management Capacity

2.4.1 Project Governance Structures (CCB, G4.1)

The Uákiry REDD+ AUDD Grouped Project has three types of governance structures, such as shown below on Figure 21.

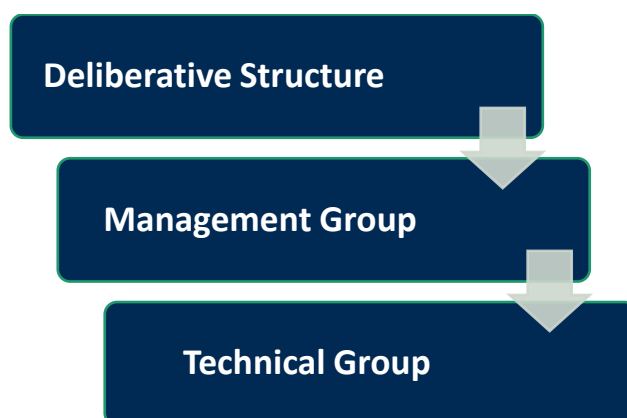


Figure 21 - Governance structure.

The Deliberative Structure bunch contains BRC, represented by Yaco2 Gestão and BRC, as project's proponent. This bunch are responsible for project decision taking.

The Management Group contains the BRC team and are responsible for receiving prior Deliberative Group decisions, designing actions, planning, and guiding the team and Technical Group on developing and accomplishing project process.

The Technical Group contains the BRC team, contractors and might around communities representant, who are responsible for project activities execution such as field monitoring and visits, studies, training, and reporting.

2.4.2 Required Technical Skills (VCS, 3.19; CCB, G4.2)

BRC staff team has more than fifteen years of experience in AFOLU carbon project, including methodological tools, community engagement, biodiversity assessment, carbon measurement and monitoring, besides PD and MR elaboration.

The skills necessary to implement the project are related to GIS, spatial modeling, forest inventory, biomass inventory, social engagement, sustainable development projects implementation, biodiversity surveys and assessment, land use change monitoring, biodiversity monitoring, social aspects monitoring. BRC complete team is listed and described in (APPENDIX_REQUIRED-TECHNICAL-SKILLS).

2.4.3 Management Team Experience (VCS, 3.19; CCB, G4.2)

Not necessary. As described in the previous section, BRC employees have all the necessary skills to conduct the Uákiry REDD+ AUDD Grouped Project, from primary data collection to the preparation of PDs and MRs, social engagement, implementation, and monitoring of project activities.

2.4.4 Project Management Partnerships and Team Development (VCS, 3.19; CCB, G4.2)

Not necessary. As mentioned above, BRC employees have all the essential skills and knowledge to conduct the Uákiry REDD+ AUDD Grouped Project, ranging from primary data collection to the design, implementation, and monitoring of project activities.

2.4.5 Financial Health of Implementing Organization(s) (CCB, G4.3)

BRC's financial data is sensitive and confidential information. To ensure the integrity and security of this data, it is essential that it is only shared with relevant stakeholders by signing a non-disclosure agreement. This agreement stipulates reciprocal confidentiality obligations, ensuring that such information is adequately protected and that access to it is restricted to authorized individuals only.

2.4.6 Avoidance of Corruption and Other Unethical Behavior (VCS, 3.19; CCB, G4.3)

The BRC directors are not involved in any form of corruption, which is enforceable by the company's social statue and by brCarbon's Ethics and Conduct Code (APPENDIX_ETHICS-AND-CONDUTE-CODE).

Any kind of involvement, direct or indirectly, in bribery, embezzlement, fraud, favoritism, cronyism, nepotism, extortion, and collusion are discouraged and disliked by Conduct Code and are punishable by expulsion or other measurements.

2.4.7 Commercially Sensitive Information (VCS, 3.5.2 – 3.5.4; CCB Rules, 3.5.13 – 3.5.14)

For all project activities, commercially sensitive information shall be considered to be any contract signed between BRC and landowners, commercial, financial, business, scientific, technical or other information, the disclosure of which could reasonably be expected to result in a material financial loss or gain, compromising the contractual terms, agreements or other negotiations declared by the project proponent.

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

The Project proponent is committed to meet all applicable Brazilian laws and applicable regulations. On this sense, BRC follow consultant lawyers to observe local laws and applicable regulations.

The following laws, statutes or regulatory frameworks are the most relevant to the project activities. The presentation order will follow the governance level, from national to municipality laws, other regulatory frameworks, and follow chronology.

2.5.1.1 Federal Legal Instruments

In a wide overview of legal aspects, applied to all PAIs included in this grouped project, the project observes the principles established in the Federal Constitution (1964), as per article 225, by contribution to ecologically well-balanced environment and article 224, paragraph 1, items (I) and (III) by contribution to conservation and restoration of essential ecological processes, while supporting the preservation of attributes beyond the legal environment requirements. Along with the previously presented articles, the Federal Law number 9,985 from 2000, regulates the Federal Constitution's art. 225, § 1, items I, II, III and VII, and institutes the National System of Nature Conservation Units (SNUC) and makes other provisions.

The GPD as whole is encompassed by principles established in the National Environmental Policy - law number 6,938 from August 31st, 1981, which declares the objectives of conservation, improvement and recovery of environmental quality that is conducive to life, having among its principles the protection of ecosystems with the conservation of relevant areas (article 2, IV) and the protection of areas threatened by degradation (item IX). Another important law for the development of the project is the Federal Law number 12.305, from August 2nd of 2010, that institutes the National Policy of Solid Residue.

Regarding the land use national policy, the most important regulatory framework is the Vegetation Code, law number 12,651, from May 2012. This law states not only the forest category that must be conserved inside the private properties, but also the maximum area of native vegetation that can be converted in other land uses for economic purposes.

The project is also in straight line with the Federal Law no. 12,187, of December 29th, 2009, which instituted the National Climate Change Policy, the project is also in line with Law 14,119 / 2012, which defines concepts, objectives, guidelines, actions, and criteria for the implementation of the National Policy for Payment for Environmental Services (PNPSA), institutes the National Register of Payment for Environmental Services (CNPSA) and the Federal Program for Payment for Environmental Services (PFPSA) and provides on payment contracts for environmental services. Furthermore, these laws go according to the National Environmental Council / Federal Environmental Ministry resolution number 237, from 2017, that provides for environmental licensing.

According to the Law 14,119/2021 carbon sequestration is considered an ecosystem service with relevant benefits to society in terms of maintenance, recovery, or improvement of environmental conditions (Art.2, § II) and the Federal Program for Payment for Environmental Services (PFPSA) will promote actions of maintenance of areas covered by native vegetation that would be subject to suppression authorization for alternative and economic use (Art. 7, § VII).

The project is also based on the Federal Laws number 5.197, from January 3rd of 1967, that regulates fauna protection and provides other provisions, and number 9.605, from February 12nd of 1998, that provides for criminal and administrative sanctions derived from the environment's harmful conduct and activities, and other provisions.

Finally, despite not encompassing indigenous areas, or natural areas used for these people, has applied to the project area residents, the consultation free, prior, and informed consent approach (FPIC), as per the Convention no. 169 of the ILO and the United Nations Declaration on the Rights of Indigenous People. This population participated from the project assembling since its beginning, as presented in the community section.

2.5.1.2 Acre Legal Instruments

State Law number 2,308, from 22nd October 2010 - Creates the State System of Incentives for Environmental Services - SISA, the Incentive Program for Environmental Services - ISA Carbon, and other Environmental Services and Ecosystem Products Programs of the State of Acre and provides for other measures.

State Law number 3,496, from 2nd August 2019 - Establishes the Labor Fund and creates the Council State of Labor, Employment, and Income.

State Law number 4,076, from 28th December 2022 - Provides for the creation of the State Week for the Valorization of Health Workers.

State Law number 1,693, from 21st December 2005 - Creates the Agroforestry Poles and Agroforestry Backyards Programs (PQA) and authorizes the Executive Branch to grant, under resolute condition, concession of real right to use rural public lands located in agroforestry poles, for the implementation of the Sustainable Development Policy of the State of Acre.

State Law number 2,841, from 8th January 2014 - Authorizes the Executive Power to dispose, through assignment or donation conditioned with charges, movable assets destined to the strengthening of family agriculture.

State Law number 1,373, from 2nd March 2001 - Creates the Land Institute of Acre – ITERACRE and take other measures.

State Law number 1,904, from June 5th, 2007, which instituted the Ecological Economic Zoning of the State of Acre - ZEE / AC and the guidelines of the State Policy for the Valorization of Forest Environmental Assets.

State Law number 2,308, from October 2010 which instituted the State System of Incentives for Environmental Services – SISA and the Incentive Program for Environmental Services - ISA Carbon.

State Law number 2,693, from 17 th January 2013 which follows the federal definition for the CAR, with the purpose of integrating environmental information from rural properties and possessions, composing the database for control, monitoring, environmental planning and economic, in addition to combating deforestation and burned. In addition, amends Laws nos. 1904, of June 5, 2007, that “Institutes the Ecological Zoning Economic State of Acre – ZEE”; and 2025, from October 20, 2008,” which creates the Program State Certification of Production Units Relatives of the State of Acre” and gives other providences.

Publication of **State Decree No. 7,734/2014** that regulates the CAR in the State of Acre and creates the CAR and PRA Technical Management Office, a body shared between the Secretary of State for the Environment - SEMA and the Institute of Environment of Acre – IMAC.

State Law number 3,278, from 20th July 2017 - Adds and amends provisions of state Laws nº 1,373, of March 2, 2001; 2,840 of January 8, 2014; and 1,693 of December 21, 2015.

State Law number 3,349 from 18 th December 2017 Establishes the Environmental Regularization Program for rural properties and possessions within the State of Acre.

State Law number 3,973, from 25th July 2022 - Ratifies the Protocol of Intentions signed between the States of Acre, Alagoas, Amapá, Bahia, Distrito Federal, Espírito Santo, Goiás, Maranhão, Mato

Grosso, Mato Grosso do Sul, Minas Gerais, Pará, Paraíba, Paraná, Pernambuco, Piauí, Rio Grande do Norte, Rio Grande do Sul, Roraima, Santa Catarina, São Paulo, Sergipe and Tocantins, for the constitution of the Interstate Consortium with the objective of promoting the confrontation with the adverse effects of climate changes in Brazil.

State Law number 3,963, from 8th July 2022 - Classifies the Institute of Agricultural and Forestry Defense (IDAF), as of public interest and essential service in the health area.

2.5.1.3 Sena Madureira Municipality Legal Instruments

Municipal Law number 133, from 23rd September 1999 - Provides for the Municipal Environmental Policy and establishes the Municipal Environmental System.

Municipal Law number 746, from 29th March 2023 - Provides for the Single System of Social and Political Assistance.

Municipal Law number 643, from 29th March 2019 - Establishes general rules for bidding and contracting public-private partnerships within the scope of public administration.

2.5.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18.2; CCB, G3.11)

The project proponent is committed to fully complying with all Brazilian laws and regulations pertaining to workers' rights. In addition, it will ensure that all workers are duly informed of their rights, as established in Brazilian labor legislation, prior to their employment.

The following is a list of Brazil's relevant laws and regulations applicable to worker's rights:

The Brazilian Constitution, Chapter II-Social Rights, Articles 7- 11 which addressed: i) Minimum wage, ii) Normal working hours, iii) Guidance on vacation and weekly leave iv) Guidance on maternity and paternity leave v) Recognition of collective bargaining vi) Prohibition of discrimination¹⁸. In addition to the Constitution, there are two additional decrees related to Brazilian labor laws.

Labor Federal Law (Consolidate of Working Laws (CLT)) DECRETO-LEI N. ° 5.452, DE 1° DE MAIO DE 1943 ¹⁹. This decree gives more clarification on the subjects: i) Hourly, daily, weekly, and monthly work hours, ii) Employment of minors and women, iii) Establishes a minimum wage, iv) Worker safety and safe working environments, v) Defines penalties for employer's non-compliance, vi) Establishes a judicial work-related process for addressing all worker related issues. FEDERAL LAW N° 5,889, June 8th of 1973 (Establishes Regular Norms for Rural Workers). This is a complementary law to the 1943 decree, once prior to 1973, rural workers did not have the same rights as urban workers. In 1973, this law was established to specify the equality between urban and rural workers, along with compensation for overtime.

Complementary Law 123/2006 of December 14, 2006. Establishes the National Statute of Microenterprise and Small Business; change devices of the Laws in 8,212 and 8,213, both dated July 24, 1991, of the Consolidation of Labor Laws - CLT, approved by Decree-Law no. 5,452, of 1st of May 1943, of Law no. 10,189, of February 14th of 2001, of the Complementary Law in the 63, of January 11, 1990; and revokes the Laws in the 9,317, of the 5th of December 1996, and 9841 of October 5, 1999. Art. 3 For the purposes of this Complementary Law, are considered micro-enterprises or companies small business company, simple society, the individual company of limited liability and the entrepreneur who refers to art. 966 of the Law in 10,406 of the 10th of January 2002 (Civil Code), duly registered in the Registry of Merchant Companies or in the Civil Registry of Legal Entities.

2.5.3 Human Rights (VCS, 3.19)

BRC acknowledges and fully respects the rights of community members residing within the project areas. The project does not intend to interfere with the way of life of these community members, nor to remove them from the areas they inhabit.

Although the project areas do not accommodate indigenous peoples, BRC fully recognizes and respects their rights. The company is committed to ensuring that all activities conducted uphold the human, cultural, and territorial rights of local communities, promoting a dialogue-based and cooperative approach.

BRC and PP are committed to working in partnership with local communities, valuing their traditions, knowledge, and ways of life, and ensuring they are fairly and sustainably benefited by the ongoing projects.

To guarantee the respect to Human Rights by all the BRC collaborators, there are some internal norms and practices that need to be followed, as Ethic Code (APPENDIX_ETHICS-AND-CONDUTE-CODE), Social Statute (APPENDIX_SOCIAL-STATUTE), and the internal policies of the Human Resources sector (APPENDIX_HR-INTERNAL-POLICIES).

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

The project is committed to preserving the traditions and way of life of community members residing within the project areas. The project does not intend to interfere with local cultures or the customs of the communities. BRC is committed to preserving the traditions and way of life of community members residing within and around the project areas, without intending to interfere with the culture or customs of the involved communities.

In addition to respecting and preserving existing traditions, BRC seeks to promote the recognition of traditional customs as important assets for the sustainable development of the region in collaboration with local communities. PP's approach prioritizes open dialogue and cooperation, always seeking to benefit communities and respect local cultures and identities.

Anyway, there is no indigenous people inside the project area, and in its immediate neighborhood.

2.5.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

The Uákiry Grouped Project REDD+ AUDD initially consists of **4 rural properties** belonging to the **Junior Galvane Batista** and **BATISTA E CIA LTDA** company. This project has multiple proponents, they are **BRCarbon** and **YACO2** where the first organization has overall control and responsibility for the project, and the second has the project ownership, has the legal right to control and operate project activities, as previously demonstrated above.

The project will be developed in these properties, which are called project activity instances (**PAI# 1 - PAI# 4**). The project areas are in the municipality of Sena Madureira, and PAI #1 is located on the border of the municipalities of Sena Madureira and Rio Branco.

BRC, is in charge of developing and implementing the project activities. It has conducted a rigorous land tenure due diligence with the legal staff team and all the proprieties meet the internal requirements and the VCS and CCB standards regarding proof the rights of the carbon credits. It's important to highlight that exists an internal check list that requires to the landowner a total of fifteen documents and each propriety has a specific legal report according to the juridic feasibility, all this information is available as project annex.

As a grouped project, new areas must be added to the project in the future verifications, once it is proven the land regularization is fully documented.

The geographic region of this grouped project is in the state of Acre, which is composed by private and public areas. Public lands are subdivided into indigenous lands, accounting for approximately 14.85% of the state; protected areas, covering about 10.49%; sustainable use conservation units, representing 20.49%; and settlements, with an extension of 10.50%. Private areas are distributed among those registered with the National Institute for Colonization and Agrarian Reform (INCRA) and the Rural Environmental Registry (CAR), comprising 39.24%. The hectare values assigned to each land category correspond to the calculated areas from shapefiles provided in public and governmental databases, such as those supplied by MMA, FUNAI, and INPE. Percentage values were calculated based on the total area of the state. The sum of values for each category may exceed the total area of the state due to overlaps between conservation units. This territorial division of the state of Acre can be seen in Figure 22.

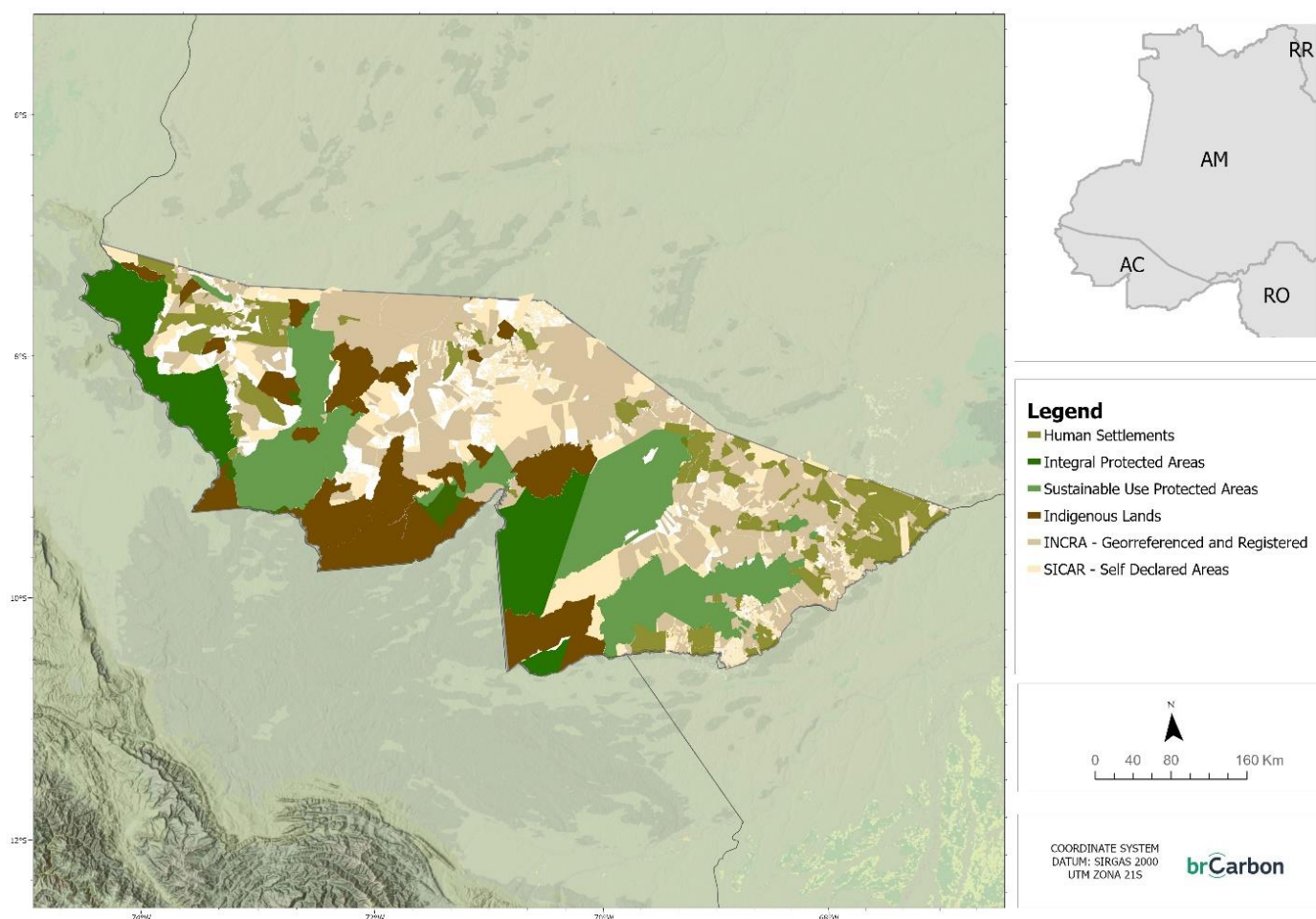


Figure 22. Land distribution in the state of Acre.

Table 15. Land Category

Land Category	Area (ha)	%
Conservation Units with Integral Protection	1,722,236.75	10.49%
Conservation Units with Sustainable Use	3,363,746.95	20.49%
Human Settlements	1,723,561.91	10.50%
Indigenous Lands	2,437,710.91	14.85%
Protected Area	9,247,256.52	56.33%
Total Private Areas	6,441,784.609	39.24%
Acre State	16,417,400.16	100%

BRC has conducted a proper investigation, in collaboration with the legal team, into the ownership of the land in the project areas, ensuring that it complies with the internal requirements and the rules

of the CCB and VCS, in order to guarantee the rights to carbon credits. It is important to note that landowners are asked for a list of documents, which can be found in “ANNEX_LAND-OWNERSHIP”.

2.5.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

The BRC demonstrates a commitment to recognizing and respecting the traditional communities present in the area. As part of its initiatives to strengthen the rights and recognition regarding land use by these communities, the decision was made to exclude from the project scope areas currently used for pasture and agriculture by community members. This measure aims not only to protect the livelihoods of these communities but also to ensure that their land use rights are fully respected and preserved. The exclusion of these areas was previously mapped, and the DETER-B data was used as an indicator of degradation and deforestation. All new project activities will be mapped using the same approach to ensure that the customary rights of the communities allow access to the natural resources they rely on.

Another measure to recognize the rights of traditional communities was taken by the legal owner of the area, involving the donation of land parcels to the families residing in the Porongaba PAI#01. Documents regarding these donations and the allocation of parcels per family are available for consultation in the (ANNEX_LAND-TENURE). This initiative not only strengthens the bond between the community and the landowner but also aims to deter potential encroachers from areas contributing to deforestation and the displacement of local communities. In addition to securing land ownership rights for beneficiary families, this measure plays an important role in acknowledging their rights to the land.

2.5.7 Free, Prior and Informed Consent (VCS, 3.18; CCB, G5.2)

During participatory workshops between July 21st and 25th, 2021, as part of the second phase of social engagement, prior consent was obtained to implement the project. BRC provided a Project Concept Summary, discussed costs, risks, and benefits, collected suggestions, and answered community questions, including a dedicated session for women. While workshops in Potiguar, Palmares, and Katianã were unanimously approved, those in Porongaba were approved by the majority. BRC acknowledges distrust due to the project's new and complex nature, understanding concerns of non-approved groups. BRC recognizes that the FPIC process is ongoing throughout the project lifecycle and intends to hold further meetings.

Description of process for obtaining consent

The consent for the project was obtained through meetings and participatory workshops with the community of Porongaba. Initially, the project's objectives were explained, along with the potential benefits for the involved communities. Subsequently, participants had the opportunity to express their concerns regarding their rights and how they would be affected by the project. A survey of all identified demands was conducted, and later, the community voted on the project's implementation and their desire to be beneficiaries of it.

Outcome of FPIC process	During the vote to obtain project consent, some members of the community expressed concern about it. BRC understands that a certain level of distrust is natural, especially regarding a new and complex initiative for the communities, and that this distrust sometimes manifests as disapproval, particularly concerning the assurance of their lands and relocation. BRC clarified all doubts and ensured that no local residents would be relocated, and there was also no obligation to accept the project and become beneficiaries of it. The vote was approved by a large majority.
Description of process for obtaining consent	The consent for the project was obtained through meetings and participatory workshops with the Katianã community. Initially, the project's objectives were explained, along with the potential benefits for the involved communities. Subsequently, participants had the opportunity to express their concerns regarding their rights and how they would be affected by the project. A survey of all identified demands was conducted, and later, the community voted on the implementation of the project and their desire to be beneficiaries of it.
Outcome of FPIC process	BRC clarified all doubts and ensured that no local resident would be relocated, while also making it clear that there was no obligation to accept the project. As a result, the community unanimously voted in favor of implementing the project.
Description of process for obtaining consent	The consent for the project was obtained through meetings and participatory workshops with the Potiguar community. Initially, the project's objectives were explained, along with the potential benefits for the involved communities. Subsequently, participants had the opportunity to express their concerns regarding their rights and how they would be affected by the project. A survey of all identified demands was conducted, and later, the community voted on the implementation of the project and their desire to be beneficiaries of it.
Outcome of FPIC process	BRC clarified all doubts and ensured that no local resident would be relocated, while also making it clear that there was no obligation to accept the project. As a result, the community unanimously voted in favor of implementing the project.
Description of process for obtaining consent	The consent for the project was obtained through meetings and participatory workshops with the Palmares community. Initially, the project's objectives were explained, along with the potential benefits for the involved communities. Subsequently, participants had the opportunity to express their concerns regarding their rights and how they would be affected by the project. A survey of all identified demands was conducted, and later, the community voted on the implementation of the project and their desire to be beneficiaries of it.
Outcome of FPIC process	BRC clarified all doubts and ensured that no local resident would be relocated, while also making it clear that there was no obligation to accept

the project. As a result, the community unanimously voted in favor of implementing the project.

2.5.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

Process used to design the benefit sharing plan	The project conducts regular participatory workshops and workshops with the beneficiary communities. During these meetings, community members present their demands, which are evaluated in accordance with the project's priorities. The sharing of benefits is transformed into actions requested by the communities themselves in participatory meetings.
Summary of the benefit sharing plan	All actions involving benefits transformed into actions for the communities are communicated during regular participatory meetings. Any action or information that groups wish to keep private is agreed upon during these regular meetings.
Approval and dissemination of benefit sharing plan	In all meetings, summaries of the project monitoring results are provided. These reports contain all the actions that benefited the communities, which were agreed upon in participatory meetings.

2.5.9 Property Rights Protection (VCS, 3.18, 3.19; CCB, G5.3)

The project's activities do not involve the displacement of people, as demonstrated in this document by the detailed description of the planned actions, as well as by direct on-site observations and interviews with community members. On the contrary, the project has been designed to respect the traditional communities' areas of use, as evidenced by the boundaries of the project area, which exclude adjacent pasture and forest areas.

2.5.10 Illegal Activity Identification (VCS, 3.19; CCB, G5.4)

Illegal deforestation is a reality in the eastern region of the state of Acre. It is therefore extremely important that conservation actions focus on preventing illegal deforestation, guaranteeing the effective protection of the region's natural resources and biodiversity. To this end, it is essential to implement strict inspection and monitoring measures, as well as promoting sustainable land use practices and encouraging the engagement of local communities in environmental preservation.

The process of identifying threats and risks to the preservation of carbon stocks will be carried out during the initial analyses in a GIS environment. Using monitoring data sources such as the Amazon Deforestation Calculation Program (PRODES), the General Coordination of Earth Observation (DETER), the Annual Mapping of Land Cover and Land Use in Brazil Project (MAPBIOMAS) and remote sensing information (including Landsat, Sentinel and Google Earth images), it will be possible to map the dynamics of land use across the entire rural property. This methodology for monitoring illegal activities in the area is described in detail in (APPENDIX_BRC-SOP-MONITORING).

2.5.11 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

During the stages of social engagement, land conflicts were identified at the Porongaba farm, referred to as PAI#01. These conflicts are characterized by the illegal sale and purchase of plots within the farm, with most of these transactions involving trespassers, acting as a significant driver of deforestation in the region. Due to this illegal occupation, there is being growing a community in the project area called Pau Forte, where are concentrated the bad faith squatters, as can be seen in the map (APPENDIX_PAU-FORTE-MQP). This new group is becoming really organized and they have even founded a representative association to try to make them legitim in the territory, and to weaken the representativity of the landowner and the project itself. The Rural Producers Association of Ramal Nova Olinda – ASPROLINDA, was founded on May 17th, 2023, as can be seen at (<https://cnpj.biz/51283163000160>).

To address these identified conflicts on-site, the owner opted to donate plots to families residing there, particularly those identified as traditional communities that have been residing in the area for generations, the bona fide squatters. This measure aims to acknowledge traditional occupation and legitimize these communities while removing trespassers from the land. The formalization and process of plot donations can be reviewed in (ANNEX_LOT-DONATIONS-PORONGABA).

Another measure to combat deforestation and the invasion of areas was the implementation of a coordinated command and control action, requested by the landowner. This operation has been jointly conducted by law enforcement agencies, including the Federal Police and the Military Police, as well as environmental agencies such as IBAMA and IMAC. The operation and its outcomes can be viewed in (ANNEX_CONTROL-ACTION-REPOSSESSION).

2.5.12 Approvals (CCB, G5.7)

The UÁKIRY REDD+ AUDD Grouped Project is a collaborative initiative aimed at developing conservation projects to prevent illegal deforestation in the region.

At the federal level, there is a technical note from the Ministry of the Environment (MMA) recognizing the role of REDD+ projects in the voluntary market, through the Secretariat of the Amazon and Environmental Services and the Department of Forest Conservation and Environmental Services. In this official document providing guidance on the Federal Government's Forest Plus Initiative (Floresta+), the voluntary market is referred to as follows: "Considering the full potential for mitigation in the land use change and forestry sector in Brazil, this context presents a valuable opportunity to attract significant investments in projects that operate directly in the territory, providing environmental monitoring, surveillance, and protection services in areas, transforming the local reality and promoting conservation and restoration of native vegetation on a large scale, an essential contribution to reducing illegal deforestation, supporting sustainable development, and ensuring the long-term conservation of Brazilian forests."

Thus, as the GP fulfils all the requirements of the technical note described above, the grouped project is perfectly aligned with the host country's conservation strategy. On a state scale, the BRC is strengthening communication between the different State Agencies.

It is important to emphasize that the grouped project is structured on the voluntary market in private areas, where approval for the implementation of project activities depends solely on the communities and the landowner. This means that it is not necessary to obtain any national, state or municipal approvals to start its activities. However, BRC adopts a direct approach to informing and inviting executive bodies to join the initiative and contributes, within its remit, to strengthening the project's positive impacts on communities and the environment.

2.5.13 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB G5.9)

2.5.13.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes ☒ No

2.5.13.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

2.5.13.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

2.5.14 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.5.14.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

2.5.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.5.14.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”?

☐ Yes ☒ No

2.6 Additional Information Relevant to the Project

2.6.1 Leakage Management (VCS, 3.11, 3.15)

To manage project leakage, it is essential to adopt a comprehensive management plan to address the different ways in which leakage can occur. Leakage mitigation actions will be related to rigorous monitoring and assessment to identify potential leakage areas and understand land use patterns in areas adjacent to project zones.

It is essential to actively involve local communities and relevant stakeholders in the design and implementation of conservation strategies. This includes establishing partnerships with landowners and local communities to promote sustainable land use practices and provide viable economic alternatives to deforestation activity.

To prevent the geographical displacement of deforestation activities, control and monitoring of areas adjacent to conservation zones will be carried out. To deal with land use substitution, economic incentive strategies will be implemented to promote sustainable forest management practices and the conservation of natural ecosystems.

To mitigate the increase in demand for natural resources, it is important to promote economic diversification and the development of sustainable value chains that reduce dependence on destructive activities such as subsistence farming or illegal logging.

To summarize, the management plan involves monitoring measures, community engagement and regional coordination to ensure that conservation efforts are effective and sustainable in the long term.

2.6.2 Further Information

All relevant information regarding the project has already been described in the preceding sections.

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology (VCS, 3.1)

The project is based on the following methodology and tools.

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0015	VM0015 Methodology for Avoided Unplanned Deforestation	1.2
Tool	VT0001	VT0001 Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities	3.0
Manual	-	Social and Biodiversity Impact Assessment (SBIA) Manual for REDD+ Projects – Core Guidance for Project Proponents	Part 1
Tool	-	AFOLU Non-Permanence Risk Tool	4.2

3.1.2 Applicability of Methodology (VCS, 3.1)

Reference ID/Title	Applicability condition	Justification of conformance
VM0015 / Methodology for Avoided Unplanned Deforestation	a) Baseline activities include planned or unplanned logging for timber, fuel-wood collection, charcoal production, agricultural or grazing activities as long as the category is unplanned deforestation according to the most recent <i>VCS Standard</i> .	Uákiry REDD+ AUDD GDP baseline scenario encompasses unplanned deforestation for cattle and agriculture purposes, as well as timber and fuelwood
VM0015 / Methodology for Avoided Unplanned Deforestation	b) Project activities include one or a combination of the eligible categories defined in the description of the methodology scope (table 1 and Appendix 1 of the VM0015).	According to Table 1 of the VM0015, the project activities fit in scenario “D”, Avoided Deforestation with Logging in the Baseline and Project Cases (Figure “D” in Appendix 1, page 159).
VM0015 / Methodology for Avoided Unplanned Deforestation	c) The project area can include one or a combination of forest types, such as, but not limited to, old growth forest, degraded forest, secondary forests, planted forests and agroforestry systems meeting the definition of “forest”.	The first project activities instances area encompasses mainly two types of representative forest (1) Dense Ombrophilous Forest: Lowland (Db) and 2) Open Ombrophilous Forests: Alluvial (Aa) and Lowlands (Ab). These types represent % of total project area as follow: Dense Ombrophilous – 17.2%; Open Lowlands – 78.2%; Open Alluvial - 4.6%.
VM0015 / Methodology for Avoided Unplanned Deforestation	d) At the project start, the project area includes only land qualifying as “forest” for a minimum of 10 years prior to the project start date.	For baseline modelling and climate benefits (VCUs) it is considered land categorized as “forest” for a minimum of 10 years prior to the project start date. For more information refer to section 3.1.4, below.
VM0015 / Methodology for Avoided Unplanned Deforestation	e) The project area may include forested wetlands (such as bottomland forests, floodplain forests, mangrove forests) where they do not grow on peat. In this methodology peat comprised organic soils with at least 65% organic matter and minimum depth of 50 cm.	The project does not include peat forest in its baseline, nor in the project activities, only the Dense Ombrophilous Forest, that is a floodplain forests.
VT0001 / Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities	a) AFOLU activities the same or like the proposed project activity on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project shall not lead to violation of any applicable law even if the law is not enforced.	The project activities are based in conservation of the Legal Reserve ²⁰ that include monitoring against forest degradation and deforestation, promote training and capacitation focus on sustainable activities in the communities in the project zone. Neither of these activities will lead to violation of any applicable law.

²⁰ Law 12.651/2012 available at https://www.planalto.gov.br/ccivil_03/_ato2011-2014/2012/lei/l12651.htm

Reference ID/Title	Applicability condition	Justification of conformance
VT0001 / Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities	b) The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario. Project proponent(s) proposing new baseline methodologies shall ensure consistency between the determination of a baseline scenario and the determination of additionality of a project activity.	The Uákiry REDD+ AUDD Grouped Project made use of the approved VCS Methodology VM0015 - Methodology for Avoided Unplanned Deforestation, version 1.2 - and the module Estimation of Emissions Reductions from Avoiding Unplanned Deforestation (AUDef), version 0.2, April 12nd 2023. The applied methodology includes all the steps to define the most plausible baseline scenario, that will be provided by VERRA.

3.1.3 Project Boundary (VCS, 3.12)

Table 16 – GHG sources of Uákiry REDD+ Grouped project.

Source	Gas	Included?	Justification/explanation of choice
Baseline	CO ₂	No	Counted as carbon stock change
	CH ₄	Yes	Fire is used as an instrument for converting the forest to pasture or other land used in the baseline scenario. In the process of legal forest suppression, commercial wood is extracted, and the remaining woody material is felled and burned on site. CH ₄ is expected to be emitted due to woody biomass burning in the baseline scenario.
	N ₂ O	Yes	According to VM0015, section 5.4, only when it's significant.
	CO ₂	No	Not a significant source
	CH ₄	No	Conservatively excluded
	N ₂ O	No	Conservatively excluded
Project	CO ₂	No	Counted as carbon stock change
	CH ₄	Yes	Fire is used as an instrument for converting the forest to pasture or other land used in the baseline scenario. In the process of legal forest suppression, commercial wood is extracted, and the remaining woody material is felled and burned on site. If a fire occurs in the project scenario the CH ₄ will be accounted for as project emissions.
	N ₂ O	Yes	According to VM0015, section 5.4, only when it's significant.
	CO ₂	No	Not a significant source
	CH ₄	No	Conservatively excluded
	N ₂ O	No	Conservatively excluded
	CO ₂	No	According to VM0015 it is reasonable to assume that the project activity, including when harvest activities are planned (such as logging for timber, fuel-wood collection, and charcoal production), produces less emissions of GHG than the baseline activities implemented prior and after deforestation on the deforested lands.

Table 17 – Carbon pools included in Uákiry REDD+ Grouped project.

Carbon Pool		Included?	Justification/Explanation
Above ground	Tree	Yes	Main source of project emissions, through deforestation. Carbon stock changes in this pool are always significant
	Non tree	Yes	Inclusion is to be decided by the project proponent and must be included in the case where categories in the baseline final land cover is perennial crop. Part of the forest types in the project area are characterized by the occurrence of bamboo, vines, and shrubs. The inclusion is a conservative project approach and doesn't impact in over estimation of the GHG baseline emissions.
Below Ground	Tree	Yes	Inclusion is optional per VM0015, section 5.3. Carbon stocks are significant in this pool in tropical forests, representing 10-30% of the aboveground tree woody biomass (Nogueira et al., 2008).
	Non tree	Yes	Inclusion is optional per VM0015, section 5.3. Carbon stocks are significant in this pool in tropical forests, part of the forest types in the project area are characterized by the occurrence of bamboo, vines, and shrubs.
Dead Wood		Yes	Inclusion is to be decided by the project proponent per VM0015 section 5.3. Otherwise, is mandatory if this carbon pool is greater in the baseline final land cover (post-deforestation) than project scenario, and unless demonstrated insignificant. The inclusion is a conservative project approach and doesn't impact in over estimation of the GHG baseline emissions.
Litter		Yes	Inclusion is optional per VM0015 section 5.3
Soil Organic Carbon		No	Not to be measured in conversions to pastures or perennial crop according to the latest version of the VCS Methodology Requirements.
Harvested Wood Products		No	According to VM0015 section 5.3 to be included when is significant.

The following Figure 23 is intended to demonstrate project boundaries, physical locations of various management activities. For that, it aggregates different points previously presented, such as communities, leakage areas, field transect used by the biodiversity and forest inventory teams, location of lodging and finally the project core areas limits, where are the forest conservation and maintenance. These locations show the set of relevant points, where the various project implementation measures have been taking place.

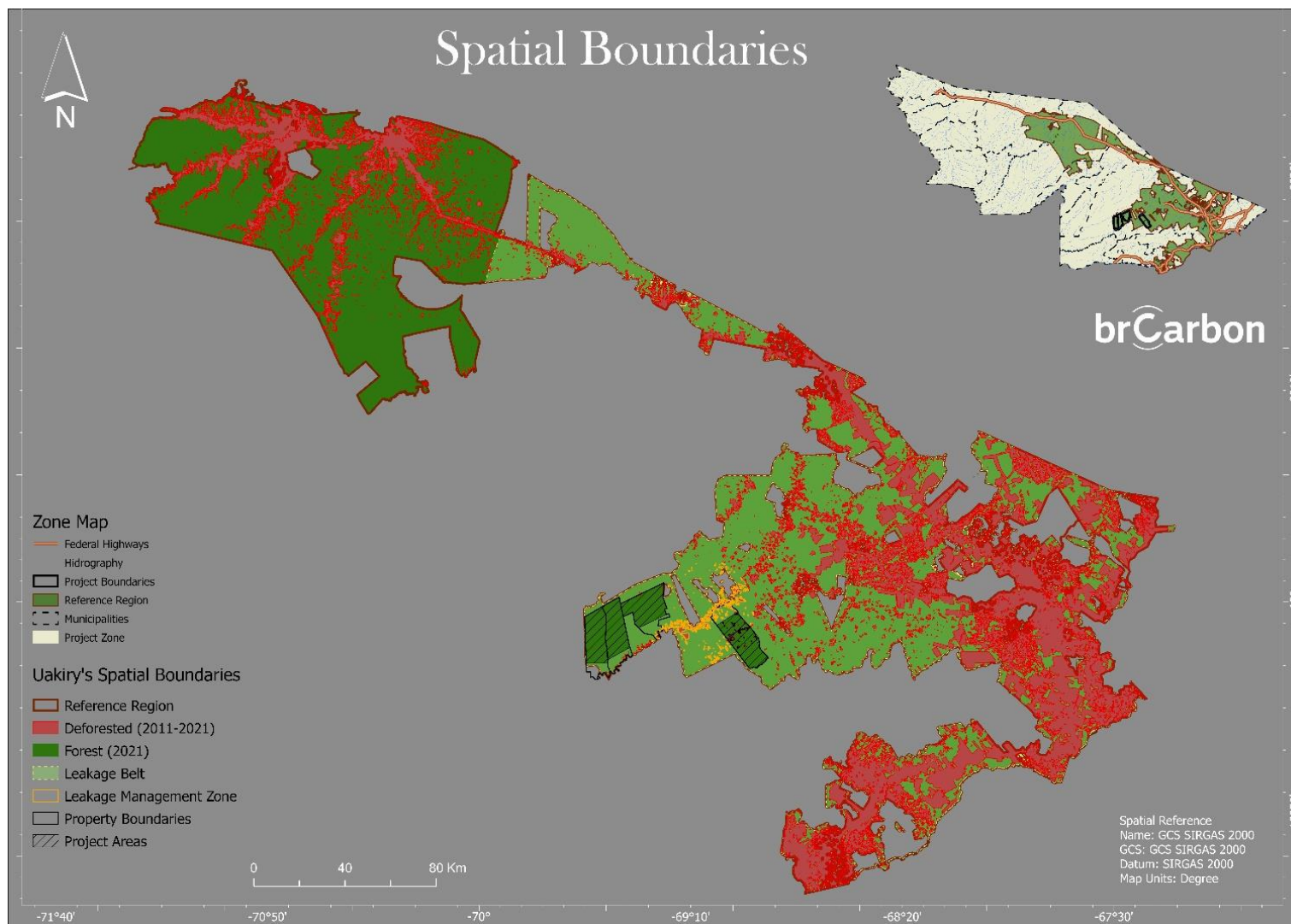


Figure 23 - Map of the project boundary.

3.1.4 Baseline Scenario (VCS, 3.13)

It was followed the methodology VM0015 v 1.2 to establish the Uákiry REDD+ GPD baseline scenario. Following the methodological approach, this analysis is divided in 4 steps.

3.1.4.1 Reference Region

To define the reference area, the project utilized political limits such as municipal and watershed boundaries. This approach allows for the integration of the region's social, economic, and environmental intricacies, encompassing factors such as deforestation catalysts, landscape layouts, ecological status, as well as socio-economic and cultural elements (VM0015 v1.2, section 5.1.1).

The Reference Region was created from the sum of the project areas (85,739.82 ha) and based on the criteria established in VM00015 version 1.2 item 2 that for projects smaller than 100,000 ha, the Reference Region should be 20 to 40 times larger than the project area. Once the maximum size of the Reference Region had been established, areas similar to those of the project were selected, which in this case would be private areas with 90% of the ecological conditions and landscape configuration, such as: soil classes, vegetation, elevation, slope, average annual rainfall, soil and socio-economic and cultural conditions. Based on the criteria set out above, government territories (Conservation Units, Indigenous Lands, Rural Settlements and Non-Destined Public Forests) were excluded, as well as carbon projects already underway. As a result, a continuous area of 3,424,389.51ha was established (40 times of project area) and fits the criteria of 90% of similarity in ecological conditions and landscape (Table 2).

3.1.4.2 Project Area

The definition of project areas is based on legal principles stipulated by the Brazilian forest code. All areas correspond to the surplus vegetation legally established as a legal reserve (RL) verified from monitoring data carried out by the federal government. Federal and state laws support between 50% and 80% of the RL on the properties chosen for the project. Landscape and ecological elements in these areas correspond to the specifications of the methodology (VM00015 v1.2, section 5.1.2).

Table 18 shows the correspondence between the project areas and the reference region.

Table 18 – Similarity analysis of the reference region (RR) and project area (PA).

Landscape configuration and ecological conditions: area in hectares and percentage				
Soil types	RR		PA	
TXo - Luvissolo Háplico Órtico	979,133.1	28.6%	13,024.9	15.2%
PVAva - Argissolo Vermelho-Amarelo Ta Alumínico	821,321.8	24.0%	68,805.7	80.2%
TCo - Luvissolo Crômico Órtico	551,855.8	16.1%		
PVAd - Argissolo Vermelho-Amarelo Distrófico	385,970.2	11.3%		
GXve - Gleissolo Háplico Ta Eutrófico	259,380.3	7.6%	3,908.7	4.6%
LVAd - Latossolo Vermelho-Amarelo Distrófico	162,121.0	4.7%		
PVAa - Argissolo Vermelho-Amarelo Alumínico	139,623.7	4.1%		
PVe - Argissolo Vermelho Eutrófico	38,128.0	1.1%		
PVd - Argissolo Vermelho Distrófico	27,173.7	0.8%		
LVd - Latossolo Vermelho Distrófico	20,820.6	0.6%		
FTd - Plintossolo Argilóvico Distrófico	16,432.3	0.5%		
PAva - Argissolo Amarelo Ta Alumínico	10,717.4	0.3%		
LAa - Latossolo Amarelo Alumínico	5,508.4	0.2%		
Others (water, etc.)	4,289.7	0.1%		
Total	3,422,476.099	100.0%	85,739.3	100.0%
Vegetation	RR		PA	
Open Ombrophylous Forest (Aa, Ab, Ap)	3,042,350.1	88.9%	70,426.6	82.1%
Dense Ombrophilous Forest (Da, Db)	362,828.8	10.6%	15,312.7	17.9%
Secondary Vegetation	12,973.8	0.4%		
Others (Water, etc.)	4,323.4	0.1%		
Total	3,422,476.1	100.0%	85,739.3	100.0%
Rainfall	RR		PA	
Af – Tropical	1,377,439.7	40.2%	37,517.9	43.8%
Am - Tropical	2,045,036.4	59.8%	48,221.4	56.2%
Total	3,422,476.1	100.0%	85,739.3	100.0%
Elevation	RR		PA	
< 50				
50-100				
100-150	71,226.0	2.1%		
150-200	1,170,671.1	34.2%	536.8	0.6%
200-250	1,858,434.3	54.3%	61,762.9	72.0%
250-300	317,955.5	9.3%	23,409.5	27.3%
300-350	6,091.9	0.2%	30.1	0.0%

350-400	8.2	0.0%		
Total	3,424,386.9	100.0%	85,739.3	100.0%
Slope	RR	PA		
Plan	322,208.9	9.4%	2,442.4	2.8%
Smooth Wavy	1,534,711.2	44.8%	30,363.6	35.4%
Wavy	1,440,754.7	42.1%	49,637.8	57.9%
Strong wavy	126,151.5	3.7%	3,295.5	3.8%
Mountainous	562.2	0.0%		
Steep	0.9	0.0%		
Total	3,424,389.5	100.0%	85,739.3	100.0%

3.1.4.3 Leakage belt

In the state of Acre, livestock farming has shown significant growth until 2021, characterized by an increase in both pasture areas and cattle herds. According to data from the Image Processing and Geoprocessing Laboratory (LAPIG), pasture expansion grew from 380,659 hectares in 1985 to 2,129,004 hectares in 2021. This represents a significant leap, from 2.32% to 13% of the state's territory occupied by pasture during this period.

At the same time, the cattle herd also grew remarkably, from 266,000 Animal Units (AU) in 1985 to 2.9 million AU in 2021. This consistent increase reflects the expansion and intensification of cattle ranching in the state of Acre over the decades, highlighting the significant role played by the sector in the region's economy and rural landscape (Figure 24).

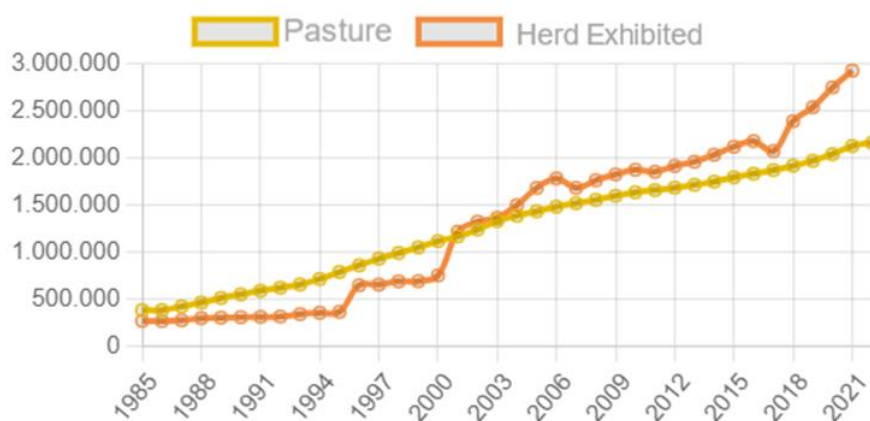


Figure 24 - Pasture and Cattle Headcount Graph from LAPIG (2021) for Acre state.

The data presented on the growth of pasture areas and cattle herds in the state of Acre reflects a process of modernization and intensification of livestock farming in the region. This phenomenon is in line with the trend observed throughout the Legal Amazon, as mentioned by Freitas Junior and

Barros in their 2021 study. It is important to note that the annual growth of pastures is directly linked to deforestation in the region. The increase in pasture areas often occurs at the expense of converting forests into pastures, which represents the main form of environmental degradation associated with cattle ranching in the Amazon. This phenomenon is widely documented in scientific literature, as pointed out by Barona et al. in 2010.

To highlight the predominant economic activity and meet the requirements of Option I, annual Land Use and Land Cover (LULC) classification data provided by Mapbiomas (Souza Jr et al., 2020) for the period 2011 to 2021 was overlaid with annual deforestation monitoring from PRODES - INPE from 2010 to 2022. This overlap made it possible to calculate the conversion rate of forest areas into pastures, which was done by intersecting the deforestation of one year with the LULC data of the subsequent year ($Conversion\ Rate = Deforestation_{year} \cap LULC_{year+1}$).

This methodology provides a detailed understanding of changes in land cover over time, allowing for a precise analysis of the conversion of forest areas into pastures, a crucial aspect for understanding the impacts of the expansion of cattle ranching in the AMACRO region and its environmental consequences. The data for the historical reference period indicates that the conversion rate from forest to pasture was 79.9% between 2011 and 2021. This means that almost 80% of the areas cleared during this period were converted into pasture. In the last five years, from 2017 to 2021, this rate has increased to 83.1%. These figures clearly show that the economic forces associated with livestock production are driving regional deforestation. The high rate of conversion from forest to pasture reflects the intensification of cattle ranching in the region, which has contributed significantly to the increase in deforestation, as shown in Table 19.

Table 19 - Conversion Rate of Deforestation into Pasture (2011 – 2021) for RR and LB

Year	Deforestation (ha)	Pasture (ha)	Conversion Rate
2011	12,424.4	8,595.8	69.2%
2012	12,078.0	7,306.8	60.5%
2013	8,470.7	6,338.5	74.8%
2014	14,588.3	11,176.9	76.6%
2015	9,305.0	7,069.4	76.0%
2016	9,305.0	7,469.3	80.3%
2017	10,528.6	8,086.1	76.8%
2018	16,595.0	13,771.8	83.0%
2019	28,609.8	24,230.2	84.7%

2020	23,832.1	20,293.5	85.2%
2021	35,996.2	30,837.2	85.7%
Total	181,733.2	145,175.5	79.9%

In order to define the distance and opportunity cost between the meatpacking plants and the areas that impact the forest, a vector analysis was carried out among the state meatpacking plants registered with the SIE (State Inspection Service), located in the municipalities of Porto Acre, Rio Branco, Senador Guimard, Xapuri and Sena Madureira. The criteria for selecting the slaughterhouses were based on the density of these facilities in the municipalities and their proximity to the BR-364 highway, which cuts through Acre from east to west, providing access to the states of Rondônia and Amazonas. The specific location of the meatpacking plants was obtained from the IMAZON website: www.imazongeo.org.br.

This analysis makes it possible to identify the meatpacking plants with the greatest potential to influence cattle ranching in the region and, consequently, the deforestation associated with this activity. The proximity of these meatpacking plants to cattle production areas may indicate a direct relationship between the demand for meat and the expansion of pasture areas, often resulting in deforestation to create new pasture areas.

The delimitation of the distance of the opportunity cost used a radius of 150 km from the meatpacking plants located in the municipalities mentioned above. This methodology is in line with the study published by IMAZON²¹, which indicates the distance from the meat-packing plants' purchasing area as a relevant measure for understanding the impact of cattle ranching on deforestation in the Amazon. After applying this criterion, the forest areas within this established limit were selected. The resulting area, known as the Leakage Belt, totaled 1,254,145.1 hectares. The Figure 25 highlights the spatial limits of the leakage belt.

²¹ <https://imazon.org.br/os-frigorificos-vao-ajudar-a-zerar-o-desmatamento-da-amazonia/>

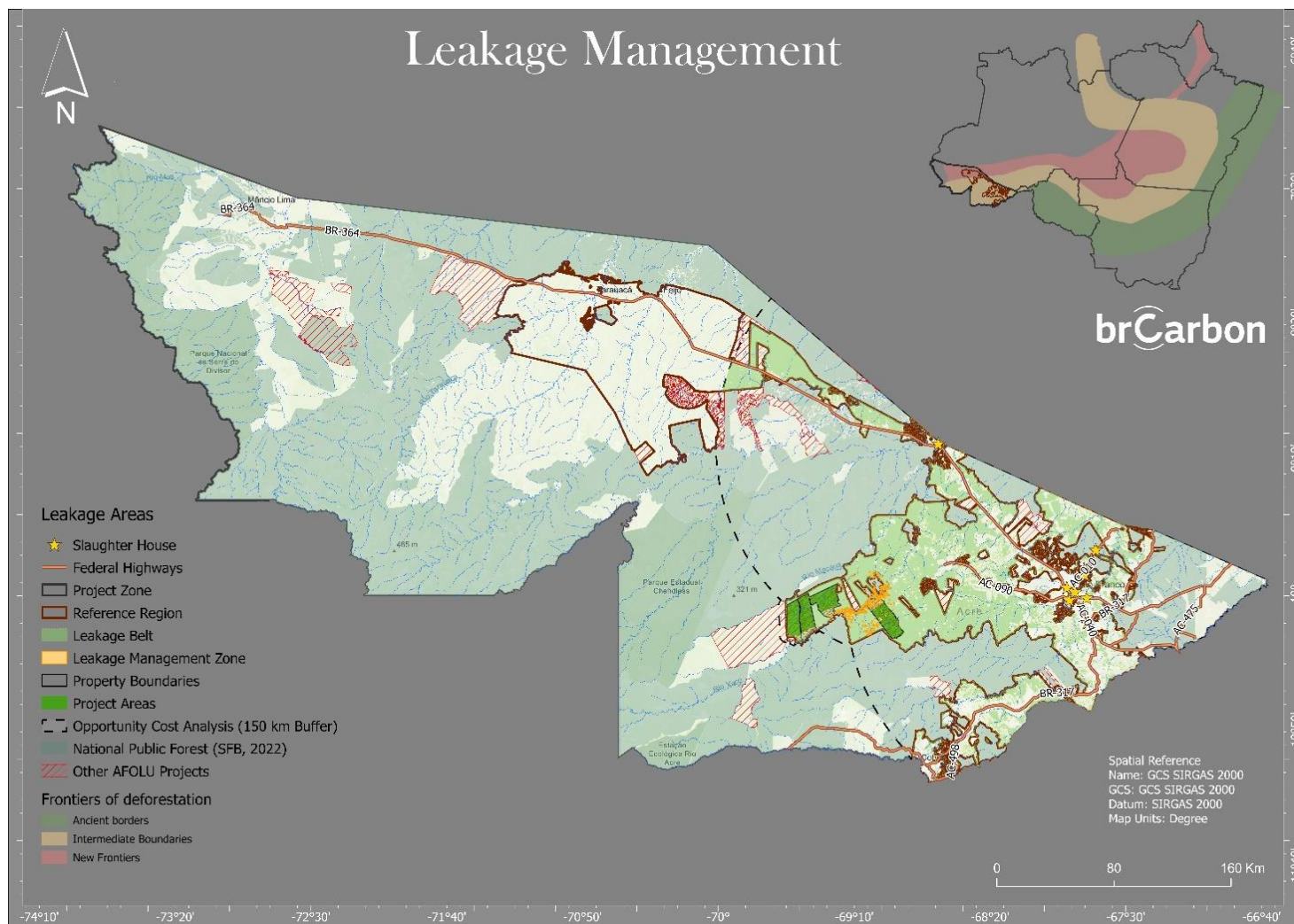


Figure 25 - Spatial explicit boundaries of Leakage Belt and Leakage Management Zones. Reference Region and Project Areas – PAI#s.

3.1.4.4 Leakage Management Zones (LMZ)

To delimit the Leakage Management Zones (LMZ), PP followed the criteria in section 5.1.4 of VM0015, which states that the limits should be defined according to the Leakage Management Plan (LMP). The LMP aims to mitigate risks and prevent the displacement of deforestation activities, focusing on improving agricultural and livestock practices. This investment aims to generate added value for each deforested productive area, in full compliance with environmental regulations and the current Forest Code (Brazilian Forest Law).

One of the fundamental pillars of the project is to train local actors, who play a crucial role in forest conservation and sustainable production. The field team will implement specific training programs, offering courses aimed at facilitating the adoption of sustainable agricultural practices. The aim is to ensure that local producers are familiar with methods that benefit both their livelihoods and the environment. The focus of these initiatives is to reduce the need to expand productive areas on forest fragments, prioritizing productivity gains in already established areas and improving the performance of these activities.

The field team will take the initiative to implement specialized training programs. This will cover specialized instruction for firefighting brigades, ensuring they have essential skills to preserve the region's forest resources. At the same time, the creation of new production chains for non-timber forest products will be promoted, boosting the development of the bioeconomy.

The plan is comprehensive, incorporating promotional and educational strategies. The field team will actively engage with the regional community through a variety of channels, such as agricultural events and local communications, highlighting the benefits of forest conservation and sustainable management. This will be done through examples of successful actions carried out at LMZ. In this way, the plan not only seek to comply with the prescribed methodology, but also contributes to the broader objective of sustainable land management and forest conservation.

Based on these objectives, the LMZ was defined. These areas meet the criteria of methodology VM0015, which states that they should be characterized as non-forest, as well as areas that have characteristics compatible with the implementation of activities outlined in the LMP, to provide a better chance of success for the actions.

The LMZ was delineated using an intermediate point between the project areas, which serves as a focal point for deforestation within the Leakage Belt. Starting from this reference point, a 25-kilometer buffer was computed to identify areas with agricultural and livestock activities. This region was recognized as spaces devoid of forest cover based on PRODES data. The overall extent of the LMZ amounts to 8,664.50 hectares. The objective is to identify areas with characteristics and interests suitable for implementing the goals of the LMP. Additionally, by presenting the methods and results

to other local producers within the LMZ to find new participants. Efforts aim to improve agricultural and livestock practices, thereby reducing the need for deforestation in new areas.

3.1.4.5 Forest

To assess the Forest Coverage Benchmarks and confirm the existence of the forest 10 years before the start of the project in the selected areas were used Landsat 5 images (TM sensor, Thematic Mapper) from 2011 and images from Sentinel 2 (MSI sensor, Multispectral Instrument) from 2020 for the project start date. A set of maps was created for each farm, displaying both the images used in the classification and the results of that classification.

All methodological details are presented in the document entitled "*Image acquisition, processing, and classification procedure for defining project areas*" presented in "APPENDIX_GIS-DATA". This document details the data sources and pre-processing, data classification and post-processing, and classification accuracy assessment, the metadata shows the folders with each category of data. In summary, the images are selected and downloaded; at the same time, atmospheric correction and level conversion are performed. Following download, images were reprojected, bands were stacked, project areas were clipped, and samples were selected. The classification is performed by a Machine Learning algorithm (Random Forest) with tested accuracy. Finally, the classification results are visually validated with high-resolution images and adjusted. Afterwards, the Project Areas classification was incorporated into PRODES (see Step 2) to produce detailed information for modeling.

The incorporation of the Random Forest method in the processing of Remote Sensing data emerges as pivotal in the nuanced classification of land use and occupation. To achieve this, PP opted for the utilization of the RStoolbox package in R, harnessing the capabilities of the superClass function. The classification process, employing the Random Forest model, undergoes systematic adjustment and validation phases, outlined in the subsequent steps:

1. **Model Training:** Definition of the proportion of samples allocated to training and validation; Verification of distance between training and validation samples; Collection of data on each sample polygon, proportional to its size.
2. **Data Extraction:** Values are extracted from the images based on the samples used for obtaining the training data.
3. **Model Fit:** The model is fitted using the sampled training data.
4. **Classification and Evaluation:** Assignment of classes to all pixels in the image based on the final model; Model validation with independent data (validation samples) and generation of a confusion matrix and cross-validation.

In the methodology for assessing classification accuracy, the classification result yields a comprehensive dataset, encompassing a confusion matrix, statistical metrics, model particulars, and a map illustrating the classification of land uses. The confusion matrix serves as a succinct summary of the classifier's performance against validation data, scrutinizing how effectively the Random Forest algorithm distinguishes between forested and non-forested areas. Evaluation metrics including Accuracy, the Kappa index, and the McNemar test are used.

3.1.4.6 Temporal boundaries

3.1.4.6.1 Starting date and end date of the historical period

The historical reference period used for the model calibration is from 2011 until 2015, while for the model validation, the year of 2020 was used.

3.1.4.6.2 Starting date of the project crediting period of the AUD project activity

The VCS project crediting period starts on 02nd August 2021 and ends on 01st August 2121, in a total of 100 years, the maximum granted by the VCS standard v.4.5. As the project start date is based on the date that the conservation agreement was signed between the landowner and BRC, how it was presented on Section 2.1.9 - Project Start Date.

PAIs #1-4 will generate GHG emission reductions eligible for issuance as VCU's for 40 years. A legal agreement (APPENDIX_LEGAL-AGREEMENTS) was signed with the landowner to continue the management practices that will lead to the project area conservation for at least 40 years, thus, the project activity instances longevity is 40 years. New PAIs will have no different crediting period starting dates, based on the conservation agreements signed by the landowner.

3.1.4.6.3 Starting date and end date of the first fixed baseline period

The fixed baseline period starts on 02nd August 2021 and ends on 1st August 2026, in a total of six years as stated in the VM0015 section 5.2 and VCS 4.5 section 3.2.5.

3.1.4.7 Analysis of historical land-use and land-cover change

3.1.4.7.1 Collection of appropriate data sources

To define land-use and land-cover change analysis, data were provided by the Brazilian Amazon Forest Monitoring Program by Satellite (PRODES), a system established by Brazil's National Institute for Space Research (INPE) to monitor deforestation through clear-cutting in the Legal Amazon since 1988. PRODES utilizes images captured by the NASA/USGS Landsat series satellites, which offer a spatial resolution of 20 to 30 meters and revisit the same area every 16 days. Irrespective of the imaging instrument used, PRODES maps a minimum area of 6.25 hectares.

This system has proven to be of great importance for public policy actions and planning in the Amazon region. Recent results, based on analyses conducted by independent experts, indicate an impressive

level of accuracy, approaching 95%. PRODES data is freely distributed through the TerraBrasilis portal (<http://terrabrasilis.dpi.inpe.br/downloads/>), a web platform developed by INPE for accessing, consulting, analyzing, and disseminating geographic data related to native vegetation monitoring projects.

Therefore, the data used to analyze changes in land use and land cover during the historical reference period within the reference region and project area correspond to the years 2001, 2015, and 2020. Table 20 provides details of the image data used for forest and non-forest classification layers for each of these years.

Table 20 – Data sources used by PRODES/INPE.

Vector (Satellite or Airplane)	Sensor	Resolution		Coverage	Acquisition date	Scene or Point identifier	
		Spatial	Spectral (μm)	(km^2)	(DD/MM/YY)	Path / Latitude	Row / Longitude
Satellite	LANDSAT 5/TM	30 x 30	0.450 - 2.35 μm	202,204.2	05/09/2011	1	67
				200,910.4	11/08/2011	2	66
				1,788,014.3	11/08/2011	2	67
				128,773.5	11/08/2011	2	68
				708,883.8	18/08/2011	3	66
				314,538.8	18/08/2011	3	67
				952,462.5	09/08/2011	4	66
				202,204.2	23/08/2012	1	67
				200,910.4	20/07/2012	2	66
				1,788,014.3	20/07/2012	2	67
				128,773.5	10/09/2012	2	68
				708,883.8	08/08/2012	3	66
				314,538.8	14/07/2012	3	67
				952,462.5	06/08/2012	4	66
	LANDSAT 8/OLI	30 x 30	0.450 - 2.29 μm	202,204.2	08/07/2013	1	67
				200,910.4	15/07/2013	2	66
				1,788,014.3	31/07/2013	2	67
				128,773.5	31/07/2013	2	68
				708,883.8	08/09/2013	3	66
				314,538.8	22/07/2013	3	67
				952,462.5	29/07/2013	4	66
				202,204.2	12/08/2014	1	67
				200,910.4	19/08/2014	2	66
				1,788,014.3	03/08/2014	2	67
				128,773.5	19/08/2014	2	68
				708,883.8	10/08/2014	3	66
				314,538.8	10/08/2014	3	67
				952,462.5	17/08/2014	4	66
				202,204.2	30/07/2015	1	67
				200,910.4	06/08/2015	2	66
				1,788,014.3	21/07/2015	2	67
				128,773.5	21/07/2015	2	68
				708,883.8	28/07/2015	3	66
				314,538.8	21/07/2015	3	67
				952,462.5	19/07/2015	4	66
				202,204.2	22/01/2016	1	67
				200,910.4	18/04/2016	2	66
				1,788,014.3	18/04/2016	2	67
				128,773.5	23/07/2016	2	68
				708,883.8	30/07/2016	3	66
				314,538.8	30/07/2016	3	67
				952,462.5	06/08/2016	4	66

Vector (Satellite or Airplane)	Sensor	Resolution		Coverage	Acquisition date	Scene or Point identifier	
		Spatial	Spectral (μm)	(km^2)	(DD/MM/YY)	Path / Latitude	Row / Longitude
				202,204.2	19/07/2017	1	67
				200,910.4	27/08/2017	2	66
				1,788,014.3	26/07/2017	2	67
				128,773.5	26/07/2017	2	68
				708,883.8	02/08/2017	3	66
				314,538.8	02/08/2017	3	67
				952,462.5	24/07/2017	4	66
				202,204.2	23/08/2018	1	67
				200,910.4	14/08/2018	2	66
				1,788,014.3	29/07/2018	2	67
				128,773.5	29/07/2018	2	68
				708,883.8	06/09/2018	3	66
				314,538.8	20/07/2018	3	67
				952,462.5	27/07/2018	4	66
				202,204.2	10/08/2019	1	67
				200,910.4	01/08/2019	2	66
				1,788,014.3	01/08/2019	2	67
				128,773.5	17/08/2019	2	68
				708,883.8	30/07/2019	3	66
				314,538.8	24/08/2019	3	67
				952,462.5	30/07/2019	4	66
				202,204.2	27/07/2020	1	67
				200,910.4	18/07/2020	2	66
				1,788,014.3	18/07/2020	2	67
				128,773.5	19/08/2020	2	68
				708,883.8	10/08/2020	3	66
				314,538.8	10/08/2020	3	67
				952,462.5	01/08/2020	4	66

3.1.4.7.2 Definition of classes of land-use and land-cover

Forest and non-Forest were the only classes used to describe land use and land cover (LU/LC). Based on Landsat images, whose characteristics were discussed previously, PRODES defines deforestation polygons as regions with a total area greater than 6.25 ha. The Non-Forest class corresponds to areas of deforestation detected by PRODES, the complete removal of forest cover in a short period of time. In this process, the forest cover is completely removed and replaced by other covers and uses (agricultural, pasture, urban, hydroelectric, etc.), or it can even be abandoned and enter a regeneration process. Deforestation can be seen as a single process, which starts with the forest intact and ends with the conversion of the original forest to another land cover. When PP consider deforestation as a single process, it is necessary to include not only the extremes of this process, which are more obvious and easier to identify but also the forest degradation gradient produced over the course of the deforestation process, which can occur slowly over time, due to the continuous logging and successive occurrences of forest fire. Table 21 presents the list of all land use and land cover classes and uses.

Table 21 – List of all land use and land cover classes existing at the project start date within the reference region.

Class identifier		Trend in carbon stock	Presence in	Baseline activity			Description (including criteria for unambiguous boundary definition)
IDcl	Name			LG	FW	CP	
1	Forest	Constant	RR, PA, LK	Yes	Yes	Yes	Remaining forest
2	Non-Forest	Constant	RR, PA, LK	No	No	No	Deforestation areas to cattle ranching and agriculture

3.1.4.7.3 Definition of categories of land-use and land-cover change

For activity data PP applied the Method 1. Thus, the change categories within the project area and leakage belt that could occur during the project crediting period **are not applicable** (Tables 6 and 7 of VM0015 v 1.2 were suppressed. See Appendix 6: List of Tables).

3.1.4.7.4 Analysis of historical land-use and land-cover change

3.1.4.7.4.1 Pre-processing

The images used are of the Landsat class, that is, they have a spatial resolution of around 30 meters, and a revisit rate of around 16 days. The images are available from their providers already orthorectified, with geometric system correction refined using control points and digital terrain elevation models. With this level of correction, the images are ready to be used. Images are taken in such a way as to highlight deforestation areas.

The PRODES use as reference the calendar year of deforestation refers to the period from August 1st of one year to July 31st of the following year. It is assumed that most deforestation occurs during the dry season. The date of the dry season for that location was established for each image based on climatological parameters. Unobserved areas due to cloud cover must be considered when calculating the estimated increment for each image. In cases of high cloud coverage, images from multiple satellites (or dates) can be used.

3.1.4.7.4.2 Interpretation and classification

The PRODES methodology uses an incremental map, which hides the areas that have been deforested in previous years by using an exclusion mask. Only parts of the image from the reference year containing primary forest are used during the interpretation. This mask prevents the possibility of old deforestation being mapped again. The exclusion mask also includes areas where there is no natural occurrence of forests. Additionally, deforestation increments through photo interpretation by specialists to identify the pattern of change from forest cover to clear cutting based on the three main elements observable in the images: tonality, texture, and context.

3.1.4.7.4.3 Post-processing

PRODES data are distributed in a matrix format with the presentation of all classes defined by PRODES: annual deforestation (accumulated up to 2007 and annual from 2008), inconsistencies in the classification of deforestation, cloud cover, water bodies, and naturally occurring areas. not forested. In the GIS environment, PP reclassified these classes to obtain forest and non-forest class maps for the years 2011 to 2020, as previously described. All this information is also organized in vector data in shapefile format.

3.1.4.7.4.4 Map accuracy assessment

To validate the accuracy of the PRODES classification across the entire reference region, random points were generated (150 points for each year). The accuracy of the classification was evaluated using a confusion matrix, based on Landsat images and the PRODES classification, providing the assessment of accuracy. Table 22 provides a summary of the accuracy observed for each class, as well as a global analysis.

Table 22 – Results of the accuracy assessment of the land use data used in the reference period.

Class	2011	2015	2020
Forest	0.9768	0.831	0.8860
Non-Forest	0.8819	0.9452	0.9577
Global	0.9466	0.9015	0.9073

3.1.4.7.4.5 Preparation of a methodology Appendix to PD

See the GIS methodology in the project description (APPENDIX_GIS-DATA).

3.1.4.8 Analysis of agents, drivers and underlying causes of deforestation and their likely future development

3.1.4.8.1 Identification of agents of deforestation

Acre has a history of land occupation and use of natural resources that spans different periods and contexts. The earliest period of commercial use of resources comes with the rubber cycle, where the extraction of latex from the rubber tree led to the migration of workers and the establishment of rubber plantations, resulting in significant changes to the landscape and the way of life of local communities. In recent decades, the region has seen an expansion of farming and cattle raising, with the advance of the agricultural frontier over forest areas. Deforestation for agricultural, livestock and infrastructure purposes has been one of the main causes of environmental degradation and loss of biodiversity. Logging also plays an important role in the local economy. Between 2019 and 2021, Acre

logged a total of 981,959 m³ of timber²² and 38,341 hectares, especially in the municipalities of Porto Acre, Feijó, Sena Madureira, Acrelândia and Bujari, almost all of which was logged on private properties²³.

The main indicators of human impact on forest regions are deforestation, and in the last 10 years, the reference region has lost around 181,733 hectares. This figure represents the total area of forest that has been converted to other uses, such as agriculture, livestock or urbanization. Along with deforestation, hotspots are often used for clearing vegetation after deforestation. Another factor to consider is forest degradation, which in most cases stems from predatory logging, causing a loss of diversity, fragmentation of the habitat of plant and animal species, as well as reducing humidity and thus leaving the forest more vulnerable to forest fires.

Roads are necessary for access to areas of forest massif, transporting wood, animals and people. The road network in the reference region includes approximately 2,000 kilometers of roads, of which around 50% are unofficial paths. Road construction can facilitate human access to previously remote areas, increasing pressure on natural resources and increasing the risk of deforestation and environmental degradation. In addition, it is important to note that the main official highways, such as BR-364, BR-317 and the state highway AC-090 (Transaccreana), can play a significant role in accessing and exploiting forest areas, contributing to the impacts mentioned above.

The main economic sector in the reference region is livestock farming, which currently requires around 2.2 million hectares to maintain production levels in 2021 (Mapbiomas), with pasture occupying 80% of the entire area deforested in the last 10 years (2011 to 2021). An important aspect to highlight is that during the reference period the cattle herd (data from SIDRA/IBGE) increased in all 14 municipalities that are part of the reference region, with the biggest increase being identified in Manuel Urbano with a 210% increase in relation to the cattle herd in 2011. In Sena Madureira, the number of animals increased from 214,368 in 2011 to 385,448 in 2021, an increase of almost 80%.

Although Acre is the state with the lowest rate of deforestation compared to its neighbors, Rondônia and Amazonas, the pressure on the forest has been growing and advancing into territories that should be protected and developing sustainable activities such as the extraction of natural products. In this context, the Chico Mendes RESEX, a federal Sustainable Use Conservation Unit, has a territorial area of approximately 1 million hectares, covers the municipalities of Rio Branco, Xapuri, Capixaba, Epitaciolândia, Brasília, Assis Brasil and Sena Madureira²⁴, and is located very close to the area of the projects and consequently within the boundaries of the Reference Region. In the last 10 years,

²² [Tabela 289: Quantidade produzida e valor da produção na extração vegetal, por tipo de produto extrativo \(ibge.gov.br\)](#)

²³ [Simex - Imazon](#)

²⁴ [ICMBio – Instituto Chico Mendes de Conservação da Biodiversidade \(www.gov.br\)](#)

between 2011 and 2021, the Resex lost 36,620 hectares of forest, making it one of the 5 most threatened Conservation Units in the Amazon²⁵, replacing forests with pasture.

Based on this scenario, it is possible to observe the dynamics of land use in the region, and even though some activities take place on private properties in a legal manner, as is the case with logging, other activities such as illegal deforestation to raise cattle is a common factor in protected and private areas that need protection and a sustainable economic alternative capable of providing conditions to maintain biodiversity and the population that depends on the forest for their survival.

Table 23 – Summary of the main deforestation agent in the reference region.

Name of the deforestation agent	Illegal ranching activities, timber extraction and land grab
Brief description of the main social, economic, cultural and other relevant features	In the 1960s and 1970s, the opening of highways, colonization projects, tax incentives and other government efforts to integrate the Amazon region with the rest of the country attracted large population groups to the Brazilian Amazon. This migration accelerated the population growth of many municipalities in the North region, especially those located along the newly created highways. In the context of Acre, the construction of roads that connect the main municipalities located in the project region, such as AC-405 and AC-90, as well as interconnecting AMACRO's border states, facilitates the flow and intensification of wood extraction as well as the conversion of forest to agricultural areas, mainly on the border with the state of Rondônia. The dynamics of deforestation in this region have historical roots due to the promotion of colonization that occurred during the dictatorship. At this time the federal government supported the occupation of land by settlers who after the opening of forest areas received the title of the area. Thus, the current context of deforestation in the reference region originates from the creation of the BR-364 in a pattern of “fishbone”. The dynamics of deforestation is now updated since the weakness in the public surveillance and forest monitoring which has been contributing to occur the action of the three main agents (logger, land grabber and rancher) that in most situations the same stakeholder assumes the role of the three agents of deforestation. For example, by extracting timber from illegal areas and with revenue from illegal logging, the farmer invests in grazing and buying cattle. After the degradation of the pasture, with low productivity the rancher looks for new areas to maintain the deforestation dynamics.
Brief assessment of the most likely development of the population size	In the continuity from the existing land use practices, it is anticipated that areas with high opportunity costs, particularly those associated with livestock activities, will experience heightened forest pressure. This attraction towards deforestation is driven by the collective actions of three key stakeholders, leading to an increase in population size within the reference region. Conversely, in regions with lower potential for transition but still susceptible to change, where deforestation dynamics are fueled by land speculation and acquisition, population growth is not expected to surge.

²⁵ [Terrabrasilis – Plataforma de dados geográficos \(inpe.br\)](https://terraviva.org.br/pt-br/territorio/territorio-tema/territorio-tema-1)

Statistics on historical deforestation attributable

According to the Mapbiomas data all the land use after the deforestation in the reference region is the pasture. So, all the deforestation that occurred in the reference period in the reference region can be attributable to the deforestation dynamics of the three main deforestation agents, as it was described above.

3.1.4.8.2 Identification of deforestation drivers

See Table 24, below, where the factor maps identified as deforestation drivers and were used in the land use change model are described according to the VM0015 requirements.

Table 24 – Identification of deforestation drivers

Deforestation driver	Evidence that the identified variables have been a driver for deforestation during the historical reference period	Driver variables will impact on each agent group's decision to deforest.	Likely future development
<i>Distance of the accumulated deforestation in 2020</i>	The reason for choosing this variable is the characteristic of deforestation in the region known as border deforestation (the reference region is located in the "arc of deforestation"). In these locations, forest cover in the vicinity of an agricultural / urban landscape matrix has a higher opportunity cost for conservation than in isolated and / or hard to reach regions. Thus, productive areas tend to increase, following the logic of "puxadinho" over conserved areas.	Illegal timber and Cattle ranching	The distance from deforestation is one of the main variables in the deforestation arc. It aggregates different vectors, as it is the transition from the forest class to the non-forest class. For example, the deforestation that occurs due to the opening of a road / branch leads to the phenomenon of the "fish bone" that is nothing more than the distance from the deforestation that occurred around a variable explaining the change in land use, which makes forest areas close to the road / branch with the greatest potential for transition to a non-forest class. Another example, common in the reference region, is related to the opening of the "puxadinhos" of unproductive pasture areas, since the degradation of pastures by the region's rudimentary livestock activity and low technical qualification makes extensive livestock farming a common practice, which results in a low average density of cattle per hectare of 0.5 heads. So, when there is a greater demand from the consumer market for meat, the rancher opens new areas to expand the herd and meet market demand; soon opening livestock areas in the forest areas on the edge of degraded pastures. Thus, it is to be expected that the distance from deforestation will be an important deforestation driver in the reference region,

Deforestation driver	Evidence that the identified variables have been a driver for deforestation during the historical reference period	Driver variables will impact on each agent group's decision to deforest.	Likely future development
			displacing the agents mentioned above in the southeast direction of the state.
<i>Distance of roads</i>	Roads are known and consolidated deforestation vectors on a global scale and are responsible for deforestation in "fishbone" format.	Illegal timber and Cattle ranching	According to the VM0015, section 6.2, the most likely future development is not applied to spatial variables, such slope, elevation etc.
<i>Distance of the rivers</i>	This variable was chosen because the areas close to the watercourses are areas most used for various agricultural activities. Mainly for the easy accessibility to the resource and also for being in the riparian zone where are more fertile soils, with high presence of organic material. They are also a common means of transport in the Amazon region, facilitating the flow of illegal timber products	Illegal timber	According to the VM0015, section 6.2, the most likely future development is not applied to spatial variables, such slope, elevation etc.

Deforestation driver	Evidence that the identified variables have been a driver for deforestation during the historical reference period	Driver variables will impact on each agent group's decision to deforest.	Likely future development
	and access to remote regions.		
<i>Distance of communities</i>	Locations with clustering of people, with little infrastructure and poor access to public policy, find themselves in a state of neglect by public governance, especially regarding environmental command / control. Forested areas have a great vulnerability around the localities that demand wood (energy, construction and commercialization), which initially selectively extract, then cut and burn for the implantation of low productivity and low value pastures. aggregate.	Cattle ranching	According to the VM0015, section 6.2, the most likely future development is not applied to spatial variables, such slope, elevation etc.

3.1.4.8.3 Identification of underlying causes of deforestation

The deforestation agents and the event related underlying causes identified for this project are presented on following Table 25.

Table 25 – The deforestation agents and the event related underlying causes.

Underlying causes of deforestation	Briefly description	Likely future development	Project measures
<i>Land uses policies and their enforcement</i>	According to official data (INPE/PRODES), in 2008, Rio Branco and Sena Madureira together had 6,366 hectares of deforested area (clear cutting and degradation), equivalent to 20% of its total area. Despite the efforts that contributed to the deforestation reduction in the AMACRO region, in the last 5 years (reference period) all the reference region has been facing in the advance of deforestation. This situation has become more evident since the last federal administration, through the weakening of inspection bodies, lack of punishment for environmental crimes and a significant reduction in immediate actions to combat and control of illegal activities in the region.	The weakening of environmental management, as discussed in the previous sections, at different levels from the promotion of sustainable activities, environmental regularization, supervision and command and control actions have medium and long-term impact on combating deforestation. Thus, it is expected that even in the scenario of changing environmental management especially in the strengthening of actions to Prevent and Combat Deforestation, effectively the fall in the upward curve of deforestation rates will happen in an interval of 2-4 years. It's important to highlight that the actions to stop illegal deforestation in Brazil is linked with NDCs goals to 2030.	The project aims to promote a valuation of environmental assets, contributing to the decarbonization of society and, thus, initiating a transition to a low-carbon economy. In this scenario, the project encourages the environmental conservation of forest fragments, both in the area and in the project zone. The project aims to promote a value of environmental assets, contributing to a decarbonization pathway of the society and thus initiating a transition to a low carbon economy. In this scenario, the project promotes the environmental conservation of forest fragments, both in the project area and in the project zone. It is worth mentioning that in parallel BRC (PP) is involved in different spaces of discussion, networks, and initiatives. These advocacy efforts aim to contribute, especially, with the adjustment and improvement of public policies, among them stand out: Climate Observatory, Forest Code Observatory and also in the REDD Alliance Network.

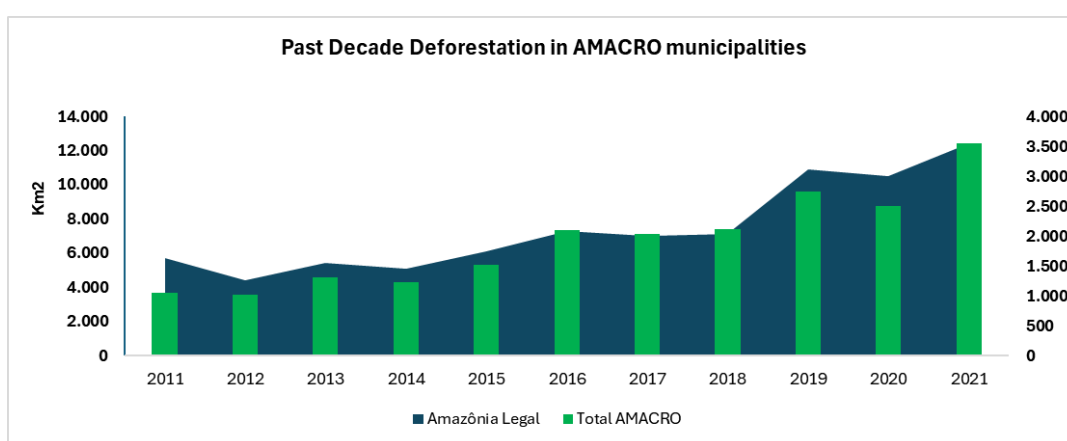
Underlying causes of deforestation	Briefly description	Likely future development	Project measures
<i>Population pressure and land speculation</i>	The population growth rates consistently contribute to deforestation across several levels. At the macro/international level, as consumer demand grows for agricultural commodities, particularly those in which Brazil plays a significant role in trade relations, pressures mount for increased production. This demand, driven by population growth in economically powerful regions, indirectly affects the opportunity cost of land. Consequently, areas with low opportunity costs, such as forested regions, face increased demand for conversion to more productive uses. While this cause-and-effect relationship holds true in certain contexts within the Legal Amazon, the southeastern region of the state of Acre presents two potential drivers of deforestation. Firstly, the actions of illegal squatters and land grabbers who encroach upon vulnerable lands where legal oversight is limited, typically remote areas near the Iaco River. Within this region, where the project is situated, evidence of deforestation exists both within the project area and its surrounding reference region. Secondly, there are buyers from other regions of the country, such as São Paulo and Paraná, who purchase land with the intention of converting forests into soybean plantations and cattle ranches. This trend not only exacerbates land speculation but also serves as a pull factor for additional buyers seeking market opportunities, particularly in beef production. Consequently, residents and small landowners occasionally sell their land to these new buyers and migrate to urban centers, where access to essential services provided by public authorities is more reliable.	The tendency is for increased pressure on forest ecosystems due to population growth, bearing in mind that the municipalities that are part of the reference region have low demographic density values. Other information that corroborates to this scenario is the historic population growth data. In this scenario of potential population growth, especially in rural areas close to forest areas, where they are isolated places with low access to basic public services about birth control, thus with the increase and growth of the population. Added to this scenario is the absence of rural extension and technical assistance, which results in open areas with low productivity. Therefore, with the increase in population and unproductive open areas, the Forest in the reference region is under pressure from this deforestation agent	The project will promote different technical assistance and rural extension activities in the project zone, these actions will aim to train rural producers in the management of productive areas, aiming at recovery and increased productivity, with this it is expected that the pressure on forest areas will decrease.

Analysis of chain of events leading to deforestation

The dynamics of deforestation is a complex and multifaceted phenomenon which is dependent on a series of agents and vectors, which are driven by aspects: social, cultural, and economic (micro and macro). The AMACRO region encompasses the states of Amazonas, Acre and Rondônia and is considered the newest deforestation frontier in Legal Amazon with accelerated growth since 2014. This region covers an area of 454,220 km², which includes 32 municipalities, with significant amounts of deforested areas in recent years²⁶. The municipalities within the reference region experienced a notable rise in local deforestation rates. Similarly, in the AMACRO municipalities, there was an increase in deforestation, particularly pressing towards the southeast of Acre, notably in the municipalities of Sena Madureira, Acrelândia and Plácido de Castro, where growth forest losses were concentrated along the frontier with the state of Rondônia and Amazonas.

In the reference period analyzed by the Uákiry REDD Grouped Project (2011-2021), the municipalities in the reference region showed an increase in local deforestation rates, but with an oscillatory trend detected. This situation has become a reality in the Brazilian Legal Amazon because of the weakening of environmental management, the loosening of environmental inspection and monitoring and the total suspension of command-and-control actions promoted by the federal administration of (2018-2021). Despite the change that occurred in 2022 the impacts of mistakes made in recent years lasted for a significant period. Figure 26 shows that the states surrounding AMACRO have a historical trend of deforestation likewise the boundaries of the Legal Amazon, which supports the evidence that certain mechanisms for incentives providing irregular practices were implemented during this period.

Figure 26. Comparison of Legal Amazon states and AMACRO trends of deforestation.



The road infrastructure projects can be seen as a catalyst for the migration of large populations to the region, thereby promoting the exploitation of timber as the primary economic endeavor. This kickstarts a cycle leading to forest degradation. In certain cases, forest exploitation coincides with land conversion, particularly for livestock farming, sometimes funded by proceeds from illegal logging

²⁶ [Resumo - Zona de Desenvolvimento Sustentável Abunã-Madeira — Superintendência do Desenvolvimento da Amazônia \(www.gov.br\)](http://www.gov.br)

operations. In 2021, the construction of a bridge spanning the Abuanã River, connecting the states of Acre and Rondônia, has eased the transportation of timber and made the region more appealing for the conversion. In some situations, forest exploitation occurred concomitantly with the conversion of land, mainly for livestock activity, including financing the implementation of pasture with resources obtained through illegal logging. This development has led to the establishment of urban centers, stimulating migration to the area, thereby increasing its population density and the demand for goods and services in the newly productive zones.

It's important to note that in numerous instances, human-induced disturbances disrupt the carbon cycle within forest ecosystems, occurring repeatedly over time and significantly impacting the structure and biodiversity of the forest. Predatory logging, coupled with frequent droughts, forest fires, and the expanding influence of forest edges, ease the forest's vulnerability to conversion for land use.

The disturbances have a series of causes linked by different agents, vectors of deforestation and Underlying causes, from: the demand for wood, expansion of agriculture and livestock, the offer of agricultural credit without environmental conditions and the establishment of roads and cities. These agents/vectors act motivated by market demand for agricultural products, lack of governance and corruption, population growth and migration.

3.1.4.8.4 Conclusion

Thus, today, due to the historical location of wood extraction, together with the entire logistics and productive chain installed, the forest areas are under intense pressure, mainly for charcoal and wood, associate with the demand to new pasture areas.

The expansion of grain production on the border surrounding the state of Rondônia has intensified the conversion of forests to be occupied by pasture areas, configuring the context of the new arc of consolidated deforestation in the AMACRO region. Additionally, the state of Acre has made substantial investments in infrastructure, particularly in enhancing access to the Chico Mendes vicinity road. This road intersects with the RESEX (Extractive Reserve) of the same name and serves as a crucial land route to the project area. Moreover, the BR-317 serves as the primary access point to the RESEX vicinity and plays a vital role in facilitating the transportation of beef within the region.

With this, it is expected that the baseline scenario established based on the dynamics of deforestation that occurred over the reference period will continue in the coming years and, only, the sum of efficient actions and strategies in the control, combat, monitoring deforestation, allied to the promotion of actions to value environmental assets and bioeconomy products and the promotion of sustainable activities, will succeed in reversing this scenario of increased deforestation rates in the reference region.

So, based in the steps previously performed it is possible to conclude that the most-likely future deforestation trend can remain oscillatory, and continue increasing the pressure in the next years

until the joint actions to avoid the deforestation can get to the objectives to effectively decline the deforestation rates in the legal Amazon. Also, it's important to highlight that the relationship between the agents' groups, deforestation drivers, underlying causes and historical levels of deforestation was verified and as it was explained the main causes of the deforestation dynamics.

3.1.4.9 Projection of future deforestation

3.1.4.9.1 Projection of the quantity of future deforestation

3.1.4.9.1.1 Selection of the baseline approach

According to the selection criteria, it was identified that deforestation rates measured in different historical periods (2011-2020) in the reference region do not reveal any trends, therefore, the baseline approach selected to project the project's future deforestation was the historical average approach. Conclusive evidence of the agents and drivers that explain different historical rates of deforestation is presented in section 3.1.4.3.

The deforestation rates measured between 2011 and 2020 show normal behavior (Shapiro-Wilk normality test p-value > 0.05), therefore, the non-parametric Mann-Kendall test was used to analyze the significance of the trend. This means that it does not assume a specific distribution of data and is not affected by extreme values. It provides three main types of information: the Kendall rank correlation coefficient (Tau), indicating an increasing or decreasing trend (varies between -1 and 1); the Sen slope (S), which estimates the overall slope of the time series as the median of the slopes between each pair of points in the series; and significance, indicated by the p-value, where a trend is considered statistically significant if the p-value is less than 0.05.

After performing the Mann-Kendall test, a p-value of 0.0491 was obtained, with an S value of 23 and a Tau of 0.5111. These results indicate a positive upward trend. However, it is important to highlight that this trend does not reach statistical significance.

3.1.4.9.1.2 Quantification projection of future deforestation

According to the analysis presented, the selection of the baseline approach and quantitative projection of future deforestation was the historical average approach. Under this approach, the rate of baseline deforestation is assumed to be a continuation of the average annual rate measured during the historical reference period in the reference region. The annual baseline deforestation area that applies at year t to stratum i within the reference region is calculated as follows:

$$ABSLRR_{i,t} = ARR_{i,t-1} \times RBSLRR_{i,t}/100$$

Where:

$ABSLRR_{i,t}$ = Annual area of baseline deforestation in stratum i within the reference region at year t (ha/yr);

$ARR_{i,t-1}$ = Area with forest cover in stratum i in the reference region at year $t - 1$ (ha);

$RBSLRR_{i,t}$ = Deforestation rate applicable to stratum i within the reference region at year t (%);

$t = 1, 2, 3, \dots, T$, a year of the proposed project crediting period (dimensionless);

$i = 1, 2, 3, \dots, IRR$, a stratum within the reference region (dimensionless).

For calculation of the deforestation rate ($RBSLRR_{i,t}$) was employed the method proposed by Puyravaud, J. P. (2003), as recommended by VM0015. The annual rate of change in forest cover can be determined using the following equation:

$$RBSLRR_{i,t} = \left(\frac{1}{t_2 - t_1} \right) \times \ln \left(\frac{A_2}{A_1} \right)$$

Where:

$t_2 - t_1$ = represents the time interval between the final time (t_2) and the initial time (t_1);

A_2 and A_1 = denote the forested area in the reference region at time (t_2) and (t_1), respectively.

The forest area by year and the corresponding deforestation rate for the historical period (2011-2020) are presented in following tables.

year	$ARR_{i,t-1}$ (ha)	$RBSLRR_{i,t}$
2011	2,430,093.3	-0.0051
2012	2,418,015.4	-0.0050
2013	2,409,544.7	-0.0035
2014	2,394,956.3	-0.0061
2015	2,385,651.3	-0.0039
2016	2,369,224.0	-0.0069
2017	2,358,695.4	-0.0045
2018	2,342,100.4	-0.0071
2019	2,313,490.6	-0.0123
2020	2,289,658.6	-0.0104
$RBSLRR_{i,t}$ (2011-2020)		-0.0066

Table 26 – Annual areas of baseline deforestation in the reference region (RR).

Area deforested per forest class icl within the reference region				Total baseline deforestation in the reference region	
IDic>	1	2	3		
Name >	Open alluvial forest (Aa)	Open lowland forest (Ab)	Dense Ombrophilous forest (Db)	ABSLRRt	ABSLRR
				annual	cumulative
Project year t	ha	ha	ha	ha/yr	ha
2021	1,700.6	13,870.6	2,395.7	19,987.9	19,987.9
2022	1,700.6	13,870.6	2,395.7	19,988.9	39,976.8
2023	1,700.6	13,870.6	2,395.7	19,989.9	59,966.7
2024	1,700.6	13,870.6	2,395.7	19,990.9	79,957.6
2025	1,700.6	13,870.6	2,395.7	19,991.9	99,949.5
2026	1,700.6	13,870.6	2,395.7	19,992.9	119,942.4

Table 27 – Annual areas of baseline deforestation in the project area (PA).

Area deforested per forest class icl within the project area				Total baseline deforestation in the project area	
IDicI>	1	2	3		
Name >	Open alluvial forest	Open lowlands forest	Dense Ombrophilous forest	ABSLPA _t	ABSLPA
				annual	cumulative
Project year t	ha	ha	ha	ha/yr	ha
2021	25.90	439.90	101.3	567.10	567.1
2022	25.90	439.90	101.3	567.10	1,134.2
2023	25.90	439.90	101.3	567.10	1,701.3
2024	25.90	439.90	101.3	567.10	2,268.4
2025	25.90	439.90	101.3	567.10	2,835.5
2026	25.90	439.90	101.3	567.10	3,402.6

Table 28 – Annual areas of baseline deforestation in the leakage belt (LB).

Area deforested per forest class icl within the leakage belt				Total baseline deforestation in the leakage belt	
IDicl>	1	2	3		
Name >	Open alluvial forest (Aa)	Open lowland forest (Ab)	Dense Ombrophilous forest (Db)	ABSLKt	ABSLK
				annual	cumulative
Project year t	ha	ha	ha	ha/yr	ha
2021	929.6	5,727.3	1,336.7	7,993.6	7,993.6
2022	929.6	5,727.3	1,336.7	7,993.6	15,987.2
2023	929.6	5,727.3	1,336.7	7,993.6	23,980.8
2024	929.6	5,727.3	1,336.7	7,993.6	31,974.4
2025	929.6	5,727.3	1,336.7	7,993.6	39,968.0
2026	929.6	5,727.3	1,336.7	7,993.6	47,961.6

3.1.4.9.2 Projection of the location of future deforestation

3.1.4.9.2.1 Preparation of factor maps

All factor maps that were generated are presented below, see Table 29, it obtained all the spatial data related to each of the four variables analyzed, and it created digital maps representing the Spatial Features of each variable. The LCM requires producing Distance Maps from the mapped features (e.g., distance to roads or distance to already deforested lands) and this data was created using the Empirical approach which is suggested by the VM0015 (section 8.1.3.1).

Table 29 – Factor maps selected to the model selected for the baseline scenario.

Factor Map		Source	Variables represent		Meaning of the categories or pixel value		Other Maps and Variables used to create factor map		Algorithm or Equation used	Comments
ID	File Name		Unit	Description	Range	Meaning	ID	File Name		
1	DIST_UTM_RST_DEFORESTATION.tif	The cartographic base used was generated with the 2018 data from the PRODES INPE Amazon Monitoring System.	meters	Distance of the accumulated deforestation in 2020.	0 – 10,000	Values close to 0 are closer to the deforestation.	1A	SHP_RST_DEFORESTATION.tif	The classes accumulated by deforestation, forests, and hydrography in raster format using R. Subsequently, the Euclidean distance was calculated only for deforestation in the QGIS 3.28 software.	The reason for choosing this variable is the characteristic of deforestation (the reference region is located in the "arc of deforestation"). In these locations, forest cover in the vicinity of an agricultural / urban landscape matrix has a higher opportunity cost for conservation than in isolated and/or hard to reach regions. Thus, productive areas tend to increase, following the logic of "puxadinho" over conserved areas.
2	DIST_UTM_RST_ROADS.tif	The cartographic base used was provided by Imazon and the National Transport and Infrastructure Department (DNIT). Refers to the systematic mapping of branches, secondary and primary roads, in a 1: 50000 scale.	meters	Distance of main and secondary roads	0 – 5,000	Values close to 0 are closer to main roads.	2A	SHP_RST_ROADS.tif	The data was clipped to the reference region using QGIS 3.28 software. The information was converted to raster, then the Euclidean road distances within the reference region were calculated.	Roads are known and consolidated deforestation vectors on a global scale and are responsible for deforestation in "fishbone" format.
3	DIST_UTM_RST_WATER.tif	The cartographic base used was generated with the hydrographic data of the National Water and Basic Sanitation Agency (ANA), on a scale of 1: 250,000.	meters	Distance of the rivers	0 – 5,000	Values close to 0 are closer to hydrography.	3A	SHP_RST_WATER.tif	The spatial information of the hydrography was grouped and reference region using QGIS 3.28 software. The information was converted to raster, and the Euclidean hydrographic distances within the RR were calculated.	This variable was chosen because the areas close to the watercourses are areas most used for various agricultural activities. Mainly for the easy accessibility to the resource and also for being in the riparian zone where are more fertile soils, with high presence of organic material. They are also a common means of transport in the Amazon region, facilitating the flow of illegal timber products and access to remote regions.
4	DIST_UTM_RST_COMMUNITY.tif	The cartographic base used was generated based on IBGE (2010) spatial information, in 1: 250000 scale. Also communities mapped in	meters	Distance of communities	0 – 5,000	Values close to 0 are closer to municipalities headquarters.	4A	SHP_RST_COMMUNITY.tif	It was clipped to the reference region using QGIS 3.28 software. The information was converted to raster, and the Euclidean communities distances within the reference region were calculated.	Locations with clustering of people, with little infrastructure and poor access to public policy, find themselves in a state of neglect by public governance, especially with regard to environmental command / control. Forested areas have a great vulnerability around the localities that demand wood (energy, construction and commercialization), which initially selectively extract, then cut and burn for the implantation of low productivity and low value pastures. aggregate.

3.1.5 Additionality (VCS, 3.14)

According to the VM0015: “Additionality of the proposed AUD project activity must be demonstrated using either the most recent VCS-approved VT0001 Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities.”

Based on that, the following steps of VCS-approved VT0001 Tool, were assessed:

STEP 1. Identification of alternative land use scenarios to the AFOLU project activity.

STEP 2. Investment analysis to determine that the proposed project activity is not the most economically or financially attractive of the identified land use scenarios; or

STEP 3. Barrier's analysis; and

STEP 4. Common practice analysis.

3.1.5.1 Regulatory Surplus (VCS, 3.14)

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☐ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☐ No ☒ Not applicable

3.1.5.2 Additionality Methods (VCS, 3.14)

STEP 1. Identification of alternative land use scenarios to the AFOLU project activity

Sub-step 1a. Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity:

Based in the economic and political trends and regional business as usual, two credible alternative land use scenarios were identified for the project area, as follow:

Forest cover maintenance, i.e.: through conservation activities resulting from incentives other than the REDD project.

Illegal deforestation for pasture (cattle raising) and agriculture purposes or simply for land grabbing and real estate speculation.

The two land use scenarios identified above occur in the municipalities encompassed by the grouped project, the same is observed for the surrounding municipalities. In addition, they are also credible and in line with legislation or with historical land uses, practices, and economic trends in the region. For additional information regarding project region historic land use dynamic, please refer to section 3.1.4 - *Baseline scenario*.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations.

According to this section, the following step must be applied:

Demonstrate that all land use scenarios identified in sub-step 1a follow all mandatory applicable legal and regulatory requirements.

Considering the project activities will be developed in the Legal Reserve of private properties, only scenario 1 is in accordance with current laws and regulations. Even not being in line with the mandatory applicable legislation and regulations, the illegal deforestation for cattle and agriculture purposes is widely observed and common carried out in the project region, thus scenario 2 is credible and very much in line with the common land use practice observed in the region. Therefore, the uncontrolled land occupation by land-grabbers and independent producers, followed by illegal deforestation, is a credible scenario for the entire property, including Legal Reserve and Permanent Preservation Area. It's worth noting that this land use pattern is observed in the surrounding municipalities, as well, encompassing the entire southeast of Acre, where applicable mandatory legal

or regulatory requirements are systematically not enforced and non-compliance with those requirements is widespread.²⁷

Given that the Acre state exhibits the second-highest rate of native forest coverage within the Legal Amazon, with approximately 50% of its territory designated as protected natural areas and Conservation Units geared toward sustainable use, such as the RESEX Chico Mendes situated near the project boundaries, illegal activities has been reported. Based on an extensive study conducted by IMAZON in 2013, it was found that a significant portion of the Amazon territory lacks effective land control by the Brazilian government. The research revealed that approximately 158 million hectares (32% of the territory) are purportedly privately owned areas without official verification or validation by government institutions. Furthermore, a survey conducted by the MapBiomass Alert Platform in 2020 highlighted that nearly 98% of the deforestation recorded in Brazil exhibited signs of irregularity. This irregularity stemmed from deforestation activities carried out without legal authorization or occurring within protected areas, such as conservation units, indigenous lands, or Permanent Preservation Areas (APPs). The absence of adequate inspection further exacerbates the issue of deforestation. A study revealed that only 1.3% of the 115,688 deforestation alerts in the Amazon, as published by the MapBiomass Alerts platform between 2019 and 2020, underwent inspection resulting in embargoes or infractions by IBAMA (Coelho-Junior et al., 2022). This accounted for merely 6.1% of the total deforested area detected. Notably, the state of Acre ranked fourth in terms of the highest number of validated deforestation alerts identified by the platform.

Based in the above mentioned, Scenarios 1 and 2 are plausible alternative land use to the VCS AFOLU project activity, where scenario 1 is deemed plausible, once forest conservation of Legal Reserve follows all mandatory legislation and regulations, while scenario 2 is deemed plausible considering the widespread BAU of non-enforcement and non-compliance with the land use legislation and regulations. It's worth mentioning that incompliance with environmental laws and legal requirements for land use is common in the Amazon and can be found in many relevant articles and studies about the region. Please refer to APPENDIX_ADDITIONALITY and APPENDIX_CARBON-POOLS.

Sub-step 1c. Selection of the baseline scenario

The historical patterns of land use and occupation in the Amazon strongly suggest that deforestation is the most probable outcome for forested areas within the project's boundaries. According to data from the Brazilian National Space Agency (INPE)²⁸, approximately 729,000 km² of the Amazon biome have already been deforested, accounting for 17% of the total biome. Over the past two decades alone, 300,000 km² of forest have been cleared. Moreover, predictive models forecasting deforestation afterwards (one to three years ahead), based on deforestation rates reaching peak

²⁷ <https://mapbiomas.org/dados-sobre-fiscalizacao-mostram-que-impunidade-ainda-predomina-no-combate-ao-desmatamento>

²⁸ <http://www.inpe.br/faq/index.php?pai=6>

levels observed in the last decade, highlight Acre as the fourth most at-risk state for deforestation in 2023. This projection estimates deforestation covering an area of 1,269 km², equivalent to 11.75% of all threatened territories within the Amazon. These data are from PrevisIA²⁹, a platform that uses artificial intelligence to indicate areas at risk of deforestation in the Brazilian Amazon and which, in 2022, showed an accuracy of almost 80% in predicting deforestation. In a BAU scenario, the tool predicts a loss of over 11,805 km² of native forest in 2023.

The PrevisIA platform enables the visualization of a heat map illustrating the likelihood of forest destruction during the deforestation season starting in 2023. Below (Figure 27), PP present the map generated by the platform, emphasizing the regions where project areas are situated, all falling within high-risk deforestation zones as heat colors. The highlighted areas on the map depict the greatest risk of deforestation, with significant expansion evident within the project boundaries.

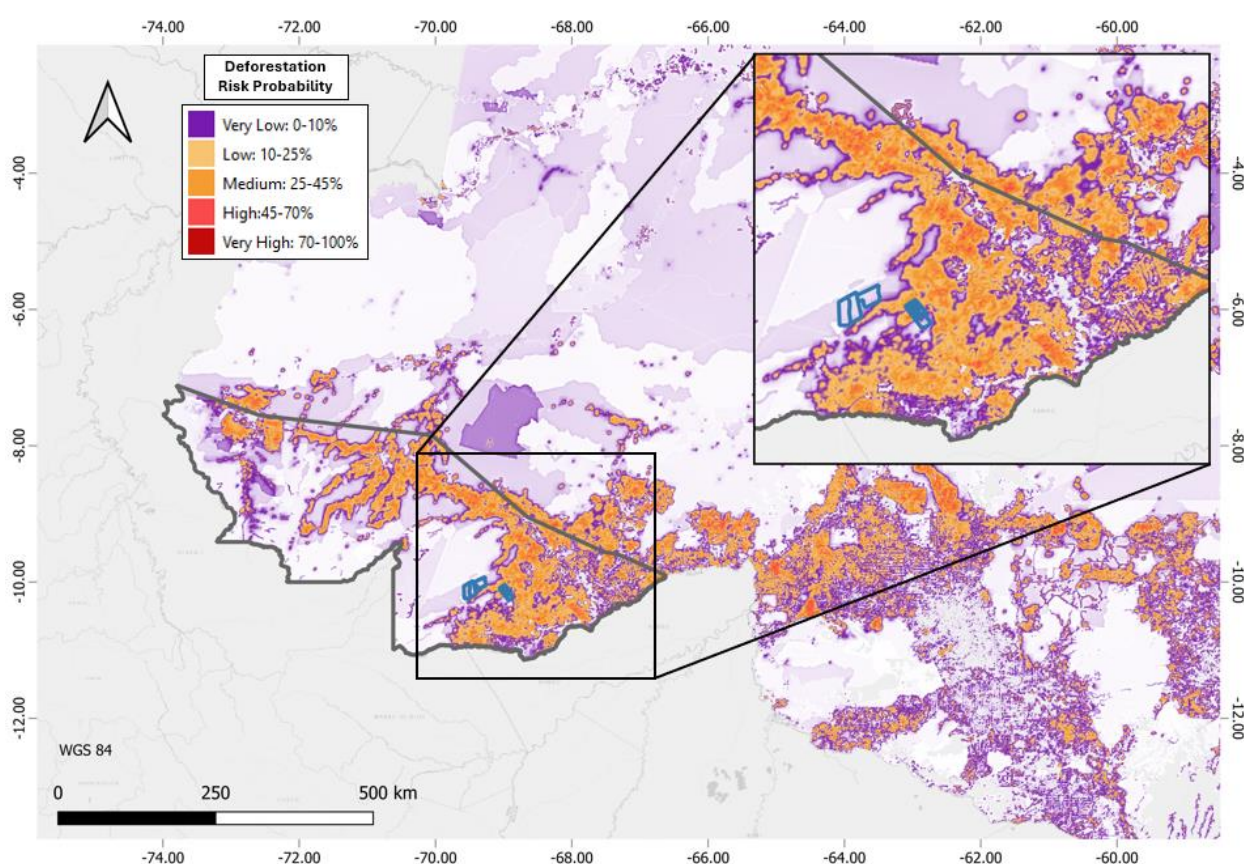


Figure 27 – Probability of projected deforestation risk for the Brazilian Amazon, with emphasis on Acre state and project boundaries, according to projections on the PrevisIA platform.

Analyzing the Table 30 below that presents the deforested area in the ten municipalities that make up the project in the Reference Region, PP observe that the municipality of Plácido de Castro has the lowest forest cover among the municipalities (21.91%) and has the highest rate of deforestation, as the accumulated deforestation until 2021 (78.09%). According to the most recent data, after one decade of drop in the deforestation rates in AMACRO states context, it was observed a shifting in this

²⁹ PrevisIA was created in partnership with the Amazon Institute of Man and Environment (Imazon), Microsoft, and Fundo vale. The tool analyzes a set of variables to indicate the areas at most significant risk of deforestation in the amazon biome.

trend from 2016 on, where the annual deforested area and fire outbreaks have shown a significant increasing (Figure 28). This new trend has become more evident between 2018 until 2022, where the deforestation and fire events have boosted up by changes in the Brazilian government environmental policy, after election in 2018. Between 2018 and 2022, a noticeable increase in deforestation and wildfires occurred, largely attributed to shifts in Brazil's governmental environmental policies following the 2018 election. The administration during this period displayed minimal dedication to environmental concerns, favoring instead close ties with industries like cattle ranching, mining, and agribusiness. This resulted in significant reductions in funding for IBAMA, the primary agency responsible for monitoring and enforcing environmental regulations, under the leadership of both the president and the Minister of the Environment at that time.

During the same period, infrastructure projects were implemented, such as the bridge connecting the states of Rondônia and Acre³⁰, which could serve as a facilitator for the transportation of timber and contribute to expansion of deforestation in the southwest region of the state. Please refer to APPENDIX_ADDITIONALITY.

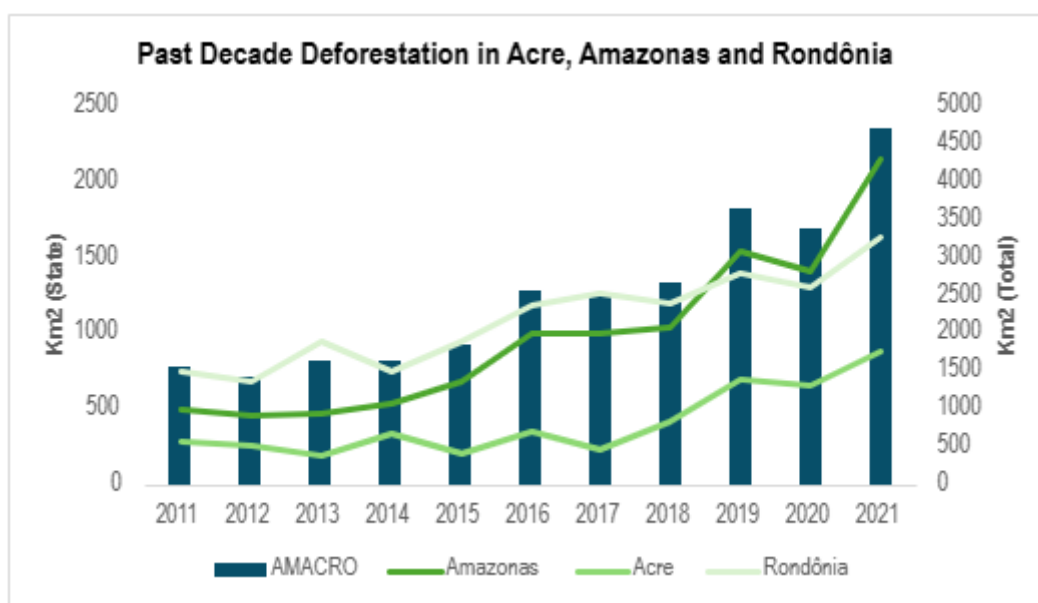


Figure 28 – Annual deforested area for Amazon biome from 2008 until 2022 - km² (INPE). Context of AMACRO states.

As mentioned in section 2.2.2, the Acre state is responsible for beef production of 290 tons since 2020 and driven more than 4,279 square kilometers of deforestation from 2011 to 2021 (PRODES/INPE). A correlation pattern between deforestation and cattle activity is observed in most regions of Amazon Forest frontier. Therefore, the most likely scenario for the project area, in the absence of the project activities, is option 3 (Illegal deforestation for cattle production and agriculture purposes or simply for land grabbing and real estate speculation). Even considering several

³⁰ [Ponte do Abunã é inaugurada em Rondônia e vai facilitar o acesso ao Acre | Rondônia | G1 \(globo.com\)](https://g1.globo.com/rondonia/noticia/infraestrutura/ponte-do-abuna-e-inaugurada-em-rondonia-e-vai-facilitar-o-acesso-ao-acre-rondonia-g1-2023-07-11.g1)

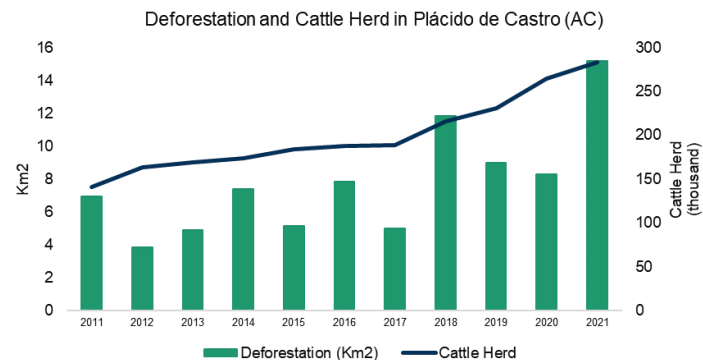
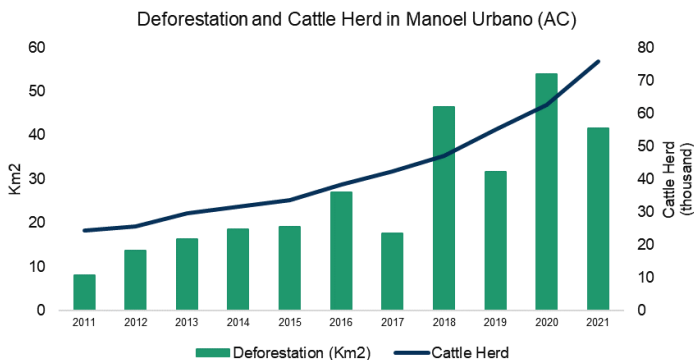
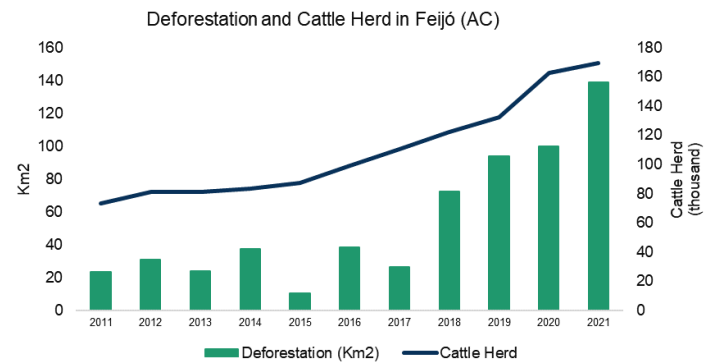
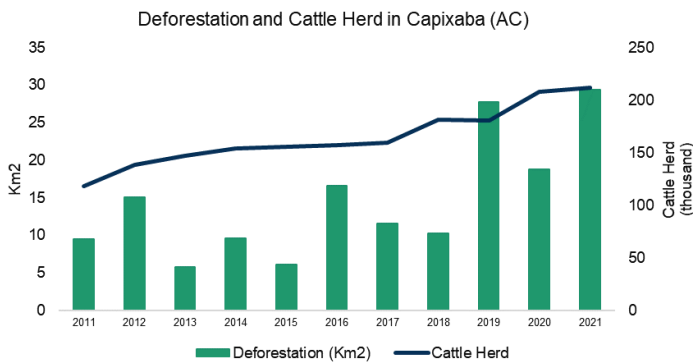
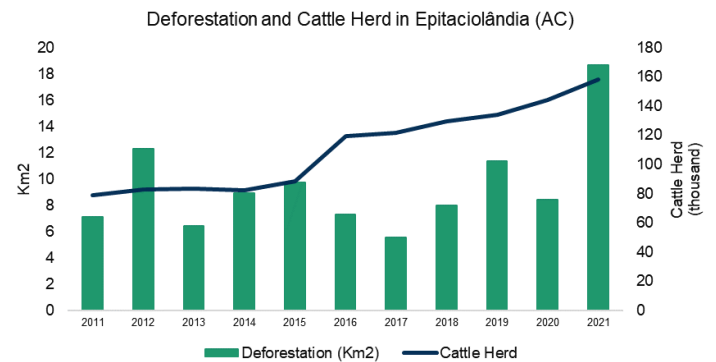
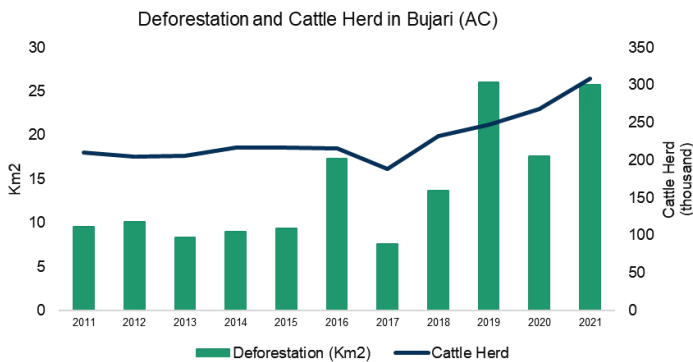
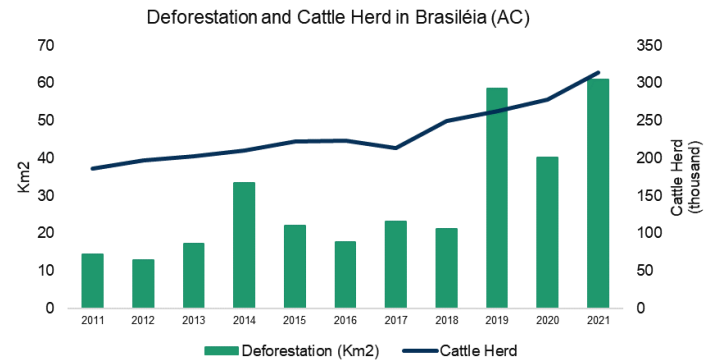
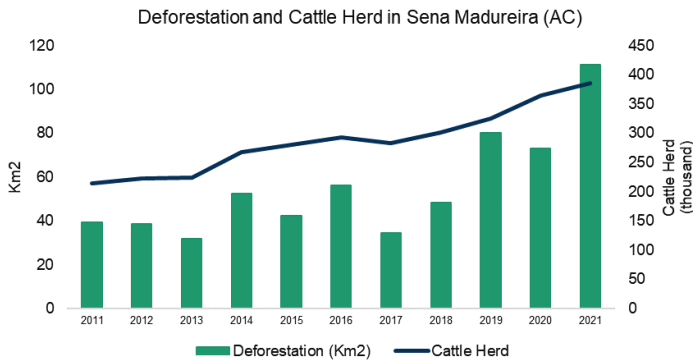
applicable laws that mandate forest conservation, the current political and economic context, as mentioned above, suggests a systematic lack of enforcement of applicable laws and regulations.

Table 30 – Accumulated deforestation, deforestation increment and forest cover by municipality.

Municipalities	Area (Km ²)	Accumulated deforestation until 2021 (Km ²)	Increment of Deforestation last two years (km ²)	Forest Cover 2021 (km ²)
Brasiléia	3,927.00	1,589.25 (40.47 %)	101.32 (2.58%)	2,337.6 (59.53 %)
Bujari	3,017.00	1,328.57 (44.04 %)	43.45 (1.45%)	1,688.2 (55.96 %)
Capixaba	1,716.00	1,002.23 (58.41 %)	48.20 (2.81%)	713.8 (41.60 %)
Epitaciolândia	1,659.00	923.57 (55.67 %)	27.10 (1.63%)	735.4 (44.33 %)
Feijó	27,964.00	1,953.4 (6.99 %)	238.89 (0.86%)	25,993.1 (92.95 %)
Manoel Urbano	10,647.00	564.7 (5.30 %)	95.54 (0.90%)	10,034.6 (94.25 %)
Plácido de Castro	1,936.00	1,511.85 (78.09 %)	23.47 (1.21%)	424.2 (21.91 %)
Porto Acre	2,618.00	1,410.9 (53.89 %)	51.01 (1.95%)	1,207.1 (46.11 %)
Rio Branco	8,822.00	3,017.16 (34.20 %)	153.37 (1.74%)	5,804.7 (65.80 %)
Santa Rosa do Purus	6,131.00	112.24 (1.83 %)	9.26 (0.15%)	5,984.1 (97.60 %)
Tarauacá	20,226.00	1,827.24 (9.03 %)	155.77 (0.77%)	18,384.5 (90.90 %)
Senador Guiomard	2,330.00	1,761.15 (75.59 %)	30.02 (1.29%)	568.9 (24.42 %)
Sena Madureira	23,731.00	2,247.41 (9.47 %)	184.31 (0.78%)	21,480.4 (90.52 %)
Xapuri	5,351.00	1,582.65 (29.58 %)	98.09 (1.84%)	3,768.4 (70.42 %)

The histograms presented below illustrate the dynamics of historical data on accumulated deforestation versus cattle herd for the 14 project municipalities. Highlight for the municipality of Rio Branco, today considered the main financial center of the state of Acre when it comes to agribusiness, where the largest cattle producers in the region are located. The municipality contributes with the largest agro GDP and is the second in the deforestation ranking with Feijó as the first, having surpassed the mark of 100 km² deforested in 2021. The municipalities in Figure 29 signs a growth future trend in cattle population between 2017-2021 and a positive relationship with deforestation area. Nowadays, an expressive growth rate encompasses the states of the North: Rondônia, Tocantins and Acre, and 234,4 million of cattle head in 2022 equivalent to 4,3% in comparison to 2021 for the whole country context³¹.

³¹ [Com rebanho de 234,4 milhões, Brasil tem mais gado que pessoas, mostram dados do IBGE | CNN Brasil](#)



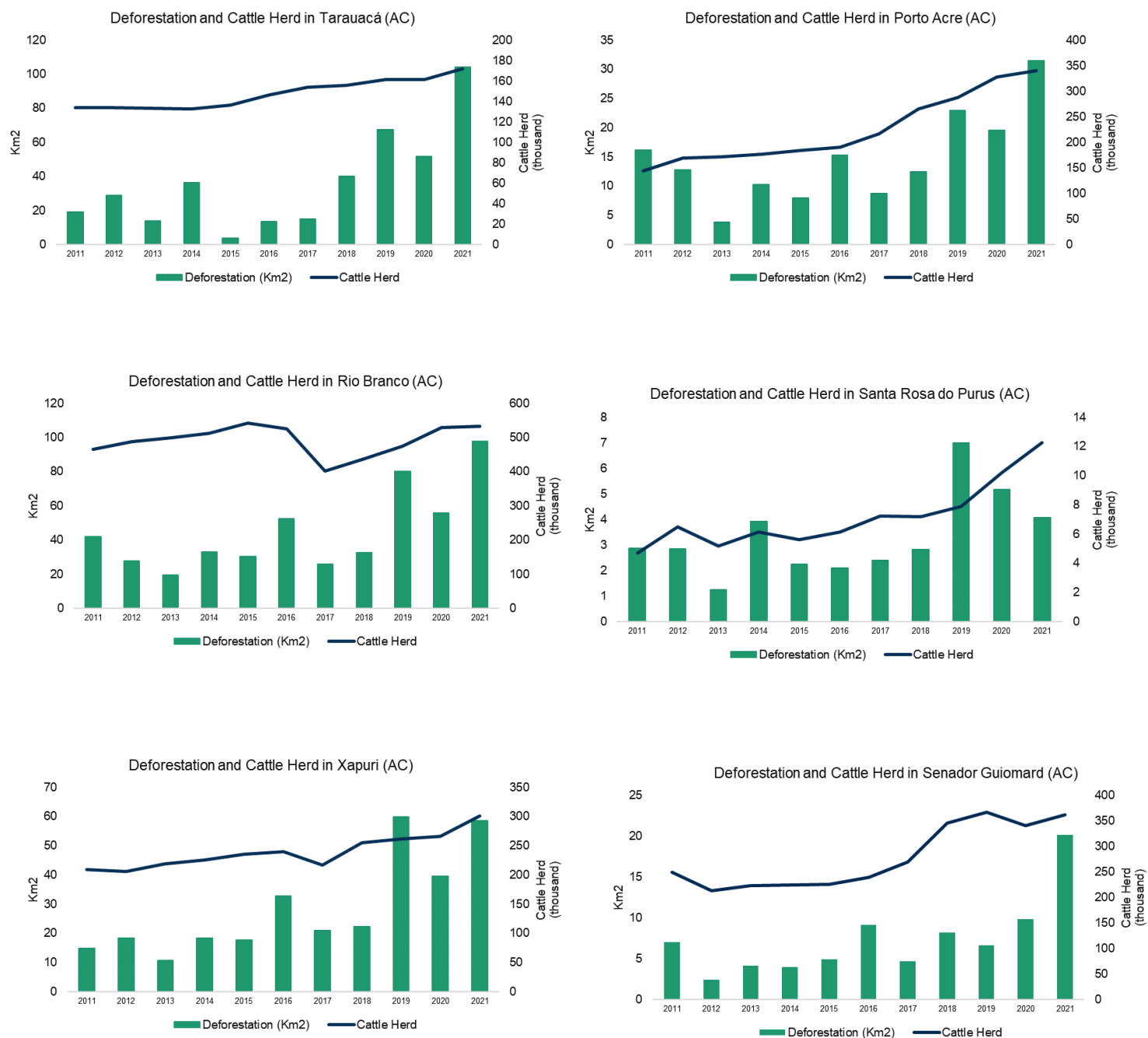


Figure 29 – Deforestation (Km²) and Head of Cattle in the six Municipalities of the Project (IBGE).

STEP 2. Investment analysis

Sub-step 2a. Determine appropriate analysis method.

Considering that the VCS AFOLU grouped project generates no financial or economic benefits other than VCS related income, the VT0001 v.3 states that the simple cost analysis (Option I) must be used.

Sub-step 2b. Option I. Apply simple cost analysis.

Considering that project activity cannot be associated to financial revenues, the simple cost analysis (option I) was chosen.

Despite the economic potential of non-timber forest products, landowners in the region do not make use of this potential for several reasons. The most important is the regional economic model, strongly based on livestock and agribusiness, and the lack of a structured market that guarantees flow and good prices. In addition, the lack of research for the use and processing methods still demands a lot of work. Rare are the non-timber forest products in large quantities and generate considerable values, such as açai and Brazil nuts, which together in 2021 generated almost R\$ 233.754 mil reais in the main municipalities of the project's micro-region (IBGE, 2021). In general, very little is known about the forest's potential for food, medicine, fiber, oils, and extracts. Despite the enormous known variety of species with non-timber economic potential, the lack of investment in research, mapping of individuals, storage and processing techniques, logistical restrictions, and technical assistance, added to a still incipient consumer market, means that many species of flora, with the possibility of generating alternative income for families, is not commercially exploited. On the other hand, logging, charcoal, and livestock activities end up consolidating themselves as the main activities associated to deforestation, since all 3 have a well-consolidated market and technological packages. In the case of livestock activity, in addition to the widespread culture and structured market with good liquidity, it also has a series of incentives and financing lines. Thus, in the scenario with the project, where activities are implemented to reduce illegal deforestation and promote the maintenance of Legal Reserve areas, is not plausible to be associated with any source of income, other than that arising from the sale of reduced emissions credits (VCU).

Finally, when it's analyzed the only products from extractive action that have any relevance in the project region, the açai and Brazil nut, it is worth mentioning that even in a scenario without a project, the Brazil nut trees and açai trees will remain in the pasture.

STEP 3. Barrier analysis

As stated in the VT0001, barrier analysis may be performed instead of or as an extension of investment analysis. If this step is used, determine whether the proposed project activity faces barriers that:

- a) Prevent the implementation of this type of proposed project activity without the revenue from the sale of GHG credits; and
- b) Do not prevent the implementation of at least one of the alternative land use scenarios. Use the following sub-steps:

Sub-step 3a. Identify barriers that would prevent the implementation of the type of proposed project activity.

The following barriers were identified, as those able to prevent the project activities in a scenario without any kind of PES mechanism, as carbon credits (VCUs). If not alone, at least in conjunction, these barriers can prevent the implementation of the project activities.

I. Institutional barriers

This refers to the risks related to changes in government policies or laws, as well as the weakening of enforcement of forest and land-use-related legislation. Despite the legal restriction on deforestation in RL and APP areas. In practice, this restriction depends on law enforcement, which in some political contexts ends up not happening, on the contrary, in recent years there has been a movement to dismantle the inspection apparatus motivated by political actors at different levels, including the federal executive branch, as presented in APPENDIX_ADDITIONALITY

II. Technological barriers

Absence of facilities to convert, store and add value to production, in other words, this barrier refers to the lack of equipment and infrastructure for NTFP processing, as laboratories, bio factories, fruit dryers, warehouses, freezers, etc.

III. Barriers related to local tradition

Despite some local people keeping some ethnobotanical knowledge, most communities face significant technical assistance restriction and do not practice most of this knowledge, that could lead to an income generation based in NTFP. In addition, this knowledge has not been taught to new generations in a systematic way.

IV. Barriers due to social conditions and land-use practices

This is assumed as a potential barrier, once the illegal practices assumed as the baseline scenario (illegal logging and illegal deforestation, followed by cattle raising, as well as crimes related to land disputes and land grabbing) are associated with a quick return, in addition of being widespread in the region. This, associated with the low-income profile of most population, ends up boosting the pressure over the forest remnants.

V. Barriers relating to markets, transport, and storage

Firstly, the distance from project activities and undeveloped roads and infrastructure that incur large transport costs to the final customer³², eroding the competitiveness and profitability of timber and non-timber products in the project area. And then, absence of facilities to convert, store and add value to production.

³² Considering that most of the NTFP produced in the Project has its final customer in big cities out of the Northern region. (e.g.: vegetal oils, cocoa, nuts and açai)

For documented evidence regarding the above-mentioned barriers, see the references in APPENDIX_ADDITIONALITY.

Sub-step 3b. Show that the identified barriers would not prevent the implementation of at least one of the alternative land use scenarios (except the proposed project activity).

I. Institutional barriers

Although there are institutional restrictions (laws and inspection agencies) for opening new areas of forest for livestock activities, the field reality does not reflect this constraint. The systematic lack of resources, personnel and infrastructure, associated with corruption involving public servants, landowners, political, land grabbers and ranchers (please refer to APPENDIX_ADDITIONALITY) lead to a scenario of little effectiveness of the command-and-control apparatus against deforestation, so that, institutional barriers, although they exist in theory, in practice they are unable to curb illegal deforestation for livestock occupation and grabbing purposes.

II. Technological barriers

The PP were not able to identify any technological barrier that could prevent the livestock activity in the project region, in the contrary, this activity, as this is conducted in the region, does not require much technological resources.

III. Barriers due to local ecological conditions

Livestock activity is very flexible and not affected by local ecological conditions. Only severe droughts are supposed to disturb the activity, but considering the high precipitation index in the region, it does not figure a barrier for the baseline activity.

IV. Barriers related to local tradition.

The PP were not able to identify any local tradition barrier that could prevent livestock activity in the project region. Cattle raising activity is very common and widespread in the region, furthermore the cattle raising works as a high liquidity savings for locals.

V. Barriers due to social conditions and land-use practices

The PP were not able to identify any social conditions or land-use practices barrier that could prevent the livestock or agriculture activity in the project region once the cattle rising and agriculture activity is common and widespread in the region.

VI. Barriers relating to markets, transport and storage.

The livestock activity has good liquidity and despite market prices fluctuation, does not face real market restriction. Herd transportation and storage can be a problem due to the road's conditions. However, this does not figure a real restriction once the herd can be commercialized to local slaughterhouses and be transported by road in the dry season.

STEP 4. Common practice analysis

According to step 4, the project proponent must provide an analysis to which extent similar activities to the one proposed as the VCS AFOLU project activity have been implemented previously or are currently underway.

Similar activities are defined as that which are of similar scale, take place in a comparable environment, inter alia, with respect to the regulatory framework and are undertaken in the relevant geographical area, subject to further guidance by the underlying methodology. Other registered VCS AFOLU project activities shall not be included in this analysis.

3.1.6 Methodology Deviations (VCS, 3.20)

According to the VCS v.4.5, section 3.20, projects may deviate from the procedures related to measurements or monitoring set out in methodologies where the proposed deviation leads to the same level of accuracy or is more conservative than what is set out in the methodology. This GPD has one methodology deviations:

1. *CAB_{tree,i,t} - carbon stock in aboveground biomass trees was estimated from LiDAR data through local drone surveys. The proposed methodology based on wall-to-wall procedure is considered as the most reliable applications with remote sensing for carbon estimations. The advantages are in mapping and detail more precisely large forest areas despite ground-based forest inventory. The carbon stock provided by this approach does not distinguish among specialized literature review and the last released 4th Emissions Inventory from MCTIC (2020) in Brazil.*

Thus, with the proposed deviation, the project will achieve the same level of accuracy in GHG estimates. The deviation will not negatively impact the conservativeness of the quantification of GHG emission reductions.

3.2 Quantification of Estimated GHG Emission Reductions and Removals

3.2.1 Baseline Emissions (VCS, 3.15)

The procedures for quantifying baseline emissions are aligned with VM0015. In step 5 of the methodology, baseline activity data for forest to non-forest change was established (see definition of LU/LC change component of baseline, section 8.1.4).

This section outlines the changes in carbon stocks across the pools within the baseline scenario. Initial stocks are derived from direct measurements conducted in the project areas within the Sena Madureira region, on private properties of Katianã, Porongaba, Palmares and Potiguar rubber

plantations. UAV-LiDAR surveys for aboveground carbon estimations, data from local studies following the TIER 3 IPCC quality instructions, peer-reviewed literature, and IPCC data parameters, while post-deforestation stocks are sourced from peer-reviewed literature were applied.

This section is derived from step 6 of VM0015 (section 8.1.5), aiming to estimate changes in baseline carbon stocks and baseline non-CO2 emissions resulting from forest fires utilized for deforestation.

The average carbon stocks were estimated for each LU/LC class. The Uákiry REDD+ Grouped project collected field data necessary to estimate carbon stocks in all pools within the forest class considered within the project boundaries. All procedures are based on Standard Operating Procedures (SOP), and detailed results can be found in the forestry measurement report (APPENDIX_CARBON-POOLS).

3.2.1.1 Initial Forest class

3.2.1.1.1 Overall pools

Forest carbon stock in aboveground tree biomass (CAB_{tree,bsl,i}) was determined using UAV-LiDAR surveys for estimates in a large fixed sampled areas, employing representative random sampling per forest classes, as per VM0015 requirements. The carbon stock in belowground tree biomass (CBB_{tree,bsl,i}) was calculated based on root-to-shoot ratios sourced from peer-reviewed literature (Nogueira et al., 2008). Similarly, carbon stock in aboveground non-tree biomass (CAB_{non-tree,bsl,i}) was estimated according to the findings of same peer-reviewed literature (Nogueira et al., 2008). Palms and vines are accounted for as 1.9% and 3.4%, respectively, of the aboveground tree biomass in dense tropical forests, and as 8.6% and 10.7%, respectively, in open tropical forests.

Forest carbon stock in dead wood (CDW_{bsl,i}) was estimated based on field measurements of fixed area plots, considering standing dead trees and lying dead wood, according to VM0015 criteria. Forest carbon stock in litter (CLI_{bsl,i}) is estimated according to peer reviewed literature through root-to-shoot ratios of 4.1% (Nogueira et al., 2008). Within the project areas, no evidence of management activities has occurred before calculating the baseline carbon stock changes. Thus, three forest classification were considered on estimation of carbon stocks: Open alluvial forest, Open lowland forest and Dense Ombrophilous forest. The following tables shows the values for the different forests in LB and PA.

Table 31 - Carbon stocks per hectare of initial forest classes icl existing in the project area and leakage belt ID icl=1 (Table 25 from VM0015 v1.2).

Project year t	Initial forest class icl													
	Name	Open alluvial forest (Aa)												
	IDicl	1												
	Average carbon stock per hectare +-90% CI													
	Cabicl		Cbbicl		Cdwicl		Clicl		Cab_nontreeicl		Cbb_nontreeicl		Ctoticl	
	C stock (t CO2e/ ha)	+90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+90% CI (t CO2e/ ha)	C stock (t CO2e/ ha)	+90% CI (t CO2e/ ha)
0	370.05		37.01		14.20		9.49		39.60		7.92		478.26	
1		395.95		39.60		18.46		10.25		42.33		8.47		511.74
2														
3														
...		344.15		34.41		9.94		8.73		36.86		7.36		444.78
T														

Table 32 - Carbon stocks per hectare of initial forest classes icl existing in the project area and leakage belt ID icl=2 (Table 25 from VM0015 v1.2).

Project year t	Initial forest class icl													
	Name	Open lowland forest (Ab)												
	IDicl	2												
	Average carbon stock per hectare +-90% CI													
	Cabicl		Cbbicl		Cdwicl		Clicl		Cab_nontreeicl		Cbb_nontreeicl		Ctoticl	
	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)
0	402.11		40.21		9.64		9.64		43.03		8.61		513.23	
1		430.26		43.03		12.53		10.41		45.99		9.21		549.16
2														
3														
..		373.96		37.40		6.75		8.87		40.06		8.00		477.31
T														

Table 33 - Carbon stocks per hectare of initial forest classes icl existing in the project area and leakage belt ID icl=3 (Table 25 from VM0015 v1.2).

Project year t	Initial forest class icl														
	Name	Dense Ombrophilous forest (Db)													
	IDicl	3													
	Average carbon stock per hectare +-90% CI														
	Cabicl		Cbbicl		Cdwicl		Clicl		Cab_nontreeicl		Cbb_nontreeicl		Ctoticl		
	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	C stock (t CO2e/ha)	+90% CI (t CO2e/ha)	
0	470.43		145.93		44.37		19.07		24.93		4.99		709.72		
1		503.36		156.15		57.68		20.59		26.65		5.34		759.40	
2															
3															
...		437.50		135.72		31.06		17.54		23.21		4.64		660.04	
T															

3.2.1.1.2 Wood Products

Not applicable. Throughout the development of the Project Description, the PP meticulously examined all carbon pools, and it was determined that in this first approach the values attributed to harvested wood products were deemed insignificant.

3.2.1.1.3 Post-deforestation Class – Pasture

Post-deforestation carbon stock from land-use changes to pasture is assumed to be the long-term average stocks in a period of deforestation. Carbon stock in pasture is estimated based on parameters from local studies, literature and IPCC. It was decided to adopt the nominal error established in the IPCC reference (75%), from which the data were extracted. The PP understands that adopting the IPCC nominal error, the estimates for the post deforestation class (grassland) stay in line with the database from which it was originated, in addition, it presents much more conservative values, avoiding any overestimation. For more information, detailed methodology, parameter tables and total sampling used in database, see project APPENDIX CARBON-POOLS.

Table 34 - Long-term (20-years) average carbon stocks per hectare of post-deforestation LU/LC classes present in the reference region (Table 27 from VM0015 v1.2).

Zone		Post-deforestation LU/LC-class <i>fcl</i>							Area weighted long-term (20 years) average carbon stocks per zone <i>z</i>					
		Name: Pasture <i>fcl</i> =1												
		<i>Cab_{fcl}</i>	<i>Cbb_{fcl}</i>	<i>Cdw_{icl}</i>	<i>Cab_{nontree}</i>	<i>Cbb_{nontree}</i>	<i>Clfcl</i>	<i>Cab_z</i>	<i>Cbb_z</i>	<i>Cdw_{icl}</i>	<i>Cab_{nontree z}</i>	<i>Cbb_{nontree z}</i>	<i>Clfcl z</i>	<i>Ctot z</i>
		<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>	<i>C stock</i>
<i>ID_z</i>	<i>Nam_e</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>	<i>t CO₂e ha⁻¹</i>
1	RR	19.31	50.15	0.00	0.00	0.00	0.00	19.31	50.15	0.00	0.00	0.00	0.00	69.46

3.2.1.1.4 Calculations of emissions factors

The current methodology VM0015 v1.2 section 8.1.5.1 for AFOLU uses default linear functions as premises for decay in carbon stocks based on initial forest classes (*icl*) and afterwards an increase in carbon stock in post-deforestation classes (*fcl*) for all included pools described in PD section 3.1.3. The subsequent tables provide the flux of how the carbon emissions factors based on initial classes on LU/LC procedures.

Table 35 - Emission Factors for Initial Forest Classes icl = 1 (Method 1) (Table 33 from VM0015 v1.2)

Open alluvial forest							
Year after deforestation		$\Delta C_{ab,ctz,t}$	$\Delta C_{bb,ctz,t}$	$\Delta C_{dw,ctz,t}$	$\Delta C_{ab,nontree,ctz,t}$	$\Delta C_{bb,nontree,ctz,t}$	$\Delta C_{Li,ctz,t}$
1	t^*	-370.1	-3.7	-1.4	-39.6	-0.8	-9.5
2	t^*+1	0.0	-3.7	-1.4	0.0	-0.8	0.0
3	t^*+2	0.0	-3.7	-1.4	0.0	-0.8	0.0
4	t^*+3	0.0	-3.7	-1.4	0.0	-0.8	0.0
5	t^*+4	0.0	-3.7	-1.4	0.0	-0.8	0.0
6	t^*+5	0.0	-3.7	-1.4	0.0	-0.8	0.0
7	t^*+6	0.0	-3.7	-1.4	0.0	-0.8	0.0
8	t^*+7	0.0	-3.7	-1.4	0.0	-0.8	0.0
9	t^*+8	0.0	-3.7	-1.4	0.0	-0.8	0.0
10	t^*+9	0.0	-3.7	-1.4	0.0	-0.8	0.0
11	t^*+10	0.0	0.0	0.0	0.0	0.0	0.0
12	t^*+11	0.0	0.0	0.0	0.0	0.0	0.0
13	t^*+12	0.0	0.0	0.0	0.0	0.0	0.0
14	t^*+13	0.0	0.0	0.0	0.0	0.0	0.0
15	t^*+14	0.0	0.0	0.0	0.0	0.0	0.0
16	t^*+15	0.0	0.0	0.0	0.0	0.0	0.0
17	t^*+16	0.0	0.0	0.0	0.0	0.0	0.0
18	t^*+17	0.0	0.0	0.0	0.0	0.0	0.0
19	t^*+18	0.0	0.0	0.0	0.0	0.0	0.0
20	t^*+19	0.0	0.0	0.0	0.0	0.0	0.0
21	t^*+20	0.0	0.0	0.0	0.0	0.0	0.0

Table 36 - Emission Factors for Initial Forest Classes icl = 2 (Method 1) (Table 33 from VM0015 v1.2)

Open lowland forest							
Year after deforestation		$\Delta Cab_{ctz,t}$	$\Delta Cbb_{ctz,t}$	$\Delta Cdw_{ctz,t}$	$\Delta Cab_{nontree,ctz,t}$	$\Delta Cbb_{nontree_{ctz,t}}$	$\Delta CLI_{ctz,t}$
1	t^*	-402.1	-4.0	-1.0	-43.0	-0.9	-9.6
2	t^*+1	0.0	-4.0	-1.0	0.0	-0.9	0.0
3	t^*+2	0.0	-4.0	-1.0	0.0	-0.9	0.0
4	t^*+3	0.0	-4.0	-1.0	0.0	-0.9	0.0
5	t^*+4	0.0	-4.0	-1.0	0.0	-0.9	0.0
6	t^*+5	0.0	-4.0	-1.0	0.0	-0.9	0.0
7	t^*+6	0.0	-4.0	-1.0	0.0	-0.9	0.0
8	t^*+7	0.0	-4.0	-1.0	0.0	-0.9	0.0
9	t^*+8	0.0	-4.0	-1.0	0.0	-0.9	0.0
10	t^*+9	0.0	-4.0	-1.0	0.0	-0.9	0.0
11	t^*+10	0.0	0.0	0.0	0.0	0.0	0.0
12	t^*+11	0.0	0.0	0.0	0.0	0.0	0.0
13	t^*+12	0.0	0.0	0.0	0.0	0.0	0.0
14	t^*+13	0.0	0.0	0.0	0.0	0.0	0.0
15	t^*+14	0.0	0.0	0.0	0.0	0.0	0.0
16	t^*+15	0.0	0.0	0.0	0.0	0.0	0.0
17	t^*+16	0.0	0.0	0.0	0.0	0.0	0.0
18	t^*+17	0.0	0.0	0.0	0.0	0.0	0.0
19	t^*+18	0.0	0.0	0.0	0.0	0.0	0.0
20	t^*+19	0.0	0.0	0.0	0.0	0.0	0.0
21	t^*+20	0.0	0.0	0.0	0.0	0.0	0.0

Table 37 - Emission Factors for Initial Forest Classes icl = 3 (Method 1) (Table 33 from VM0015 v1.2)

Dense Ombrophilous Forest 3							
Year after deforestation		$\Delta Cab_{ctz,t}$	$\Delta Cbb_{ctz,t}$	$\Delta Cdw_{ctz,t}$	$\Delta Cab_{nontree,ctz,t}$	$\Delta Cbbnontree_{ctz,t}$	$\Delta CLI_{ctz,t}$
1	t^*	-470.4	-14.6	-4.4	-24.9	-0.5	-19.1
2	t^*+1	0.0	-14.6	-4.4	0.0	-0.5	0.0
3	t^*+2	0.0	-14.6	-4.4	0.0	-0.5	0.0
4	t^*+3	0.0	-14.6	-4.4	0.0	-0.5	0.0
5	t^*+4	0.0	-14.6	-4.4	0.0	-0.5	0.0
6	t^*+5	0.0	-14.6	-4.4	0.0	-0.5	0.0
7	t^*+6	0.0	-14.6	-4.4	0.0	-0.5	0.0
8	t^*+7	0.0	-14.6	-4.4	0.0	-0.5	0.0
9	t^*+8	0.0	-14.6	-4.4	0.0	-0.5	0.0
10	t^*+9	0.0	-14.6	-4.4	0.0	-0.5	0.0
11	t^*+10	0.0	0.0	0.0	0.0	0.0	0.0
12	t^*+11	0.0	0.0	0.0	0.0	0.0	0.0
13	t^*+12	0.0	0.0	0.0	0.0	0.0	0.0
14	t^*+13	0.0	0.0	0.0	0.0	0.0	0.0
15	t^*+14	0.0	0.0	0.0	0.0	0.0	0.0
16	t^*+15	0.0	0.0	0.0	0.0	0.0	0.0
17	t^*+16	0.0	0.0	0.0	0.0	0.0	0.0
18	t^*+17	0.0	0.0	0.0	0.0	0.0	0.0
19	t^*+18	0.0	0.0	0.0	0.0	0.0	0.0
20	t^*+19	0.0	0.0	0.0	0.0	0.0	0.0
21	t^*+20	0.0	0.0	0.0	0.0	0.0	0.0

3.2.1.1.5 Calculation of Baseline Carbon Stock Changes

The VM0015 section 8.1.4.2 provides two methods to project the LU/LC classes for carbon stock changes that will replace forests in the baseline historical period. Thus, the *Method 1* was applied in which assumes a representative future trends for post-deforestation. The result of each class and carbon pools calculations is reported from

Table 38 to Table 44 for the project area.

Table 38 - Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) (Table 36 from VM0015 v1.2).

Carbon Stock Changes in Aboveground Biomass per Initial Forest Class <i>icl</i>				Total Carbon Stock Change in Aboveground Biomass of Initial Forest Classes in the Project Area		Carbon Stock Changes in Aboveground Biomass Per Post-deforestation Zone <i>z</i>		Total Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Project Area		Total Net Carbon Stock Change in Aboveground Biomass of Post-deforestation Zones in the Project Area	
ID _{icl}	1	2	3	ΔCab BSLPA _{icl} ,t Annual	ΔCab BSLPA <i>icl</i> Cumulative	ID _{iz}	1	ΔCab BSLPA _z ,t Annual	ΔCabBSLPA _z Cumulative	ΔCab BSLPA _t Annual	ΔCab BSLPA Cumulative
Name	Open alluvial	Open lowland	Dense Ombrophilous			Name	Open alluvial				
Project year	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	Project year	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e
2021	- 9,584.30	-176,888.19	-47,654.90	-234,127.38	-234,127.38	2021	1,095.21	1,095.21	1,095.21	-233,032.17	-233,032.17
2022	- 9,584.30	-176,888.19	-47,654.90	-234,127.38	-468,254.76	2022	2,190.42	2,190.42	3,285.63	-231,936.96	-464,969.13
2023	- 9,584.30	-176,888.19	-47,654.90	-234,127.38	-702,382.14	2023	3,285.63	3,285.63	6,571.26	-230,841.75	-695,810.89
2024	- 9,584.30	-176,888.19	-47,654.90	-234,127.38	-936,509.52	2024	4,380.84	4,380.84	10,952.09	-229,746.54	-925,557.43
2025	- 9,584.30	-176,888.19	-47,654.90	-234,127.38	-1,170,636.90	2025	5,476.05	5,476.05	16,428.14	-228,651.33	-1,154,208.76
2026	- 9,584.30	-176,888.19	-47,654.90	-234,127.38	-1,404,764.28	2026	6,571.26	6,571.26	22,999.40	-227,556.12	-1,381,764.89

Table 39 - Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) (Table 36 from VM0015 v1.2).

Carbon Stock Changes in Below ground Biomass per Initial Forest Class icl				Total Carbon Stock Change in Below ground Biomass of Initial Forest Classes in the Project Area		Carbon Stock Changes in Below ground Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Below ground Biomass of Post-deforestation Zones in the Project Area		Total Net Carbon Stock Change in Below ground Biomass of Post-deforestation Zones in the Project Area	
ID icl	1	2	3	ΔC_{bb} BSLPA _{icl,t} Annual	ΔC_{bb} BSLPA _{icl} Cumulative	ID z	1	ΔC_{bb} BSLPA _{z,t} Annual	ΔC_{bb} BSLPA _z Cumulative	ΔC_{bb} BSLPA _t Annual	ΔC_{bb} BSLPA Cumulative
Name	Aa	Ab	Db			Name					
Project year	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	Project year	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e
2021	-95.84	-1,768.88	- 1,478.30	-3,343.03	-3,343.03	2021	2,844.01	2,844.01	2,844.01	-499.02	-499.02
2022	- 191.69	-3,537.76	- 2,956.61	-6,686.06	- 10,029.09	2022	5,688.02	5,688.02	8,532.04	-998.04	-1,497.05
2023	- 287.53	-5,306.65	- 4,434.91	- 10,029.09	- 20,058.18	2023	8,532.04	8,532.04	17,064.07	- 1,497.05	-2,994.11
2024	- 383.37	-7,075.53	- 5,913.22	- 13,372.12	- 33,430.30	2024	11,376.0 5	11,376.0 5	28,440.12	- 1,996.07	-4,990.18
2025	- 479.21	-8,844.41	- 7,391.52	- 16,715.15	- 50,145.44	2025	14,220.0 6	14,220.0 6	42,660.18	- 2,495.09	-7,485.27
2026	- 575.06	- 10,613.29	- 8,869.83	- 20,058.18	- 70,203.62	2026	17,064.0 7	17,064.0 7	59,724.25	- 2,994.11	- 10,479.37

Table 40 - Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) (Table 36 from VM0015 v1.2).

Carbon Stock Changes in Dead wood Biomass per Initial Forest Class icl				Total Carbon Stock Change in Dead wood Biomass of Initial Forest Classes in the Project Area		Carbon Stock Changes in Dead wood Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Dead wood Biomass of Post-deforestation Zones in the Project Area		Total Net Carbon Stock Change in Dead wood Biomass of Post-deforestation Zones in the Project Area	
ID icl	1	2	3	ΔCdw BSLPAicl,t Annual	ΔCdw BSLPA icl Cumulative	ID z	1	ΔCdw BSLPAz,t Annual	ΔCdwBSLRR z Cumulative	ΔCdw BSLPA t Annual	ΔCdw BSLPA Cumulative
Name	Open alluvial	Open lowland	Dense Ombrophilous			Name	Open alluvial				
Project year	t CO2e	t CO2e	t CO2e	t CO2e	t CO2e	Project year	t CO2e	t CO2e	t CO2e	t CO2e	t CO2e
2021	-36.78	-424.06	-449.43	-910.28	-910.28	2021	-	-	-	-910.28	-910.28
2022	-73.56	-848.13	-898.87	-1,820.55	-2,730.83	2022	-	-	-	-1,820.55	-2,730.83
2023	-110.33	-1,272.19	-1,348.30	-2,730.83	-5,461.66	2023	-	-	-	-2,730.83	-5,461.66
2024	-147.11	-1,696.25	-1,797.74	-3,641.10	-9,102.76	2024	-	-	-	-3,641.10	-9,102.76
2025	-183.89	-2,120.32	-2,247.17	-4,551.38	-13,654.14	2025	-	-	-	-4,551.38	-13,654.14
2026	-220.67	-2,544.38	-2,696.61	-5,461.66	-19,115.79	2026	-	-	-	-5,461.66	-19,115.79

Table 41 - Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) (Table 36 from VM0015 v1.2).

Carbon Stock Changes in litter Biomass per Initial Forest Class icl				Total Carbon Stock Change in litter Biomass of Initial Forest Classes in the Project Area		Carbon Stock Changes in litter Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in litter Biomass of Post-deforestation Zones in the Project Area		Total Net Carbon Stock Change in litter Biomass of Post-deforestation Zones in the Project Area	
ID icl	1	2	3	ΔCli BSLPAicl,t Annual	ΔCli BSLPA icl Cumulative	ID z	1	ΔCli BSLPAz,t Annual	ΔCli BSLPAz Cumulative	ΔCli BSLPA t Annual	ΔCli BSLPA Cumulative
Name	Open alluvial	Open lowlands	Dense Ombrophilous			Name	Open alluvial				
Project year	t CO2e	t CO2e	t CO2e	t CO2e	t CO2e	Project year	t CO2e	t CO2e	t CO2e	t CO2e	t CO2e
2021	-245.79	-4,240.64	-1,931.45	-6,417.88	-6,417.88	2021	-	-	-	-6,417.88	-6,417.88
2022	-245.79	-4,240.64	-1,931.45	-6,417.88	-12,835.76	2022	-	-	-	-6,417.88	-12,835.76
2023	-245.79	-4,240.64	-1,931.45	-6,417.88	-19,253.64	2023	-	-	-	-6,417.88	-19,253.64
2024	-245.79	-4,240.64	-1,931.45	-6,417.88	-25,671.52	2024	-	-	-	-6,417.88	-25,671.52
2025	-245.79	-4,240.64	-1,931.45	-6,417.88	-32,089.40	2025	-	-	-	-6,417.88	-32,089.40
2026	-245.79	-4,240.64	-1,931.45	-6,417.88	-38,507.28	2026	-	-	-	-6,417.88	-38,507.28

Table 42 - Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) (Table 36 from VM0015 v1.2).

Carbon Stock Changes in Aboveground nontree Biomass per Initial Forest Class icl				Total Carbon Stock Change in Aboveground nontree Biomass of Initial Forest Classes in the Project Area		Carbon Stock Changes in Aboveground nontree Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in Aboveground nontree Biomass of Post-deforestation Zones in the Project Area		Total Net Carbon Stock Change in Aboveground nontree Biomass of Post-deforestation Zones in the Project Area	
ID icl	1	2	3	Δcab_ntree BSLPAicl,t Annual	Δcab_ntree BSLPAicl Cumulative	ID z	1	Δcab_ntree BSLPAz,t Annual	Δcab_ntree BSLR Rz Cumulative	Δcab_nontr ee BSLPA Annual	Δcab_ntree BSLPA Cumulative
Name Project year	Open alluvial t CO2e	Open lowlands t CO2e	Dense Ombrophilous t CO2e	t CO2e	t CO2e	Name Project year	Open alluvial t CO2e	t CO2e	t CO2e	t CO2e	t CO2e
2021	-1,025.52	-18,927.04	-2,525.71	- 22,478.27	-22,478.27	2021	-	-	-	-22,478.27	-22,478.27
2022	-1,025.52	-18,927.04	-2,525.71	- 22,478.27	-44,956.53	2022	-	-	-	-22,478.27	-44,956.53
2023	-1,025.52	-18,927.04	-2,525.71	- 22,478.27	-67,434.80	2023	-	-	-	-22,478.27	-67,434.80
2024	-1,025.52	-18,927.04	-2,525.71	- 22,478.27	-89,913.06	2024	-	-	-	-22,478.27	-89,913.06
2025	-1,025.52	-18,927.04	-2,525.71	- 22,478.27	-112,391.33	2025	-	-	-	-22,478.27	-112,391.33
2026	-1,025.52	-18,927.04	-2,525.71	- 22,478.27	-134,869.59	2026	-	-	-	-22,478.27	-134,869.59

Table 43 - Baseline Carbon Stock Change for Selected Carbon Pools in the Project Area (Method 1) (Table 36 from VM0015 v1.2).

Carbon Stock Changes in belowground nontree Biomass per Initial Forest Class icl				Total Carbon Stock Change in belowground nontree Biomass of Initial Forest Classes in the Project Area		Carbon Stock Changes in belowground nontree Biomass Per Post-deforestation Zone z		Total Carbon Stock Change in belowground nontree Biomass of Post-deforestation Zones in the Project Area		Total Net Carbon Stock Change in belowground nontree Biomass of Post-deforestation Zones in the Project Area	
ID icl	1	2	3	Δcbb_ntree BSLPAicl,t Annual	Δcbb_ntree BSLPA icl Cumulative	ID z	1	Δcbb_ntree BSLPAz,t Annual	$\Delta cbb_ntreeBSLP$ Az Cumulative	Δcbb_ntree BSLPA Annual	Δcbb_ntree BSLPA Cumulative
Name	Open alluvial	Open lowlands	Dense Ombrophilous			Name	Open alluvial				
Project year	t CO2e	t CO2e	tCO2e	t CO2e	t CO2e	Project year	t CO2e	t CO2e	t CO2e	t CO2e	t CO2e
2021	-20.51	-378.54	-50.51	-449.57	-449.57	2021	-	-	-	-449.57	-449.57
2022	-41.02	-757.08	-101.03	-899.13	-1,348.70	2022	-	-	-	-899.13	-1,348.70
2023	-61.53	-1,135.62	-151.54	-1,348.70	-2,697.39	2023	-	-	-	-1,348.70	-2,697.39
2024	-82.04	-1,514.16	-202.06	-1,798.26	-4,495.65	2024	-	-	-	-1,798.26	-4,495.65
2025	-102.55	-1,892.70	-252.57	-2,247.83	-6,743.48	2025	-	-	-	-2,247.83	-6,743.48
2026	-123.06	-2,271.24	-303.09	-2,697.39	-9,440.87	2026	-	-	-	-2,697.39	-9,440.87

The total Net Carbon Stock Change in **ALL POOLS** of Post-deforestation Zones in the Project Area is summarized in Table 44.

Table 44 - Baseline Carbon Stock Change in all Carbon Pools in the Project area (Method 1)

Project year	Total Net Carbon Stock Change in ALL POOLS of Post-deforestation Zones in the Project Area	
	Δ CBSLPA Annual	Δ CBSLPA Cumulative
	t CO ₂ e	t CO ₂ e
2021	-263,787.18	-263,787.18
2022	-264,550.83	-528,338.00
2023	-265,314.48	-793,652.48
2024	-266,078.12	-1,059,730.60
2025	-266,841.77	-1,326,572.38
2026	-267,605.42	-1,594,177.80

3.2.1.1.6 Baseline Non-CO₂ Emissions from Forest Fires

For deforested areas from forest fires as a source of non-CO₂ gases emissions when an important component of the baseline, PP used the section 8.1.5.2 that describes the steps of methodology to calculation. Below PP present Table 45 and Table 46 related to non-CO₂ emissions.

Table 45 - Parameters Used to Calculate Non-CO2 Emissions from Forest Fires. (Table 41 from VM015 v1.2)

Initial Forest Class		% Fburnt icl	tCO2e ha-1 Cab	tCO2e ha-1 Cdw	tCO2e ha-1 CI	% Pburnt ab, icl	% Pburnt dw, icl	% Pburnt I, icl	% CE ab, icl	% CE dw, icl	% CE I, icl	tCO2e ha-1 ECO2-ab	tCO2e ha-1 ECO2-dw	tCO2e ha-1 ECO2-I	tCO2e ha-1 EBBCO2-tot	tCO2e ha-1 EBBN20 icl	tCO2e ha-1 EBBCH4 icl	tCO2e ha-1 EBBtot icl
IDcl	Name																	
1	Open alluvial forest	24%	370.05	14.20	9.49	52%	40%	100%	50%	50%	50%	23.57	0.69	1.15	25.41	0.24	2.33	2.56
2	Open lowland forest	24%	402.11	9.64	9.64	52%	40%	100%	50%	50%	50%	25.61	0.47	1.17	27.25	0.25	2.50	2.75
3	Dense Ombrophilous forest	24%	470.43	44.37	19.07	52%	40%	100%	50%	50%	50%	29.96	2.15	2.32	34.43	0.32	3.15	3.48

Table 46 - Baseline Non-CO2 Emissions from Forest Fires in the Project Area. (Table 42 from VM0015 v1.2)

	Emissions of non-CO2 gasses from baseline forest fires						Total baseline non-CO2 emissions from forest fire in the project area	
	ID icl = 1		ID icl = 2		ID icl = 3			
Project year t	ABSLPA icl,t	EBBBSLtot icl	ABSLPA icl,t	EBBBSLtot icl	ABSLPA icl,t	EBBBSLtot icl	Annual EBBBSLPAt	Cumulative EBBBSLPA
	ha	CO2e ha-1	ha	CO2e ha-1	tCO2e	tCO2e	tCO2e	tCO2e
2021	25.90	66.43	439.90	1,209.93	101.30	352.03	1,628.39	1,628.39
2022	25.90	66.43	439.90	1,209.93	101.30	352.03	1,628.39	3,256.77
2023	25.90	66.43	439.90	1,209.93	101.30	352.03	1,628.39	4,885.16
2024	25.90	66.43	439.90	1,209.93	101.30	352.03	1,628.39	6,513.54
2025	25.90	66.43	439.90	1,209.93	101.30	352.03	1,628.39	8,141.93
2026	25.90	66.43	439.90	1,209.93	101.30	352.03	1,628.39	9,770.32

3.2.2 Project Emissions (VCS, 3.15)

According to the VM0015 section 8.2.1 (step 7), future carbon stock changes and non-CO2 emissions from forest fires under the project scenario must be estimated ex-ante to assist in guiding the optimal implementation of emission reduction measures and to allow reasonable projections of revenue to be made. In this regard, three approaches are stated in the methodology:

- 1) Planned activities within the project area.
- 2) Unplanned deforestation that cannot be avoided.
- 3) Possible future catastrophic events – which are unpredictable and are therefore excluded from the ex-ante assessment.

3.2.2.1 Ex Ante Estimation of Actual Carbon Stock Changes due to Planned Activities

As per this section of methodology, certain areas of forest within the project area might be subject to project activities that will change the carbon stocks of these areas compared to the baseline, like:

- I. Planned deforestation to build project infrastructure.
- II. Planned degradation for timber logging, fuel-wood collection, and charcoal production.
- III. Protection without harvesting leading to carbon sequestration in forest classes that at project start are below their carbon stock potential at maturity in situ.

There is no planned deforestation and degradation during the crediting period in all PAI.

3.2.2.1.1 Emissions through Planned Deforestation

Table 47 shown the emissions that will be resulting from this planned deforestation activity. No planned deforestation has been identified in the project area.

Table 47 - Ex ante estimated actual carbon stock decrease due to planned deforestation in the project area
(Table 43 from VM0015, v1.2).

Project year t	Areas of planned deforestation x Carbon stock change (decrease) in the project area				Total carbon stock decrease due to planned deforestation	
	IDct = 1		IDct = 2		annual	cumulative
	APDPA _{icl,t}	C _{toticl,t}	APDPA _{icl,t}	C _{toticl,t}	ΔCPDdPA _t	ΔCPDdPA
	ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0.00	0.00	0.00	0.00	0	0
2022	0.00	0.00	0.00	0.00	0	0
2023	0.00	0.00	0.00	0.00	0	0
2024	0.00	0.00	0.00	0.00	0	0
2025	0.00	0.00	0.00	0.00	0	0
2026	0.00	0.00	0.00	0.00	0	0

3.2.2.1.2 Emissions through Planned Degradation

Logging activities result in a reduction of carbon stocks, attributable to the biomass loss of the extracted timber and the biomass loss in the forest damaged in the process of timber extraction. The decrease in carbon stock due to logging followed the expression below. The calculated values of emissions are presented in the Table 48. No emissions from planned deforestation have been identified.

$$\Delta C_{tot_{icl,t}} = \left((Vol_{icl,t} * D_{mm} * CF) + (Vol_{icl,t} * LDF) \right) * \frac{44}{12}$$

Where:

$\Delta C_{tot_{icl,t}}$ = Carbon emission per hectare due to timber harvests in the project area in stratum *icl* in year t (t CO₂e.ha⁻¹)

$Vol_{icl,t}$ = Volume of timber per hectare extracted from within the project area in stratum *icl* in year t (m³.ha⁻¹) from PMFS.

D_{mm} = Mean wood density (g cm⁻³). 0.67 from Fearnside (1997)

CF = Carbon fraction of biomass (t C t d.m.⁻¹) 0.485 from Nogueira et al (2008)

LDF= Logging damage factor (t C m-3). 0.71 from Person et al (2014)

Table 48 - Ex Ante Estimated Actual Carbon Stock Decrease Due to Planned Logging Activities in the Project Area. (Table 44 from VM0015 v1.2).

Project year t	Areas of planned logging activities x Carbon stock change (decrease) in the project area				Total carbon stock decrease due to planned logging activities	
	<i>IDct</i> = 1		<i>IDct</i> = 2		annual	cumulative
	APLPAicl,t	ΔCtoticl,t	APLPAicl,t	ΔCtoticl,t	ΔCPLdPA _t	ΔCPLdPA
	ha	tCO ₂ -e ha ⁻¹	ha	tCO ₂ -e ha ⁻¹	tCO ₂ -e	tCO ₂ -e
2021	0.00	0.00	0.00	0.00	0.00	0.00
2022	0.00	0.00	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00	0.00	0.00
2026	-	-	-	-	-	-

The total Carbon Stock Decrease due to Planned Activities in the Project Area is shown in Table 49 below. There are no emissions from planned activities.

Table 49 - Ex Ante Carbon Stock Decrease Due to Planned Activities in the Project Area (Table 46 from VM0015 v1.2).

Project year t	Total carbon stock decrease due to planned deforestation		Total carbon stock decrease due to planned logging activities		Total carbon stock decrease due to planned activities	
	annual	cumulative	annual	cumulative	annual	cumulative
	ΔCPDdPA _t	ΔCPDdPA _t	ΔCPLdPA _t	ΔCPLdPA _t	ΔCPAdPA _t	ΔCPAdPA _t
	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	0.00	0.00	0.00	0.00	0.00	0.00
2022	0.00	0.00	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00	0.00	0.00

3.2.2.1.3 Ex-ante estimation of carbon stock changes due to unavoidable unplanned deforestation within the project area.

According to the VM0015, to allow *ex ante* projections to be made, the project proponent shall make a conservative assumption about the effectiveness of the proposed project activities and estimate an Effectiveness Index (*EI*) between 0 (no effectiveness) and 1 (maximum effectiveness). The estimated value of *EI* is used to multiply the baseline projections by the factor $(1 - EI)$ and the result shall be considered the *ex-ante* estimated emissions from unplanned deforestation in the project case.

$$\Delta CUDdPA_t = \Delta CBSL_t * (1 - EI)$$

Where:

$\Delta CUDdPA_t$	Total ex-ante actual carbon stock change due to avoided unplanned deforestation at year <i>t</i> in the project area; tCO ₂ -e
$\Delta CBSL_t$	Total baseline carbon stock change at year <i>t</i> in the project area; tCO ₂ -e
<i>EI</i>	Ex ante estimated Effectiveness Index; % (70%)
<i>t</i>	1, 2, 3 ... <i>T</i> , a year of the proposed project crediting period; dimensionless

The project proponent believes that under the REDD project governance it will be possible to reduce in 70% the deforestation in the project area compared to the baseline scenario, therefore the *ex-ante* Effectiveness Index (*EI*) adopted for the first baseline period is 0.7.

The Ex Ante Estimated Net Actual Carbon Stock Changes in the Project Area is summarized in Table 50.

Table 50 - Ex Ante Estimated Net Carbon Stock Change in the Project Area Under the Project Scenario. (Table 47 from VM0015 v1.2).

Project year	Total carbon stock decrease due to planned logging activities		Total carbon stock increase due to growth without harvest		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock change in the project case	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	ΔCPLdP_{A_t}	ΔCPLdPA	ΔCPLiP_{A_t}	ΔCPLiPA	ΔCUDdP_{A_t}	ΔCUDdPA	ΔCPSP_{A_t}	ΔCPSPA
	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	-	0	0	0	79,136	79,136	79,136	79,136
2022	-	0	0	0	79,365	158,501	79,365	158,501
2023	-	0	0	0	79,594	238,096	79,594	238,096
2024	-	0	0	0	79,823	317,919	79,823	317,919
2025	-	0	0	0	80,053	397,972	80,053	397,972
2026	-	0	0	0	80,282	478,253	80,282	478,253

3.2.2.2 Ex Ante Estimation of Actual Non-CO₂ Emissions from Forest Fires

As forest fires have been included in the baseline scenario, non-CO₂ emissions from biomass burning is included in the project scenario. This is done by multiplying the baseline emissions EBBPSPAt by the factor $(1 - EI)$, which is 70%. The results are presented in Table 51.

Table 51 - Total Ex Ante Estimated Actual Emissions of Non- CO₂ Gases Due to Forest Fires in the Project Area (Table 48 from VM0015 v1.2).

Project year	Total baseline non-CO ₂ emissions from forest fire in the project area	
	Annual EBBBSLPAt	Cumulative EBBBSLPA
	tCO ₂ e	tCO ₂ e
2021	488.52	488.52
2022	488.52	977.03
2023	488.52	1,465.55
2024	488.52	1,954.06
2025	488.52	2,442.58
2026	488.52	2,931.09

3.2.3 Leakage Emissions (VCS 2.5, 3.2, 3.6, 3.15, 4.3)

According to the VM0015, two sources of leakage are considered in this methodology and must be addressed:

3.2.3.1 Decrease in carbon stocks and increase in GHG emissions associated with leakage prevention measures.

In the first case, the PP BRCs commits to invest part of the annual budget generated through the sales of VCU in mitigation measures to guarantee that the offsite carbon stocks will not decrease. Such investments will reinforce a holistic approach and strategy, e.g., rural technical assistance, leadership capacity-building, social empowerment, education etc. Focusing on supporting leakage management areas through the intensification of sustainable territorial management and monitoring illegal practices (logging, grazing, land grabbing and mining) in the leakage belt. For this reason, it was assumed that the implementation of the Uákiry REDD+ Project will not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Notwithstanding, it is not expected that the implementation of project activities will generate any offsite decreases in carbon stocks. In fact, the project implementation is expected to additionally reduce deforestation outside the project boundaries, as compared to the baseline scenario. Therefore, the anticipated decrease in carbon stocks and increase in GHG emissions associated with leakage prevention measures are estimated to be zero (0).

3.2.3.2 Decrease in carbon stocks and increase in GHG emissions associated with activity displacement leakage.

The second source of leakage considered by the methodology refers to activities that cause deforestation within the project area in the baseline scenario and could be displaced outside the project boundary due to the implementation of the AUD project activity. VM0015 states that if carbon stocks in the leakage belt area decrease more during project implementation than projected in the baseline case, this will be an indication that leakage due to displacement of baseline activities has occurred.

According to VM0015, section 8.3.1.2, for the leakage ex-ante calculation, if deforestation agents do not participate in leakage prevention activities and project activities, the Displacement Factor shall be 100%. Where leakage prevention activities are implemented, the factor shall be equal to the proportion of the baseline agents estimated to be given the opportunity to participate in leakage prevention and project activities.

The project foresees that 50% of the potential leakage agents are contemplated, or will be given the opportunity, to participate in the leak mitigation activities, as per the initiative developed in the

leakage management areas, therefore the “Displacement Leakage Factor” (DLF) assumed for ex-ante leakage calculations is 0.5.

Since emissions from forest fires have been included in the baseline, the ex-ante emissions from forest fires due to activity displacement leakage will be calculated by multiplying baseline forest fire emissions in the project area by the same DLF used to estimate the reduction in carbon stocks.

Table 52 - Ex Ante Estimated Leakage due to Activity Displacement. (Table 61 from VM0015 v1.2).

Year	Total ex ante estimated decrease in carbon stocks due to displaced deforestation		Total ex ante estimated increase in GHG emissions due to displaced forest fires	
	annual	cumulative	annual	cumulative
	Δ CADLKt tCO ₂ -e	Δ CADLK tCO ₂ -e	EADLKt tCO ₂ -e	EADLK tCO ₂ -e
2021	39,568.08	39,568.08	244.26	244.26
2022	39,682.62	79,250.70	244.26	488.52
2023	39,797.17	119,047.87	244.26	732.77
2024	39,911.72	158,959.59	244.26	977.03
2025	40,026.27	198,985.86	244.26	1,221.29
2026	40,140.81	239,126.67	244.26	1,465.55

3.2.3.3 Ex Ante Estimation of Total Leakage

Table 53 - Ex Ante Estimated Total Leakage. (Table 62 from VM015 v1.2).

Project year	Total Ex Ante GHG Emissions from Increased Grazing Activities		Total Ex Ante Increase in GHG Emissions due to Displaced Forest Fires		Total Ex Ante Decrease in Carbon Stocks due to Displaced Deforestation		Carbon Stock Decrease due to Leakage Prevention Measures		Total Net Carbon Stock Change due to Leakage		Total Net Increase in Emissions due to Leakage	
	annua l	cumul ative	annual	cumulative	annual	cumulative	annua l	cumu lative	annual	cumulative	annual	cumulative
	EgLKt	EgLK	EADLKt	EADLK	ΔCADLKt	ΔCADLK	ΔCLP MLKt	ΔCLP MLK	ΔCLK _t	ΔCLK	ELKt	ELK
	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ - e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	-	-	244.26	244.26	39,568.08	39,568.08	-	-	39,568.08	39,568.08	244.26	244.26
2022	-	-	244.26	488.52	39,682.62	79,250.70	-	-	39,682.62	79,250.70	244.26	488.52
2023	-	-	244.26	732.77	39,797.17	119,047.87	-	-	39,797.17	119,047.87	244.26	732.77
2024	-	-	244.26	977.03	39,911.72	158,959.59	-	-	39,911.72	158,959.59	244.26	977.03
2025	-	-	244.26	1,221.29	40,026.27	198,985.86	-	-	40,026.27	198,985.86	244.26	1,221.29
2026	-	-	244.26	1,465.55	40,140.81	239,126.67	-	-	40,140.81	239,126.67	244.26	1,465.55

The total Ex Ante Estimation of Total Leakage is summarized in **Erro! Autoreferência de indicador não válida.**

3.2.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals (VCS, 3.15, 4.1)

3.2.4.1 Significance Assessment

The evaluation of greenhouse gas emissions from sources, carbon storage reductions, and any relevant leakage emissions associated with this Project Activity has been carried out using the "Tool for assessing the significance of GHG emissions in A/R CDM project activities (Version 01)". Despite some aggregates comprising less than 5% of the total emission reductions, they have been incorporated as outlined in Appendix 2 of VM0015 v1.2 (please refer the calculus spreadsheet in APPENDIX CARBON-POOLS).

3.2.4.2 Calculation of Ex Ante Total Net GHG Emissions Reductions

The net anthropogenic GHG emission reduction of the proposed AUD project activity is calculated as follows:

$$\Delta REDD_t = (\Delta CBSLPA_t + EBBBSLPA_t) - (\Delta CPSPA_t + EBBPSPA_t) - (\Delta CLK_t + ELK_t)$$

Where:

$\Delta REDD_t$: Ex post estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t (tCO₂e);

$\Delta CBSLPA_t$: Sum of baseline carbon stock changes in the project area at year t (tCO₂e);

$EBBBSLPA_t$: Sum of baseline emissions from biomass burning in the project area at year t (tCO₂e);

$\Delta CPSPA_t$: Sum of ex post actual carbon stock changes in the project area at year t (tCO₂e);

$EBBPSPA_t$: Sum of ex post actual emissions from biomass burning in the project area at year t (tCO₂e);

ΔCLK_t : Sum of ex post leakage net carbon stock changes at year t (tCO₂e);

ELK_t : Sum of ex post leakage emissions at year t (tCO₂e);

t: 1, 2, 3 ... T, a year of the proposed project crediting period (dimensionless).

The project emission reduction is calculated based in the baseline land use change model and biomass carbon stock per hectare. For additional information, please refer section 3.2.1 above. For project emission, please refer to section 3.2.2. For information regarding leakage emission and leakage management please refer to section 3.2.3.

3.2.4.3 Calculation of Ex Ante Verified Carbon Units (VCUs)

The equation for calculating VCUs during the monitoring period is presented below. The risk factor used to calculate VCS buffer credits (VBC) is 12%, as calculated in Non-permanence Risk Report. The results are presented in Table 54.

$$VCU_t = \Delta REDD_t - VCB_t$$

$$VCB_t = (\Delta CBSLPA_t - \Delta CPSPA_t) - RF_t$$

AUD

VCU_t: Number of Verified Carbon Units that can be traded at time t (tCO₂e);

ΔREDD_t: Ex post net anthropogenic greenhouse gas emission reduction attributable to the project activity at year t (tCO₂e);

VBC_t: Number of Buffer Credits deposited in the VCS Buffer at time t (t CO₂-e);

ΔCBSLPA_t: Sum of baseline carbon stock changes in the project area at year t (tCO₂e);

ΔCPSPA_t: Sum of ex post actual carbon stock changes in the project area at year t (tCO₂e);

RF_t: Risk factor used to calculate VCS buffer credits (%);

t: 1, 2, 3 ... T, a year of the proposed project crediting period (dimensionless).

Table 54 - Ex Ante Estimated Net GHG Emission Reductions: ($\Delta REDD_t$) and Verified Carbon Units (VCUt). (Table 64 from VM0015 v1.2)

Project year	Baseline carbon stock changes		Baseline GHG emissions		Ex ante project carbon stock changes		Ex ante project GHG emissions		Ex ante leakage carbon stock changes		Ex ante leakage GHG emissions		Ex ante net anthropogenic GHG emission reductions		Ex ante VCUs tradable		Ex ante buffer credits	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	ΔCBS_{LPA_t}	ΔCBS_{LPA_t}	$EBBBS_{LPA_t}$	$EBBBS_{LPA_t}$	ΔCPS_{PA_t}	$\Delta CPSP_{PA_t}$	$EBBP_{SPAt}$	$EBBPS_{PA_t}$	ΔCL_{K_t}	ΔCLK_t	ELK_t	ELK_t	$\Delta REDD_t$	$\Delta REDD_t$	VCU_t	VCU_t	VBC_t	VBC_t
	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	263,787.2	263,787.2	1,628.4	1,628.4	79,136.2	79,136.2	488.5	488.5	39,568.1	39,568.1	244.3	244.3	145,978.6	145,978.6	128,461.1	128,461.1	17,517.4	17,517.4
2022	264,550.8	528,338.0	1,628.4	3,256.8	79,365.2	158,501.4	488.5	977.0	39,682.6	79,250.7	244.3	488.5	146,398.6	292,377.1	128,830.7	257,291.9	17,567.8	35,085.3
2023	265,314.5	793,652.5	1,628.4	4,885.2	79,594.3	238,095.7	488.5	1,465.5	39,797.2	119,047.9	244.3	732.8	146,818.6	439,195.7	129,200.3	386,492.2	17,618.2	52,703.5
2024	266,078.1	1,059,730.6	1,628.4	6,513.5	79,823.4	317,919.2	488.5	1,954.1	39,911.7	158,959.6	244.3	977.0	147,238.6	586,434.3	129,570.0	516,062.2	17,668.6	70,372.1
2025	266,841.8	1,326,572.4	1,628.4	8,141.9	80,052.5	397,971.7	488.5	2,442.6	40,026.3	198,985.9	244.3	1,221.3	147,658.6	734,092.9	129,939.6	646,001.7	17,719.0	88,091.1
2026	267,605.4	1,594,177.8	1,628.4	9,770.3	80,281.6	478,253.3	488.5	2,931.1	40,140.8	239,126.7	244.3	1,465.5	148,078.6	882,171.5	130,309.2	776,310.9	17,769.4	105,860.6

State the non-permanence risk rating (%)	12%
Has the non-permanence risk report been attached as either an appendix or a separate document?	☒ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	Not applicable.
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☒ No Not applicable.
State, in tCO ₂ e, the expected total GHG benefit to date.	882,171.5
Is the number of GHG credits issued below the LTA?	☐ Yes ☒ No Not applicable.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
02-Aug-2021 to 02-Aug-2026	9,770.3	2,931.1	1,465.5	105,860.6	882,171.5	Not applicable	776,310.9
Total	9,770.3	2,931.1	1,465.5	105,860.6	882,171.5	Not applicable	776,310.9

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation (VCS, 3.16)

According to the VM0015 v1.2 net anthropogenic GHG emission reduction of the proposed AUD project activity is calculated as follows:

$$\Delta REDD_t = (\Delta CBSLPA_t + EBBBSLPA_t) - (\Delta CPSPA_t + EBBPSPA_t) - (\Delta CLK_t + ELK_t)$$

Where:

$\Delta REDD_t$	<i>Ex ante</i> estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t ; tCO ₂ e
$\Delta CBSLPA_t$	Sum of baseline carbon stock changes in the project area at year t ; tCO ₂ e
Note:	The absolute values of $CBSLPA_t$ shall be used in equation 19.
$EBBBSLPA_t$	Sum of baseline emissions from biomass burning in the project area at year t ; tCO ₂ e
$\Delta CPSPA_t$	Sum of <i>ex ante</i> estimated actual carbon stock changes in the project area at year t ; tCO ₂ e
Note:	If $CPSPA_t$ represents a net increase in carbon stocks, a negative sign before the absolute value of $CPSPA_t$ shall be used. If $CPSPA_t$ represents a net decrease, the positive sign shall be used
$EBBPSPA$	Sum of (<i>ex-ante</i> estimated) actual emissions from biomass burning in the project area at year t ; tCO ₂ e
CLK_t	Sum of <i>ex ante</i> estimated leakage net carbon stock changes at year t ; tCO ₂ e
Note	If the cumulative sum of CLK_t within a fixed baseline period is > 0 , CLK_t shall be set to zero.
ELK_t	Sum of <i>ex ante</i> estimated leakage emissions at year t ; tCO ₂ e 1, 2, 3 ... T , a year of the proposed project crediting period; dimensionless

The project emission reduction is calculated based in the baseline land use change model and biomass carbon stock per hectare. For additional information, please refer section 3.2.2, above. For information regarding leakage emission and leakage management please refer to section 3.2.3.

Data / parameter	CF
Data unit	Dimensionless
Description	Default value of carbon fraction in biomass
Source of data	Value from the literature (Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories").
Value applied	0.485
Justification of choice of data or description of measurement methods and procedures applied	Default factor calculated by Brazilian central Amazon.
Purpose of data	
Comments	-

Data / parameter	44/12
Data unit	Dimensionless
Description	Carbon mass to CO ₂ e mass conversion factor
Source of data	From scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU
Value applied	44/12
Justification of choice of data or description of measurement methods and procedures applied	Conversion from C to CO ₂ based on molecular weights
Purpose of data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	-

Data / parameter	R
Data unit	Dimensionless
Description	Root to shoot ratio appropriate to forest type.
Source of data	Value from the literature (Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories").

Value applied	Dense Ombrophilous Forest: 0.31 Open Ombrophilous Forest: 0.1
Justification of choice of data or description of measurement methods and procedures applied	Eco-region-specific root to shoot ratio is used for accuracy.
Purpose of data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	-

Data / parameter	CI
Data unit	Dimensionless
Description	Litter fraction relative to stand biomass of trees ≥ 10 cm dbh
Source of data	Value from the literature (Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories").
Value applied	0.041
Justification of choice of data or description of measurement methods and procedures applied	Eco-region-specific litter fraction is used for accuracy
Purpose of data	• Calculation of baseline emissions • Calculation of project emissions • Calculation of leakage
Comments	-

Data / parameter	Cab_nontree
Data unit	Dimensionless
Description	Non-tree fraction relative to stand biomass of trees ≥ 10 cm dbh
Source of data	Value from the literature (Nogueira et al 2008 - "Estimates of forest biomass in the Brazilian Amazon: New allometric equation and adjustments to biomass from wood-volume inventories").
Value applied	0.053
Justification of choice of data or description of measurement methods and procedures applied	Eco-region-specific non-tree fraction stock is used for accuracy
Purpose of data	• Calculation of baseline emissions • Calculation of project emissions

	Calculation of leakage
Comments	-

Data / parameter	Cbb_nontree
Data unit	Dimensionless
Description	Non-tree below ground fraction relative to stand non-tree biomass
Source of data	Value from the literature (IPCC, 2006b. Chapter 2 Generic Methodologies Applicable to Multiple Land Use Categories 1 59.).
Value applied	0.2
Justification of choice of data or description of measurement methods and procedures applied	Default value from IPCC.
Purpose of data	- Calculation of baseline emissions - Calculation of project emissions - Calculation of leakage
Comments	-

Data / parameter	Dnm
Data unit	g cm-3
Description	Mean wood density
Source of data	Value from the literature (Fearnside 1997 - " Wood density for estimating forest biomass in Brazilian Amazonia ").
Value applied	0.67
Justification of choice of data or description of measurement methods and procedures applied	Eco-region-specific value of mean wood density is used for accuracy.
Purpose of data	Calculate the decrease in carbon stock due to logging
Comments	-

Data / parameter	LDF
Data unit	t C m-3
Description	Logging damage factor

Source of data	Value from the literature (Pearson et al 2014 – “Carbon emissions from tropical forest degradation caused by logging”)
Value applied	0.71
Justification of choice of data or description of measurement methods and procedures applied	The Pearson et al 2014 study was identified as the most authoritative study on this topic relating to sustainable forest management for Brazil.
Purpose of data	Calculate the decrease in carbon stock due to logging
Comments	-

Data / parameter	DLF
Data unit	Dimensionless
Description	Displacement Leakage Factor
Source of data	Local assessment
Value applied	0.5
Justification of choice of data or description of measurement methods and procedures applied	The project foresees that 50% of the potential leakage agents are contemplated, or will be given the opportunity, to participate in the leak mitigation activities, as per the initiative developed in the leakage management zones, therefore the “Displacement Leakage Factor” (DLF) assumed for ex-ante leakage calculations is 0.5.
Purpose of data	Ex-ante leakage calculations
Comments	-

Data / parameter	EI
Data unit	Dimensionless
Description	Ex ante estimated Effectiveness Index
Source of data	Local assessment
Value applied	0.70
Justification of choice of data or description of measurement methods and procedures applied	The project proponent believes that under the REDD project governance it will be possible to reduce in 70% the deforestation in the project area compared to the baseline scenario, therefore the ex-ante Effectiveness Index (EI) adopted for the first baseline period is 0.7.
Purpose of data	Ex ante estimated net carbon stock change in the project area under the project scenario
Comments	-

3.3.2 Data and Parameters Monitored (VCS, 3.16)

Data / parameter	ACPAicl,t
Data unit	ha
Description	Annual area of initial forest classes icl within the Project Area affected by catastrophic events at year t
Source of data	Remote sensing data and GIS.
Description of measurement methods and procedures to be applied	DETER and subsequent visual validation with high-resolution images will be used.
Frequency of monitoring/recording	At each time a significant catastrophic event occurs.
Value applied	0 (no significant catastrophic event occurred)
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Purpose of data	Calculation of project emissions
Calculation method	Remote sensing and GIS
Comments	-

Data / parameter	Aforaget
Data unit	ha
Description	Area under forage above the baseline in leakage management zones
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.

Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	APFPA icl,t
Data unit	ha
Description	Annual area of planned fuel- wood and charcoal activities in forest class icl at year t in the project area
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	There are no significative planned fuel-wood and charcoal activities in forest class icl at year t in the project area.

Data / parameter	APNiPAicl,t
Data unit	ha
Description	Annual area of forest class icl with increasing carbon stock without harvest at year t in the project area
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	0

Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	The Uákiry REDD+ project has opted to does not account or monitor the carbon change in forest that was in a disturbed condition prior to the project start date, (with the potential to remove carbon during the regeneration process), following a conservative approach.

Data / parameter	APSLKfcl,t
Data unit	ha
Description	Annual area of class fcl with decreasing carbon stock in leakage management zones in the project case at year t
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	AUFPAicl,t
Data unit	ha
Description	Annual area of initial forest classes icl within the Project Area affected by forest fires at year t

Source of data	Remote sensing data and GIS
Description of measurement methods and procedures to be applied	Using Sentinel 2A images will be quantified the area of forest fires within the Project Area. The dNBR index will be used for spatial evidence of spectral change and, respectively, remote detection of burned areas. Such an automated process on the Google Earth Engine platform is however validated with visual interpretation and high-resolution spatial images by experienced remote sensing analyst. In this context, the area of fire, which has been reduced in the biomass stock due to fire, were measured.
Frequency of monitoring/recording	Annually
Value applied	0 (no significant forest fire event occurred)
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Purpose of data	Calculation of project emissions
Calculation method	Remote sensing and GIS
Comments	-

Data / parameter	ABSLKicl,t			
Data unit	ha			
Description	Area of initial (pre deforestation) forest class icl deforested at time t within the leakage belt.			
Source of data	Remote sensing data and GIS			
Description of measurement methods and procedures to be applied	For the annual quantification of the land use change, "Forest to Non-Forest" in the leakage belt was used the PRODES monitoring system. Detailed information of this remote monitoring using orbital sensors is available at: METADATA PRODES - INPE (inpe.br) and PRODES - INPE (inpe.br) methodology.			
Frequency of monitoring/recording	Annually			
Value applied		Open alluvial forest (Aa)	Open lowland forest (Ab)	Dense Ombrophilous forest (Db)
	2022	929,6	5727,3	1336,7
	2023	929,6	5727,3	1336,7
Monitoring equipment				

QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Calculation method	Calculation of leakage.
Comments	Analysis of satellite images and maps

Data / parameter	ABSLPA _{icl,t}						
Data unit	ha						
Description	Area of initial (pre deforestation) forest class icl deforested at time t within the project area.						
Source of data	Remote sensing data and GIS						
Description of measurement methods and procedures to be applied	For the annual quantification of the land use change, "Forest to Non-Forest" in the project area was used the PRODES monitoring system. Detailed information of this remote monitoring using orbital sensors is available at: METADATA PRODES - INPE (inpe.br) and PRODES - INPE (inpe.br) methodology.						
Frequency of monitoring/recording	Annually						
Value applied	Project year t	Stratum i of the reference region in the project area				Total	
		1	2	...	IRR	annual	cumulative
		ABSLPA _{i,t}	ABSLPA _{i,t}	ABSLPA _{i,t}	ABSLPA _{i,t}	ABSLPA _{i,t}	ABSLPA _{i,t}
		ha/yr	ha/yr	ha/yr	ha/yr	ha/yr	ha/yr
	2022	567,1				567,1	1134,2
	2023	465,8				465,8	1600
Monitoring equipment	Remote sensing and GIS						
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.						
Purpose of data	Calculation of project emissions.						
Calculation method	Analysis of satellite images and maps						
Comments	-						

Data / parameter	APDPA _{icl,t}
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Data unit	ha
Description	Areas of planned deforestation in forest class icl at year t in the project area
Source of data	Annual post-harvesting report; Remote sensing and GIS
Description of measurement methods and procedures to be applied	The planned deforestation activities in the project area that resulted in carbon stock decrease were subject to monitoring.
Frequency of monitoring/recording	Annually
Value applied	2022: 0 ha 2023: 0 ha
Monitoring equipment	-
QA/QC procedures to be applied	Best practices in remote sensing.
Purpose of data	Calculation of project emissions.
Calculation method	Sum of the number of timber yards times their dimensions plus the width of open roads times their length.
Comments	Planned deforestation mainly includes implementation of the forest management infrastructure, such as opening of main and secondary roads, skidding trails, and timber yards in each annual production unit within the project area.

Data / parameter	APLPAicl,t
Data unit	ha
Description	Areas of planned logging activities in forest class icl at year t in the project area
Source of data	Annual post-harvesting report; Remote sensing and GIS
Description of measurement methods and procedures to be applied	The planned logging activities in the project area that resulted in carbon stock decrease were subjected to monitoring.
Frequency of monitoring/recording	Annually
Value applied	2022: 0 ha 2023: 0 ha
Monitoring equipment	-
QA/QC procedures to be applied	Best practices in remote sensing.
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	-

Data / parameter	ΔCADLKt		
Data unit	t CO ₂ -e		
Description	Total decrease in carbon stocks due to displaced deforestation at year t		
Source of data	Remote sensing data and GIS		
Description of measurement methods and procedures to be applied	Deforestation in the leakage belt area was considered activity displacement leakage. Activity data for the leakage belt area was determined using the same methods applied to monitoring		
Frequency of monitoring/recording	annually		
Value applied	2022	39682,62	
	2023	39902,72	
Monitoring equipment	Remote sensing data and GIS		
QA/QC procedures to be applied	Best practices in remote sensing		
Purpose of data	Calculation of leakage.		
Calculation method	Emissions from deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock in year t per unit area.		
Comments	The monitored deforestation in the leakage belt was lower than projected in the baseline, hence there was no leakage.		

Data / parameter	$\Delta\text{CFCdPat}$		
Data unit	t CO ₂ -e		
Description	Total decrease in carbon stock due to forest fires and catastrophic events at year t in the project area		
Source of data	Remote sensing data and GIS		
Description of measurement methods and procedures to be applied	Annual area of initial forest classes icl within the Project Area affected by catastrophic events at year t times the decrease in carbon stock due these events.		
Frequency of monitoring/recording	Annually		
Value applied	2022: 0 2023: 0		
Monitoring equipment	Remote sensing and GIS		
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.		
Purpose of data	Calculation of project emissions.		

Calculation method	Remote sensing and GIS
Comments	No significant catastrophic event or forest fires occurred

Data / parameter	$\Delta\text{CFCiPat}$
Data unit	t CO ₂ -e
Description	Total increase in carbon stock due to forest fires and catastrophic events at year t in the project area.
Source of data	Annual area of initial forest classes icl within the Project Area affected by catastrophic events at year t times the increase in carbon stock due these events.
Description of measurement methods and procedures to be applied	Annually
Frequency of monitoring/recording	0
Value applied	Remote sensing and GIS
Monitoring equipment	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
QA/QC procedures to be applied	Calculation of project emissions.
Purpose of data	Remote sensing and GIS
Calculation method	No significant catastrophic event or forest fires occurred
Comments	Annual area of initial forest classes icl within the Project Area affected by catastrophic events at year t times the increase in carbon stock due these events.

Data / parameter	$\Delta\text{CLPMLKt}$
Data unit	t CO ₂ -e
Description	Carbon stock decrease due to leakage prevention measures
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	0
Monitoring equipment	-

QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	ΔCPAdPA_t
Data unit	tCO ₂ e
Description	Total decrease in carbon stock due to all planned activities at year t in the project area
Source of data	Annual post-harvesting report; Remote sensing and GIS
Description of measurement methods and procedures to be applied	The planned activities in the project area that resulted in carbon stock decrease were subjected to monitoring.
Frequency of monitoring/recording	Annually
Value applied	2022: 0 tCO ₂ e 2023: 0 tCO ₂ e
Monitoring equipment	-
QA/QC procedures to be applied	Best practices in remote sensing
Purpose of data	Calculation of project emissions.
Calculation method	This parameter is the sum of: carbon stock decrease due to planned deforestation, carbon stock decrease due to planned logging activities, and carbon stock decrease due to planned fuel wood and charcoal activities.
Comments	-

Data / parameter	ΔCPAiPA_t
Data unit	t CO ₂ -e
Description	Total increase in carbon stock due to all planned activities at year t in the project area
Source of data	-

Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Increase in carbon stock was ignored ex ante by the PP, and no credit was claimed for carbon stock increases in deforested areas. This aligns with the guidelines outlined in section 8.1.5 of VM0015, maintaining a conservative approach.

Data / parameter	ΔCPDdPA_t
Data unit	t CO ₂ -e
Description	Total decrease in carbon stock due to planned deforestation at year t in the project area
Source of data	Annual post-harvesting report; Remote sensing and GIS
Description of measurement methods and procedures to be applied	The planned activities in the project area that resulted in carbon stock decrease were subjected to monitoring.
Frequency of monitoring/recording	annually
Value applied	2022: 0 2023: 0
Monitoring equipment	-
QA/QC procedures to be applied	Best practices in remote sensing
Purpose of data	Calculation of project emissions.
Calculation method	Area deforested due to planned deforestation times t CO ₂ -e ha in initial forest class incl.
Comments	-

Data / parameter	ΔCPFIP_t
Data unit	t CO ₂ -e

Description	Total increase in carbon stock due to planned fuel-wood and charcoal activities at year t in the project area
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	No production of fuel wood or charcoal occurred within the project area during this monitoring period.

Data / parameter	ΔCPLdPA_t
Data unit	t CO ₂ -e
Description	Total decrease in carbon stock due to planned logging activities at year t in the project area
Source of data	Annual post-harvesting report; Remote sensing and GIS
Description of measurement methods and procedures to be applied	The planned logging in the project area that resulted in carbon stock decrease were subjected to monitoring.
Frequency of monitoring/recording	annually
Value applied	2022: 0 2023: 0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	Area subjected to logging times t CO ₂ -e ha explored.
Comments	-

Data / parameter	ΔCPLiPA_t
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Data unit	t CO ₂ -e
Description	Total increase in carbon stock due to planned logging activities at year t in the project area
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Increase in carbon stock was ignored ex ante by the PP, and no credit was claimed for carbon stock increases in deforested areas. This aligns with the guidelines outlined in section 8.1.5 of VM0015, maintaining a conservative approach.

Data / parameter	$\Delta\text{CPNiPat}$
Data unit	t CO ₂ -e
Description	Total increase in carbon stock due to planned protection of growing forest classes in the project area at year t
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-

Comments	The Uákiry REDD+ project has opted to does not account or monitor the carbon change in forest that was in a disturbed condition prior to the project start date, (with the potential to remove carbon during the regeneration process), following a conservative approach.
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Data / parameter	ΔCPSLKt
Data unit	t CO ₂ -e
Description	Total annual carbon stock change in leakage management zones in the project case.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	ΔCPSPAt
Data unit	t CO ₂ -e
Description	Total project carbon stock change within the project area at year t.
Source of data	Remote sensing data and GIS.
Description of measurement methods and procedures to be applied	Sum of ex post actual carbon stock changes in the project area at year t.
Frequency of monitoring/recording	annually

Value applied	2022: 79,365 2023: 79,805
Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Purpose of data	Calculation of project emissions.
Calculation method	Remote sensing and GIS.
Comments	-

Data / parameter	$\Delta CUCiPat$
Data unit	t CO ₂ -e
Description	Total increase in carbon stock in areas affected by catastrophic events (after such events) at year t in the project area
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Increase in carbon stock was ignored ex ante by the PP, and no credit was claimed for carbon stock increases. This aligns with the guidelines outlined in section 8.1.5 of VM0015, maintaining a conservative approach.

Data / parameter	$\Delta CUDdPat$
Data unit	tCO ₂ e
Description	Total actual carbon stock change due to unavoided unplanned deforestation at year t in the project area

Source of data	Remote sensing and GIS
Description of measurement methods and procedures to be applied	To measure forest cover change due to unplanned deforestation in the project area was used the PRODES monitoring system. Detailed information of this remote monitoring using orbital sensors is available at: METADATA PRODES - INPE (inpe.br) and PRODES - INPE (inpe.br) methodology.
Frequency of monitoring/recording	Annually
Value applied	2022: 79,365 2023: 79,805
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	To control and guarantee the quality of spatial data of to unavioded unplanned deforestation, it has been validated with visual interpretation with experienced analyst of monitoring systems using high spatial resolution (Planet) and temporal images (Sentinel 2A).
Purpose of data	Calculation of project emissions.
Calculation method	
Comments	Emissions from deforestation at each forest class are quantified by multiplying the detected area of forest loss by the average forest carbon stock per unit area.

Data / parameter	$\Delta\text{CUF}_{\text{dPA}t}$
Data unit	t CO ₂ -e
Description	Total decrease in carbon stock due to unplanned (and planned – where applicable) forest fires at year t in the project area
Source of data	Remote sensing data and GIS
Description of measurement methods and procedures to be applied	Using Sentinel 2A images will be quantified the area of forest fires within the Project Area. The dNBR index will be used for spatial evidence of spectral change and, respectively, remote detection of burned areas. Such an automated process on the Google Earth Engine platform is however validated with visual interpretation and high-resolution spatial images by experienced remote sensing analyst. In this context, the area of fire, which has been reduced in the biomass stock due to fire, were measured.
Frequency of monitoring/recording	annually
Value applied	2022: 0 2023: 0
Monitoring equipment	Remote sensing data and GIS
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation by experienced analysts the monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Purpose of data	Calculation of baseline emissions. Calculation of project emissions.

Calculation method	Project area affected by forest fire times carbon stock decrease due to forest fire.
Comments	No significant forest fire event occurred

Data / parameter	$\Delta\text{CUFiPat}$
Data unit	t CO ₂ -e
Description	Total increase in carbon stock in areas affected by forest fires (after such events) at year t in the project area
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-
Comments	Increase in carbon stock was ignored ex ante by the PP, and no credit was claimed for carbon stock increases. This aligns with the guidelines outlined in section 8.1.5 of VM0015, maintaining a conservative approach.

Data / parameter	ΔREDDt
Data unit	t CO ₂ -e
Description	Net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	2022: 146,398.6 2023: 147,205.6
Monitoring equipment	-

QA/QC procedures to be applied	-
Purpose of data	Calculation of VCUs generated by the Project.
Calculation method	See in section 8.4.1.3.
Comments	-

Data / parameter	EADLKt
Data unit	t CO ₂ -e
Description	Total ex ante increases in GHG emissions due to displaced forest fires at year t
Source of data	Remote sensing and GIS
Description of measurement methods and procedures to be applied	Forest fires in the leakage belt area can be considered activity displacement leakage. GHG emissions due displaced forest fires shall be subjected to monitoring, when significant.
Frequency of monitoring/recording	annually
Value applied	2022: 244.26 2023: 244.26
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing
Purpose of data	Calculation of leakage.
Calculation method	Emissions at each forest class are estimated by multiplying the detected area of forest loss (ha) by the total GHG emission from biomass burning in forest class i at year t (tCO ₂ e/ha)
Comments	PP assumed that during the process of converting forests to non-forest, burning the forest after clearing it for conversion to pasture is a common practice in the region.

Data / parameter	EBBPSPAt
Data unit	tCO ₂ e
Description	Sum of (or total) actual non-CO ₂ emissions from forest fire at year t in the project area.
Source of data	Remote sensing and GIS.
Description of measurement methods and procedures to be applied	Non-CO ₂ emissions were subjected to monitoring and accounting.
Frequency of monitoring/recording	Annually

Value applied	2022: 488.52 2023: 488.52
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation with experienced analyst of monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Purpose of data	Calculation of project emissions.
Calculation method	Using Sentinel 2A images there was quantified the area of forest fires within the Project Area. The dNBR index were used for spatial evidence of spectral change and, respectively, remote detection of burned areas. Such an automated process on the Google Earth Engine platform was however validated with visual interpretation and high-resolution spatial images by experienced remote sensing analyst.
Comments	No forest fire was detected in Project area.

Data / parameter	EBBBSLtoticl
Data unit	t CO2-e
Description	Sum of (or total) actual non-CO2 emissions from forest fire at year t in strata i in forest class icl.
Source of data	Remote sensing and GIS.
Description of measurement methods and procedures to be applied	Non-CO2 emissions were subjected to monitoring and accounting.
Frequency of monitoring/recording	annually
Value applied	2022: 0 2023: 0
Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation with experienced analyst of monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Purpose of data	Calculation of project emissions.
Calculation method	Using Sentinel 2A images there was quantified the area of forest fires within the Project Area. The dNBR index were used for spatial evidence of spectral change and, respectively, remote detection of burned areas. Such an automated process on the Google Earth Engine platform was however validated with visual interpretation and high-resolution spatial images by experienced remote sensing analyst.
Comments	No forest fire was detected in Project area.

Data / parameter	EBBCH4icl
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Data unit	t CO ₂ -e.ha-1
Description	CH ₄ emission from biomass burning in forest class icl.
Source of data	Hughes et al 2000 - "Fire in the Brazilian Amazon: biomass, nutrient pools, and losses in slashed primary forests" and IPCC values.
Description of measurement methods and procedures to be applied	CH ₄ emissions were subjected to monitoring and accounting.
Frequency of monitoring/recording	annually
Value applied	Open alluvial forest: 2.33 Open lowland forest: 2.50 Dense Ombrophilous forest: 3.15
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions. Calculation of project emissions. Calculation of leakage.
Calculation method	EBBCH ₄ icl, t = EBBCH ₄ icl, t * 12/44 * ERCH ₄ *16/12*GWPC ₄ EBBCH ₄ icl, t = Per hectare CO ₂ emission from biomass burning in slash and burn in forest class icl at year t; tCO ₂ -e ha-1
Comments	-

Data / parameter	EBBN ₂ Oicl
Data unit	t CO ₂ -e.ha-1
Description	N ₂ O emission from biomass burning in forest class icl.
Source of data	Hughes et al 2000 - "Fire in the Brazilian Amazon: biomass, nutrient pools, and losses in slashed primary forests" and IPCC values.
Description of measurement methods and procedures to be applied	N ₂ O emissions were subjected to monitoring and accounting.
Frequency of monitoring/recording	annually
Value applied	2022: 0 2023: 0
Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation with experienced analyst of monitoring systems used with high-spatial resolution (Planet) and temporal (Sentinel 2A) images.
Purpose of data	Calculation of project emissions.

Calculation method	Using Sentinel 2A images there was quantified the area of forest fires within the Project Area. The dNBR index were used for spatial evidence of spectral change and, respectively, remote detection of burned areas. Such an automated process on the Google Earth Engine platform was however validated with visual interpretation and high-resolution spatial images by experienced remote sensing analyst.
Comments	-

Data / parameter	EBB _{total}
Data unit	t CO ₂ -e
Description	Total GHG emission from biomass burning in forest class icl.
Source of data	Hughes et al 2000 - "Fire in the Brazilian Amazon: biomass, nutrient pools, and losses in slashed primary forests" and IPCC values.
Description of measurement methods and procedures to be applied	Total GHG emissions were subjected to monitoring and accounting.
Frequency of monitoring/recording	annually
Value applied	Open alluvial forest: 2.56 Open lowland forest: 2.75 Dense Ombrophilous forest: 3.48
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of baseline emissions. Calculation of project emissions. Calculation of leakage.
Calculation method	$EBB_{total,t} = EBB_{N2O,icl,t} + EBB_{CH4,icl,t}$
Comments	-

Data / parameter	ECH _{4fermt}
Data unit	t CO ₂ -e
Description	CH ₄ emissions from enteric fermentation at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0

Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	ECH4mant
Data unit	t CO2-e
Description	CH4 emissions from manure management at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	EdirN2Omant
Data unit	t CO2-e
Description	Direct N2O emissions from manure management at year t.

Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	EgLKt
Data unit	t CO2-e
Description	Emissions from grazing animals in leakage management zones at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification,

	fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.
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Data / parameter	EindNOmant
Data unit	t CO2-e
Description	Indirect N2O emissions from manure management at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	ELKt
Data unit	t CO2-e
Description	Sum of ex post estimated leakage emissions at year t.
Source of data	Remote sensing and GIS.
Description of measurement methods and procedures to be applied	Emissions from leakage were subjected to monitoring and accounting.
Frequency of monitoring/recording	annually
Value applied	2022: 244.26 2023: 244.26
Monitoring equipment	-

QA/QC procedures to be applied	Best practices in remote sensing and GIS
Purpose of data	Calculation of leakage.
Calculation method	$ELK_t = EADLK_t (\text{baseline}) - EADLK_t (\text{ex-post})$
Comments	-

Data / parameter	ELPMLK _t
Data unit	t CO ₂ -e
Description	Annual total increase in GHG emissions due to leakage prevention measures at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	EN20mant
Data unit	t CO ₂ -e
Description	N ₂ O emissions from manure management at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-

Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	j
Data unit	dimensionless
Description	number of organic fertilizer types.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	Populationt
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Data unit	number of heads
Description	Equivalent number of forage- fed livestock at year t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	0
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of leakage.
Calculation method	-
Comments	During this first crediting period (2021-2026), the Uákiry REDD Project conducted planned interventions, which did not result in a decrease in carbon stocks or an increase in GHG emissions within the leakage management area. The measures implemented by the project to prevent leakage did not involve agricultural intensification, fertilization, fodder production, or other actions aimed at enhancing cropland and grazing land areas.

Data / parameter	VBCt
Data unit	t CO2-e
Description	Number of Buffer Credits deposited in the VCS Buffer at time t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	2022: 17,567.8 2023: 17,618.2
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of VCU's generated by the Project.
Calculation method	See in section 8.1.4.3.

Comments	-
Data / parameter	VCUt
Data unit	t CO2-e
Description	Number of Verified Carbon Units (VCUs) to be made available for trade at time t.
Source of data	-
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	annually
Value applied	2022: 128,830.7 2023: 129,200.3
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of VCUs generated by the Project.
Calculation method	See in section 8.4.1.3.
Comments	-

Data / parameter	VOLicl
Data unit	m ³ .ha ⁻¹
Description	Volume of timber per hectare extracted from within the project area in stratum icl in year t.
Source of data	Annual post-harvesting report.
Description of measurement methods and procedures to be applied	After felling and sectioning the tree, the actual volume value of the logs is obtained.
Frequency of monitoring/recording	Annually
Value applied	14.09
Monitoring equipment	The same equipment applied in the forest inventory
QA/QC procedures to be applied	Control procedures applied to forest inventory
Purpose of data	Calculation of project emissions.

Calculation method	After harvesting operations, each harvested timber log is measured (diameter and height) to determine de volume.
Comments	-

Data / parameter	RFt
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Source of data	- VCS Non-Permanence Risk Report (v4.2) - Remote sensing data and GIS, - Literature data.
Description of measurement methods and procedures to be applied	All sources of data from the VCS Non-Permanence Risk Report were used to measure the various risk factors.
Frequency of monitoring/recording	Annually
Value applied	12%
Monitoring equipment	Remote sensing and GIS.
QA/QC procedures to be applied	Best practices in remote sensing and GIS.
Purpose of data	This parameter represents the non-permanence risk rating of the project during the current monitoring period, which was used to determine the number of buffer credits that shall be deposited into the AFOLU pooled buffer account.
Calculation method	This parameter was calculated using the AFOLU Non-Permanence Risk Tool.
Comments	-

3.3.3 Monitoring Plan (VCS, 3.16, 3.20)

The monitoring plan of the project follows Section 9.2 of the VM0015 methodology, which consists of three distinct tasks. Task 1 is related to monitoring actual carbon stock changes and GHG emissions periodically in the project area; Task 2 is related to revisiting the baseline projections and monitoring the leakage zones; Task 3 calculate the Ex post of net GHG emission reductions. In this monitoring period, only Task 1 was applied since no alterations in baseline are needed. All subtasks applied in this monitoring period are described below, following the aspects:

- Technical description of the monitoring tasks.
- Data collected.

- Overview of data collection procedures.
- Quality control and quality assurance procedures.
- Data storage (archiving).
- Organization and responsibilities of the parties involved in all the above.

3.3.3.1 Monitoring of actual carbon stock changes and GHG emissions within project area

1) Monitoring project implementation

- This subtask evidence all actions and activities directly provided by the project on an annual basis. Specifically report the main objective of combating and reducing deforestation and forest degradation (removal of illegal wood, forest fires, use of wood for firewood, and other uses) and Sustainable Forest Management in the Project Area and Leakage Belt. A detailed description of the forest measurement of carbon pools measured in the project is available in APPENDIX CARBON-POOLS.
- Such deforestation monitoring activities (ABSLPAi,t) and forest degradation applied remote sensing techniques to detect illegal deforestation and forest degradation activities. Several geospatial databases that measure forest loss with different methodologies and sensors and different objectives were also systematized. Monitoring systems were used to detect illegal activities in near real-time (DETER, SAD, GLAD), and for the annual measurement and national reporting of GHG emissions (PRODES).
- Geospatial information was made available openly and free on the web. It was downloaded and systematized in a Geographic Information Systems environment (GIS) and is available in project APPENDIX GIS-DATA.
- To control and guarantee the quality of spatial data of deforestation and forest degradation, PP validated with visual interpretation with experienced analysts of these monitoring systems using high spatial resolution (Planet) and temporal images (Sentinel 2A).
- Digital documentary storage was carried out in the cloud and on Hard Disks and analog documents at the company's headquarters, located in Piracicaba. The evidence is widely available in any project verification action if access by the audit team is required. The systematization of this temporal database was the responsibility of BRC.

BRC was responsible for the implementation of all activities mentioned above.

2) Monitoring land use and land-cover change

- For the annual quantification of the land use change, "Forest to Non-Forest," the PRODES monitoring system was used. Detailed information on this remote monitoring through orbital sensors is available at: PRODES METADATA - INPE (inpe.br) and PRODES - INPE methodology (inpe.br). To quantify the change in land use from "Forest to Forest Decrease / Biomass Increment," DETER was used. Detailed DETER information is available at: DETER - B (mma.gov.br) and DETER Methodology (inpe.br³³).
- The main parameter measured was the change of the use class "Forest" to other land uses, and forest with biomass decrease (Degraded Forest and forest fires and use for agricultural purposes) as suggested by Table 51 of VM00015. Increase in biomass due to natural regeneration was ignored by PP to maintain a conservative approach.
- Cartographic information was made available openly and free of charge on the web. It was downloaded and systematized in a Geographic Information Systems environment. In addition, actions and monitoring activities of the project were evidenced through digital spatial data, satellite images, and others.
- To control and guarantee the quality of spatial data of land use change, PP validated with visual interpretation with experienced analysts of these monitoring systems used with high spatial resolution and temporal images (Sentinel 2A).
- Digital documentary storage was carried out in the cloud and on Hard Disks and analog documents, at the company's headquarters, located in Piracicaba. The evidence is widely available in any project verification action if access by the audit team is required.

BRC was responsible for the implementation of all activities mentioned above.

3) Monitoring the change in carbon stocks and non-CO2 emissions from forest fires

- Using Sentinel 2A images, the area of forest fires within the Project Area was quantified. The dNBR index was used for spatial evidence of spectral change and, respectively, remote detection of burned areas. Such an automated process on the Google Earth Engine platform was, however, validated with visual interpretation and high-resolution spatial images by experienced remote sensing analysts. In this context, the area of fire, which had been reduced in the biomass stock due to fire, was measured.
- When significant, the biomass loss due to forest fires was monitored by burned area (AUFPAicl,t). Representing the change of the use class "Forest" to other land uses, forest with

³³ <https://doi.org/10.1109/JSTARS.2015.2437075>

biomass decrease to forest fires, as suggested by Table 51 of VM00015. Other parameters monitored were EBB_{total},t/EBBN₂₀,t/EBBCH₄,t to quantify non-CO₂ emissions.

- Selection of cloudless Sentinel 2A images before and after the fire event with NIR and SWIR bands to calculate the NBR (Normalized Burn Ratio) ex ante and ex post, and later calculate the difference for the index obtaining the Delta NBR index.
- For control and quality assurance of spatial data of forest fires, BRC validated with visual interpretation with experienced analysts of these monitoring systems used with high-spatial resolution and temporal (Sentinel 2A) images.
- Digital documentary storage was carried out in the cloud and on Hard Disks and analog documents, at the company's headquarters, located in Piracicaba. The evidence is widely available in any project verification action if access by the audit team is required.

BRC was responsible for the broad implementation of all activities mentioned above.

4) Monitoring the Impacts of natural disturbances and other extreme events

- For quantification of forest cover loss due to natural disturbances and extreme events, when significant, DETER and subsequent visual validation with high-resolution images were used. Deter's detailed information was available at: DETER - B (mma.gov.br) and DETER Methodology (inpe.br).
- The main parameter that was measured was ACPA_{cl,t}. Representing the change of the use class "Forest" to other land uses due to extreme events. The change occurred for the forest class with biomass decreases and later increase in biomass, as suggested by Table 51 of VM00015.
- Cartographic information was made available openly and free of charge on the web. It was downloaded and systematized in GIS. In addition, actions and monitoring activities of the project were evidenced through digital spatial data, satellite images, and others.
- To control and guarantee the quality of spatial data of land use change, BRC validated with visual interpretation with experienced analysts of these monitoring systems used with high spatial resolution and temporal images (Sentinel 2A).
- Digital documentary storage was carried out in the cloud and on Hard Disks and analog documents, at the company's headquarters, located in Piracicaba. The evidence is widely available in any project verification action if access by the audit team is required.

BRC was responsible for the broad implementation of all activities mentioned above.

3.3.3.2 Monitoring Leakage Belt

For monitoring the Leakage Belt, the same tasks and cartographic basis for monitoring the Project Area, described earlier in section 3.3.3.1, were useful. As suggested by VM00015 in section 8.3. The ABSLLKt, the parameter for the geographic boundaries of the Leakage Belt, will be monitored.

3.3.3.3 Ex-post net anthropogenic GHG emission reduction

With the achievement of steps 1.1 and 1.2 of Task 1 in section 8.4 of VM00015, the reduction of greenhouse gas emissions was calculated due to the reduction of deforestation and forest degradation within the Project Area and Leakage Belt. In this context, step c) of Monitoring Task 1 was completed. The parameters required to perform monitoring, collection and measurement procedures, quality control and assurance, systematization/storage of spatial data, and responsibility of these subtasks were described earlier in section 3.3.3.1 for Project Area, which were likewise to Leakage Belt monitoring.

3.3.4 Dissemination of Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

The monitoring plan and monitoring results will be available on Verra's website, through the full monitoring reports documents. The communities will directly be involved in the project and surrounding stakeholders the activities developed and take access to important documents. All information used is open and widely reported by BRC and verifiable in external audit processes of project verifications.

3.4 Optional Criterion: Climate Change Adaptation Benefits

3.4.1 Regional Climate Change Scenarios (CCB, GL1.1)

Not applicable.

3.4.2 Climate Change Impacts (CCB, GL1.2)

Not applicable.

3.4.3 Measures Needed and Designed for Adaptation (CCB, GL1.3)

Not applicable.

4 COMMUNITY

4.1 Without-Project Community Scenario

4.0.1 Descriptions of Communities at Project Start (CCB, CM1.1)

According to the latest socioeconomic diagnosis conducted by the social team of BRC in 2022, there are approximately 76 families residing on the properties, but only PAI#1 having population both by the river and in the center, which is the most populous area of the property, achieving almost 90% of the total of people in the properties, what makes Porongaba the main area. The total of 76 families mapped reached around 265 people spread around the Uakiry project's territory.

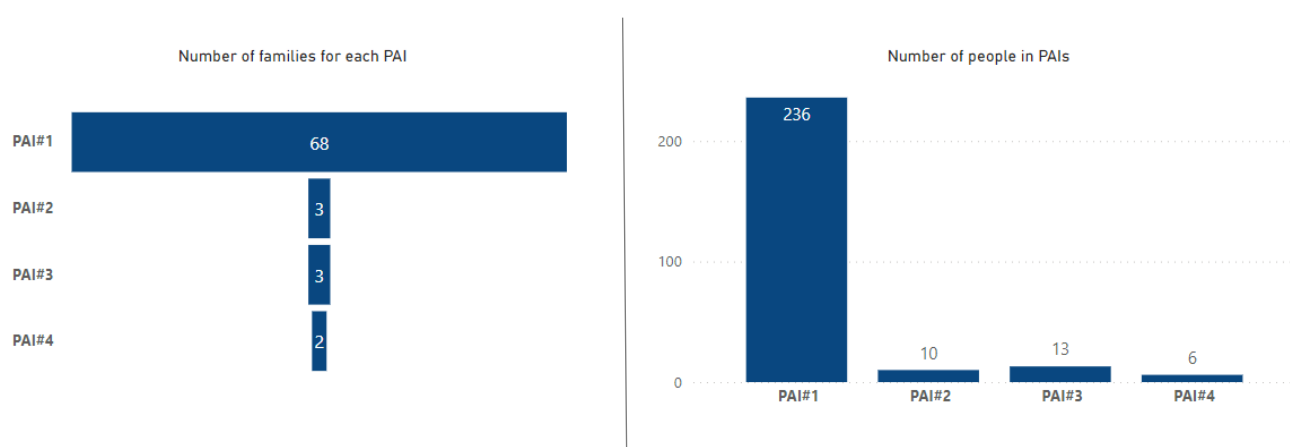


Figure 30. Population density in the project areas

Most of these people are descended from rubber tappers who settled in the region during the rubber cycle in Brazil and can be understood as riverside populations or family farmers and squatters. They are Portuguese speakers; other languages speakers were not mapped. About religiosity most of them are Christian, divided by catholic (60%), or evangelic (17%). The presence of the catholic church in the territory is deep and made a profound impact on the, and few of them has no religion at all (12%) and some people did not inform their faith (7%), while the rest has cited the following of catholic and evangelical faith (4%) in a typical syncretic way. It is important to highlight that historically people history. To picture it, is enough to check the HCV mapped, which one is directly related to the catholic church – the nun's grave.

Related to social organization, there is no traditional authority structures nor representative institutions, a pre-existent community association is not operational.

The main economic activity is cattle breeding, and the commercialization of brazil nut can occur as a complementary activity. The production of rice, beans, corn, cassava, chicken, pigs, and other animals exists only for subsistence. Fishing and hunting for food are common. In general, the family leaders do not know their own household income.



Figure 31. Cracking nuts and raising cattle in one Acrean rubber farm

Communities live in isolation, with low access to public policies, goods, or services. The education offered is of low quality, the schools have a deficient infrastructure, the teaching is multi-serial and offered by a few teachers, as a result, there is a high degree of illiteracy and functional illiteracy.

There are two state schools in PAI#01, “Iolanda Pereira Nunes State Elementary School” and “Manoel Rufino State Elementary School” are operational and serves the local and surrounding communities. The graph below shows the level of education found during the socioeconomic survey in the project areas. As can be seen, most have incomplete elementary education, others do not study or are not applicable (in the case of children).

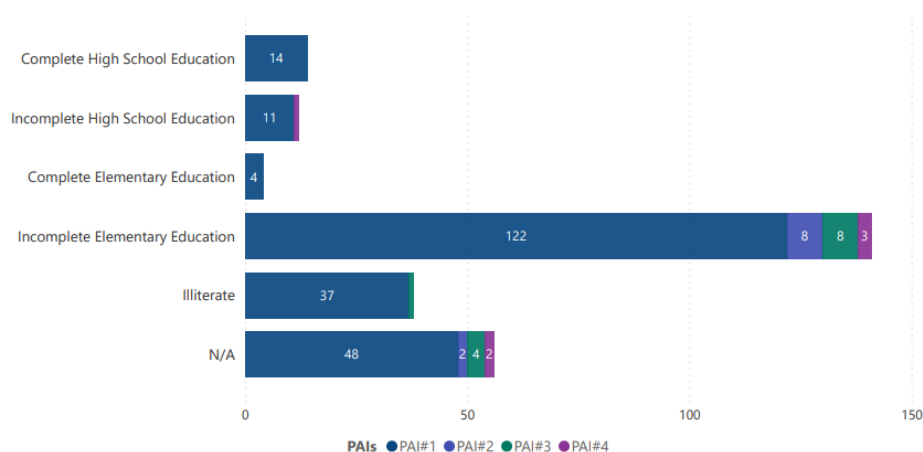


Figure 32. Educational Level

Seeking to the population characteristics in the gender issue, there is a men majority (53%). About the age, most of them are child, young or young adults, between 0 to 32 years. In percentage, around

30% are till 12 years old (childhood), while almost 42% is aged between thirteen and thirty-two years (adolescence and young adults), what compounds more than 70% of the whole population. Meanwhile, few people are older than 65 years, in a number, less than 5%, once, they generally, after retiring, use to move to the city headquarters, mainly Sena Madureira and Rio Branco, where public services are more accessible.

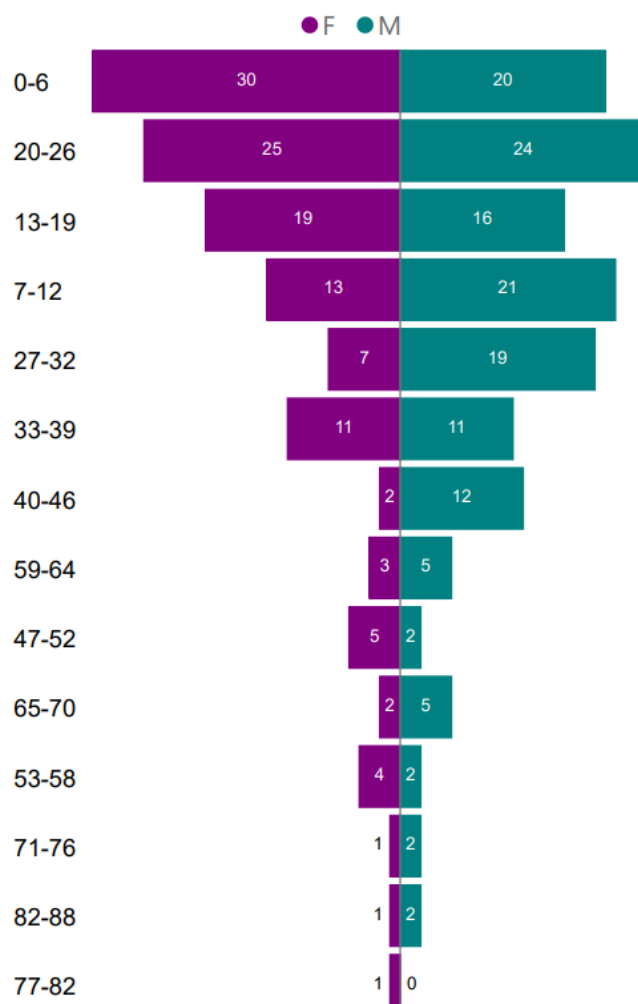


Figure 33. Age pyramid and gender distribution in PAIs.

The nearest hospital stays in Rio Branco, depending on the means of transport, the trip can take two-days along the Iaco River or a one-day trip by Transacarana road, which accentuates their vulnerability.

Mostly water is collected from springs, but not just, as it is collected too from the rain and from the river itself, and is used for many functions, as drinking, preparing food, washing dishes, clothes, and bathing. In accord to a water potability analysis, made in 2021, there is water contamination in every spring tested. In many situations, animals have free access to the same source of water. To try to avoid health problems, some families treat the water through using sodium hypochlorite or a filter, however because of the distance to the cities and costs, the access to these treatments are unusual.

As can be imagined, the sanitation condition is very peccary. Analyzing the whole city of Sena Madureira, about two thirds of the population has no access to piped water, while there even no data about sanitary sewage (Sistema Nacional de Informações sobre Saneamento – SNIS, 2021)³⁴. In the rural areas, as where are the project area, this picture is even worse, in general the houses have only rudimentary structures, as inappropriate cesspool, or even no structures at all, what means that feces and urine go directly to the soil, potentializing the spread of diseases. Furthermore, the low level of formal education makes the local people has a lack of good practices to domestic sanitation.

The garbage situation is challengeable as well once the traditional practice on the territory is to burn or simply throw it away in an open gully.

In terms of houses structures, they are very simple, generally made with local materials, like wood and fibers, mixing these materials with industrialized or zinc tiles, just a few residencies are made with a straw roof.



Figure 34. House's structure.

In sum, the communities' condition is in general precarious, being a picture of the vulnerability condition of most of the families, reflecting the lack of public services and the distance from the city's headquarters, which makes the access to basic services and housing conditions.

³⁴ <https://www.aguaesaneamento.org.br/municipios-e-saneamento/ac/sena-madureira>

4.0.2 Interactions between Communities and Community Groups (VCS, 3.19; CCB, CM1.1)

As part of the project's long-term objectives, there is a focus on increased interaction with the communities through participatory processes, where everyone is invited to participate at different times and locations. This approach aims to reach as many residents, community members, and community groups as possible within the project area. Furthermore, these differences and particularities are considered in the design and implementation of project activities.



Figure 35 - Public meetings and interactions with communities.

Within the project's scope, there are no community groups identified, because local social organization is absent in the territory. Although there was once an association, it disbanded prior to the project's inception due to organizational deficiencies, as agreed upon by certain community members. The majority does not perceive this as problematic, resulting in a lack of concerted efforts towards reactivation. However, it's essential to acknowledge the challenges in doing so, particularly given the dispersed nature of dwellings, making meetings and gatherings logistically difficult and costly. Additionally, there are differences in cultural practices among the communities found in the project areas. For instance, there are variations in religious groups, access to basic services, and modes of transportation (either by land or boat). Generally, PAI#1 has a higher proportion of outsiders and better access to education. In contrast, the more remote areas, such as PAI#2 to PAI#4, face greater difficulties accessing basic resources and leaving the cities. Therefore, some communities have more access to schools, education, and healthcare due to their proximity to urban centers and roads than others located farther away.

Nevertheless, local inhabitants engage in social interactions primarily through local festivals and religious activities. Additionally, exchanges occur within commercial spheres or through certain social activities that yield mutual benefits, such as collaborative efforts in infrastructure development or

other specific tasks. Another form of social interaction among community members involves hiring one another for daily wage labor on particular projects.

As part of the conservation project aimed at preventing illegal deforestation, project proponents proactively take measures to ensure that activities do not adversely impact the natural environment or communities. Beyond actively identify and address any potential adverse socio-economic and environmental impacts, this also includes assessing risks to stakeholder well-being and participation, such as trade-offs with food security, land loss, and identify any negative impacts on livelihoods throughout the territory. Inside the territory there is no local social organization, it used to have one association, but before the project started, in accord with some communarians, it was shut down because of lack of organization. Most of them do not see it as a problem, thus there is no real effort been made to reactivate it. However, it is necessary to recognize the difficulty to do it, once the reality is that most of the houses are far from each other, what makes meetings and reunions very hard to make – it is expensive to walk around the territory and it takes time to do so.

Besides that, the local inhabitants have its own social interaction, been mainly through local festivals and religious activities. Beyond that, there are exchanges in a commercial sphere or in some social activities which creates mutual benefits, like the opening of a branch road or other kinds of specific tasks. Another form of social interaction between community members is the hiring of one another to pay daily wages for specific projects.

As part of the conservation project aimed at preventing illegal deforestation, project proponents proactively take measures to ensure that activities do not adversely impact the natural environment or communities. They actively identify and address any potential adverse socio-economic and environmental impacts. This includes assessing risks to stakeholder well-being and participation, such as trade-offs with food security, land loss, and negative impacts on livelihoods. Furthermore, the project upholds human rights, including equal opportunities, the prohibition of forced labor, and the protection of the rights of marginalized groups. Measures are implemented to preserve cultural heritage and property rights, including securing the rights of Indigenous Peoples and local communities and preventing forced displacement. Additionally, the project has no negative impacts on biodiversity and ecosystems, and measures are taken to prevent the introduction of invasive species or the degradation of natural ecosystems. These actions are outlined in the project's social engagement protocols, aiming to ensure responsible and sustainable implementation.

4.1.3 High Conservation Values (CCB, CM1.2)

According to the assessment carried out in the field, based on interviews and participatory approach, the local communities had huge difficulties to recognize any High Conservation Values (HCVs) in the project area. The concept of HCV was approached with communities during the first and second stages of social engagement. At first, communities were encouraged to identify HCVs in face-to-face

meetings held door-to-door. In a second moment, the communities were encouraged to draw on a map the location and extent of possible HCVs, according to their own perspective.

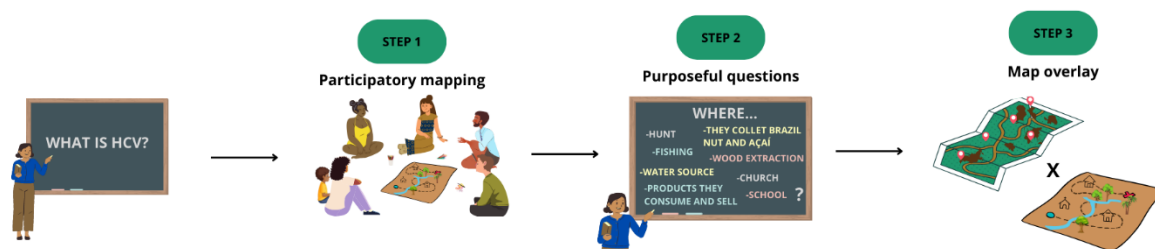


Figure 36. Identification of High Conservation Value approach, at social engagement.

On none of the occasions a good understanding of the concept was characterized so that the mapping in question could not be considered satisfactory, requiring new strategies to make the concept clear and touchable for the local people. Once BRC understands that the concept of HCV is too abstract for communities. Thus, to develop the perception and the comprehension of this HCV complex conception by the communities and communitarians, there will be used tools of participative approaches to achieve the understanding of it, and then analyze the existence or not of any HCV in the territory.

To do it, so, the strategy will be through an applying of a participatory field mapping may be carried out in the future occasions for an adequate demarcation of possible HCVs.

In accord to the Social Engagement Protocol (APPENDIX_SOCIAL-ENGAGEMENT-PROTOCOL), the use of the Integration of Ecosystem Services / The Economics of Ecosystem and Biodiversity (ISE/TEEB) approach enables the understanding of the complexity of HCV. For it, this tool serves to make clear dependencies and impacts – positives or negatives – of human activities upon the ecosystem and the biodiversity, and more, to map the risks, conditions, tendencies, and tradeoffs, thus it makes possible the elaboration of actions to minimize the negative effects and potentialize the positive effects of these activities upon the well-being dimension. This tool is based on a participatory approach made to work alongside young people and adults, which makes possible reflection and the achieve of common solutions, as it subsidizes use plan agreements and action plans, giving inputs to make a management adaptative project.

The implementation of this participatory approach is a short-term action; however, it has a medium- and long-term impact. In this sense, the PP intends to stimulate and support the discussion in schools and other gathering opportunities to deep the understanding about themes as ecology, water cycle, carbon cycle, nutrients, ecosystems, and biodiversity, as well as ecosystem services, in face to opportunism access to basic themes and concepts which involves the importance to maintain the forest and its biodiversity alive through sustainable practices to promote improvement on local.

Having said that, BRC identified potential HCVs in the project areas that are related to community livelihoods (Table 55 and Table 56).

Table 55 - Areas that are fundamental for the livelihoods of communities (HCV).

High Conservation Value	Areas that are fundamental for the livelihoods of communities
Qualifying Attribute	The project zone provides essential food, fuel, fodder, medicines, and building materials without readily available alternatives. Communities hunt, fish, use non-timber forest products as medicine and wood to build houses and boats. All these inputs come from the surroundings of their homes, in the project area, and are supported in a broad spectrum by the continuous landscape.
Focal Area	The project area is here defined as the focal area that needs to be managed to maintain or enhance this HCV. To this end, the project will invest in actions related to fauna monitoring with a focus on game species, technical assistance and rural extension related to sustainable livestock and sustainable forest management.

Table 56 - Areas that are critical for the traditional cultural identity of communities (HCV).

High Conservation Value	Areas that are critical for the traditional cultural identity of communities.
Qualifying Attribute	Porongaba cemetery, nun Dulce tombstone and the house of pray in Baixo Alegre, may be considered as important for the traditional cultural identity of communities in Porongaba. Cemeteries become part of the collective memory of the place, important for the construction of identity and social reproduction. There is a legend about nun Dulce who lived in this area in the 19th century, and was buried in a marble and iron tombstone, which rarely is found in a remote place like that. The prayer houses are important places for the cultural identity and association of communities in Porongaba.
Focal Area	At Porongaba Farm (PAI#2), the families at Porongaba and Santa Luzia settlements recognized the cemeteries located in their respective locations, as areas of special value, as they have cultural and religious significance for these families. The project will provide for cleaning access to these locations and renovations to prayer houses.



Figure 37 - Areas that are critical for the traditional cultural identity of communities (HCV). Nun Dulce tombstone.



Figure 38 - Areas that are critical for the traditional cultural identify of communities (HCV): Praying house.

4.1.4 Without-Project Scenario: Community (CCB, CM1.3)

In the without-project land use scenario, what is expected is the maintenance of the conditions found during social baseline, where a low level of education, lack of access to health programs, lack of basic sanitation, lack of solid waste proper management and the maintenance of livestock production-based economy, can be observed.

Therefore, the traditional communities described above, the condition would remain in the sense of lack of access to some services as internet, and technical assistance and rural extension, or the improvement of some public services, as the quality of education, or even some goods, as solar power kit. Although, if the perspective is related to a rise of deforestation coming from outsiders, new squatters who are pressuring the forest in the region and bringing insecurity to the traditional inhabitants, the without-project scenario could transform the reality in a fast way to a much worse scenario.

Thus, the inexistence of the project would be a bad scenario to the population who lives there for a long already, while could be a great opportunity to land grabbers and all the risks it can bring to the territory.

4.2 Net Positive Community Impacts

4.2.1 Expected Community Impacts (CCB, CM2.1)

The anticipated impacts resulting from project activities under the with-project scenario can be found in this section. BRC will bear all costs associated with the development, implementation,

monitoring, and certification of the project. No cost will be passed on to traditional communities. The execution of the project does not pose risks to traditional communities.

All the expected benefits are related to the Section 2.1.16 – Project Activities and Theory of Change.

Community group	Traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Digital inclusion and solar energy
Type of benefit/cost/risk	Direct actual benefit: internet and solar energy is already installed and operational in schools in Porongaba. Direct predicted benefit: these goods will be installed in other schools in different communities and possibly at homes. Indirect predicted benefit: education, health, trade, and citizenship conditions improvement. Livelihood and well-being condition improvement.
Change in well-being	The project will directly impact community's livelihood and well-being through digital inclusion. The internet will alleviate the vulnerable condition in which communities find themselves, and improve conditions for access education, health, trade, and citizenship.

Community group	Traditional communities of Porongaba
Impact(s)	Education
Type of benefit/cost/risk	Direct actual benefit: renovation school buildings, internet installation and purchase of equipment (TV, computer, and printer) for schools in Porongaba. Direct predicted benefit: students and teachers with more conditions and resources. The internet will enable the practice of distance learning, with a greater number of teachers. Indirect predicted benefit: improvement in the quality of teaching and education of traditional communities in Porongaba.
Change in well-being	The improvement in the quality of teaching and education in the traditional communities of Porongaba, through an improvement in available resources and distance learning practice.

Community group	Traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Health
Type of benefit/cost/risk	Direct actual benefit: access to health services through a partnership for raise the length of the itinerant health campaign beyond the original scope; improvement on access to public services. Direct predicted benefit: Increasing on access to public services and improvement on local Governance. Indirect predicted benefit: improvement on access to basic rights.
Change in well-being	The lack of state presence is one of the biggest problems in territories to far from city headquarters and through public and private partnerships, it will make the population closer to the state and its benefits, what will make the citizenship more accessible and

	practicable to the local population, indirectly improving well-being and opportunities.
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Community group	Traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Water
Type of benefit/cost/risk	Direct actual benefit: water potability analysis, distributing water filters in Porongaba. Installation of water pumps, water tanks and hydraulic systems to supply homes and schools. Direct predicted benefit: improvement on water quality for human consume. Indirect predicted benefit: improved on general health conditions.
Change in well-being	Increasing access to quality water and reduction on water transmitted diseases.

Community group	Traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Training, employment
Type of benefit/cost/risk	Direct actual benefit: Community training and employment generate in activities related to forest inventory and others project activities. Direct predicted benefit: Increased number and amount of training. Indirect predicted benefit: Improvements in skills and employment opportunities among communities.
Change in well-being	Trainings will increase the communities' skills and create job opportunities for interested parties.

Community group	Traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Training and livelihood
Type of benefit/cost/risk	Direct actual benefit: Direct participation on planning, implementing, and monitoring the project actions, beyond participation on capacity courses. Direct predicted benefit: access to new knowledges and social tools to strengthen the social organization, at management and bureaucrat challenges. Development on individual and collective skills throughout participatory and dialogical approaches for organizing and decision making in the organizations, promoting improvement on collective well-being. Indirect predicted benefit: fostering citizenship through the appreciation of the social organizations.
Change in well-being	More organized people and territory to demand rights, beyond improvement on local autonomy to implement better conditions in the communities.

Community group	Traditional communities of Porongaba, Palmares, Potiguar and Katiana
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Impact(s)	Education
Type of benefit/cost/risk	Direct actual benefit: Improvement on environmental educational access through didactic and pedagogical materials for understanding of sustainable practices and its consequences to the socioenvironmental issues. Direct predicted benefit: a better understand of the options to subside the decision making to improve the livelihood through forest conservation and sustainable use of its resources. Indirect predicted benefit: better formed and informed people can understand in a more conscious way the causal relations between the human activities and the impacts that comes from it, making possible to make better choices to improve the wellbeing and to the citizenship practice.
Change in well-being	The improvement of knowledge creates a capacity to promote changes and better choices on the progress of collective well-being. The understand of biodiversity and ecosystem services has a direct relation to the human well-being, and it's a key to a sustainable and prosperous future.

Community group	Traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Livelihoods, training, and employment
Type of benefit/cost/risk	Direct actual benefit: development of low-carbon, inclusive and sustainable value chains. Direct predicted benefit: Development of low-carbon, inclusive and sustainable value chains, from the production to the commercialization, what helps to a better local governance through a participatory and transparent process. And it makes an increase of family income and improvement in their livelihoods and the presence on production of women and youth people. Indirect predicted benefit: Knowledge improvement and development of personal and collective skills focused on value chains and territory planning, on social organization, productive management and access to market. Fostering the arise of women and young people leadership, throughout the development of better citizenship practices.
Change in well-being	Strengthen and improvement on sustainable productive activities and local productive arrangements instead of productive activities deforesting and degradation. The benefits of the improvement on ecosystem through fostering agroforest systems, promotes well-being, food security, income increase, and gender equity.

Community group	Young people of Traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Training and livelihoods
Type of benefit/cost/risk	Direct actual benefit: Engagement of youth for program participation; access to knowledge and development personal and collective skills to management and monitoring the actions. Direct predicted benefit: structuring/designing the program of socio-environmental agents/monitors and people trained to work in

	planning and monitoring project actions, value chain activities and organization management. Indirect predicted benefit: Development of skills to act in local development plans and organization management, value chains and local productive arrangements.
Change in well-being	Development of skills and capacities to recognize of social and environmental in planning, decision making and actions to local sustainable development.
Community group	Women of traditional communities of Porongaba, Palmares, Potiguar and Katiana
Impact(s)	Education and livelihoods
Type of benefit/cost/risk	Direct actual benefit: Activities that promoted engagement and inclusion of women in the project, such as workshops, training and courses. Direct predicted benefit: Courses and training for women to generate income. Workshops to dialogue groups on topics related to female empowerment (women's rights, domestic violence, social division of labor, women's entrepreneurship) Indirect predicted benefit: Inclusion, empowerment and autonomy of women
Change in well-being	Awareness and knowledge on important topics, which can contribute to their engagement in the project and empowerment.
Community group	Traditional communities of Porongaba
Impact(s)	Livelihoods and well-being
Type of benefit/cost/risk	Direct actual benefit: Development of activities that strengthen local culture, such as sporting and religious celebration and the renovation of the community chapel. Direct predicted benefit: Promoting community engagement in project activities and well-being.
Change in well-being	Engagement of people through activities that protect cultural and spiritual values and improve their well-being.
Community group	Traditional communities of Porongaba
Impact(s)	Livelihoods and well-being
Type of benefit/cost/risk	Indirect actual benefit: Make it possible to these families access new forms of public policies and increasing their security with their land rights guaranteed. Indirect predicted risk: increase in governance and legal sanctions, potentially leading to local conflicts.
Change in well-being	The project will bring an increase in territorial governance and dialogue with responsible environmental agencies, which may bring an increase in legal sanctions for those who are commenting on land grabbing. As a mitigation measure, the project will move actions for land title regularization, recognizing the property rights of those who traditionally exercise their right of ownership.

4.2.2 Negative Community Impact Mitigation (VCS, 3.19; CCB, CM2.2)

Project activities are not designed to negatively impact traditional communities' well-being. The project may, however, increase the level of governance and dialogue with government agencies responsible for inspection, which may increase the incidence of legal sanctions, affecting land grabbers. As a mitigation measure for traditional communities, BRC will work with land regularization activity, where applicable, recognizing the property rights of those groups that traditionally exercise the right of tenure. This activity type is complex and will be conducted with due care, considering the continuous communication channels and grievance mechanisms created by the project.

More than that, during the process of land donation some questions can emerge, mainly related to the use of the land, to the traditional practice of open new areas annually for the agricultural production, and the later transformation of the area into pasture, once with the donation they will be limited to the regularized area, preventing the traditional practices. To avoid bigger problems, the PP will provide technical assistance and rural extension to give them the knowledge and tools to change their way of production to a more productive and sustainable form.

The identified HCVs will be improved by the implementation of project activities. The areas that are fundamental for the livelihoods of communities will be improved by the additional conservation of forest cover promoted by the implementation of the project activities, while the areas that are critical for the traditional cultural identity of communities will be improved by the renovation of prayer houses and cleaning access to the cemetery.

It is worth to cite that the BRC Social Engagement Protocol has in it stages the approach the ISE/TEEB mentioned above (Section 4.1.3), and this technic will subside the PP and the population to understand the human-nature relationship and, from it, avoid or mitigate potential negative impacts. The use of this tool will make the analyze participatory and, although, more legitim.

4.2.3 Net Positive Community Well-Being (VCS, 3.19; CCB, CM2.3, GL1.4)

The net well-being impacts of the project are positive for all identified and legitim (bona fide squatters) community groups. This can be demonstrated by comparing the social baseline (without project land use scenario) with the project scenario considering the planned activities and the theory of change matrix. In this sense, BRC expects to cause substantial changes in communities well-being, through digital inclusion, wildfires reduction, improvements in health and education, improvement in access to water and water quality, improvement on citizenship, improvements in food and nutrition security, training, qualification and generation of employment and household income.

4.2.4 High Conservation Values Protected (CCB, CM2.4)

As already presented above, BRC has identified areas that are fundamental for the livelihoods of communities (project area itself) and areas that are critical for the traditional cultural identity of communities (prayer houses, cemetery) as HCVs. The measures to avoid deforestation relates directly to the forest protection, and indirectly this action helps to maintain and reinforce the area where some of the needs for the population is found, been thus a way to protect the HCV. While the prayer house and the cemetery will be renovated and maintained clear, and accessible to be used as an HCV for the population.

Additional measures related to forest protection, the prayer houses and the cemetery will be thought to improve the conditions of the identified HCVs, thus there will be a constant analyze to define and redefine the best strategies to not just improve but value the HCVs. BRC does not recognize negative impacts of the project on the identified HCVs.

4.3 Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.1)

The project activities will generate many positive impacts for other stakeholders, such as the Municipal Health Department of Sena Madureira through the Itinerant Health Program. BRC will support the institution in carrying out the action in the territory, assisting in its organization, and financing the logistic of transportation and implementation in the areas. A negative impact that can be reported is the potential for communication failures that may occur due to changes caused by elections, staff turnover, or lack of clarity in intentions on the part of BRC.

Considering these factors, the project aims to develop health and education actions in the territory by formalizing the partnership with the necessary institutions and jointly defining a work plan. This is to contribute to expanding the institutions' work with the support in the territory. Other actions will be developed in partnership with public organizations, as the project aims to contribute to the increased effectiveness and scale of public policies especially in the project's area.

In relation to the recent invaders and land grabbers, the command-and-control actions are consequences of the invasions, and they can have an impact on the territory, as well as on the families of those who came in a non-legitimate form.

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

Frequent meetings will be held with the aim of avoiding communication failures with other stakeholders, where project objectives will be clarified, intentions and partnership possibilities discussed, and project summaries made available. Additionally, the communication channel "Alô brCarbon" will be introduced for suggestions, questions and complaints regarding the project and the

company's activities. In relation to the recent invaders, the owner of the areas has been taking the necessary legal and juridical measures to address the issue.

4.3.3 Net Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

The project aims to generate only positive impacts for all stakeholders, with all risks and potential negative impacts identified and mitigated. To generate positive impacts, diagnostics and meetings will be conducted with all involved stakeholders to align expectations regarding the project and discuss a joint work plan if necessary, aiming for the engagement and empowerment of all necessary stakeholders for the development of each activity.

4.4 Community Impact Monitoring

4.4.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The community monitoring plan presents the predicted activities of the project, the benefited communities and other stakeholders involved, the sampling methods used, dates, frequency (when it is possible) of each one of them.

Monitoring is a continuous process of planning, recording, and evaluation. The theory of change involves different implementation stages of the project activities, and relates to many thematic, which makes it complex, once a lot of tools are needed to analyze and act upon the project proposes.

To guarantee the proper comprehension of the data to build the project baseline and monitor the project activities, BRC has developed a platform (app) to collect and analyze this information in a way that will be possible to record and manage all the needs to the development of the project. This system will help the PP to check what has been effective and what is not, and from it adapt and improve the activities. But, not just that, this strategy is going to be useful to avoid loss of data from applied questionnaires and interviews, fostering in a digital way all the information demanded to be reported.

The table below presents the project activities related to Theory of Change, benefited communities and other stakeholders involved, sampling methods used, dates, frequency, and results.

Table 57 - Community monitoring plan

Activities	Benefited communities and other stakeholders involved	Sampling methods used, dates and frequency	Results
Installation of internet towers and solar energy equipment	Porongaba, Palmares, Potiguar and Katianã communities	The number of towers installed, solar panels deployed, batteries bank deployed, and the amount of optic fiber installed. Date: - Frequency: once	Planning phase
Renovation and equipment of schools located in Seringal Porongaba	Porongaba community	The number of schools renovated, and the number of equipment delivered. Date: - Frequency: once	Planning phase
Articulation with the Municipal / State / Federal Public Power and private initiative to support public policies and projects/program in the project area.	Municipal Education Department of Sena Madureira (AC) Municipal Health Department of Sena Madureira (AC) Municipal Social Service Department of Sena Madureira (AC)	The number of meetings with public power Date: - Frequency: annual	Planning phase
Training and partnerships to strengthen social organization	Porongaba community	The number of trainings on social organization Date: - Frequency: biannual	Planning phase
Development of inclusive and low-carbon value chains	Porongaba, Palmares, Potiguar and Katianã communities	The number of plans and reports prepared Date: - Frequency: biannual	Planning phase
Carbon pedagogy lectures	Porongaba, Palmares, Potiguar and Katianã communities	The number of lectures conducted Date: - Frequency: on demand	Planning phase

Activities	Benefited communities and other stakeholders involved	Sampling methods used, dates and frequency	Results
Socio-environmental monitor program	Porongaba community	The number of public meetings and workshops conducted Date: - Frequency: biannual	Planning phase
Technical Assistance and Rural Extension focused on low-carbon livestock and agroforestry systems	Porongaba and Palmares communities	The number of workshops and visits made Date: - Frequency: biannual	Planning phase
Recognition and inclusion of women in the productive and social activities	Porongaba, Palmares, Potiguar and Katianã communities	The number of workshops conducted Date: - Frequency: annual	Planning phase
Sustainable Forest Management	Porongaba, Palmares, Potiguar and Katianã communities	The number of trainings conducted, forest management plan prepared, and safety equipment delivered. Date: - Frequency: every five years	Planning phase
Promotion of improvements in water quality and access to water	Porongaba, Palmares, Potiguar and Katianã communities	The number of water potability analysis conducted, water filter delivered and water pumps installed Date: - Frequency: on demand	Planning phase
Support for religious, cultural and sports structures, equipment, materials and traditional celebration	Porongaba, Palmares, Potiguar and Katianã communities	Number of religious, cultural and sports structures, equipment, materials and traditional celebration supported Date: - Frequency: annual	Planning phase
Hiring of community personnel	Porongaba, Palmares, Potiguar and Katianã communities	The number of employments generated by the project Date: - Frequency: on demand	Planning phase

4.4.2 Monitoring Plan Dissemination (CCB, CM4.3)

As per practice, required by the Verra's rules, the monitoring plan and monitoring results will be available on Verra's website through the GPD and monitoring reports full documents. Beyond that, the project activities will be presented to the local stakeholders through meetings and informs, what is previewed to happen during the fourth social engagement stages.

Furthermore, BRC has planned to create a semiannual newsletter, which will be disseminated in printed version to the communities and other stakeholders that BRC visit, as well as digitally via email and WhatsApp. It will include a summary of the main project activities in each community and municipality, allowing everyone to stay informed and reach out for any questions and/or suggestions.

4.5 Optional Criterion: Exceptional Community Benefits

4.5.1 Exceptional Community Criteria (CCB, GL2.1)

4.5.2 Short-term and Long-term Community Benefits (CCB, GL2.2)

4.5.3 Community Participation Risks (CCB, GL2.3)

4.5.4 Marginalized and/or Vulnerable Community Groups (CCB, GL2.4)

4.5.5 Net Impacts on Women (CCB, GL2.5)

4.5.6 Benefit Sharing Mechanisms (CCB, GL2.6)

4.5.7 Benefits, Costs, and Risks Communication (VCS, 3.18; CCB, GL2.7)

4.5.8 Governance and Implementation Structures (CCB, GL2.8)

4.5.9 Smallholders/Community Members Capacity Development (CCB, GL2.9)

5 BIODIVERSITY

5.0 Without-Project Biodiversity Scenario

5.0.1 Existing Conditions (VCS, 3.19; CCB, B1.1)

The project area is in Acre state, in the southwest part of the Amazon domain. With more than 80% forest cover, Acre ranks as the second state with the highest proportion of forest in Brazil, second only to Amazonas (MapBiomass, 2023). The predominant vegetation type in the region is the Dense Ombrophylous Forest and Open Ombrophylous Forest, specifically, the Open Ombrophylous Forest with Bamboo and Palm Trees (Acre, 2006). There are 1,400 vertebrate species with occurrence

confirmed for Acre state, of which 769 are birds, 203 are mammals, 111 amphibians, 91 reptiles and 320 fishes (Acre, 2006). The studies indicates that approximately 40% of Brazil's mammals and 45.8% of Brazil's birds occur in Acre, highlighting its significance for the country's biodiversity conservation efforts.

However, the biodiversity and ecological importance of the region is threatened by deforestation and forest degradation. The highest deforestation rates of Acre state are concentrated in the municipalities of Rio Branco and Sena Madureira, where the project areas are located. The main threat to region biodiversity is the expansion of deforestation over the project areas, caused by invaders and squatters. This region has a significant portion of agrarian settlements occupied by small family farmers whose primary activity include livestock breeding, animal rearing, and the production of milk and cassava flour. In this scenario, the need for actions to conserve forest cover and to carry out biodiversity monitoring studies is imperative.

A diagnostic campaign was conducted to monitor four groups of vertebrates: amphibians, birds, mammals, and reptiles. During this diagnosis, fauna specialists collected data on species richness, endemic species, and threatened species.

The total species richness found was 222 (32 amphibians, 161 birds, 18 mammals and 11 reptiles), of which 92 species are endemic to the Amazon. Among these species, eight are classified as Vulnerable (VU) by the IUCN Red List and one mammal species is classified as Endangered (EN). All information regarding the fauna survey campaign, such as description of the methodology used, list of species and photographic records, is available in "APPENDIX_BIODIVERSITY". The table below summarizes some information about the threats to species and habitats located in this project, demonstrating that the planned activities do not foresee negative impacts on them.

Table 58 - Summary of threats to species and habitats found on the project area, as well as contributions to their conservation.

Threatened species and habitats located in this project	Project Impacts
Threatened birds: <i>Cnipodectes superrufus</i> , <i>Pionites leucogaster</i> , <i>Primolius couloni</i> and <i>Tinamus tao</i>	Four bird species found in the project area have been categorized on the IUCN Red List as "VU – Vulnerable", indicating a significant risk of extinction in the near future if threats to their survival and reproduction are not adequately addressed. Among the main threats to these birds are habitat loss and predatory hunting (Bird et al., 2011; Collar et al., 2020; Soares-Filho et al., 2006). The project activities aim to prevent illegal deforestation, thereby safeguarding the forest cover, that serves as habitat for these species. Additionally, the project aims to monitor the occurrence of these birds in the region, thus contributing to their conservation efforts.
Threatened mammals: <i>Ateles chamek</i> , <i>Alouatta puruensis</i> ,	The primate <i>Ateles chamek</i> is categorized on the IUCN Red List as "EN – Endangered", indicating that the species will probably become extinct in the near future. The other mammal species listed here are categorized as "VU – Vulnerable" which also indicates that they are at significant risk of

<i>Cebus unicolor</i> and <i>Tapirus terrestris</i>.	becoming extinct in the near future, if threats to their survival and reproduction are not properly addressed. Among the main threats to these mammals are habitat loss and predatory hunting (Aquino et al., 2013; Soares-Filho et al., 2006). The project activities aim to prevent illegal deforestation, thus preserving the forest cover that serves as habitat for these species. Additionally, the project aims to monitor the occurrence of these mammals in the region, thus contributing to the conservation of this group.
Threatened reptiles: <i>Podocnemis unifilis</i>	<i>P. unifilis</i> is classified on the IUCN Red List as "VU - Vulnerable", indicating a significant risk of extinction in the near future if threats to its survival and reproduction are not adequately addressed. Among the main threats to the species are habitat pollution, natural and anthropogenic nest predation and hunting for human consumption (Andrade, 2008). The preferred habitat of this species is rivers and lakes, but during river floods, they go to flooded forests in search of protection from predators and to feed on fruits and seeds that become available (Fachín-Terán et al., 1995). The project activities aim to prevent illegal deforestation, thus preserving forest cover and contributing to the quality of the aquatic ecosystems used by the species. Additionally, the project aims to monitor the occurrence of <i>P. unifilis</i> in the region, contributing to the species conservation.
Threatened habitat: Open Bamboo Forest	The state of Acre has a forest typology dominated by Bamboo with a predominance of species of the genus <i>Guadua</i> (Silva et al., 2019). This plant typology adds more diversity to the ecosystem of these forests and concentrates species such as <i>Cnipodectes superrufus</i>, which are specialists in this habitat, making them more sensitive to degradation (Tobias et al., 2008). Therefore, the activities of this project aim to contribute to the maintenance of forest cover by combating illegal deforestation.

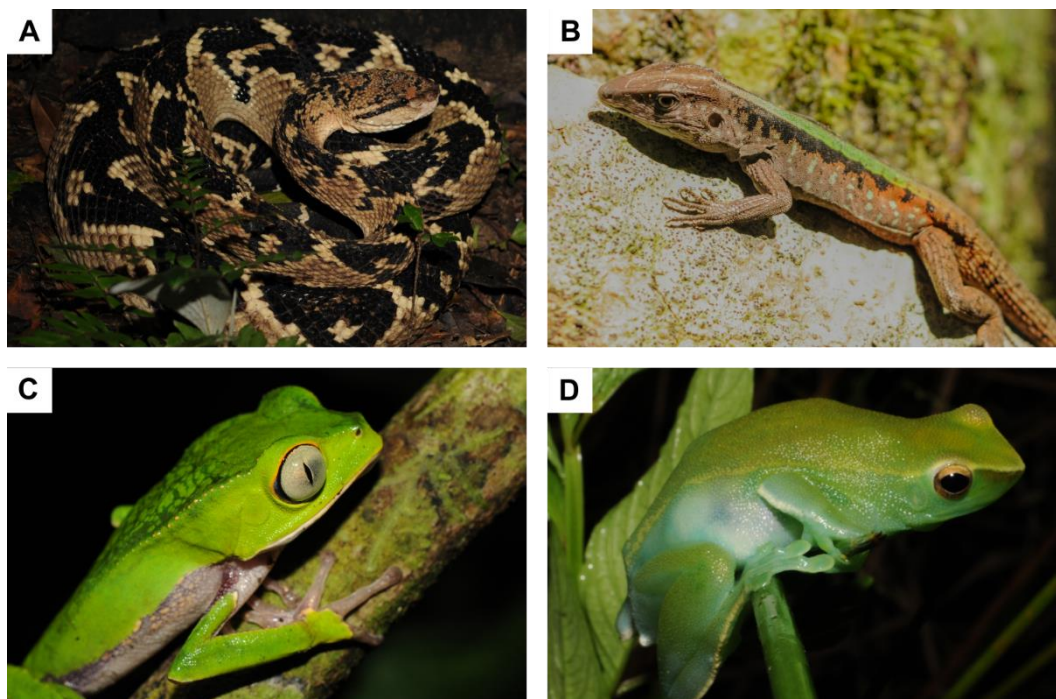


Figure 39 – A) *Lachesis muta*; B) *Kentropyx cf. calcarata*; C) *Phyllomedusa vaillanti*; D) *Sphaenorhynchus lacteus*.

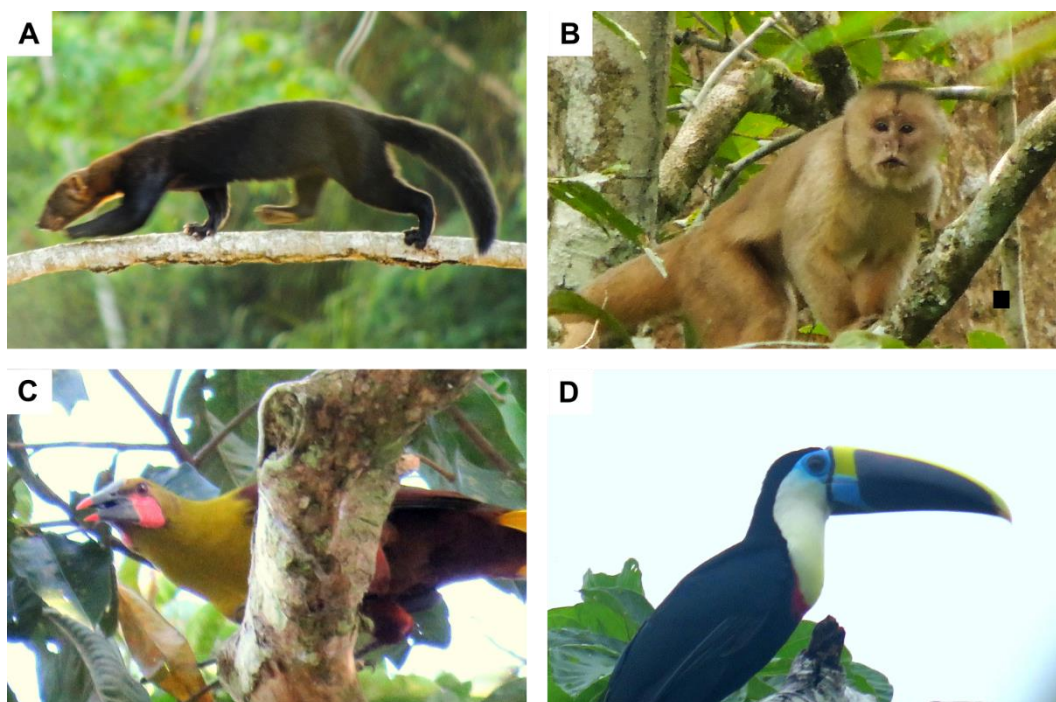


Figure 40 - A) *Eira Barbara*; B) *Cebus albifrons*; C) *Psarocolis bifasciatus*; D) *Ramphastus tucanus*.

5.0.2 High Conservation Values (CCB, B1.2)

The project zone is located in a region of global importance, which concentrates significant biological diversity (including endemic, rare and threatened species) and provides ecosystem services that contribute to the balance of natural processes at a global level. These attributes ensure the presence of viable populations of most, if not all, naturally occurring species, maintaining natural patterns of distribution and abundance. Three categories of HCVs related to biodiversity were identified in the project zone: HCV 1 – Species diversity; HVC 2 – Landscape-level ecosystems, ecosystem mosaics and IFL and HCV 3 – Ecosystems and habitats.

The HCV 1 was identified by the existence of: i) globally significant concentration of endemic species and ii) globally significant concentration of threatened species. The HCV 2 was identified by the existence of a large global landscape-level ecosystem. And HCV 3 was identified by the presence of threatened and rare habitats. The attributes of HCVs are described in the tables below.

Table 59 - HCV attributes: Significant concentrations of endemic species

High conservation value	HCV 1: Significant concentrations of endemic species
Qualifying attribute	The initial fauna survey carried out in the project area revealed the existence of 92 species endemic to the Amazon region, some with distribution restricted to Acre state. These data show the relevance of project area and the importance of implementing conservation measures to safeguard the survival these endemic species. The project area is located in the Inambari Endemism Center, one of the largest centers of endemism in the Amazon. Spanning 1,326,684

	km ² (Da Silva, 2013), this region possesses exceptional biodiversity resulting from a combination of characteristics from the Amazon plain and typical high-altitude species from the Andes Mountain range (Cracraft, 1985). Nevertheless, although the Inambari Endemism Center stands as one of the best-preserved areas in the Amazon, a large portion of its territory consists of unprotected forest areas (Rodrigues and Gaston, 2001). This highlights the importance of activities aimed at forest conservation in the region.
Focal area	The project intends to maintain this HCV through the conservation of forest ecosystems that provide shelter, food, and a variety of complex ecological interactions for endemic species.

Table 60 - HCV attributes: Significant concentrations of threatened species

High conservation value	HCV 1: Significant concentrations of threatened species
Qualifying attribute	Nine species classified as Vulnerable or Endangered, according to the IUCN Red List, were identified in the project area. These species are extremely sensitive to habitat destruction and hunt pressure, therefore, conservation actions in the region are extremely important. Habitat loss, environmental degradation and climate change represent real threats to the survival of countless species. For this reason, effective conservation measures are necessary to mitigate possible irreversible impacts on local biological diversity.
Focal area	The project intends to maintain this HCV through the conservation of forest ecosystems that provide shelter, food, and a variety of complex ecological interactions for threatened species.

Table 61 - HCV attributes: Global significant large landscape-level ecosystem.

High conservation value	HCV 2: Global significant large landscape-level ecosystem
Qualifying attribute	The project zone it is located within the Amazon Forest, the biggest tropical forest in the world that houses thousands of species and provides different ecosystem services to humanity. It is estimated that the Amazon houses approximately 425 mammal species, 1300 bird species, 371 reptiles and 427 amphibians (Mittermeier et al., 2003)
Focal area	The project zone represents the conservation of thousands of hectares of native forests located within the legal reserve that suffer highly illegal deforestation pressure. Through the conservation of the legal reserve, the project aims to prevent habitat loss, fragmentation, and forest conversion, which are the primary drivers of biodiversity loss in the Amazon.

Table 62 - HCV attributes: Threatened and rare habitats.

High conservation value	HCV 3: Threatened and rare habitats
Qualifying attribute	The project zone is located in a unique portion of the Amazon biome, characterized by the naturally and restrict occurrence of Open

	Bamboo Forests. Around 59% of Acre state is covered by this vegetation type. The bamboo forests are biologically relevant due to its widespread occurrence in this region (Carvalho et al., 2013; Ferreira, 2014; Silveira, 2005) and high concentration of endemic species.
Focal area	Patches of Open Bamboo Forests are found within the project areas; therefore, the project activities will maintain this HCV, through the combat of illegal deforestation.

5.0.3 Without-project Scenario: Biodiversity (CCB, B1.3)

Since the common practice in the region involves slash-and-burn agriculture and intensive cattle-raising (Vieira et al., 2008), without the project, the illegal deforestation pressure would gradually expand within the project area. The deforestation would encroach and reduce the species habitat, negatively affecting its survival. Habitat loss is one of the main causes of species extinctions, however, other human activities such as forest degradation caused by selective logging and predatory hunting also pose severe threats to species survival.

5.1 Net Positive Biodiversity Impacts

5.1.1 Expected Biodiversity Changes (VCS, 3.19; CCB, B2.1)

The with-project scenario in the project zone and over the project lifetime predicts only positive impacts to biodiversity. The conservation activities aim to promote habitat conservation and avoid predatory hunting. The table below shows the expected biodiversity changes.

Table 63 - Expected biodiversity changes.

Biodiversity element	Amphibians, birds, mammals, and reptiles
Estimated change	Predicted positive indirect impact: maintenance of species richness and conservation
Justification of change	The project activities will promote the maintenance of forest cover, thereby ensuring the maintenance of suitable habitat for various species. Furthermore, the project will promote the continuous monitoring focusing on endangered, endemic, and game species.

5.1.2 Mitigation Measures (VCS, 3.19; CCB, B2.3)

Negative impacts on biodiversity are not expected due to project activities implementation neither within nor outside the project area. All project activities are designed to conserve the forest cover and avoid habitat loss within the project area. The threatened and endemic species identified (HCV) will be positively impacted by those activities.

5.1.3 Net Positive Biodiversity Impacts (CCB, B2.2, GL1.4)

Considering the project initial scope, the project activity implementation will avoid the deforestation of hundreds of native forests hectares throughout its lifetime, thus directly contributing to the maintenance of habitat for hundreds of species in the project zone. The project's anticipated net impacts on biodiversity in the project zone will be positive compared with conditions under the without-project land use scenario.

5.1.4 High Conservation Values Protected (CCB, B2.4)

The project activities do not foresee to negatively affect the HCVs related to biodiversity. The implementation of project activities will result in the direct conservation of forest cover in private areas in the project zone, thus contributing to the maintenance of habitats, species diversity, protection of threatened and endemic species and landscape integrity.

5.1.5 Species Used (VCS, 3.19; CCB, B2.5, B2.6)

This conservation project does not include the use of any species. However, if initiatives arise in collaboration with local communities that involve the planting of plant species, native species will be carefully selected and methodologies appropriate to the area will be implemented, ensuring that there is no harm to local species and that the principles of environmental conservation are respected.

5.1.6 Invasive Species (VCS, 3.19; CCB, B2.5)

This project does not include the introduction of species in any of the planned activities.

5.1.7 GMO Exclusion (CCB, B2.7)

The project activities do not foresee the introduction of any species including genetically modified organisms. All activities planned are directly linked to forest conservation aiming the long-term health and resilience of the ecosystem.

5.1.8 Inputs Justification (VCS, 3.19; CCB, B2.8)

This project does not include the use of chemical inputs, such as fertilizers, and pesticides, in any of the proposed activities. However, it is important to emphasize that, if local traditional communities express interest in starting agricultural practices linked to this project, sustainable and natural approaches will be adopted to fertilize the soil.

5.1.9 Waste Products (VCS, 3.19; CCB, B2.9)

The implementation of project activities is not expected to result in significant waste generation. However, during field trips some conventional waste may be generated, which will be properly separated for appropriate disposal. Organic waste will be processed locally, following appropriate

disposal protocols, while other waste, whether recyclable or not, will be collected and sent to the nearest city, ensuring its correct disposal.

5.2 Offsite Biodiversity Impacts

5.2.1 Negative Offsite Biodiversity Impacts (CCB, B3.1) and Mitigation Measures (CCB, B3.2)

This project aims to conserve forest cover through the implementation of measures against illegal deforestation. Considering that the main threats to biodiversity are associated with the loss and degradation of habitats, PP emphasize that the activities of this project will not result in negative impacts on biodiversity, both inside and outside the area covered by the project.

5.2.2 Net Offsite Biodiversity Benefits (VCS, 3.19; CCB, B3.3)

No negative impact on biodiversity is expected, both inside and outside the project area. Considering the efforts to promote forest cover and species conservation, the net benefits for biodiversity will always be positive.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

This section outlines the biodiversity monitoring plan, providing details on the variables to be monitored, sampling sites, methods employed, and the frequency of monitoring.

Biodiversity variables

The biodiversity variables to be monitored include species richness and the number of endemic, threatened and game species within four groups of vertebrates: amphibians, birds, mammals, and reptiles. Species richness will be calculated based on the number of species registered during the monitoring campaigns. Endemic species will be identified based on distribution and occurrence data available in databases, scientific data, and specialized literature.

The level of threat will be assessed at a global level following the IUCN Red List classification. Game species will be mapped considering the habits and practices of local communities that interact with biodiversity, as well as information available in scientific literature on hunting and fishing in the region. Through this monitoring, the project aims to contribute to the maintenance of biological diversity, identifying and mitigating possible threats to priority species and habitats for conservation.

Sampling sites

Monitoring surveys will take place within the project activity instances, preferably along the transects and plots previously installed for the forest carbon inventory.

Sampling methods

Specific methodologies will be applied to monitor each group. The birds sampling will be conducted through the active search methodology in two periods of the day: the first starting from sunrise until 1 pm, and the second starting at 3 pm until dusk. In this methodology, the specialist walks at a constant speed through the sampling points, recording the presence of the identified species (Vielliard and Silva, 1990). All visual and/or auditory sightings will be considered, and binocular and audio recorders might be used to assist in the species identification and enhance accuracy.

The sampling of amphibians and reptiles will be carried out through the active search methodology, at three different times of the day: in the morning, at dusk and at night. This method involves visually searching for individuals in microhabitats, such as: litter, fallen logs, stones, aerial roots, holes in the ground or trees, aquatic environments, termite mounds, among others. (Heyer et al., 1994). All visual and/or auditory sightings will be recorded and, when necessary, the vocalizations of anuran amphibians will be recorded with the aid of a digital recorder for later analysis.

For mammals, the survey will be conducted through active search and camera traps. Active search will be conducted in two periods: from sunrise until 1pm and from early afternoon until dusk. This methodology aimed to record direct (sightings and vocalizations) and indirect (footprints, feces, bones, and others) observations of species occurrence in project area. All sightings will be documented in field records and, whenever possible, supported by photographic evidence. The camera traps will be positioned in tree trunks at 30 cm above from the ground and will be in operation 24h a day during the sampling period (Cutler and Swann, 1999 cited by Vogliotti, 2003). This method is effective for species with low population density, as it enables the study of animals' natural behavior without causing interference (Vogliotti, 2003).

Monitoring frequency

To monitor biodiversity variables, two monitoring campaigns will be carried out in each PAI: one during the dry season and another during the rainy season. This seasonal approach aims to better understand the species composition throughout the year. After this initial monitoring, campaigns will be repeated every five years, providing an ongoing, long-term assessment of changes in biodiversity and the effectiveness of conservation measures implemented.

5.3.2 Biodiversity Monitoring Plan Dissemination (CCB, B4.3)

Information about the project's biodiversity monitoring plan will be publicly available through VERRA website, where it can be accessed through this document as well as the monitoring reports. Moreover, highlights of biodiversity conservation activities will be shared on BRC's social media platforms. A summary of the primary monitoring findings will be compiled into simplified reports, which will be distributed to the property owners participating in the project. Lastly, to enhance transparency and foster community involvement in project activities, PP plan to regularly disseminate monitoring results through public meetings and biodiversity workshops throughout the project's duration.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

5.4.1 High Biodiversity Conservation Priority Status (CCB, GL3.1)

Not applicable. The project does not seek to be validated to the Gold Level for exceptional biodiversity benefits.

5.4.2 Trigger Species Population Trends (CCB, GL3.2, GL3.3)

Not applicable. The project does not seek to be validated to the Gold Level for exceptional biodiversity benefits.

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